

PHENOM™
BY EMBRAER



PHENOM 300

EASA

AIRPLANE FLIGHT MANUAL

EMBRAER S.A.

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NOTE: PHENOM 300 IS A COMMERCIAL DESIGNATION USED IN THIS MANUAL TO REFER TO THE EMB-505 AIRPLANES.

ANAC APPROVAL: _____

ADEMIR ANTÔNIO DA SILVA
GERENTE-GERAL DE CERTIFICAÇÃO DE
PRODUTO AERONÁUTICO

DATE: _____

29 April 2010

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REVISION APPROVAL

AFM-2666 – EMB-505 – EASA

REVISION: 27

DATE: NOVEMBER 17, 2025

Approval n° 1105/2025/GTPR/GCPP/SAR-ANAC:

AFFECTED INFORMATION	DESCRIPTION
5-10 – Code 01	THRUST SETTING TABLES Update Thrust Setting Tables values.
5-15 – Code 01	SIMPLIFIED TAKEOFF ANALYSIS Update Simplified Takeoff Analysis Tables values.
5-15 – Code 02	
5-15 – Code 03	
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5-16 – Code 01	TAKEOFF 2ND SEGMENT NET GRADIENT (%) Update Takeoff 2 ND Segment Net Gradient (%) Tables values.
5-16 – Code 02	
5-16 – Code 03	
5-16 – Code 04	
5-18 – Code 02	DRIFTDOWN TABLES (NET GRADIENT) Update Driftdown Tables (Net Gradient) Tables values.
5-20 – Code 01	APPROACH CLIMB GRADIENT Update OEI – Approach Climb Gradient Tables values.
5-20 – Code 02	
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AFFECTED INFORMATION	DESCRIPTION
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5-25 – Code 02	
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5-25 – Code 01	FACTORED LANDING DISTANCE Update Factored Landing Distance Tables values.
5-25 – Code 02	
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5-25 – Code 01	CORRECTED LANDING DISTANCE – WET RUNWAY AND DRY OVERSPEED RUNWAY Update Corrected Landing Distance Tables values.
5-25 – Code 02	
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5-25 – Code 01	CORRECTED LANDING DISTANCE – CONTAMINATED RUNWAYS Update Corrected Landing Distance Tables values.
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INTRODUCTION

This Airplane Flight Manual (AFM) meets certification requirements set by CS-23 and provides the approved information necessary to safely operate the PHENOM 300.

For additional information, please contact:

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APPLICABILITY

The AFM cover presents the manual identification according to its applicability as follows:

- AFM 2666-01: applicable to EMB-505 equipped with PW535E engines and using the Metric Units.
- AFM 2666-02: applicable to EMB-505 equipped with PW535E engines and using the Imperial Units.
- AFM 2666-03: applicable to EMB-505 equipped with PW535E1 engines and using the Metric Units.
- AFM 2666-04: applicable to EMB-505 equipped with PW535E1 engines and using the Imperial Units.



REVISIONS

Embraer may periodically revise this manual as required to update information or provide information not available at the time of printing. Revised data may result from Embraer approved airplane modifications or from improved techniques gained through operational experience. Changes to the text are indicated by a vertical line on the outside margin of the page.

A vertical line adjacent to the page number will indicate relocated or rearranged text or illustrations.

A Revision Approval, at the beginning of this manual, shows the approval date and the authority signature, as well as a brief description of the last revision modifications.

The basic issue date of this Manual is presented on the title page. The revisions to the basic Manual will be numbered sequentially (Rev. 1, 2, 3 and so forth). This also applies to pages revised by an earlier revision.

TEMPORARY REVISIONS

Temporary Revisions may be issued when the need arises. The Temporary Revision will be clearly identified and will be replaced in the next Revision. Temporary Revisions have the same force as a Revision. Temporary Revisions will not be included in the List of Effective Pages. The last pages of the Revision Approval, at the beginning of this manual, provide a brief description of the Temporary Revision modifications.

LIST OF EFFECTIVE PAGES

A List of Effective Pages for all the Sections is located at the beginning of this Manual. This list presents the date of issue for each page and it is revised with the Manual. Its purpose is to verify that the Manual is current.

ORGANIZATION

This Airplane Flight Manual is divided as follows:

- Section 1 - General
- Section 2 - Limitations
- Section 3 - Normal Procedures
- Section 4 - Emergency and Abnormal Procedures
- Section 5 - Performance
- Section 6 - Weight and Balance
- Supplements
- Appendices

The Sections and their use are explained below.

SECTION 1 – GENERAL

This Section contains general information pertaining to the Manual, such as the revision rules, its organization and a definition of terms.

SECTION 2 – LIMITATIONS

This Section contains airworthiness certification limitations.

The limitations restrict airplane operation in accordance with the airworthiness certificate.

SECTION 3 – NORMAL PROCEDURES

This Section contains the required normal procedures.

SECTION 4 – EMERGENCY AND ABNORMAL PROCEDURES

This Section contains the actions to be performed in the event of failures.

SECTION 5 – PERFORMANCE

This Section contains performance data and general information and configurations.



SECTION 6 – WEIGHT AND BALANCE

This Section provides instructions referring to the weighing and loading of the PHENOM 300 airplane.

SUPPLEMENTS

The Supplements provide the necessary additional information for the airplane operation, when equipped with optional systems and equipment not provided with the standard airplane, or when specific operational information is applicable to the airplane (e.g. ferry flights, special operation etc). Supplements may modify or complete the limitations, procedures or performance data of the basic AFM.

APPENDICES

Appendices may be incorporated in the AFM to provide additional information for the operation of the airplane, in a format that may not be the same as the basic AFM.

GUIDANCE INFORMATION

Some portions of the AFM are provided as guidance information and were not applied for a certification process, being presented in the AFM for the operator convenience. Such portions are presented in segregated Sections, Supplements or Appendices, and their pages do not bear the notation "ANAC APPROVED".



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DEFINITION OF TERMS

The following definitions apply to the terms below:

WARNING: OPERATING PROCEDURES, TECHNIQUES AND OTHER RELATED INFORMATION WHICH MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE, IF NOT FOLLOWED.

CAUTION: OPERATING PROCEDURES, TECHNIQUES AND OTHER RELATED INFORMATION WHICH MAY RESULT IN DAMAGE OR DESTRUCTION OF EQUIPMENT, IF NOT FOLLOWED.

NOTE: Operating procedures, techniques and other related information which are considered essential to emphasize.

LAND AS SOON AS POSSIBLE

An emergency will be declared. A landing should be accomplished at the nearest suitable airfield considering the severity of the emergency, weather conditions, field facilities, ambient lighting, airplane gross weight and ATC guidance. Depending on the situation criticality and considering the occupants safety, a landing off the runway may have to be attempted.

LAND AS SOON AS PRACTICAL

The emergency conditions are less urgent, and although the flight is to be terminated, the degree of the emergency is such that an immediate landing at the nearest adequate airfield may not be necessary.

NO ICING

Atmospheric conditions are such that no ice formation is predicted or there is no perceptible ice formation on airplane surfaces.

IN ICING/WITH ICE

Atmospheric conditions may lead to ice formation or there is ice accreted on airplane surfaces.



General

CLUB SEAT

The Club Seat consists of four face-to-face seats on passenger cabin.



ABBREVIATIONS AND ACRONYMS USED

ABBREVIATIONS OR ACRONYMS	MEANING
°C	Degrees Celsius
°F	Degrees Fahrenheit
ft	Feet
hPa	Hectopascal
kg	Kilogram
kt	Knot
ℓ	Liter
lb	Pound
m	Meter
mph	Miles per Hour
psi	Pound per Square Inch
sec	Second
VA	Volt-Ampere
AC	Advisory Circular
ACJ	Advisory Circular Joint
ADC	Air Data Computer
ADF	Automatic Direction Finder
ADS	Air Data System
ADS-B	Automatic Dependent Surveillance Broadcast
ADS-C	Automatic Dependent Surveillance Contract
AEO	All Engines Operating
AFCS	Automatic Flight Control System
AFM	Airplane Flight Manual
AGL	Above Ground Level



ABBREVIATIONS OR ACRONYMS	MEANING
AHRS	Attitude and Heading Reference System
A-I	Anti-Ice
ALT	Altitude
AMS	Air Management System
ANAC	Agência Nacional de Aviação Civil
AOA	Angle of Attack
AP or A/P	Autopilot
ARINC	Aeronautical Radio, Incorporated
ARM	Armed
ASTM	American Society for Testing and Materials
ATC	Air Traffic Control
ATN	Aeronautical Telecommunication Network
ATN B1	ATN Baseline 1
ATR	Automatic Thrust Reserve
AUX	Auxiliary
BA	Balance Arms
BC	Back Course
BRNAV	Basic Area Navigation Standards
BS	Body Station
CAS	Crew Alerting System
CB	Circuit Breaker
CDI	Course Deviation Indicator
CG	Center of Gravity
COMM	Communication
CON	Continuous
CPDLC	Controller-to-Pilot Data Link Communication



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ABBREVIATIONS OR ACRONYMS	MEANING
CRS	Course
CSC	Current Speed Control
DC	Direct Current/Digital Controller
DISAG	Disagree
DLMS	Data Link Management System
DME	Distance Measurement Equipment
DR	Dead Reckoning
EASA	European Aviation Safety Agency
EFCU	Electronic Fuel Control Unit
EIS	Engine Indication System
ELT	Emergency Locator Transmitter
ENG	Engine
FAA	Federal Aviation Administration
FADEC	Full Authority Digital Engine Control
FAF	Final Approach Fix
FANS	Future Air Navigation System
FAR	Federal Aviation Regulations
FCE	Flight Control Electronic
FCSOV	Flow Control Shutoff Valve
FIREX	Fire Extinguisher
FIS-B	Flight Information Service Broadcast
FL	Flight Level
FMA	Flight Mode Annunciation
FMS	Flight Management System
FPA	Flight Path Angle
FWD	Forward
GA	Go-Around
GFC	Garmin Flight Control
GIA	Garmin Integrated Avionics



ABBREVIATIONS OR ACRONYMS	MEANING
GND	Ground
GNSS	Global Navigation Satellite System
GP	Glide Path
GPS	Global Positioning System
GPU	Ground Power Unit
GS	Ground Speed
HDG	Heading
HSI	Horizontal Situation Indicator
HTR	Heater
ICAO	International Civil Aviation Organization
IESI	Integrated Electronic Standby Instrument
IFR	Instrument Flight Rules
ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
ISA	International Standard Atmosphere
ITT	Interturbine Temperature
KOEL	Kinds of Operation Equipment List
LDA	Localizer Type Directional Aid
LDG	Landing
LEMAC	Leading Edge Mean Aerodynamic Chord
LFE	Landing Field Elevation
LH	Left Hand
LNAV	Lateral Navigation
LOC	Localizer
LP	Localizer Performance without Vertical Guidance
LPV	Localizer Performance with Vertical Guidance
LRN	Long Range Navigation



ABBREVIATIONS OR ACRONYMS	MEANING
MAC	Mean Aerodynamic Chord
MAN	Manual
MAX	Maximum
MEA	Minimum Enroute Altitude
MEL	Minimum Equipment List
MFD	Multifunction Display
MIL	Military
MIN	Minimum
MLG	Main Landing Gear
MLS	Microwave Landing System
M _{MO}	Maximum Operating Mach
MNPS	Minimum Navigational Performance Specifications
M _{RA}	Maximum Recommended Turbulent Air Penetration Speed
N1	Fan Speed
N2	High Pressure Compressor Speed
NAS	National Airspace System
NAT	North Atlantic
NAV	Navigation
NAVCOM	Navigation and Communication
NDB	Non Directional (Radio) Beacon
NEXRAD	Next-Generation Radar
NLG	Nose Landing Gear
NOTAM	Notice to Airmen
NPRV	Negative Pressure Relief Valve
OAT	Outside Air Temperature

ABBREVIATIONS OR ACRONYMS	MEANING
OEI	One Engine Inoperative
OPERA	Optimized Performance Analyzer
OVHT	Overheat
PAX	Passenger
PBE	Protective Breathing Equipment
PERF	Performance Management
PFD	Primary Flight Display
PREDCT	Predictive
PRNAV	Precision Area Navigation
PRSOV	Pressure Regulator Shutoff Valve
PRV	Pressure Relief Valve
PWS	Predictive Windshear
QAV	Aviation Kerosene
RAIM	Receiver Autonomous Integrity Monitoring
RBHA	Regulamento Brasileiro de Homologação Aeronáutica
RCP	Required Communication Performance
RH	Right Hand
RNAV	Area Navigation System
RNP	Required Navigation Performance
RNP AR	Required Navigation Performance Authorization Required
ROAAS	Runway Overrun Awareness and Alerting System
ROL	Roll Hold
RPM	Revolutions Per Minute
RTCA	Radio Technical Commission for Aeronautics



ABBREVIATIONS OR ACRONYMS	MEANING
RVSM	Reduced Vertical Separation Minimum
SAT	Static Air Temperature
SBAS	Satellite Based Augmentation System
SDF	Simplified Directional Facility
STAB	Stabilized
STBY	Standby
SVS	Synthetic Vision System
SWPS	Stall Warning and Protection System
TA	Traffic Advisory
TAT	Total Air Temperature
TAWS	Terrain Awareness and Warning System
TFR	Temporary Flight Restriction
TGL	Temporary Guidance Leaflet
TIS	Traffic Information System
TO or T/O	Takeoff
TO/GA	Takeoff/Go-Around
TOLD	Takeoff and Landing Data
TSO	Technical Standard Order
V	Volt/Vertical
V ₁	Decision Speed
V ₂	Takeoff Safety Speed
VA	Operating Maneuvering Speed
V _{AC}	Approach Climb Speed
V _{AP}	Approach Speed
VDI	Vertical Deviation Indicator
VDL M2	VHF Digital Link Mode 2



ABBREVIATIONS OR ACRONYMS	MEANING
V _{FE}	Maximum Flap Extended Speed
VFR	Visual Flight Rules
V _{FS}	Final Segment Speed
VHF	Very High Frequency
V _{LE}	Maximum Landing Gear Extended Speed
V _{LO}	Maximum Landing Gear Operating Speed
V _{LOF}	Lift-Off Speed
V _{MC}	Minimum Control Speed
V _{MO}	Maximum Operating Speed
VNAV	Vertical Navigation
V _O	Operating Maneuvering Speed
VOR	VHF Omnidirectional Range
V _R	Rotation Speed
V _{REF}	Landing Reference Speed
V _S	Stall Speed
VSI	Vertical Speed Indicator
W&B	Weight and Balance
WAAS	Wide Area Augmentation System
WAI	Wing Anti-Ice (N1 Ice Bug)
WOW	Weight on Wheels
WNSHR	Windshear
WSHEAR	Windshear
WSHR	Windshear
WX	Weather
XBLEED	Crossbleed
XFEED	Crossfeed
YD	Yaw Damper



AIRPLANE CONFIGURATION

The airplane configuration may be identified by:

- Documentation delivered with the airplane:
 - Flight Operations Manuals Configuration.
 - Service Bulletin Report.
 - Design Change Report.
- Incorporated Service Bulletins.

SERVICE BULLETIN TABLE

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

The following table shows the Service Bulletins which affect the airplane operation and have been incorporated into AFM.

- NOTE:** - Check (✓) the accomplished Service Bulletins or equivalent modification factory-incorporated.
- An equivalent modification factory-incorporated may be identified in the documentation delivered with the airplane (Flight Operations Manuals Configuration, Service Bulletin Report, and Design Change Report).
 - When a new software version is incorporated into the airplane, either during its production or by a Service Bulletin, subsequent software versions are equivalent and have the functionalities of the previous one, except when informed otherwise.



General

S/N: 50500203

Registration Number: CSPHA

SB	SUBJECT	✓
SB 505-00-0001	Removal of icing limitation above 30000 ft.	✓
SB 505-00-0008	Operational weight increase.	✓
SB 505-24-0014	Replacement of the GCU.	✓
SB 505-24-0015	Replacement of the main battery 2.	✓
SB 505-24-0017	Installation of the new starter generator.	✓
SB 505-25-0004	Installation of the belted toilet.	✓
SB 505-27-0011	Activation of the flap FULL position.	✓
SB 505-27-0018	Installation of the steep approach.	✓
SB 505-29-0005	Replacement of the hydraulic accumulator.	✓
SB 505-31-0001	Installation of G1000 load 43 (0734.27) avionics software and CMC load 9 software.	
SB 505-31-0004	Installation of G1000 load 54 (0734.54) avionics software and CMC load 10 software.	
SB 505-31-0005	Installation of G1000 load 63.7 (0734.6D) avionics software and CMC load 11 software.	
SB 505-31-0017	Installation of G3000 load 13.25 (1633.13) avionics software.	✓
SB 505-31-0018	Installation of G1000 load 83.25 (0734.8C) avionics software.	
SB 505-33-0002	Replacement of the Lights Control Panel.	✓
SB 505-34-0010	Installation of NXT-600 Mode S/ADS-B transponder system (G1000).	
SB 505-34-0011	Installation of NXT-600 Mode S/ADS-B transponder system (G3000).	✓
SB 505-73-0001	New FADEC (Full Authority Digital Engine Control) software upgrade – version 4.3.	✓

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SB	SUBJECT	✓
SB 505-00-0030	Increase of the MZFW (Maximum Zero Fuel Weight).	✓

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GARMIN G1000 GLOBAL NAVIGATION SATELLITE SYSTEM (GPS/SBAS)

SYSTEM APPROVALS

The following table summarizes Garmin G1000 GNSS system approvals in the airplane, subject to the limitations described in section GARMIN GNSS (GPS/SBAS) NAVIGATION SYSTEM LIMITATIONS. Further details are also provided in the following paragraphs. These do not constitute operational approvals.

SYSTEM APPROVALS	GUIDANCE
GNSS Equipment	FAA AC 20-138A
RNP Operations in the U.S. NAS	FAA AC 90-105
U.S. RNAV 2 and RNAV 1	FAA AC 90-100A
RNP-10 (RNAV 10)	FAA Order 8400.12B
RNP-4	FAA Order 8400.33 FAA AC 20-138A
PRNAV/RNAV 1	TGL-10 Rev 1 FAA AC 90-96A
BRNAV/RNAV 5	EASA AMC 20-4 FAA AC 90-96A
Airplanes Pre-Mod. SB 505-31-0018: RNP APCH (Without Baro-VNAV Approach)	EASA AMC 20-27 FAA AC 90-105
Airplanes Post-Mod. SB 505-31-0018: RNP APCH (Including Baro-VNAV Approach)	EASA AMC 20-27 FAA AC 90-105 FAA AC 20-138D
LPV Approaches	EASA AMC 20-28 FAA AC 90-107
Baro-VNAV (Enroute and Terminal)	FAA AC 20-129

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GNSS EQUIPMENT (AC 20-138A)

- The Garmin GNSS navigation system installed in the airplane is a GPS system with a Satellite Based Augmentation System (SBAS) approved under TSO-C145a Class 3, TSO-C146a Class 3, and installed in accordance with AC 20-138A.
- The Garmin GNSS navigation system as installed in the airplane complies with the requirements of AC 20-138A and is approved for navigation using GPS and SBAS (within the coverage of a Satellite Based Augmentation System complying with ICAO Annex 10) for IFR enroute, terminal area, and non-precision approach operations (including those approaches titled "GPS", "or GPS", and "RNAV (GPS)" approaches). The Garmin GNSS navigation system installed in the airplane is approved for approach procedures with vertical guidance including "LPV" and "LNAV/VNAV" minimums, within the U.S. National Airspace System.

RNP OPERATIONS IN U.S. NAS (AC 90-105)

- The Garmin GNSS navigation system as installed in the airplane complies with the equipment requirements of AC 90-105 and meets the equipment performance and functional requirements to conduct RNP terminal departure and arrival procedures and RNP approach procedures without RF (radius to fix) legs. Part 91 subpart K and 135 operators require operational approval from the FAA.

RNAV 2 AND RNAV 1 (AC 90-100A)

- The Garmin GNSS navigation system as installed in the airplane complies with the equipment requirements of AC 90-100A for RNAV 2 and RNAV 1 operations. In accordance with AC 90-100A, Part 91 operators (except subpart K) following the airplane and training guidance in AC 90-100A are authorized to fly RNAV 2 and RNAV 1 procedures. Part 91 subpart K and 135 operators require operational approval from the FAA.

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RNP-10 (RNAV 10) (ORDER 8400.12B)

- The Garmin GNSS navigation system has been found to comply with the requirements for GPS Class II oceanic and remote navigation (RNP-10) without time limitations in accordance with AC 20-138A and FAA Order 8400.12B. The Garmin GNSS navigation system can be used without reliance on other long-range navigation systems. This does not constitute an operational approval.

RNP-4 (ORDER 8400.33)

- The Garmin GNSS navigation system as installed in the airplane complies with the navigation requirements for primary means of Class II navigation in oceanic and remote navigation (RNP-4) in accordance with AC 20-138A and FAA Order 8400.33. The Garmin GNSS navigation system can be used without reliance on other long-range navigation systems. Additional equipment may be required to obtain operational approval to utilize RNP-4 performance. This does not constitute an operational approval.

PRNAV (TGL-10/AC 90-96A) AND BRNAV (AMC 20-4/AC 90-96A)

- The Garmin GNSS navigation system as installed in the airplane complies with the accuracy, integrity, and continuity of function, and contains the minimum system functions required for PRNAV operations in accordance with TGL-10 Rev 1.
- The GNSS navigation system has two ETSO-145/TSO-C145a Class 3 approved, and ETSO-146/TSO-C146a Class 3. The Garmin GNSS navigation system as installed in the airplane complies with the equipment requirements for PRNAV (RNAV 1) and BRNAV (RNAV 5) operations in accordance with AC 90-96A, TGL-10 Rev 1 and AMC 20-4. This does not constitute an operational approval.

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RNP APCH WITHOUT BARO-VNAV (AMC 20-27/AC 90-105)

- For airplanes Pre-Mod. SB 505-31-0018, the Garmin GNSS navigation system as installed in the airplane complies with the system requirements for RNP Approaches (RNP APCH) without vertical guidance (LNAV only) as per AMC 20-27 (Airworthiness Approval and Operational Criteria for RNP Approach (RNP APCH) Operations Including APV BARO-VNAV Operations) and AC 90-105 (Approval Guidance for RNP Operations and Barometric Vertical Navigation in the U.S. National Airspace System). For airplanes Pre-Mod. SB 505-31-0018, the G1000 does not support baro-VNAV approaches and LNAV/VNAV approaches are SBAS based.

RNP APCH INCLUDING BARO-VNAV (AMC 20-27/AC 90-105/AC 20-138D)

- For airplanes Post-Mod. SB 505-31-0018 or with an equivalent modification factory incorporated, the Garmin GNSS navigation system as installed in the airplane complies with the system requirements for RNP Approaches (RNP APCH) as per AMC 20-27 (Airworthiness Approval and Operational Criteria for RNP Approach (RNP APCH) Operations Including APV BARO-VNAV Operations), AC 20-138D (Airworthiness Approval of Positioning and Navigation Systems), and AC 90-105 (Approval Guidance for RNP Operations and Barometric Vertical Navigation in the U.S. National Airspace System). Temperature compensation on the final approach segment is provided, as per RTCA DO-236B. Activation of approach waypoints temperature compensation must be coordinated with ATC.

LPV APPROACHES (AMC 20-28/AC 90-107)

- The Garmin GNSS navigation system as installed in the airplane complies with the system requirements for LPV Approaches as per AMC 20-28 (Airworthiness Approval and Operational Criteria for RNAV GNSS approach operation to LPV minima using SBAS) and AC 90-107 (Guidance for Localizer Performance with Vertical Guidance and Localizer Performance without Vertical Guidance Approach Operations in the U.S. National Airspace System).

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ENROUTE/TERMINAL BARO-VNAV (AC 20-129)

- G1000 Barometric VNAV function is approved for enroute and terminal descents, as per AC 20-129. Descent guidance is provided up to the FAF waypoint when there is not a procedure that provides vertical guidance following the FAF. Guidance is provided up to the waypoint preceding the FAF (FAF-1) when there is a procedure that provides vertical guidance (ILS or SBAS based) following the FAF.

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GARMIN G3000 GLOBAL NAVIGATION SATELLITE SYSTEM (GPS/SBAS)

SYSTEM APPROVALS

The Garmin G3000 GNSS (GPS/SBAS) has been demonstrated capable of, and it has been shown to meet the accuracy requirements for the following operations. These do not constitute operational approvals.

SYSTEM APPROVALS	GUIDANCE
GNSS Equipment	FAA AC 20-138C
Airplanes with G3000 system version 3305.00 and on: GNSS Equipment	FAA AC 20-138D
RNP Operations in the U.S. NAS	FAA AC 90-105
U.S. RNAV 2 and RNAV 1	FAA AC 90-100A
RNP-10 (RNAV 10)	FAA Order 8400.12C
RNP-4	FAA Order 8400.33 FAA AC 20-138C
P-RNAV (RNAV 1)	TGL-10 Rev 1 FAA AC 90-96A
B-RNAV (RNAV 5)	EASA AMC 20-4 FAA AC 90-96A
RNP APCH (including Baro-VNAV Approach)	EASA AMC 20-27 FAA AC 20-138C FAA AC 90-105
LPV Approaches	EASA AMC 20-28 FAA AC 90-107
Airplanes Pre-Mod. SB 505-31-0017: Baro-VNAV (En-route and Terminal)	FAA AC 20-138C FAA AC 90-105
Airplanes Post-Mod. SB 505-31-0017: Baro-VNAV (Climb, Cruise and Descent)	FAA AC 20-138D FAA AC 90-105

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GNSS EQUIPMENT

- The Garmin GNSS navigation system installed is a GPS system with a Satellite Based Augmentation System (SBAS) approved under TSO-C145c Class 3, TSO-C146c Class 3, and installed in accordance with AC 20-138C. It operates with the following Satellite Based Augmentation Systems (SBAS): WAAS, EGNOS and MSAS.
- For airplanes with G3000 system version 3305.00 and on, the Garmin GNSS navigation system installed is a GPS system with a Satellite Based Augmentation System (SBAS), approved under TSO-C145d Class 3, TSO-C146d Class 3, and installed in accordance with AC 20-138D. It operates with the following Satellite Based Augmentation Systems (SBAS): WAAS, EGNOS, MSAS and GAGAN. The capability to execute a direct-to an RF Leg initial fix or to intercept the middle of an RF leg was not demonstrated. Therefore, those procedures are not allowed.
- The Garmin GNSS navigation system as installed in the airplane is approved for navigation using GPS and SBAS (within the coverage of a Satellite Based Augmentation System complying with ICAO Annex 10) for IFR en-route, terminal area, and non-precision approach operations (including those approaches titled "GPS", "or GPS", and "RNAV (GPS)" approaches). The Garmin GNSS navigation system installed in the airplane is approved for approach procedures with vertical guidance including "LPV" and "LNAV/VNAV minimums", within the U.S. National Airspace System.

RNP OPERATIONS IN U.S. NAS (AC 90-105)

- The Garmin GNSS navigation system as installed in the airplane complies with the equipment requirements of AC 90-105 and meets the equipment performance and functional requirements to conduct RNP terminal departure and arrival procedures and RNP approach procedures without RF (radius to fix) legs. Part 91 subpart K and 135 operators require operational approval from the FAA.
- For airplanes with G3000 system version 3305.00 and on, the Garmin GNSS system can also perform RNP operations including RF legs.

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RNAV 2 AND RNAV 1 (AC 90-100A)

- The Garmin GNSS navigation system as installed in the airplane complies with the equipment requirements of AC 90-100A for RNAV 2 and RNAV 1 operations. In accordance with AC 90-100A, Part 91 operators (except subpart K) following the airplane and training guidance in AC 90-100A are authorized to fly RNAV 2 and RNAV 1 procedures. Part 91 subpart K and 135 operators require operational approval from the FAA.

RNP-10 (RNAV-10) (ORDER 8400.12C)

- The Garmin GNSS navigation system has been found to comply with the requirements for GPS Class II oceanic and remote navigation (RNP-10) without time limitations in accordance with AC 20-138A and FAA Order 8400.12C. The Garmin GNSS navigation system can be used without reliance on other long-range navigation systems. This does not constitute an operational approval.

RNP-4 (ORDER 8400.33)

- The Garmin GNSS navigation system as installed in the airplane complies with the navigation requirements for primary means of Class II navigation in oceanic and remote navigation (RNP-4) in accordance with AC 20-138C and FAA Order 8400.33. The Garmin GNSS navigation system can be used without reliance on other long-range navigation systems. Additional equipment may be required to obtain operational approval to utilize RNP-4 performance. This does not constitute an operational approval.

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P-RNAV (TGL-10/AC 90-96A) AND B-RNAV (AMC 20-4/AC 90-96A)

- The Garmin GNSS navigation system as installed in the airplane complies with the accuracy, integrity, and continuity of function, and contains the minimum system functions required for P-RNAV operations in accordance with TGL-10 Rev 1.
- The GNSS navigation system has two ETSO-C145c/TSO-C145c Class 3 approved, and ETSO-C146c/TSO-C146c Class 3. The Garmin GNSS navigation system as installed in the airplane complies with the equipment requirements for P-RNAV (RNAV 1) and B-RNAV (RNAV 5) operations in accordance with AC 90-96A, TGL-10 Rev 1 and AMC 20-4. This does not constitute an operational approval.

RNP APCH (AMC 20-27/AC 20-138C/AC 90-105)

- The Garmin GNSS navigation system as installed in the airplane complies with the system requirements for RNP Approaches (RNP APCH) as per AMC 20-27 (Airworthiness Approval and Operational Criteria for RNP Approach (RNP APCH) Operations Including APV BARO-VNAV Operations), AC 20-138C (Airworthiness Approval of Positioning and Navigation Systems) and AC 90-105 (Approval Guidance for RNP Operations and Barometric Vertical Navigation in the U.S. National Airspace System). Temperature compensation on the final approach segment is provided, as per RTCA DO-236B. Activation of approach waypoints temperature compensation must be coordinated with ATC.
- For airplanes with G3000 system version 3305.00 and on, the Garmin GNSS system can also perform RNP APCH operations, including RF legs.

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LPV APPROACHES (AMC 20-28/AC 90-107)

- The Garmin GNSS navigation system as installed in the airplane complies with the system requirements for LPV Approaches as per AMC 20-28 (Airworthiness Approval and Operational Criteria for RNAV GNSS approach operation to LPV minima using SBAS) and AC 90-107 (Guidance for Localizer Performance with Vertical Guidance and Localizer Performance without Vertical Guidance Approach Operations in the U.S. National Airspace System).

BARO-VNAV (AC 20-138C/AC 90-105)

- For airplanes Pre-Mod. SB 505-31-0017, G3000 Barometric VNAV function is approved for en-route/terminal descents, as per AC 20-138C. En-route/terminal descent guidance is provided up to the FAF waypoint when there is not a procedure that provides vertical guidance following the FAF. Guidance is provided up to the waypoint preceding the FAF (FAF-1) when there is a procedure that provides vertical guidance (ILS, Baro-VNAV or SBAS based) following the FAF. Temperature compensation for approach waypoints is provided, as per RTCA DO-236B. Activation of approach waypoints temperature compensation must be coordinated with ATC.

BARO-VNAV (AC 20-138D/AC 90-105)

- For airplanes Post-Mod. SB 505-31-0017, G3000 Barometric VNAV function is approved for climb, cruise and descent, as per AC 20-138D. Descent guidance is provided up to the FAF waypoint when there is not a procedure that provides vertical guidance following the FAF. Guidance is provided up to the waypoint preceding the FAF (FAF-1) when there is a procedure that provides vertical guidance (ILS, Baro-VNAV or SBAS based) following the FAF. Temperature compensation for approach waypoints is provided, as per RTCA DO-236B. Activation of approach waypoints temperature compensation must be coordinated with ATC.
- For airplanes with G3000 system version 3305.00 and on, G3000 Barometric VNAV descent guidance is provided up to FAF waypoint, when there is a procedure that provides vertical guidance (ILS, Baro-VNAV or SBAS based) following the FAF.

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ADS-B IN

Applicable for airplanes with G3000 avionics system and ADS-B In installed. Refer to POH Section 6-08-20.

The ADS-B In system has been shown to meet compliance AC-20-172B.

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SECTION 2

LIMITATIONS

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INTRODUCTION

This airplane must be operated in accordance with the limitations presented in this Section. These limitations also apply to operations in accordance with an approved Supplement or Appendix to this AFM, except as altered by such Supplement or Appendix.

The safety and integrity of the airplane and its occupants is highly dependent on the compliance with the operating limitations. Flight crew is the primary responsible for such compliance and so should have all the limitations committed to memory. Some limitations, however, may be too complex to memorize, thus requiring an AFM consultancy. Such limitations are the following:

- Limitations which are automatically complied with by the airplane systems;
- Limitations associated to more than one parameter and that constantly varies in time;
- Tables;
- Charts.



WEIGHT

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

AIRPLANE MODEL	MAX Ramp Weight (MRW) (lb)	MAX Takeoff Weight (MTOW) (lb)	MAX Landing Weight (MLW) (lb)	MAX Zero Fuel Weight (MZFW) (lb)
PHENOM 300 (1)	18497	18387	17042	14220 (3)
PHENOM 300 (2)	18078	17968	16865	13999 (3)

NOTE: 1) Applicable to airplanes Post-Mod. SB 505-00-0008 or with an equivalent modification factory incorporated.

NOTE: 2) Applicable to airplanes Pre-Mod. SB 505-00-0008.

NOTE: 3) For airplanes Post-Mod. SB 505-00-0030 or an equivalent modification factory incorporated, the MZFW is 14440 lb.

To comply with the performance and operating limitations of the regulations, the maximum allowable takeoff and landing operational weights may be equal to, but not greater than design limits.

The takeoff weight (weight at brake release or at start of takeoff run) is the lowest of the MTOW or the following weights:

- Maximum takeoff weight as calculated using the Simplified Takeoff Analysis Tables or using the OPERA software and as limited by runway length, altitude, temperature, tire speed, brake energy and climb requirements.
- Maximum takeoff weight, as limited by enroute, and landing operating requirements.

The landing weight is the lowest of the MLW or the following weight:

- Maximum approach and landing weight as calculated using the Landing Tables or using the OPERA software and as limited by runway length, altitude, temperature, tire speed, brake energy, approach climb, and landing climb requirements.



LOADING

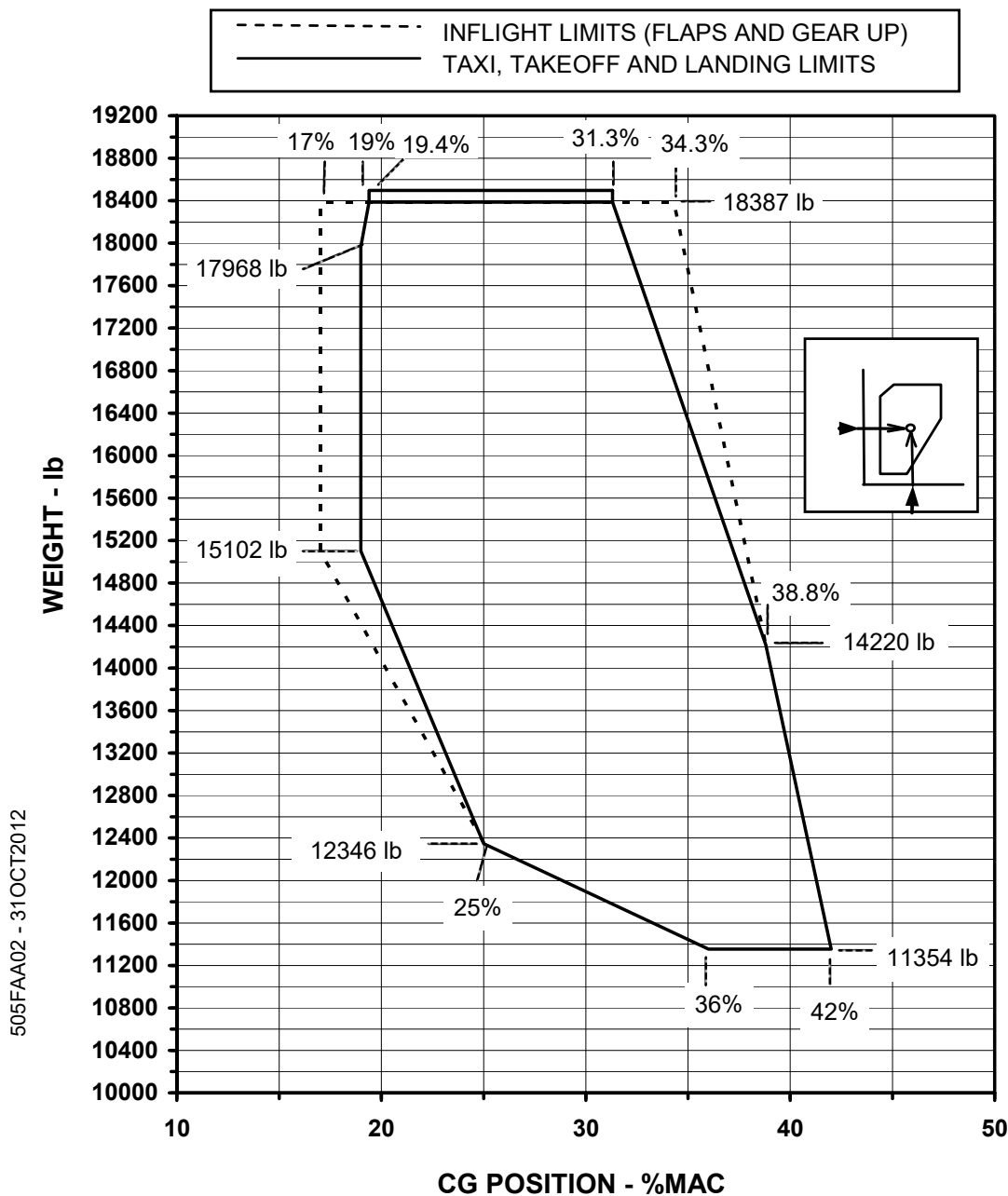
The airplane must be loaded in accordance with the information contained in the Section 6, Weight and Balance.



CENTER OF GRAVITY ENVELOPE

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

PHENOM 300



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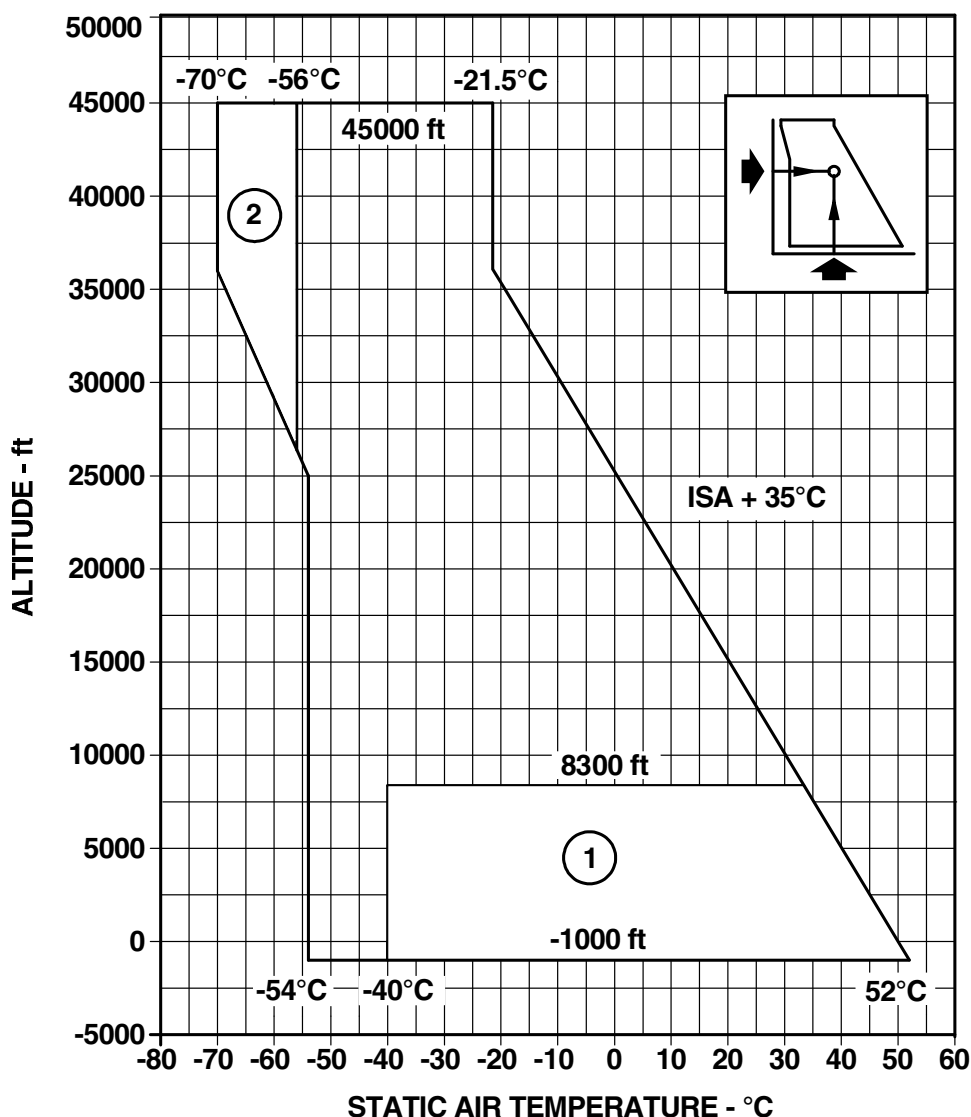
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OPERATIONAL LIMITATIONS

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

OPERATIONAL ENVELOPE



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NOTE: - In the event of a landing below -40°C , report to the maintenance personnel.

- Operations in Region 2 are limited to a minimum airspeed according to the following table.



MINIMUM OPERATION SPEED

The airplane is allowed to operate in cold environments (Region 2) respecting a Minimum Operation Speed for each Static Air Temperature range.

STATIC AIR TEMPERATURE	MINIMUM OPERATION SPEED (MACH)
From -56°C down to -60°C	0.40
From -61°C down to -65°C	0.55
From -66°C down to -70°C	0.67

CAUTION: IN CASE A DRIFTDOWN IS REQUIRED, MAINTAIN MINIMUM OPERATION SPEED UNTIL AIRPLANE IS OUTSIDE REGION 2 AND THEN PERFORM DRIFTDOWN.

CAUTION: IN CASE OF AN EMERGENCY OR ABNORMAL PROCEDURE THAT REQUIRES A REDUCTION OF THE SPEED BELOW THE MINIMUM OPERATION SPEED REQUIRED WHEN FLYING INSIDE REGION 2, USE THE SPEED INDICATED BY THE ASSOCIATED PROCEDURE AND DESCENT UNTIL AIRPLANE EXITS REGION 2.



AIRSPEEDS

LANDING GEAR OPERATION/EXTENDED SPEED (V_{LO} AND V_{LE})

V_{LO} 250 KIAS

V_{LO} is the maximum speed at which the landing gear can be safely extended and retracted.

V_{LE} 250 KIAS

V_{LE} is the maximum speed at which the airplane can be safely flown with the landing gear extended and locked.

MINIMUM CONTROL SPEEDS (V_{MC})

For takeoff:

- V_{MC} 97 KIAS

NOTE: The V_{MC} above represents the highest value to be found within the takeoff envelope. Specific V_{MC} may be obtained through the OPERA as a function of altitude, temperature, weight and according to the takeoff flaps.

For landing:

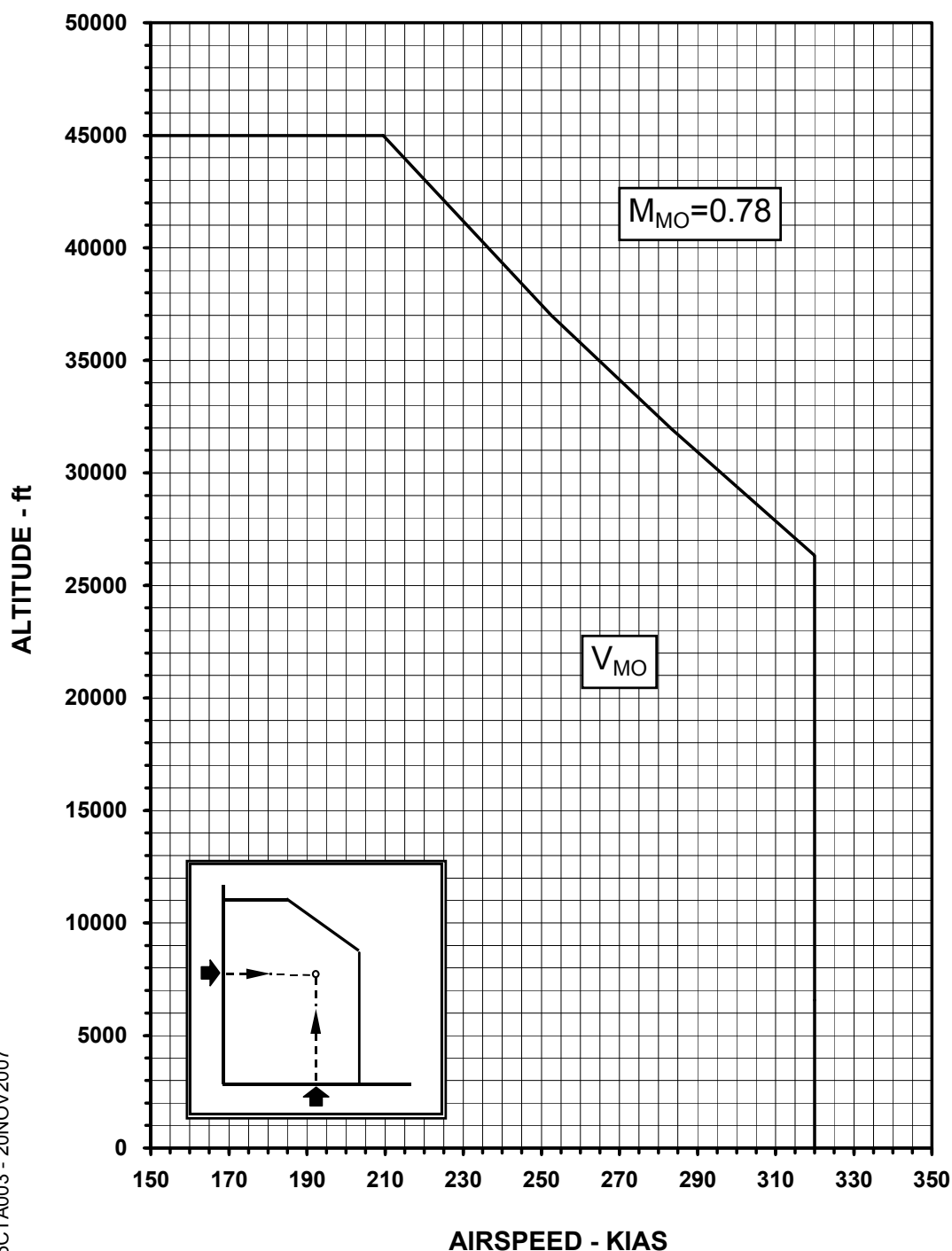
- V_{MC} (no icing conditions) 91 KIAS

- V_{MC} (icing conditions) 95 KIAS

NOTE: V_{MC} is the airspeed at which, when the critical engine is suddenly made inoperative, it is possible to maintain control of the airplane with that engine still inoperative, and thereafter maintain level flight at the same speed with an angle of bank of not more than 5 degrees.



MAXIMUM OPERATING SPEED (V_{MO}/M_{MO})



NOTE: V_{MO}/M_{MO} may not be deliberately exceeded in any regime of flight (climb, cruise or descent), unless a higher speed is authorized for flight test or pilot training.



OPERATING MANEUVERING SPEED (V_O)

V_O 205 KIAS

NOTE: - Maneuvers that involve angle of attack near stall or full application of rudder, elevator, and aileron controls should be confined to speeds below V_O . In addition, the maneuvering flight load factor limits, presented in this Section, should not be exceeded.

- Maneuvers are limited to any maneuver incident to normal flying, stalls (except whip stalls) and steep turns in which the angle of bank is not more than 60 degrees.

CAUTION: RAPID AND LARGE ALTERNATING CONTROL INPUTS, ESPECIALLY IN COMBINATION WITH LARGE CHANGES IN PITCH, ROLL, OR YAW (E.G. LARGE SIDESLIP ANGLES) MAY RESULT IN STRUCTURAL FAILURES AT ANY SPEED, EVEN BELOW V_O .



MAXIMUM FLAP EXTENDED SPEED (V_{FE})

Flap 1	180 KIAS
Flap 2	170 KIAS
Flap 3	170 KIAS
Flap FULL	160 KIAS

MAXIMUM ALTITUDE FOR FLAPS EXTENDED

Maximum Altitude for Flaps Extended 18000 ft

MAXIMUM TIRE GROUND SPEED

Maximum Tire Ground Speed..... 182 kt

MANEUVERING

No acrobatic maneuvers, including spins, are authorized.

MANEUVERING FLIGHT LOAD FACTORS

These corresponding accelerations limit bank angle during turns and limit pull-up maneuvers.

LOAD FACTOR LIMIT	FLAP 0	FLAP DOWN (1, 2, 3 AND FULL)
Positive	3.00 g	2.00 g

MINIMUM CREW

Minimum Flight Crew 1 PILOT

NOTE: - The pilot must occupy the left cockpit seat.

- Operating rules may require an operative autopilot or a second in command.

KINDS OF OPERATION

This airplane may be flown day and night in the following conditions, when the appropriate equipment and instruments required by airworthiness and operational requirements are approved, installed and in an operable condition as defined in the KINDS OF OPERATION EQUIPMENT LIST:

- Visual Flight Rules (VFR);
- Instrument Flight Rules (IFR);
- Icing conditions (below 30000 ft for airplanes Pre-Mod. SB 505-00-0001; up to 45000 ft for airplanes Post-Mod. SB 505-00-0001 or with an equivalent modification factory incorporated);
- Extended over water *.

* For extended over water operation, operating rules may require additional equipment. Such equipment must be properly approved.

KINDS OF OPERATION EQUIPMENT LIST

The following equipment list identifies the systems and equipment upon which type certification for each kind of operation was predicted. The systems and items of equipment listed must be installed and operable for the particular kind of operation indicated, unless the airplane is approved to be operated in accordance with a current Minimum Equipment List (MEL) approved by the Local Authority.

The following systems and equipment list does not include all specific flight, emergency and radio-navigation equipment required by local operating rules. It also does not include components obviously required for the airplane to be airworthy such as wings, primary flight controls, empennage, engines, control panels etc.



KINDS OF OPERATION EQUIPMENT LIST (KOEL)

OPERATION: DAY VFR

1) INSTALLATIONS

System	Function/Equipment
Environmental/Pressurization	AMS Controller
Environmental/Pressurization	Pressure Relief Valve (PRV)
Environmental/Pressurization	Negative Pressure Relief Valve (NPRV)
Environmental/Pressurization	Outflow Valve
Environmental/Pressurization	Pressurization Control
Environmental/Pressurization	Flow Control Shutoff Valves (FCSOV)
Environmental/Pressurization	Pressure Regulating Shutoff Valves (PRSOV)
Electrical	Starter Generators
Electrical	Batteries
Fire Protection	Portable Fire Extinguishers
Fire Protection	Engine Fire Detection System
Fire Protection	Engine Fire Extinguisher System
Fuel	Fuel Ejector Pumps
Fuel	Fuel Shutoff Valves
Hydraulics	Hydraulic Systems
Landing Gear	Brake Systems
Landing Gear	Steering System
Landing Gear	Landing Gear Emergency Operation System
Lights	Anti-Collision Lights



KINDS OF OPERATION EQUIPMENT LIST (KOEL)

OPERATION: DAY VFR

1) INSTALLATIONS (Continued)

System	Function/Equipment
Flight Instruments/Navigation	Air Data System (ADS)
Flight Instruments/Navigation	Attitude and Heading Reference System (AHRS)
Oxygen	Oxygen System
Miscellaneous	ELT
Miscellaneous	Seat Belts
Miscellaneous	Hand Microphone
Miscellaneous	Water Barrier

KINDS OF OPERATION EQUIPMENT LIST (KOEL)

OPERATION: DAY VFR	
2) INSTRUMENTS/INDICATIONS	
System	Function/Equipment
Environmental/Pressurization	Pressurization Indications (Cabin Altitude, Rate and Delta Pressure, Landing Field Elevation)
Electrical	Batteries Voltage Indications
Flight Controls	Flap Position Indication
Fuel	Fuel Quantity Indications
Hydraulics	Hydraulic Pressure Indications
Landing Gear	Landing Gears Position Indications
Flight Instruments/Navigation	Primary Flight Displays (PFD) (Airspeed Indication, Altitude Indication, Heading Indication, Warning Caution and Advisory Function)
Flight Instruments/Navigation	Integrated Electronic Standby Instrument (IESI) (Airspeed Indication, Altitude Indication, Heading Indication)
Flight Instruments/Navigation	Multi-Function Display (MFD)
Flight Instruments/Navigation	Magnetic Compass
Engine	Engine Indications (Oil Pressure and Temperature, Fuel Flow, ITT, N1, N2)
Warning	Aural Warning System
Warning	Takeoff Warning System
Miscellaneous	Approved Airplane Flight Manual (AFM)
Miscellaneous	Embraer Prodigy Cockpit Reference Guide



KINDS OF OPERATION EQUIPMENT LIST (KOEL)

OPERATION: NIGHT VFR

INSTALLATIONS

System	Function/Equipment
All equipment/indications required for day VFR	
Lights	Instruments Lights
Lights	Position Lights
Lights	Landing/Taxi Lights
Lights	Courtesy Lights
Lights	Flash Light
Flight Instruments/Navigation	Attitude Indication

OPERATION: IFR

INSTALLATIONS AND INDICATIONS

System	Function/Equipment
All equipment/indications required for day VFR	
All equipment/indications required for night VFR (for night flights)	
Ice Protection	Pitot/Static-ADS Heating System
Flight Instruments/Navigation	Slip-Skid Indication
Flight Instruments/Navigation	Clock



KINDS OF OPERATION EQUIPMENT LIST (KOEL)

OPERATION: ICING CONDITIONS	
INSTALLATIONS	
System	Function/Equipment
All equipment/indications required for IFR	
Environmental/Pressurization	Cockpit Fan
Ice Protection	Wing and Horizontal Stabilizer Anti-Icing System
Ice Protection	Engine Anti-Icing System
Ice Protection	Windshield Heating System
Lights	Wing Inspection Light *

* Only required for night operations.



MAXIMUM PASSENGER SEATING

Maximum passenger seating configuration, including the right cockpit seat:

SEATING CONFIGURATION	INTERIOR DESCRIPTION			MAXIMUM PASSENGER NUMBER
	7 TH SEAT	BELTED TOILET SEAT	SIDE-FACING DIVAN	
STANDARD	NO	NO	NO	7
OPTION 1	NO	YES	NO	8
OPTION 2	YES	NO	NO	8
OPTION 3	YES	YES	NO	9
OPTION 4	NO	NO	YES	9
OPTION 5	NO	YES	YES	10

NOTE: - A passenger may occupy the right cockpit seat only in single pilot operations.

- The belted toilet seat is installed through SB 505-25-0004 or is factory incorporated.
- For airplanes with belted toilet, the lavatory rigid door must be latched OPEN for taxi, takeoff and landing when occupied.
- For airplanes without belted toilet, the use of the lavatory is prohibited for taxi, takeoff and landing.
- One extra passenger oxygen mask is provided on the passenger cabin and it is accessible from any of the forward facing seat on the Club Seat.

COCKPIT AND PASSENGER CABIN

- Pilot sunvisors must be kept at the vertical position when in use and must be stowed for taxi, takeoff and landing.
- The cockpit curtain must be latched OPEN during taxi, takeoff and landing.



BAGGAGE LOADING

Maximum loading:

LH Forward Cabinet.....	44 lb
Lavatory Cabinet.....	33 lb
Aft Baggage Compartment	463 lb
Forward Baggage Compartment	110 lb
Refreshment Center	71 lb
Side-Facing Divan Stowage Compartments	36 lb

NOTE: - The maximum intensity of loading in each compartment is the following:

- Aft Baggage Compartment..... 70 lb/ft²
- Forward Baggage Compartment..... 22.1 lb/ft²
- Transportation of flammable material defined according to ICAO 9284-AN/905 Class 2.1 (flammable gases), Class 3 (flammable liquid) and Class 4 (flammable solid) is prohibited in the Forward Baggage Compartment.
- Some airplanes have stowage compartments of LH Forward Cabinet, Lavatory Cabinet, Refreshment Center and Aft Baggage Compartment with higher load capacities. Refer to their respective placards to find this information.
- The Side-Facing Divan Stowage Compartments Net must be latched during all phases of flight.

RUNWAY

Runway Slope	-2% TO +2%
Runway Surface Type.....	PAVED

WIND LIMITATIONS

Maximum Takeoff and Landing Tailwind Component	10 kt
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HYDRAULIC

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

For airplanes Pre-Mod. SB 505-29-0005, the hydraulic system accumulator pre-charge must be checked every 15 days or before next flight, whichever occurs last.

WARNING

STALL WARNING AND PROTECTION

The stall warning and protection system must be tested prior each flight.

TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

TAWS displays terrain and obstructions relative to the altitude of the airplane. The following applies:

- Navigation must not be predicated upon the use of the TAWS.

NOTE: The terrain display is intended to serve as a situational awareness tool only. It may not provide either the accuracy or fidelity, or both, on which to solely base decisions and plan maneuvers to avoid terrain or obstacles.

- To avoid giving unwanted alerts, the TAWS must be inhibited when landing at an airport that is not included in the airport database.
- Pilots are authorized to deviate from their current ATC clearance to the extent necessary to comply with TAWS warnings.
- Terrain database coverage is worldwide. However the Terrain data is not displayed when the airplane latitude is greater than 75°N or 60°S.

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- The MSL altitude indication on upper right-hand corner of terrain awareness page must not be used for navigation, especially for maintaining an assigned ATC altitude.
- ILS approaches with manually overridden frequency or course information are prohibited only for airplanes equipped with TAWS-A and G1000 system version 0734.25.

TRAFFIC COLLISION AND AVOIDANCE SYSTEM (TCAS) II

The following limitations are applicable to the Traffic Alert and Collision Avoidance System (TCAS):

- Deviation from the ATC assigned altitude is authorized only to extent necessary to comply with a TCAS Resolution Advisory (RA).
- Maneuvers must not be based solely on information presented in the traffic display.

**TRAFFIC INFORMATION SYSTEM (TIS)
(IF APPLICABLE)**

TIS is not intended to be used as a collision avoidance system and does not relieve the pilot of the responsibility to "see and avoid" other airplane.

TIS shall not be used for avoidance maneuvers during Instrument Meteorological Conditions (IMC) or when there is no visual contact with the intruder airplane.

NOTE: TIS is available only when the airplane is within the service volume of a TIS-capable terminal radar site.



SATELLITE WEATHER RADIO SYSTEM (XM WEATHER) (IF APPLICABLE)

XM Weather information must not be used for hazardous weather penetration. Weather information is provided only for hazardous weather avoidance.

NEXRAD weather data is intended for long-range planning purposes only. Due to inherent delays and relative age of the data, NEXRAD weather data should not be used for short-range avoidance of hazardous weather.

CONNEXT WEATHER (WORLDWIDE WEATHER) (IF APPLICABLE)

Connext Weather (Worldwide Weather) is intended for long-range planning purposes only. Due to inherent delays and relative age of the data, Worldwide Weather should not be used for short range avoidance hazardous weather.

SYNTHETIC VISION SYSTEM (SVS) (IF APPLICABLE)

SVS is intended to be used as an aid to situational awareness only.

Use of SVS as primary information to base decisions and/or plan maneuvers to navigate, or to avoid terrain, obstacles or traffic is prohibited.

The SVS does not assure IFR operations below published minimums.

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SVS DATABASES

The SVS databases have an area of coverage as detailed below:

- The Terrain Database has an area of coverage from North 75° Latitude to South 60° Latitude in all longitudes.
- The Airport Terrain Database coverage is worldwide. Not all airfields are depicted in the SVS.
- The Obstacle Database is limited to the United States and Europe.

The terrain data has a resolution of 9 arc-seconds, this means that the terrain elevation contours in the database are stored broken down into squares 9 arc-seconds on each side.

SVS PATHWAYS

Pathways provide a three-dimensional perspective view of the selected route of flight and its presentation is intended only to aid the pilot's awareness of the programmed flight path location relative to the airplane's current position.

The pathway is not intended for use as a primary reference in tracking the navigation path, thus, it does not substitute standard course and altitude deviation information provided by the CDI, VSI, and VDI.

The Pathway boxes are ± 100 ft in vertical plane and approximately ± 380 ft horizontally from the center of the box.

The pathway is only displayed for navigation paths that are fully defined by the sensor in use, including GPS and ILS paths.

No vertical Pathway information is presented for climbs or for navigation paths that are not fully defined by the sensor in use, like heading legs, VOR, LOC, BC or ADF segments.



ELECTRICAL

BATTERY VOLTAGE

Minimum Voltage for Engine Start..... 24 V

NOTE: Minimum GPU voltage for battery charging is 27 V.

GENERATOR LOAD

Maximum Generator Load:

On Ground 330 A EACH

In Flight 390 A EACH

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FUEL

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

AIRPLANE MODEL	PHENOM 300
Maximum usable quantity per tank	399.43 US Gal (2676.6 lb)
Unusable quantity per tank	3.75 US Gal (25.13 lb)

NOTE: - Maximum fuel capacity is 806.36 US Gal (5403 lb).

- The maximum permitted imbalance between tanks is 33 US Gal (220 lb).
- Unusable fuel increases to 6.76 US Gal (45.35 lb) in each tank if any of the associated ejector fuel pump is inoperative.
- When EIS fuel quantity is zero, any fuel remaining in the tanks can not be used safely in flight.
- The weights above have been determined using an adopted fuel density of 6.701 lb/US Gal. Different fuel densities may be used provided the volumetric limits are not exceeded.

FUEL SPECIFICATION

Brazilian Specification	CNP08-QAV-1
ASTM Specification	D1655-JET A AND JET A-1
Military Specification	MIL-DTL-83133- JP8
Russian Specification	GOST 10227-86 TS-1

NOTE: - Last issues of Standard Specifications must be used.

- For fuel additives see Aircraft Maintenance Manual.



FUEL TANK TEMPERATURE

Minimum.....-37°C

Maximum (on ground)..... 52°C

NOTE: Fuel tank temperature may be exceeded in flight, limited to 80°C.

CROSSFEED VALVE OPERATION

XFEED Knob must be set to OFF during takeoff and landing.

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POWER PLANT

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

ENGINES

Two Pratt & Whitney Canada PW535E.

OPERATIONAL LIMITS

NOTE: Applicable to airplanes Post-Mod. SB 505-73-0001 or equipped with FADEC software version 4.3 and on. For airplanes Pre-Mod. SB 505-73-0001 refer to Supplement 3.

OPERATING CONDITION		OPERATING LIMITS				
Thrust Setting	Time Limit	Maximum ITT (°C)	N2 (%)	N1 (%)	Oil Pressure (psid) (1)	Oil Temperature (°C) (5)
Maximum	10 minutes (3)	725	101	100	45 to 160	10 to 132.2
Takeoff	5 minutes (2)	700	101	100	45 to 160	10 to 132.2
Maximum Continuous (4)	N/A	680	101	100	45 to 160	10 to 132.2
Maximum Climb (4)	N/A	680	101	100	45 to 160	10 to 132.2
Ground Idle	N/A	N/A	Minimum 55.1 (OEI) Minimum 51.2 (AEO)	-	25 to 160	-40 to 132.2
Flight Idle	N/A	N/A	55.1	-	25 to 160	-
Starting	5 seconds	765	-	-	-	-40 Minimum
Transient	20 seconds	765	103	102	0 to 20	-
	200 seconds	-	-	-	-	140.5 Maximum
	400 seconds	-	-	-	20 to 270	-

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NOTE: 1) For N2 speeds above 60% the oil pressure below 45 psid is undesirable and should be tolerated only for the completion of the flight, preferably, at reduced power setting.

NOTE: 2) Normal take-off ratings are limited to 5 minutes duration.

NOTE: 3) Maximum take-off is intended for emergency situations with one engine inoperative with anti-ice on. Total of 10 minutes is allowed for any combination of maximum take-off thrust and normal take-off thrust.

NOTE: 4) Maximum climb is the maximum rating for climb phase under normal operation. Maximum continuous is automatically selected when an OEI event is detected by the FADEC.

NOTE: 5) For operation in severe cold environments, following engine start, it is permissible to operate the engine up to 70% N2, in order to warm the oil to the minimum temperature for normal operation (above 10°C).

NOTE: 6) In case of any unrecoverable engine surge lasting more than 5 seconds, a boroscope inspection to be performed in accordance to the PW535E Maintenance Manual.

CROSSWIND LIMITATION

Static takeoff power application on ground is limited to crosswind components up to 18 kt.



STARTER LIMITS

MOTORING NUMBER	COOL-DOWN TIME
1	60 seconds
2	60 seconds
3	15 minutes
4	30 minutes

NOTE: After four sequential motorings, cycle may be repeated following a 30 minute cool-down period.

OIL SPECIFICATION

Refer to the Approved Lubrication Oils list of the Pratt & Whitney Canada PW535E Maintenance Manual.



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PNEUMATIC, AIR CONDITIONING AND PRESSURIZATION

AIR CONDITIONING

For air conditioning system operation on ground the GPU must be used or one engine generator must be turned on.

PRESSURIZATION

Maximum Differential Pressure	9.4 psi
Maximum Differential Overpressure	9.6 psi
Maximum Differential Negative Pressure	-0.3 psi
Maximum Differential Pressure for Takeoff and Landing	0.2 psi

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ICE AND RAIN PROTECTION

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

Crew must activate the ice protection system when icing conditions exist or are anticipated, as per Section 3, Normal Procedures.

OPERATION IN ICING CONDITIONS

Minimum Airspeed:

- Wing and Stabilizer Anti-Ice System
Inhibited (flap and gear up) 165 KIAS
- Wing and Stabilizer Anti-Ice System
Armed (flap and gear up) 165 KIAS
- Wing and Stabilizer Anti-Ice System
Uninhibited (flap and gear up)..... 150 KIAS

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Limitations

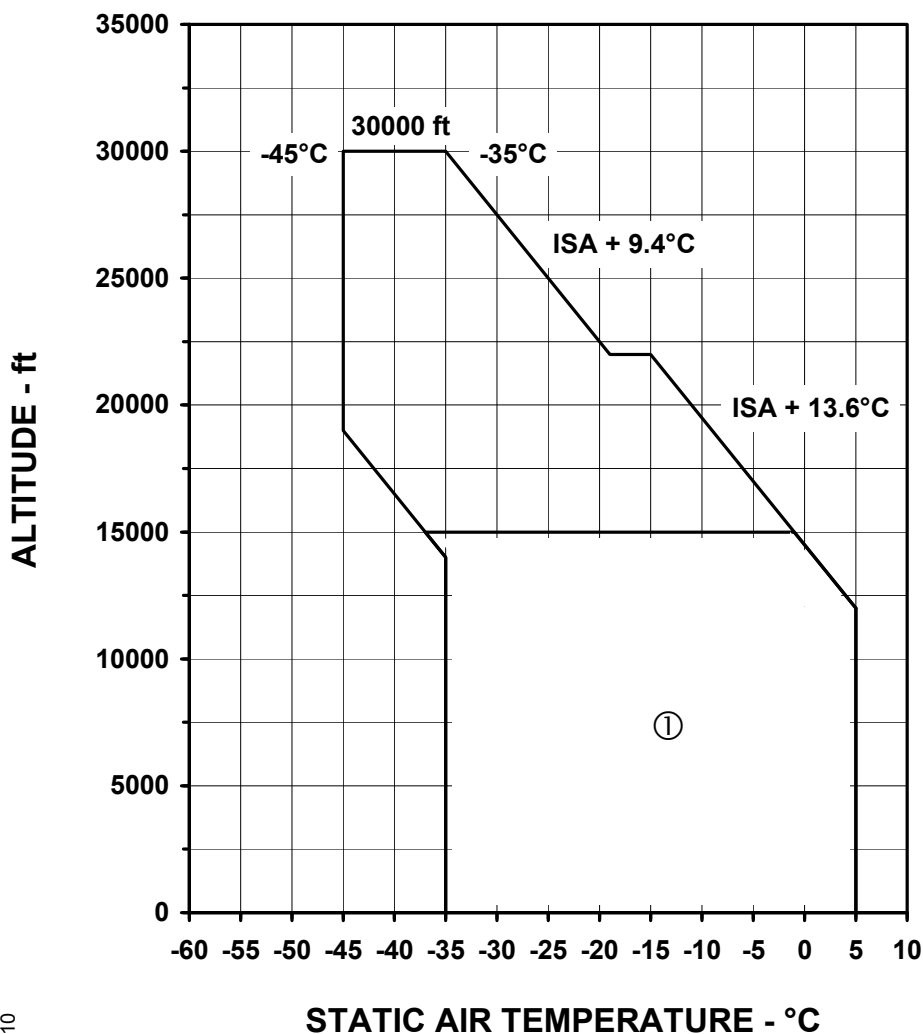
FOR AIRPLANES PRE-MOD. SB 505-00-0001

Maximum Altitude for Operation:

AEO 30000 ft
OEI 15000 ft

Minimum Temperature for Landing -33°C

WING AND STABILIZER ANTI-ICE SYSTEM OPERATIONAL ENVELOPE



ONE ENGINE INOPERATIVE (OEI) → ①

NOTE: Wing and Stabilizer Anti-Ice system is auto-inhibited outside its operational envelope.



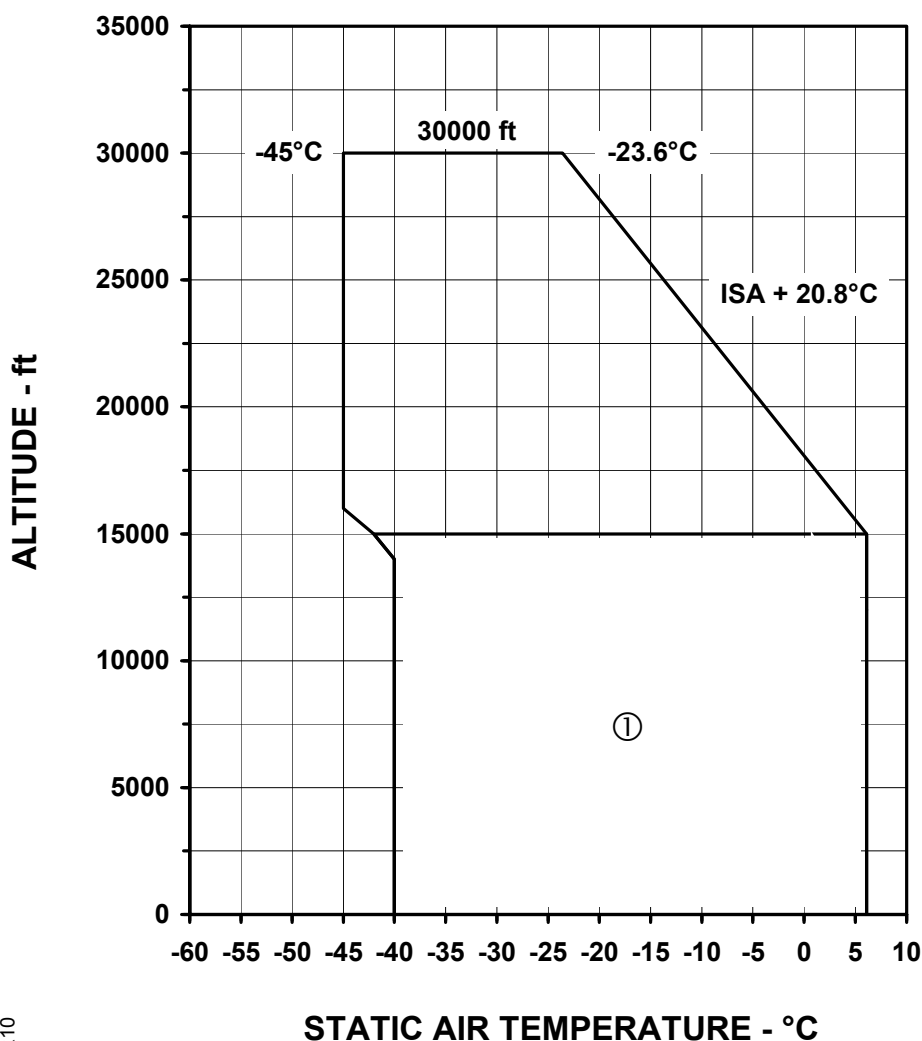
FOR AIRPLANES POST-MOD. SB 505-00-0001 OR WITH AN EQUIVALENT MODIFICATION FACTORY INCORPORATED

Maximum Altitude for Operation:

AEO 45000 ft
OEI..... 15000 ft

Minimum Temperature for Landing -38°C

WING AND STABILIZER ANTI-ICE SYSTEM OPERATIONAL ENVELOPE



ONE ENGINE INOPERATIVE (OEI) → ①

505ANAC07 - 08MAR10

NOTE: Wing and Stabilizer Anti-Ice system is auto-inhibited outside its operational envelope.



- NOTE:** - For airplanes Pre-Mod. SB 505-31-0005: If outside Wing and Stabilizer Anti-Ice system operational envelope the A-I WINGSTB INHB caution message will be presented. Follow its associated procedure (Section 4, Emergency and Abnormal Procedures).
- For airplanes Post-Mod. SB 505-31-0005 or with an equivalent modification factory incorporated:
 - If outside Wing and Stabilizer Anti-Ice system operational envelope below 30000 ft the A-I WINGSTB INHB caution message will be presented. Follow its associated procedure (Section 4, Emergency and Abnormal Procedures).
 - If above 30000 ft, the A-I WINGSTB ARM advisory message will be presented. Follow its associated procedure (Section 4, Emergency and Abnormal Procedures).
 - Icing conditions may exist whenever the Static Air Temperature (SAT) on the ground or for takeoff, or Total Air Temperature (TAT) in flight, is 10°C or below and visible moisture in any form is present (such as clouds, fog with visibility of one mile or less, rain, snow, sleet or ice crystals).
 - Icing conditions may also exist when the SAT on the ground and for takeoff is 10°C or below when operating on ramps, taxiways, or runways where surface snow, ice, standing water, or slush may be ingested by the engines, or freeze on engines, nacelles, or engine sensor probes.
 - Ice speed must not be reset until all ice has been removed.
 - In icing conditions the airplane must be operated, and its ice protection systems used as described in the operating procedures section of this manual. Where specific operational speeds and performance information have been established for such conditions, this information must be used.

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- Takeoff is prohibited with frost (polished or not), ice, snow or slush adhering to wings, control surfaces, engine inlets, or other critical surfaces.
- The airplane must exit SLD (Super Cooled Large Droplet) icing conditions environment. SLD conditions will be recognized by ice formation aft of protected surfaces or in areas that are not normally observed to collect ice.
- Flight in freezing drizzle or freezing rain is prohibited. If the airplane encounters conditions that are determined to contain freezing rain or freezing drizzle, the pilot must immediately exit the freezing rain or freezing drizzle conditions by changing altitude or course. Such conditions may be identified by the following visual cues:
 - Unusually extensive ice accreted on the airframe in areas not normally observed to collect ice.
 - Accumulation of ice on the upper surface of the wing aft of the protected area.
- There are many methods to ensure the wing is clear of ice. If visual inspection does not indicate wing contamination, a tactile (hands on surface) check of the wing leading edge and the upper surface must be accomplished prior to takeoff. The tactile check must also be performed when the holdover time is exceeded after airplane de/anti-icing fluids have been applied. This check must be performed whenever the outside temperature is less than 6°C or if it cannot be ascertained that the wing fuel temperature is above 0°C, and there is visible moisture, or:
 - Water is present on the wing; or
 - When difference between the dew point and the outside air temperature is 3°C or less; or
 - The atmospheric conditions have been conducive to frost formation.

(Continues on the next page)



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- Since the autopilot can mask tactile cues that indicate adverse changes in handling characteristics, the pilot should consider not using the autopilot when any ice is visible on the airplane. Autopilot use is prohibited in the following conditions:
 - Severe icing;
 - Unusual control force or control deflection, or unusually large control forces to move flight controls when the autopilot is disconnected periodically for checking purposes; or
 - Indications of frequent autopilot re-trimming during straight and levelled flight.

CAUTION: ON GROUND, DO NOT RELY ON VISUAL ICING EVIDENCE TO TURN ON THE ANTI-ICING SYSTEM. USE THE TEMPERATURE AND VISUAL MOISTURE CRITERIA AS SPECIFIED ABOVE. DELAYING THE USE OF THE ANTI-ICING SYSTEM UNTIL ICE BUILD-UP IS VISIBLE FROM THE COCKPIT MAY RESULT IN ICE INGESTION AND POSSIBLE ENGINE DAMAGE OR FLAMEOUT.



AUTOPILOT/YAW DAMPER

Minimum Engagement Height (dual engine).....	600 ft AGL
Minimum Engagement Height (single engine)	1000 ft AGL
Minimum Use Height (dual engine)	195 ft AGL
Minimum Use Height (single engine)	220 ft AGL
Altitude Loss (maneuvering/cruise)	160 ft

Airplane is approved for CAT I approaches. This statement does not grant operational approval to conduct CAT I operations.

Yaw Damper OFF airspeed is limited to 180 KIAS if in icing conditions.

NAVIGATION AND COMMUNICATION EQUIPMENT

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

ATTITUDE AND HEADING REFERENCE SYSTEM (AHRS)

The airplane must not operate in the regions stated on the table below:

MAGNETIC CUT-OUT REGIONS	LATITUDE	LONGITUDE
North	North of 72°N	All
	North of 70°N	Between 70°W and 128°W and Between 85°E and 114°E
	North of 65°N	Between 75°W and 120°W
South	South of 55°S	Between 120°E and 165°E
	South of 70°S	All

NOTE: Alternative procedures must be established for dispatch if the indication GEO LIMITS is displayed.



DATA LINK MANAGEMENT SYSTEM (DLMS) (IF APPLICABLE)

Pilot Flying must not use the telephone and texting functions, unless there is a problem with the radio communication.

Pilot Not Flying must not use the telephone and texting functions during abnormal/emergency operations and sterile cockpit phase, unless there is a problem with the radio communication. The sterile cockpit phase includes all ground operations involving taxi, takeoff and landing, and all other flight operations conducted below 10000 ft AGL.

GARMIN G1000 AVIONICS SYSTEM

Refer to Garmin – Embraer Prodigy Flight Deck 300 Pilot's Guide for further information on system characteristics. The Garmin G1000 avionics system has the following limitations:

- VNAV path guidance is not available for legs that do not have a defined leg distance on the horizontal plane.
- Dead Reckoning Mode use is available only in enroute (ENR) or oceanic (OCN) phases of flight. During DR the CDI is removed. The estimated navigation data supplied by the system in this condition must not be used as a sole means of navigation.
- The fuel quantity, fuel required, fuel remaining and gross weight estimate functions of the G1000 presented in AUX-WEIGHT PLANNING and AUX-TRIP PLANNING pages of MFD are supplemental information only and must be confirmed by the flight crew prior to use.



GARMIN G1000 GLOBAL NAVIGATION SATELLITE SYSTEM (GPS/SBAS)

SYSTEM LIMITATIONS

The pilot must confirm at system initialization that the Navigation database is current.

Navigation database is expected to be current for the duration of the flight. If the AIRAC cycle will change during flight, the pilot must ensure the accuracy of navigation data, including suitability of navigation facilities used to define the routes and procedures for flight. If an amended chart affecting navigation data is published for the procedure, the database must not be used to conduct the procedure.

GPS/SBAS based IFR enroute, oceanic, and terminal navigation is prohibited unless the pilot verifies and uses a valid, compatible, and current Navigation database or verifies each waypoint for accuracy by reference to current approved data.

Discrepancies that invalidate a procedure shall be reported to Garmin International. The affected procedure is prohibited from being flown using data from the Navigation database until a new Navigation database is installed in the airplane and verified that the discrepancy has been corrected.

For flight planning purposes, in areas where SBAS coverage is not available, the pilot must check RAIM availability. This requirement is not necessary if SBAS coverage is confirmed to be available along the entire route of flight.

For flight planning purposes, operations within the U.S. National Airspace System on RNP and RNAV procedures when SBAS signals are not available, the availability of GPS integrity RAIM shall be confirmed for the intended route of flight. In the event of a predicted continuous loss of RAIM of more than five minutes for any part of the intended route of flight, the flight should be delayed, canceled, or rerouted on a track where RAIM requirements can be met.

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For flight planning purposes for operations within European B-RNAV and P-RNAV airspace, if more than one satellite is scheduled to be out of service, then the availability of GPS integrity RAIM shall be confirmed for the intended flight (route and time). In the event of a predicted continuous loss of RAIM of more than five minutes for any part of the intended flight, the flight should be delayed, canceled, or rerouted on a track where RAIM requirements can be met.

For flight planning purposes, operations where the route requires Class II navigation, the airplane's operator or pilot-in-command must use the Garmin WFDE Prediction program to demonstrate that there are no outages on the specified route that would prevent the Garmin GNSS navigation system to provide primary means of Class II navigation in oceanic and remote areas of operation that requires RNP-10 or RNP-4 capability. If the Garmin WFDE Prediction program indicates fault exclusion (FDE) unavailability will exceed 34 minutes in accordance with FAA Order 8400.12B for RNP-10 requirements, or 25 minutes in accordance with FAA Order 8400.33 for RNP-4 requirements, then the operation must be rescheduled when FDE is available.

Both Garmin GPS navigation receivers must be operating and providing GPS navigation guidance to their respective PFD for operations requiring RNP-4 performance.

North Atlantic (NAT) Minimum Navigational Performance Specifications (MNPS) Airspace operations per AC 91-49 and AC 120-33 require both GPS/SBAS receivers to be operating and receiving usable signals except for routes requiring only one Long Range Navigation sensor. Each display computes an independent navigation solution based on the on-side GPS sensor. However, either display will automatically revert to the cross-side sensor if the on-side sensor fails or if the cross-side sensor is determined to be more accurate. A "BOTH ON GPS1" or "BOTH ON GPS2" message does not necessarily mean that one GPS has failed. Refer to the MFD AUX-GPS STATUS page to determine the state of the unused GPS.

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Whenever possible, RNP and RNAV routes including Standard Instrument Departures (SIDs) and Obstacle Departure Procedures (ODPs), Standard Terminal Arrival (STAR), and enroute RNAV "Q" and RNAV "T" routes should be loaded into the flight plan from the database in their entirety, rather than loading route waypoints from the database into the flight plan individually. Selecting and inserting individual named fixes from the database is permitted, provided all fixes along the published route to be flown are inserted. Manual entry of waypoints using latitude/longitude or place/bearing is prohibited.

"GPS", "or GPS", and "RNAV (GPS)" instrument approaches using the Garmin navigation systems are prohibited unless the pilot verifies and uses the current Navigation database. GPS based instrument approaches must be flown in accordance with an approved instrument approach procedure that is loaded from the Navigation database.

Not all published Instrument Approach Procedures (IAP) are in the Navigation database. Pilots planning on flying an RNAV instrument approach must ensure that the Navigation database contains the planned RNAV Instrument Approach Procedure and that approach procedure must be loaded from the Navigation database into the FMS flight plan by its name.

IFR non-precision approach approval using the GPS/SBAS sensor is limited to published approaches within the U.S. National Airspace System. Approaches to airports in other airspace are not approved unless authorized by the appropriate governing authority.

The navigation equipment required to join and fly an instrument approach procedure is indicated by the title of the procedure and notes on the IAP chart. Use of the Garmin GPS/SBAS receivers to provide navigation guidance during the final approach segment of an ILS, LOC, LOC-BC, LDA, SDF, MLS or any other type of approach not approved for "or GPS" navigation is prohibited. When using the Garmin VOR/LOC/GS receivers to fly the final approach segment, VOR/LOC/GS navigation data must be selected and presented on the CDI of the pilot flying.

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Navigation information is referenced to the WGS-84 reference system, and should only be used where the Aeronautical Information Publication (including electronic data and aeronautical charts) conform to WGS-84 or equivalent.

When using GPS/NAV mode to intercept the final approach course, monitor the flight director transition to LOC/GS. If the transition does not occur automatically, manually select HDG mode and CDI source to LOC in order to intercept the final approach course.

Localizer Performance (LP) approach procedures are not allowed.

OPERATIONAL LIMITATIONS

The Garmin – Embraer Prodigy Flight Deck 300 Cockpit Reference Guide must be available to the flight crew whenever navigation is predicated on the use of the system.

Advisory vertical guidance deviation information is only an aid to help pilots comply with altitude restrictions. When using advisory vertical guidance, the pilot must use the primary barometric altimeter to ensure compliance with all altitude restrictions, particularly during instrument approach operations.

The Flight Director must be engaged to perform Baro-VNAV operations (Enroute, Terminal and Approach).

The use of Baro-VNAV approach function is not allowed for airports located in altitudes above 5000 ft.

When using the Baro-VNAV function, the barometric altimeter must be used as the primary altitude reference for all operations, including instrument approach procedure step-down fixes.

For flight planning purposes in European airspace, flight crew should ensure sufficient means are available to navigate and land at the destination or at an alternate aerodrome in the case of loss of RNP APCH airborne capability. In particular, the flight crew should check that:

- a non-RNP APCH procedure is available at the alternate, where a destination alternate is required.
- at least one non-RNP APCH procedure is available at the destination aerodrome, where a destination alternate is not required.

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For flight planning purposes in European airspace, flight crew should ensure sufficient means are available to navigate and land at the destination or at an alternate aerodrome in the case of loss of LPV airborne capability. In particular, the flight crew should check that:

- a non-RNAV GNSS based procedure is available at the alternate (where a destination alternate is required) unless the airspace authority does not require such requirement.
- at least one non-RNAV GNSS based procedure is available at the destination aerodrome (where a destination alternate is not required) unless the airspace authority does not require such requirement.

For flight planning purposes, operations within the Brazilian airspace using GNSS requires that the availability of GPS integrity RAIM be confirmed before departure and before entrance to each phase of flight. When the RAIM is not available, other type of navigation must be used or the schedule of the flight must be changed until the RAIM is available again.

Enroute operations within the Brazilian airspace under instrument flight rules require that the airplane has the basic equipment for air navigation appropriate for the route to be flown. Such equipment shall be used compulsorily, when there is an integrity alarm and on the portion of the route where the function RAIM is supposed to be unavailable.

Airplane accomplishing GNSS SIDs or GNSS STARs within the Brazilian airspace must have their basic navigation equipment tuned on the appropriate frequencies, so as to provide fast and safe transition in the case of occurrence of RAIM alarm. In case that there is prevision of unavailability of the RAIM function during the flight period, only the basic equipment of air navigation shall be used.

For airplane accomplishing GNSS approach procedures within the Brazilian airspace the pilot must assure that the air navigation aids needed to the aerodrome operations at the alternative aerodrome are available. Pilot must designate an alternative aerodrome that offers an approach procedure in operation based in conventional air navigation aids.



HF (IF INSTALLED)

Do not rely on ADF indications shown on the PFD while transmitting in HF.

ADS-B OUT

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

Applicable for airplanes with ADS-B Out installed (Post-Mod. SB 505-34-0010 or Post-Mod. SB 505-34-0011 or with an equivalent modification factory incorporated).

The ADS-B Out system is fully compliant with the requirements of CS ACNS.D.ADSB (1090 MHz Extended Squitter ADS-B Out).

ELECTRONIC DISPLAY SYSTEM

ELECTRONIC CHARTS

The own-ship position symbol presented in the electronic charts is intended to improve pilot positional awareness. Do not use it as the basis for maneuvering.

The airplane must be dispatched with adequate backup documentation.

The electronic charts functions airworthiness approval alone does not constitute operational approval.

ELECTRONIC CHECKLIST

Operational approval is required in order to load database into the airplane and use the electronic checklist.



ENTERTAINMENT SYSTEM

ENTERTAINMENT AUDIO (IF APPLICABLE)

The crew must not use the entertainment audio at any time the main engines are running.

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SECTION 3

NORMAL PROCEDURES

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INTRODUCTION

The normal procedures contained in this manual have been developed and recommended by the manufacturer and approved by the certification authorities for use during the operation of the PHENOM 300 airplanes.

Indented explanations (lines beginning further from the margin than the others) follow previous items and provide further information.

* Items marked with an asterisk (*) need to be accomplished before the first flight of the day.

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COCKPIT/CABIN SAFETY INSPECTION

Emergency Door Locking Pin	REMOVE
Emergency Door	SECURED
Documents, Manuals and Charts	CHECK ON BOARD
Maintenance Status	CHECK
Emergency Equipment	CHECK

Verify Fire Extinguisher, Flashlight, First Aid Kit, Water Barrier and the following optional items, if installed: Protective Breathing Equipment (PBE), Smoke Goggles, Over Water Life Vest and Survival Kit.

FCE RESET Knob (if applicable)	SET TO NONE
Oxygen Bottle Valve Handle	PUSH TO RESTORE
SUPPLY CONTROL Knob	PAX AUTO
Oxygen Mask Regulator	CHECK 100%
ELECTRICAL Panel	CHECK

Check BATT 1 & 2 Switches in the OFF position. BUS TIE Knob and GEN 1 & 2 Switches in the AUTO position and set GPU Button as required and check.

BLEED 1 & 2 Switches	AUTO
XBLEED Knob	AUTO
TEST Panel	OFF
Circuit Breakers	CHECK

If any circuit breaker is not IN, investigate the cause prior to resetting.

FUEL PUMP 1 & 2 Switches	AUTO
XFEED Knob	OFF
HYD PUMP SOV 1 & 2 Switches	OPEN
ELT Switch	ARMED

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PUSHER CUTOUT Button..... PUSHED OUT

Gust Lock Pin..... REMOVE

Rudder Pedals CHECK
UNLOCKED

HEATING Panel CHECK

Check WSHLD 1 & 2 Switches in the OFF position and ADS
PROBES Knob in the AUTO position.

ICE PROTECTION Panel CHECK

Check ENG 1 & 2 Switches, WINGSTAB and INSP LIGHT
Switches in the OFF position.

LDG GEAR Lever DN

PRESSURIZATION Panel CHECK

Check MODE Switch in the AUTO position, ECS Knob in the BOTH
position and DUMP Button pushed out.

AIR CONDITIONING Panel AS REQUIRED

ENG FIRE EXTINGUISHER Panel..... CHECK

Check SHUTOFF 1 & 2 Buttons pushed out and BOTTLE Switch in
the OFF position.

Start/Stop Knobs STOP

Flap Lever ZERO

SPEED BRAKE Switch CLOSE

Parking Brake..... SET

NOTE: If parking brake pressure is suspected to be low, use wheel
chocks to secure the airplane.

Seats and Belts CONDITION

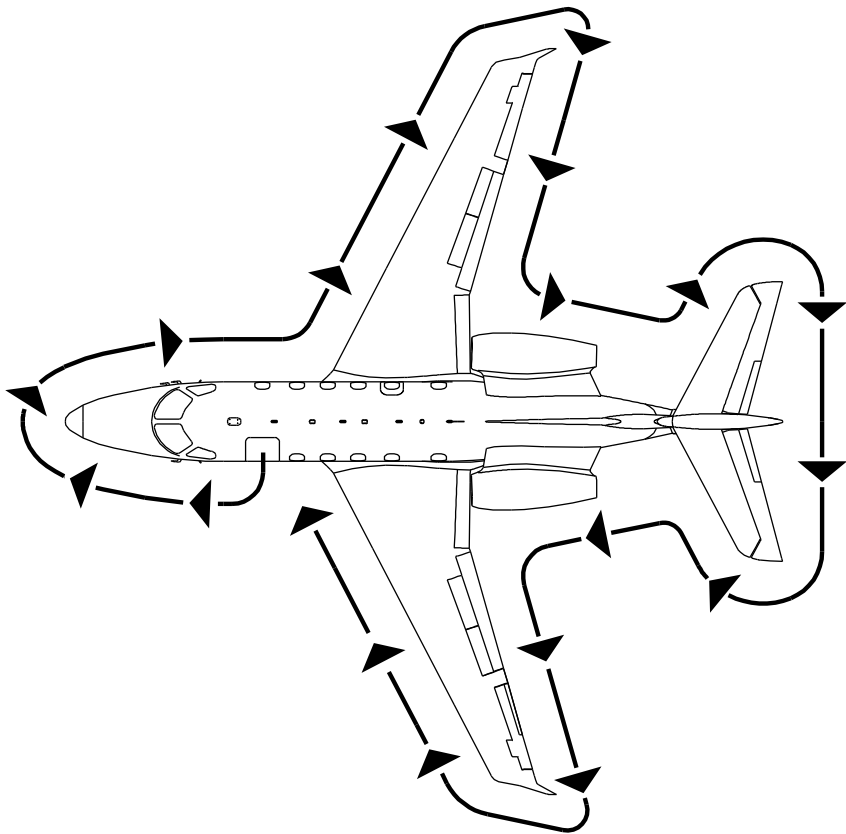


EXTERNAL INSPECTION

External Lights CHECK

Turn the lights OFF immediately after check them to avoid batteries discharge.

Recommended Walk-Around sequence:



EM500ENACOM140228B.DGN

- IASP CONDITION, NO OBSTRUCTION
- Oxygen Discharge Indicator GREEN DISK IN PLACE
- Ice Detector (if installed) CHECK
- Antennas CONDITION
- Red Beacon Light CONDITION
- NLG Doors, Wheel and Tire CONDITION

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NLG Torque Link.....	CONNECTED AND SECURED
NLG Safety Pin	REMOVED
Forward LH Baggage Compartment Door	LATCHED
Radome.....	CONDITION
Forward RH Baggage Compartment Door	LATCHED
IASP	CONDITION, NO OBSTRUCTION
Stand-By Pitot Tube.....	CONDITION, NO OBSTRUCTION
Refueling Panel.....	CLOSED
LDG/Taxi Lights	CONDITION
Fuselage Air Inlet	NO OBSTRUCTION
Engine Fan.....	CONDITION
Engine Air Inlet.....	NO OBSTRUCTION
Starter/Generator Air Inlets	CLEAR
* Fuel Drains	DRAIN AND CHECK FOR CONTAMINATION
<u>CAUTION:</u> USING AN INADEQUATE TOOL TO ACCOMPLISH THE FUEL DRAINAGE MAY CAUSE DAMAGE TO THE DRAIN VALVE.	
Fuel Drains and Dump Valves	NO LEAKS
Right Wing Leading Edge	CONDITION
MLG Doors, Wheels, Brakes and Tires	CONDITION
MLG Safety Pin	REMOVED
Fuel Cap.....	CLOSED AND LOCKED
Fuel Tank Air Inlet.....	NO OBSTRUCTION

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Navigation/Strobe Lights	CONDITION
Right Winglet	CONDITION
Right Winglet Static Dischargers	NUMBER (2) AND CONDITION
Right Aileron	CHECK FREE
Right Aileron Static Dischargers	NUMBER AND CONDITION

NOTE: Verify 2 or 3 static dischargers on each aileron, according to airplane configuration.

Right Spoilers	CONDITION
Right Flap	CONDITION
Battery Access Door	LATCHED
Cowlings	LATCHED
Engine Exhausts	CLEAR
Drain Masts	CONDITION
Engine Oil Level	CHECK
Heat Exchanger Air Exhaust	CLEAR
Pylon (air exhaust)	CONDITION
Vertical Stabilizer	CONDITION
Rudder	CONDITION
Rudder Static Dischargers	NUMBER (2) AND CONDITION
Yaw Trim Tab	CONDITION
Horizontal Stabilizer	CONDITION
Elevator	CONDITION
Pitch Trim Tab	CONDITION
Elevators Static Dischargers	NUMBER (4) AND CONDITION
Antennas	CONDITION
Ventral Rudder	CONDITION

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Normal Procedures

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Air Exhausts	NO OBSTRUCTION
Engine Exhaust.....	CLEAR
Aft Baggage Compartment Door	LATCHED
Pylon (air exhaust)	CONDITION
Cowlings.....	LATCHED
Drain Masts	CONDITION
Engine Oil Level.....	CHECK
DC Power Receptacle.....	CHECK
Hydraulic Reservoir Access Door	LATCHED
Left Flap	CONDITION
Left Spoilers	CONDITION
Left Aileron	CHECK FREE
Left Aileron Static Dischargers.....	NUMBER AND CONDITION

NOTE: Verify 2 or 3 static dischargers on each aileron, according to airplane configuration.

Roll Trim Tab.....	CONDITION
Navigation/Strobe Lights.....	CONDITION
Left Winglet	CONDITION
Left Winglet Static Dischargers.....	NUMBER (2) AND CONDITION
Fuel Tank Air Inlet.....	NO OBSTRUCTION
Left Wing Leading Edge.....	CONDITION
Fuel Cap.....	CLOSED AND LOCKED
MLG Doors, Wheels, Brakes and Tires	CONDITION
MLG Safety Pin.....	REMOVED
LDG/Taxi Lights	CONDITION
Wing Inspection Light.....	CONDITION

(Continues on the next page)



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Fuselage Air Inlet..... NO
 OBSTRUCTION

Engine Fan CONDITION

Engine Air Inlet NO
 OBSTRUCTION

Starter/Generator Air Inlets..... CLEAR

POWER UP

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

BATT 1 & 2 Switches..... ON

NOTE: - For airplanes Pre-Mod. SB 505-24-0015: If the battery has been cold soaked for 2 hours or longer at ambient temperature of -10°C (14°F) or lower, it must be pre-heated to above -10°C (14°F) prior to engine start.

- For airplanes Post-Mod. SB 505-24-0015 or with an equivalent modification factory incorporated: If the battery has been cold soaked for 2 hours or longer at ambient temperature of -18°C (-0.4°F) or lower, it must be pre-heated to above -18°C (-0.4°F) prior to engine start.

GPU Button (if applicable) AS REQUIRED

Verify AVAIL light illuminated before pushing the GPU Button in. When GPU is not available or is not necessary, maintain GPU Button pushed out.

CVDR..... TEST

Transponder/TCAS II (if applicable) AS REQUIRED

Set transponder to ALT mode or as required by local rules.

If airplane is equipped with TCAS II, verify TCAS II in STBY mode.

For operations requiring ADS-B Out, the transponder should be in ALT mode.



BEFORE START

- Oxygen Mask Flow and Microphone CHECK
Set MASK MIC Switch to the ON position and press TEST/RESET Button. Set MASK MIC Switch to the OFF position.
- TEST Panel..... TEST
Test Fire Protection, Smoke Detector and Annunciator.
- Fasten Seat Belts Signs..... ON
- External Lights AS REQUIRED
- EMER LT Switch ARM
- Fuel Quantity and Balance..... CHECK
- Oxygen Pressure CHECK MINIMUM
FOR DISPATCH
- ENG IGNITION Switches..... AUTO
- Thrust Levers IDLE
- Parking Brake..... SET
- Doors..... CLOSED

ENGINE START

- Associated Start/Stop Knob START, THEN
RUN
- Engine Parameters MONITOR
Check N2, ITT, N1 and oil pressure within operational limits.
- Repeat the sequence for the other engine.

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AFTER START

GPU (if applicable)..... DISCONNECT
ELEC EMER Button..... PUSH IN
Battery 1 & 2 Voltage..... CHECK

CAUTION: EACH BATTERY VOLTAGE MUST BE AT LEAST 23.5 VOLTS.

ELEC EMER Button..... PUSH OUT
Stall Protection System (TEST panel) TEST
AFCS Control Unit SET
Engine Takeoff Data SET
 OAT SET
 ATR..... ON OR OFF
Landing Field Elevation SET
Takeoff Speeds (V₁, V_R, V₂, V_{FS}) SET

CAUTION: SET THE SPEEDS IN SEQUENCE. IF IT IS NECESSARY TO CORRECT ONE OF THE SPEEDS OUT OF THE SEQUENCE, ALL THE OTHERS MUST BE VERIFIED.

Transponder/TCAS II (if applicable) SET

Set transponder CODE and Flight ID.

If airplane is equipped with TCAS II, verify TCAS II in STBY mode.

For operations requiring ADS-B Out, the transponder should be in ALT mode.

Altimeters (pilots and IESI) SET & X-CHECK
Flight Controls..... CHECK FREE
Trims CHECK/SET

Verify that Roll, Yaw and Pitch (NORM and BKP) trims are operating properly in both directions. Adjust Yaw and Roll trims to the neutral position and Pitch trim to Takeoff (green band).

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Flap SET FOR
TAKEOFF

External Lights AS REQUIRED

Ice Protection AS REQUIRED

**WARNING: IF ICING CONDITIONS EXIST OR ARE FORECAST,
REFER TO OPERATION IN ICING CONDITIONS
PROCEDURE.**

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BEFORE TAKEOFF

Takeoff Configuration CHECK

Press the T/O CONFIG Button on the central console and check if the aural TAKEOFF OK sounds.

CAS Messages CHECK

Fasten Seat Belts Signs ON

Yaw Damper OFF

-----SHORTLY BEFORE TAKEOFF-----

Passengers ADVISE

Lights AS REQUIRED

TAKEOFF

Thrust Levers TO/GA

NOTE: After checking thrust levers in TO/GA, check N1 equal to N1 target and green ATR indication presented on MFD if ATR ON is selected.

Engine Parameters MONITOR

 At V_R rotate the airplane according to the table below.

FLAP POSITION	1	2
PITCH ANGLE	10.5°	8°

With positive rate of climb:

LDG GEAR Lever UP

 Airspeed $V_2 + 10$ KIAS
 MINIMUM



AFTER TAKEOFF/CLIMB

LDG GEAR Lever CHECK UP

Flap ZERO

Retract flap in a speed at or below to the Maximum Flap Extended Speed (V_{FE}) for each position.

Thrust Levers CON/CLB

Altimeters (pilots and IESI) SET & X-CHECK

Yaw Damper ON

Icing Conditions VERIFY

-----ABOVE 10000 FT-----

Fasten Seat Belts Signs AS REQUIRED

External Lights AS REQUIRED

Weather Radar (if installed) AS REQUIRED

Airspeed AS REQUIRED

DESCENT

Windshield Heating ON

Pressurization CHECK LFE

Landing Speeds (V_{REF} , V_{AC} , V_{FS}) SET

CAUTION: SET THE SPEEDS IN SEQUENCE. IF IT IS NECESSARY TO CORRECT ONE OF THE SPEEDS OUT OF THE SEQUENCE, ALL THE OTHERS MUST BE VERIFIED.

Icing Conditions VERIFY

-----BELOW 10000 FT-----

Fasten Seat Belts Signs ON



APPROACH

- External Lights AS REQUIRED
- Passengers ADVISE
- XFEED Knob OFF
- Altimeters (pilots and IESI) SET & X-CHECK
- Icing Conditions VERIFY

BEFORE LANDING

- Yaw Damper OFF
- LDG GEAR Lever DN AND CHECK
- Flap SET FOR LANDING
- Airspeed V_{REF}

LANDING

- Thrust Levers IDLE
- Brakes (after touchdown) APPLY MAXIMUM

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GO-AROUND

TO/GA Button.....PRESS

NOTE: - For airplanes with G3000 system version prior to 3305.07, when the TO/GA Button is pressed in the air, the Flight Director activates GA and ROL (Wings Level) modes.

- For airplanes with G3000 system version 3305.07 and on, when pressing the TO/GA Button in the air, during an approach with navigation or approach mode, automatic activation of FMS mode occurs, provided the approach procedure is previously loaded in the flight plan. Otherwise, GA and ROL (Wings Level) modes will be engaged.

Thrust LeversTO/GA

Rotate the airplane following the flight director guidance or according to the table below.

LANDING FLAP POSITION	GO-AROUND PITCH ANGLE
3	8°
FULL	6°

Select flaps according to the table below.

LANDING FLAP POSITION	GO-AROUND FLAP POSITION
3	1
FULL	2

With positive rate of climb:

LDG GEAR LeverUP

Airspeed.....V_{AC} MINIMUM

At the acceleration altitude proceed as in a normal takeoff.



AFTER LANDING

Flap..... ZERO
Lights AS REQUIRED
Transponder AS REQUIRED

For operations requiring ADS-B Out, the transponder should be in ALT mode.

SHUTDOWN

Thrust Levers..... IDLE
Emergency/Parking Brake..... APPLY

CAUTION: MAINTAIN IDLE FOR AT LEAST 2 MINUTES PRIOR TO ENGINE SHUTDOWN.

HEATING Panel..... CHECK

Check WSHLD 1 & 2 Switches in the OFF position and ADS PROBES Knob in the AUTO position.

ICE PROTECTION Panel..... CHECK

Check ENG 1 & 2, WINGSTAB and INSP LIGHT Switches in the OFF position.

Start/Stop Knobs..... STOP

MFD Status Pages..... CHECK

Check MFD status page for engine message.

Fasten Seat Belts Signs OFF

LEAVING THE AIRPLANE

Oxygen Bottle Valve Handle..... PULL TO CUTOUT

EMER LT Switch..... OFF

BATT 1 & 2 Switches..... OFF

Gust Lock Pin INSTALL

Rudder Pedals LOCK

Lights OFF



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OPERATION IN ICING CONDITIONS

Icing conditions may exist whenever the Static Air Temperature (SAT) on the ground or for takeoff, or Total Air Temperature (TAT) in flight, is 10°C or below and visible moisture in any form is present (such as clouds, fog with visibility of one mile or less, rain, snow, sleet or ice crystals).

Icing conditions may also exist when the SAT on the ground and for takeoff is 10°C or below when operating on ramps, taxiways, or runways where surface snow, ice, standing water, or slush may be ingested by the engines, or freeze on engines, nacelles, or engine sensor probes.

The procedures below complement or change the remaining procedures presented in this Section.

WARNING: EVEN IF THE ICE DETECTOR IS INSTALLED, THE FLIGHT CREW STILL HAS THE PRIMARY RESPONSIBILITY TO DETERMINE WHEN TO ACTIVATE ICE PROTECTION SYSTEM.

NOTE: - If necessary, set the CKPT FAN Switch to HI position to avoid fog on the cockpit side window.

- The A-I WINGSTB INHB caution message and A-I WINGSTB ARM advisory message may be presented during operation in icing conditions. If the message is presented, refer to the associated procedure in Section 4, Emergency and Abnormal Procedures.

BEFORE START

ADS PROBES Knob ON

NOTE: It is recommended to turn on the system immediately before engine start.



BEFORE TAKEOFF

NOTE: Contamination check of wing upper surface and leading edge must be performed for takeoff.

With engines running check the ice protection system as follows:

WSHLD 1 and WSHLD 2 Switches ON
The CAS messages WSHLD 1 (2) HTR FAIL must not be displayed.

WSHLD 1 and WSHLD 2 Switches OFF

ENG 1 and ENG 2 Switches ON
The CAS messages A-I E1 (2) ON must be displayed (after 10 seconds).

ENG 1 and ENG 2 Switches OFF
The CAS messages must disappear.

BLEED 1 and BLEED 2 Switches AS REQUIRED

WINGSTAB Switch ON
The CAS message A-I WINGSTB ARM or A-I WINGSTB INHB may be displayed.

WINGSTAB Switch OFF
The CAS message A-I WINGSTB ARM or A-I WINGSTB INHB must disappear.

If ice detector is available:
TEST Panel Knob ICE PROT
START TEST Button PRESS
The CAS messages ICE CONDITION must be displayed.

After completing a successful test:

If SAT is between 5°C and 10°C with visible moisture:
ENG 1 and ENG 2 Switches ON
WINGSTAB Switch OFF
WSHLD 1 and WSHLD 2 Switches OFF

(Continues on the next page)



(Continued from the previous page)

If SAT is below 5°C with visible moisture:

ENG 1 and ENG 2 Switches ON

WINGSTAB Switch..... ON

WSHLD 1 and WSHLD 2 Switches..... ON

ADS PROBES Switch..... AUTO

WARNING: TAKEOFF MUST NOT BE PERFORMED WITH CAS MESSAGE A-I WINGSTB INHB PRESENTED. REFER TO THE ASSOCIATED PROCEDURE IN SECTION 4, EMERGENCY AND ABNORMAL PROCEDURES.

AFTER TAKEOFF/CLIMB, CRUISE, DESCENT OR APPROACH

The crew must activate the ice protection system when flying in icing conditions or if icing conditions are forecasted as follows:

If TAT is between 5°C and 10°C with visible moisture:

ENG 1 and ENG 2 Switches ON

WINGSTAB Switch..... OFF

WSHLD 1 and WSHLD 2 Switches..... OFF

If TAT is below 5°C with visible moisture, or at the first sign of ice accretion anywhere on the airplane, or ICE CONDITION message is displayed (if applicable), whichever occurs first:

ENG 1 and ENG 2 Switches ON

WINGSTAB Switch..... ON

WSHLD 1 and WSHLD 2 Switches..... ON

NOTE: - Check the CAS message for proper system activation.

- In-flight, the windshield is the best indication of early ice formation detection. If the airplane is not flying in icing condition, no ice is building up in the windshield and if the windshield heater is not required for defog, leave the WSHLD 1 (2) Switches OFF, turning it ON when required.

(Continues on the next page)



(Continued from the previous page)

After leaving ice conditions and if there is no ice accretion on the airplane:

ENG 1 and ENG 2 Switches..... OFF
WINGSTAB Switch OFF
WSHLD 1 and WSHLD 2 Switches OFF
WINGSTAB Switch ICE SPEED
RESET

WARNING: THE ICE PROTECTION SYSTEM MUST BE KEPT ON AND ICE SPEED MUST NOT BE RESET UNTIL CREW IS CERTAIN ALL ICE HAS BEEN REMOVED.

HOLDING CONFIGURATION

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

LDG GEAR Lever UP
Flap UP

Recommended airspeed:

Above 30000 ft..... 165 KIAS
MINIMUM
Below 30000 ft 150 KIAS
MINIMUM

DEMONSTRATED CROSSWIND

The maximum demonstrated crosswind component for takeoff is 25 kt and for landing is 28 kt. This value is not considered to be limiting.

- NOTE:** - For crosswind takeoff it may be necessary to use the lateral control to prevent the wing at the wind side from rising.
- For crosswind landings the "de-crab" technique shall be accomplished.



TURBULENT AIR PENETRATION

Turn on the fasten seat belts signs and adjust airspeed. Set thrust for penetration and avoid large thrust variations. Set trim for target speed and do not change it.

Use attitude indicator as the primary instrument. Allow altitude and airspeed to vary and maintain attitude. Avoid abrupt and large control inputs.

NOTE: Do not extend flap except for approach and landing.

MAXIMUM RECOMMENDED TURBULENT AIR PENETRATION SPEED

At or Below 32000 ft	232 KIAS
Above 32000 ft.....	232 KIAS/0.65M, WHICHEVER IS LOWER

ENGINE DRY MOTORING

Associated Thrust Lever	IDLE
Associated Start/Stop Knob.....	STOP
Associated Ignition Switch	OFF
Associated Start/Stop Knob.....	START, THEN RUN

After 30 seconds:

Associated Start/Stop Knob	STOP
Associated Ignition Switch.....	AUTO

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TRAFFIC COLLISION AND AVOIDANCE SYSTEM – TCAS II

The pilot must not initiate evasive maneuvers using Traffic Advisory information (TA) only, without visually sighting conflicting traffic.

The traffic display and advisories are intended for assistance in visually locating the indicated conflicting traffic.

Maneuvers that are in the opposite direction of the RA are extremely hazardous, especially RAs involving altitude crossing, and thus are prohibited unless it is visually determined to be the only means to assure safe separation.

WARNING: IT IS POSSIBLE IN SOME CASES TO HAVE INSUFFICIENT AIRPLANE PERFORMANCE TO FOLLOW THE TCAS COMMAND WITHOUT FLYING INTO STALL WARNING OR BUFFET. CONDITIONS WHERE THIS MAY OCCUR INCLUDE:

- **BANK ANGLE IN EXCESS OF 15 DEGREES.**
- **OPERATIONS AT AIRPORTS ABOVE 5300 FT MSL OR TEMPERATURES GREATER THAN ISA + 28°C (ISA + 50°F).**
- **ENGINE INOPERATIVE.**
- **FAILURE TO CONFIGURE THE AIRPLANE TO GO-AROUND FOLLOWING A CLIMB RA IN LANDING CONFIGURATION.**
- **FAILURE TO ADVANCE THRUST TO MAX CONTINUOUS THRUST FOLLOWING A CLIMB RA AT REDUCED THRUST.**
- **SPEEDS LESS THAN NORMAL OPERATING SPEED.**
- **ABNORMAL CONFIGURATIONS WHICH REDUCE PERFORMANCE (E.G. GEAR DOWN).**
- **TCAS COMMAND REVERSAL TO A "CLIMB-CLIMB NOW".**
- **BUFFET MARGIN LESS THAN 0.3 G.**

(Continues on the next page)

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WARNING: IF STALL WARNING OCCURS DURING AN RA MANEUVER, IMMEDIATELY ABANDON THE RA AND EXECUTE STALL RECOVERY PROCEDURES. TCAS II WILL CONTINUE TO PROVIDE RAS DURING STALL WARNING AND RECOVERY PROCEDURE.

WARNING: IF HIGH SPEED BUFFET IS ENCOUNTERED WHEN INITIALLY RESPONDING TO AN RA, RELAX PITCH FORCE AS NECESSARY TO REDUCE BUFFET, BUT STILL CONTINUE TO MANEUVER.

NOTE: - The consequences of not following an RA may result in additional RA's in which aural alert and visual annunciations may not agree with each other.

- Using every available means, clear the airspace into which you are going to maneuver. If needed, promptly and smoothly adjust the airplane's pitch to fly into the green rectangle (if displayed), and fly out of the red trapezoidal shaped avoidance zone(s). This should require no more than 0.75 g to 1.25 g maneuver (± 0.25 g).
- Exaggerated responses to TCAS RAs are not desirable or appropriate because of the other potential traffic conflicts and ATC consequences. From level flight, proper response to a TCAS RA typically results in an overall altitude deviation of 300 to 500 ft to resolve a traffic conflict.
- If a CLIMB RA is issued with the airplane in the landing configuration, a normal go-around should be initiated including the appropriate thrust increase and configuration change.
- Compliance with TCAS resolution advisories is required unless the pilot considers it unsafe to do so.
- The pilot should promptly return to the previous ATC clearance after the TCAS voice message "Clear of Conflict" is announced.
- An immediate smooth response to a RA is required to obtain maximum separations. TCAS II algorithms are based on the pilot initiating the initial maneuver within 5 seconds of the RA and within 2 1/2 seconds for additional corrective RA's (increases or reversals). Any delay in responding to RA's will reduce the separations provided.



TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

When a TAWS alert occurs, use the flight controls and thrust as necessary to correct the airplane attitude, flight path and configuration, according to the voice message presented, to provide terrain clearance.

WARNING: IF A TERRAIN AWARENESS WARNING OCCURS, IMMEDIATELY INITIATE AND CONTINUE A CLIMB THAT WILL PROVIDE MAXIMUM TERRAIN CLEARANCE UNTIL ALL ALERTS CEASE. ONLY VERTICAL MANEUVERS ARE RECOMMENDED, UNLESS OPERATING IN VISUAL METEOROLOGICAL CONDITIONS (V_{MC}), AND/OR THE PILOT DETERMINES, BASED ON AVAILABLE INFORMATION, THAT TURNING IN ADDITION TO THE VERTICAL ESCAPE MANEUVER IS A SAFER COURSE OF ACTION.

CAUTION: • FOR TAWS, THE TERRAIN DISPLAY IS INTENDED TO BE USED AS A SITUATIONAL AWARENESS TOOL ONLY AND MAY NOT PROVIDE THE ACCURACY AND/OR FIDELITY ON WHICH TO SOLELY BASE TERRAIN AVOIDANCE MANEUVERING DECISIONS.

- WHEN A TERRAIN AWARENESS CAUTION OCCURS, VERIFY THE AIRPLANE FLIGHT PATH AND CORRECT IT IF REQUIRED. IF IN DOUBT, PERFORM A CLIMB UNTIL THE CAUTION ALERT CEASES.

NOTE: It is the operator's responsibility to ensure that the database in use is valid for the planned route.



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WINDSHEAR

WINDSHEAR AVOIDANCE/PREVENTION

GO-AROUND Procedure..... ACCOMPLISH

WINDSHEAR RECOVERY

- φ Thrust Levers MAX
- φ Autopilot DISENGAGE
- φ Pitch Angle (nose up) 15°

Initially adjust the airplane pitch angle toward 15° (nose up). After that, adjust pitch attitude smoothly in order to achieve the minimum airspeed possible without stall warning activation.

NOTE: Always respect the stall warning aural alarm.

Maintain airplane configuration. Do not change gear and flap position until out of shear and terrain clearance is assured.



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SECTION 4

EMERGENCY AND ABNORMAL PROCEDURES

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INTRODUCTION

The emergency and abnormal procedures contained in this manual have been developed by the airplane manufacturer and approved by the Certification Authority for use in the operation of this airplane.

This Section provides the emergency and abnormal procedures to be performed in case of a system malfunction or failure, in order to protect passengers and/or crew from serious harm and to maintain the airworthiness of the airplane.

The Emergency and Abnormal Procedures, in this Section, are alphabetized and divided into three blocks:

- Non-Annunciated Procedures: procedures, which may not be related to a CAS message but rather to a condition presented in the airplane.
- Annunciated Procedures: procedures related to a CAS message. These procedures are grouped by system and titled with the correspondent CAS message wording.

NOTE: All Warning and Caution CAS messages have an associated procedure in the AFM Section 4. Most of the Advisory CAS messages do not have an associated procedure, as they comprise crew awareness alerts only.

- Smoke Procedures: contain all annunciated and non-annunciated smoke related procedures.

The actions identified with ϕ are recall items. They must be performed expeditiously and from memory. The other actions should be performed as soon as the condition permits.

Some procedures require to LAND AS SOON AS POSSIBLE or LAND AS SOON AS PRACTICAL, refer to Section 1, General for definition.

Conditions resulting in a red, yellow or white message must be corrected prior to flight, unless airplane dispatch is allowed by an approved MEL or another authorized special condition. If messages occur in flight and the safety conditions permit, consideration should be given to landing at an airfield where corrective maintenance can be performed.

The procedures contained herein assume that:

- Normal procedures have been properly accomplished;
- System controls are in normal condition prior to initiation of the associated procedure;
- Airplane systems are operating normally prior to the failure;
- All emergency and abnormal checklists are performed in the order they are presented in the procedure;
- Some emergency or abnormal procedures require a correction in landing distance or landing reference speed. The reference for such correction is always flap FULL, even for those airplanes that do not have flap FULL selection available;
- Aural warnings are silenced as applicable. Master Warning/Caution lights are reset as soon as the failure is recognized;
- Circuit breakers must not be reset, unless is requested by the procedures instruction.



NON-ANNUNCIATED PROCEDURES

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EMERGENCY PROCEDURES

BAGGAGE COMPARTMENT FIRE

LAND AS SOON AS POSSIBLE.

DITCHING

NOTE: The airplane is not certified for ditching. The procedure below is recommended, if required.

During descent:

Passengers.....	PREPARE/LIFE VESTS, IF AVAILABLE, DON
ATC.....	NOTIFY
Transponder	7700
Fasten Seat Belts Signs	ON
ELT	ON
Water Barrier	INSTALL

At 10000 ft, push the DUMP Button in.

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Ditching configuration:

LDG GEAR Lever UP

Flap MAX AVAILABLE

Oxygen Bottle Valve Handle PULL TO CUTOUT

If it is not possible to achieve the selected flap position, maintain
airspeed according to the following:

	FLAP POSITION	V _{REF}
NO ICING	0	V _{REF} FULL NO ICE + 27 KIAS
	1	V _{REF} FULL NO ICE + 19 KIAS
	2 and 3	V _{REF} FULL NO ICE + 6 KIAS
	FULL	V _{REF} FULL NO ICE
IN ICING/ WITH ICE	0	V _{REF} FULL NO ICE + 38 KIAS
	1	V _{REF} FULL NO ICE + 27 KIAS
	2 and 3	V _{REF} FULL NO ICE + 15 KIAS
	FULL	V _{REF} FULL NO ICE + 10 KIAS

NOTE: - If flap stops between two positions, use the minimum
airspeed associated with the next retracted position and
the V_{FE} associated with the next extended position.

- Disregard green circle indication, as it may indicate slower
speeds.

Just before water contact:

Passengers ANNOUNCE
IMPACT

After ditching:

Thrust Levers IDLE

Start/Stop Knobs STOP

Fire Extinguish System AS REQUIRED

EMERGENCY EVACUATION

Procedure ACCOMPLISH



DUAL ENGINE FAILURE

LAND AS SOON AS POSSIBLE.

- ϕ Thrust Levers..... IDLE
 - ϕ Crew Oxygen Masks DON, 100%
 - ϕ Communication..... ESTABLISH
- If both engines are restarted via autorelight:

- Engine Parameters..... MONITOR
- Monitor the engine parameters for remainder of the flight.

If no engine is restarted via autorelight:

- Altitude..... 30000 ft
MAXIMUM
- Airspeed 250 KIAS
MAXIMUM
- XFEED Knob LO2

CAUTION: IF N2 DROPS BELOW 9% IT CANNOT BE RECOVERED BY INCREASING THE AIRPLANE SPEED. IN THIS CASE, WINDMILLING IS NO LONGER POSSIBLE AND AN ASSISTED START SHOULD BE ATTEMPTED.

- NOTE:** - Start the engines one at a time, initiating the in-flight start sequence through the right engine, if possible.
- In electrical emergency condition, the 45 minutes batteries duration was demonstrated with only one assisted engine start attempt. Further assisted start attempts decrease battery duration.

If N2 is above 9% and airspeed is above 200 KIAS, a windmilling start may be accomplished as following:

- Start/Stop Knob STOP
- Start/Stop Knob RUN

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If N2 is at 9% or below accomplish an assisted start as following:

Start/Stop Knob STOP
Start/Stop Knob START THEN
RUN

If the dual engine failure condition persists:

FORCED LANDING or
DITCHING Procedure ACCOMPLISH

If at least one engine is recovered:

XFEED Knob AS REQUIRED
Engine Parameters MONITOR
Monitor the engine parameters for remainder of the flight.

If applicable:

ONE ENGINE INOPERATIVE APPROACH
AND LANDING Procedure ACCOMPLISH



EMERGENCY DESCENT

ϕ Fasten Seat Belts Signs	ON
ϕ Altitude.....	10000 ft OR MEA, WHICHEVER IS HIGHER
ϕ Thrust Levers.....	IDLE
ϕ SPEED BRAKE Switch	OPEN
ϕ Airspeed	250 KIAS/M _{MO} MAXIMUM
ϕ LDG GEAR Lever.....	DN
ϕ Transponder	7700
ϕ ATC	NOTIFY
At 10000 ft or MEA:	
SPEED BRAKE Switch.....	CLOSE

EMERGENCY EVACUATION

ϕ Thrust Levers.....	IDLE
ϕ Emergency/Parking Brake.....	ON
ϕ Start/Stop Knobs	STOP
ϕ Fire Shutoff Button	PUSH IN
ϕ PRESN MODE Switch.....	MAN
ϕ DUMP Button.....	PUSH IN
ϕ ATC	NOTIFY
ϕ Evacuation.....	PERFORM
ϕ BATT 1 & 2 Switches	OFF

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ENGINE ABNORMAL START

ϕ Start/Stop Knob (affected engine) STOP

On ground, if an engine dry motoring is considered:

ENGINE DRY MOTORING Procedure ACCOMPLISH

ENGINE FIRE, SEVERE DAMAGE OR SEPARATION

ϕ Thrust Lever (affected engine) IDLE

ϕ Start/Stop Knob (affected engine) STOP

ϕ Fire Shutoff Button (affected engine)..... PUSH IN

LAND AS SOON AS POSSIBLE.

Wait 30 seconds and if fire persists:

BOTTLE Switch DISCH

Altitude 36000 ft
MAXIMUM

ECS Knob UNAFFECTED
SIDE

Thrust Lever (operative engine)..... AS REQUIRED

Reduce airspeed as required to keep vibration within acceptable levels.

If high vibration exists:

Airspeed REDUCE
(140 KIAS
MINIMUM)

When appropriate:

ONE ENGINE INOPERATIVE APPROACH
AND LANDING Procedure ACCOMPLISH

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FORCED LANDING

This procedure is recommended for landings with all engines inoperative or for landings on unprepared surfaces. Landing on unprepared surfaces is not recommended. However, if specific circumstances render such landing inevitable, accomplish the procedures below:

- Passengers..... PREPARE
- ATC..... NOTIFY
- Transponder 7700
- Fasten Seat Belts Signs ON

- ELT ON
- Landing Data SET

At 10000 ft push in the DUMP Button and set the altimeters during approach.

For landing configuration:

- LDG GEAR Lever AS REQUIRED

NOTE: The decision to land with gear up or extended is left to pilots. The choice of configuration is based on airplane load distribution, controllability and conditions of the landing field.

- Flap..... MAXIMUM
AVAILABLE

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If it is not possible to achieve the selected flap position, maintain airspeed according to the following:

	FLAP POSITION	V _{REF}
NO ICING	0	V _{REF} FULL NO ICE + 27 KIAS
	1	V _{REF} FULL NO ICE + 19 KIAS
	2 and 3	V _{REF} FULL NO ICE + 6 KIAS
	FULL	V _{REF} FULL NO ICE
IN ICING/ WITH ICE	0	V _{REF} FULL NO ICE + 38 KIAS
	1	V _{REF} FULL NO ICE + 27 KIAS
	2 and 3	V _{REF} FULL NO ICE + 15 KIAS
	FULL	V _{REF} FULL NO ICE + 10 KIAS

NOTE: - If flap stops between two positions, use the minimum airspeed associated with the next retracted position and the V_{FE} associated with the next extended position.

- Disregard green circle indication, as it may indicate slower speeds.

Oxygen Bottle Valve Handle..... PULL TO CUTOUT

Just before touchdown:

Passengers..... ANNOUNCE
IMPACT

Reduce the vertical speed as much as possible.

After landing, if conditions permit:

Thrust Levers..... IDLE
Start/Stop Knobs..... STOP
Fire Shutoff Buttons..... PUSH IN

EMERGENCY EVACUATION

Procedure ACCOMPLISH

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FUEL LEAKAGE

LAND AS SOON AS POSSIBLE.

- XFEED Knob OFF
- Thrust Levers SAME N1 SET ON BOTH ENGINES

If N1 and fuel flow indications from both engines match, but one side is leaking:

- Asymmetric Thrust..... AS REQUIRED TO CONTROL FUEL IMBALANCE

If the fuel leakage continues and is severe:

- Affected Engine Shutdown CONSIDER

If N1 from both engines match, but fuel flow indications are discrepant:

- Affected Engine SHUTDOWN

If engine shutdown is performed:

- Affected Engine Fire Shutoff Button PUSH IN

REJECTED TAKEOFF (AT OR BELOW V₁)

- ⌀ Thrust Levers..... IDLE
- ⌀ Brakes APPLY MAXIMUM
- ⌀ Directional Control MAINTAIN

Immediately after stopping:

- ENGINE SHUTDOWN or
ENGINE FIRE, SEVERE DAMAGE OR
SEPARATION Procedure AS REQUIRED



TAKEOFF WITH ENGINE FAILURE AT OR ABOVE V_1

ϕ At V_R rotate the airplane according to following table:

FLAP POSITION	1	2
PITCH ANGLE	10.5°	8°

ϕ With positive rate of climb:

ϕ LDG GEAR Lever UP

ϕ Airspeed V_2

Maintain V_2 adjusting the pitch using CWS as required.

CAUTION: DO NOT USE FLIGHT DIRECTOR TAKEOFF MODE IN OEI CONDITION. ADJUST THE PITCH MANUALLY TO MAINTAIN V_2 AND USE CWS TO SYNCHRONIZE THE FLIGHT DIRECTOR.

At the acceleration altitude, set the flap UP position.

NOTE: For takeoff with flap 2, select the flap 1 position before setting to UP.

Accelerate the airplane to the final segment speed (V_{FS}) and set the engine thrust rate to continuous.

NOTE: After reaching the minimum airspeed, maintain the airspeed regardless of the pitch angle.

After retracting flap, accomplish the ENGINE 1 (2) FAILURE Procedure.

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ABNORMAL PROCEDURES

ABNORMAL LANDING GEAR EXTENSION

LDG GEAR Lever DN

Free Fall Handle PULL

If gear does not lock down:

Airspeed 140 KIAS
MINIMUM

Slip the airplane to attempt to lock the main gear.

Check landing gear indications. If landing gear indications are still not confirmed down, accomplish the PARTIAL OR GEAR UP LANDING Procedure.

BLANK DISPLAY UNIT WITH AUTOMATIC REVERSION

Cycle the DISPLAY BACKUP button (OUT then IN).

If the display does not revert to the normal condition or an abnormal behavior occurs, press the DISPLAY BACKUP button again.

NOTE: Only one attempt to return to a normal condition is allowed.

BLANK DISPLAY UNIT WITHOUT AUTOMATIC REVERSION

Press the DISPLAY BACKUP button and select the reversionary mode, as required.



DUAL DUCT OVERTEMPERATURE

ECS Knob OFF VENT

If above 10000 ft:

Crew Oxygen Mask DON, 100%

Communication ESTABLISH

Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

Wait 1 minute after the ECS Knob is positioned on OFF VENT.

If DUCT 1 OVERTEMP and/or DUCT 2 OVERTEMP messages still persist:

BLEED 1 & 2 Switches OFF

Leave and avoid icing conditions.

At 10000 ft or MEA:

DUMP Button PUSH IN

LAND AS SOON AS PRACTICAL.

EMERGENCY BRAKING TECHNIQUE

When applying emergency brakes, pull the handle gently until emergency/parking brake light is on. From this point, reposition of the emergency brake handle may be necessary to keep an adequate deceleration, depending on runway conditions and remaining runway length.

CAUTION: ANTISKID PROTECTION IS NOT AVAILABLE FOR EMERGENCY/PARKING BRAKE. SKIDDING THE TIRES DOES NOT DECREASE STOPPING DISTANCE AND MAY CAUSE TIRE BLOWOUT.

NOTE: The emergency/parking brake accumulator allows 6 actuations.



ENGINE ABNORMAL VIBRATION

CAUTION: IF ENGINE ABNORMAL VIBRATION IS ASSOCIATED WITH ONE OF THE FOLLOWING CAS MESSAGES: BATTERY DISCHARGE, BATTERY EXCEEDANCE, GENERATOR 1 (2) OFF BUS, PERFORM THE ASSOCIATED PROCEDURE INSTEAD OF THIS ONE.

Reduce the associated thrust lever to IDLE in order to keep vibration within acceptable levels.

NOTE: Reduce the thrust levers one at a time to attempt to identify the affected engine.

If the engine abnormal vibration persists, accomplish the ENGINE SHUTDOWN Procedure.

CAUTION: CONTINUOUS HIGH VIBRATION MAY DAMAGE THE ENGINE.

ENGINE IN-FLIGHT START

Check the ENGINE IN-FLIGHT START ENVELOPE and proceed as follows:

CAUTION: • IF N2 DROPS BELOW 9% IT CANNOT BE RECOVERED BY INCREASING THE AIRPLANE SPEED. IN THIS CASE, WINDMILLING IS NO LONGER POSSIBLE AND AN ASSISTED START SHOULD BE ATTEMPTED.

- SHOULD THE ENGINE FAIL TO LIGHT-UP WITHIN 10 SECONDS OF FUEL ON, OR ACHIEVE STABILIZED FLIGHT IDLE WITHIN 60 SECONDS FROM FUEL ON, THE ENGINE START IS DEEMED TO HAVE BEEN UNSUCCESSFUL AND MUST BE ABORTED MANUALLY.

(Continues on the next page)



(Continued from the previous page)

Assisted engine airstart:

N2 (operative engine) MIN 72%

Inoperative engine:

Thrust Lever IDLE
Start/Stop Knob STOP
Associated FUEL PUMP Switch..... ON
Start/Stop Knob START, THEN
RUN

Windmilling engine airstart:

Inoperative engine:

N2 MIN 9%
Thrust Lever IDLE
Start/Stop Knob STOP
Associated FUEL PUMP Switch..... ON
Start/Stop Knob RUN

If the engine is restarted normally:

Thrust Levers AS REQUIRED
FUEL PUMP Switches AUTO

If an abnormal start occurs:

Start/Stop Knob (affected engine) STOP
Thrust Lever (operative engine) AS REQUIRED
Fuel BALANCE

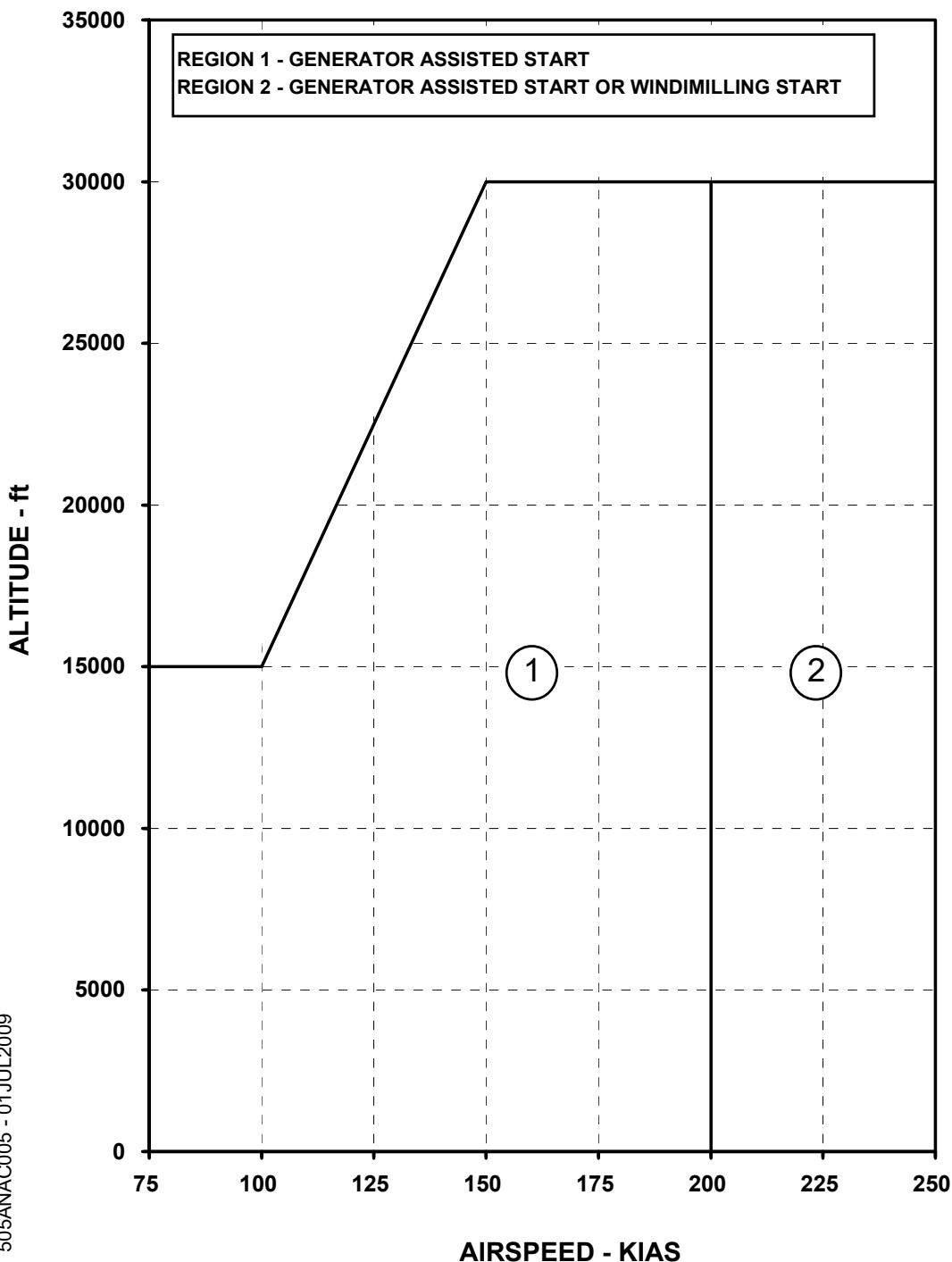
When appropriate:

ONE ENGINE INOPERATIVE APPROACH
AND LANDING Procedure ACCOMPLISH

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ENGINE IN-FLIGHT START ENVELOPE



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ENGINE ITT OVERTEMPERATURE

The ITT gray quadrant indication becomes red.

Associated Thrust Lever REDUCE

If the affected engine is still out of limits:

ENGINE SHUTDOWN Procedure ACCOMPLISH

ENGINE SHUTDOWN

LAND AS SOON AS POSSIBLE.

Thrust Lever (affected engine)..... IDLE

Start/Stop Knob (affected engine)..... STOP

Thrust Lever (operative engine)..... AS REQUIRED

Fuel BALANCE

Airspeed 140 KIAS
MINIMUM

If in icing conditions:

ONE ENGINE INOPERATIVE

IN ICING CONDITIONS Procedure ACCOMPLISH

When appropriate:

ONE ENGINE INOPERATIVE APPROACH

AND LANDING Procedure ACCOMPLISH

FUEL TEMPERATURE INDICATION FAILURE

If in flight, descend to a lower altitude if external temperature is below -37°C.

FUEL TEMPERATURE OUT OF RANGE

In flight:

If fuel temperature is below -37°C descend to lower altitudes.

If fuel temperature is above 80°C:

LAND AS SOON AS PRACTICAL.



GEAR LEVER CANNOT BE MOVED UP

If possible, wait until 400 ft height is achieved to verify if the LG WOW SYS FAIL message is presented.

If the message is not presented, simultaneously proceed:

DN LCK REL Button PRESS

LDG GEAR Lever UP

WARNING: IF CLIMB PERFORMANCE IS REQUIRED TO CLEAR OBSTACLES DURING TAKEOFF OR GO-AROUND, MOVE LANDING GEAR LEVER TO UP POSITION AT ANY TIME.

IMPAIRED OR CRACKED WINDSHIELD

Affected WSHLD Switch OFF

Airspeed 220 KIAS
MAXIMUM

Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

During descent:

PRESN MODE Switch MAN

Cabin Altitude INCREASE

Maintain cabin altitude at 10000 ft.

At or below 10000 ft:

ECS Knob OFF VENT

INADVERTENT PUSHER ACTUATION

φ PUSHER CUTOUT Button PUSH IN

Avoid side slipping and high angles of attack.

NOTE: Low Speed Awareness cues (yellow and red band) and Stall aural warning will still be available.

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LOSS OF PRESSURIZATION INDICATION

PRESN MODE Switch MAN

During pressurization manual controlling, use the table below as a reference when cabin altitude or cabin delta pressure is not being presented.

AIRPLANE ALTITUDE (ft)	CABIN ALTITUDE (ft)	DIFFERENTIAL PRESSURE (PSID)
10000	700	4.2
11000	800	4.6
12000	900	4.9
13000	1000	5.2
14000	1100	5.5
15000	1200	5.8
16000	1300	6.1
17000	1400	6.3
18000	1600	6.5
19000	1700	6.8
20000	1800	7.0
21000	1900	7.2
22000	2100	7.4
23000	2300	7.6
24000	2400	7.8
25000	2600	7.9
26000	2800	8.1
27000	3000	8.2
28000	3100	8.3
29000	3300	8.5
30000	3400	8.6
31000	3600	8.7
32000	3800	8.8
33000	4000	8.9
34000	4200	9.0
35000	4300	9.1
36000	4600	9.1
37000	4800	9.2
38000	5100	9.2
39000	5400	9.2
40000	5700	9.2
41000	6000	9.2
42000	6200	9.2
43000	6400	9.3
44000	6500	9.3
45000	6600	9.4

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ONE ENGINE INOPERATIVE APPROACH AND LANDING

During descent:

Landing Speeds (V_{REF} , V_{AC} , V_{FS})..... SET

CAUTION: SET THE SPEEDS IN SEQUENCE. IF IT IS NECESSARY TO CORRECT ONE OF THE SPEEDS OUT OF THE SEQUENCE, ALL THE OTHERS MUST BE VERIFIED.

Landing Field Elevation SET

Approach Aids SET

Altimeters..... SET/CHECK

Landing configuration:

Yaw Damper OFF

LDG GEAR Lever DN

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V_{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	3	V_{REF} FULL NO ICE + 12 KIAS	1.25
IN ICING/ WITH ICE	3	V_{REF} FULL NO ICE + 22 KIAS	1.45

(Continues on the next page)



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If a go-around is required:

TO/GA Button PRESS

NOTE: - For airplanes with G3000 system version prior to 3305.07, when the TO/GA Button is pressed in the air, the Flight Director activates GA and ROL (Wings Level) modes.

- For airplanes with G3000 system version 3305.07 and on, when pressing the TO/GA Button in the air, during an approach with navigation or approach mode, automatic activation of FMS mode occurs, provided the approach procedure is previously loaded in the flight plan. Otherwise, GA and ROL (Wings Level) modes will be engaged.

Thrust Levers TO/GA

Rotate the airplane to 8° nose up.

Flap 1

With positive rate of climb:

LDG GEAR Lever UP

Airspeed APPROACH
CLIMB SPEED



ONE ENGINE INOPERATIVE IN ICING CONDITIONS

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

XBLEED Knob AUTO

Minimum airspeed:

Wing and Stabilizer Anti-Ice System
Inhibited (flap and gear up) 165 KIAS

Wing and Stabilizer Anti-Ice System
Armed (flap and gear up) 165 KIAS

Wing and Stabilizer Anti-Ice System
Uninhibited (flap and gear up)..... 150 KIAS

Altitude 15000 ft
MAXIMUM

N1 ABOVE WAI BUG

If it is not possible to descend below 15000 ft:

Icing Conditions EXIT/AVOID

OXYGEN LEAKAGE

Leakage may be identified through the pressure indication on MFD.

Communicate the passengers that smoking is not allowed.

Altitude 14000 ft OR MEA,
WHICHEVER IS
HIGHER

Oxygen Bottle Valve Handle..... PULL TO CUTOUT

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PARTIAL OR GEAR UP LANDING

Burn off fuel to reduce touchdown speed.

Passengers NOTIFY

Prior to approach:

DUMP Button PUSH IN

LDG GEAR Lever AS REQUIRED

NOTE: The decision to land with all gear up or with any gear extended is left to the crew. The choice of configuration is based on the number of gear available, airplane load distribution, controllability and conditions of the landing field.

Just before touchdown:

Passengers ANNOUNCE

Maximum Vertical Speed 300 ft/MIN

NOTE: Reduce the vertical speed as much as possible at the touchdown.

After landing:

Thrust Levers IDLE

Start/Stop Knobs STOP

Fire Extinguishing System AS REQUIRED

EMERGENCY EVACUATION Procedure ACCOMPLISH

If a go-around is required, maintain the chosen landing gear configuration.

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STRUCTURAL DAMAGE

LAND AS SOON AS POSSIBLE.

Limit the airspeed to 170 KIAS maximum.

If the fuselage is damaged:

Altitude..... 10000 ft OR MEA,
WHICHEVER IS
HIGHER

When reaching 10000 ft:

DUMP Button PUSH IN

ECS Knob..... OFF VENT

Establish landing configuration early.

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AIRPLANE GENERAL
(DOORS, BAGGAGE COMPARTMENT
AND LIGHTING)

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DOOR EMERGENCY OPEN

Fasten Seat Belts Signs ON

LAND AS SOON AS POSSIBLE.

Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

DOOR PASSENGER OPEN

Fasten Seat Belts Signs ON

LAND AS SOON AS POSSIBLE.

Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

At 10000 ft or MEA:

DUMP Button..... PUSH IN

ECS Knob OFF VENT

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DOOR BAGGAGE AFT OPEN

LAND AS SOON AS PRACTICAL.

Airspeed REDUCE TO
160 KIAS

DOOR BAGGAGE LH (RH) OPEN

LAND AS SOON AS PRACTICAL.

Airspeed REDUCE TO
160 KIAS

EMERGENCY LIGHT NOT ARMED

EMER LT Switch ARM

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AMS (AIR CONDITIONING, PNEUMATICS AND PRESSURIZATION)

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CABIN ALTITUDE HIGH

- φ Crew Oxygen Masks DON, 100%
- φ Communication..... ESTABLISH
- φ Fasten Seat Belts Signs ON

- φ Altitude..... 10000 ft OR MEA,
WHICHEVER IS
HIGHER
- φ Thrust Levers..... IDLE
- φ SPEED BRAKE Switch OPEN
- φ Airspeed 250 KIAS/M_{MO}
MAXIMUM
- φ LDG GEAR Lever..... DN
- φ Transponder 7700
- φ ATC NOTIFY

If passenger masks are not deployed automatically or PAX OXY NO PRES message is displayed:

Oxygen SUPPLY CONTROL Knob..... PAX OVRD

At 10000 ft or MEA:

- DUMP Button..... PUSH IN
- ECS Knob..... OFF VENT
- SPEED BRAKE Switch..... CLOSE
- LDG GEAR Lever..... AS REQUIRED

If landing is not imminent, consider retracting landing gear for fuel saving.



AMS CONTROLLER FAILURE

If above 10000 ft:

Crew Oxygen Masks DON, 100%
Communication ESTABLISH
Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

Leave and avoid icing conditions.

At 10000 ft or MEA:

DUMP Button PUSH IN
ECS Knob OFF VENT

BLEED 1 (2) FAILURE

Affected BLEED Switch OFF

Wait 1 minute.

Affected BLEED Switch AUTO

If the BLEED 1 (2) FAIL message persists:

Affected BLEED Switch OFF
Altitude 36000 ft
MAXIMUM

If below 30000 ft:

Icing Conditions EXIT/AVOID

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BLEED 1 (2) LEAKAGE

Affected BLEED Switch	OFF
XBLEED Knob	OFF
ECS Knob	UNAFFECTED SIDE
WINGSTAB Switch	OFF
Altitude	36000 ft MAXIMUM

If below 30000 ft:

Icing Conditions EXIT/AVOID

If the associated BLEED OFF message is not displayed or BLEED LEAK message persists after 3 minutes:

Crew Oxygen Masks	DON, 100%
Communication.....	ESTABLISH
Associated Thrust Lever.....	IDLE
Opposite BLEED Switch.....	OFF
Altitude.....	10000 ft OR MEA, WHICHEVER IS HIGHER

At 10000 ft or MEA:

DUMP Button PUSH IN
ECS Knob..... OFF VENT

If the associated BLEED LEAK message still persists:

Associated Engine..... SHUTDOWN

When applicable:

ONE ENGINE INOPERATIVE APPROACH AND LANDING Procedure..... ACCOMPLISH



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BLEED 1 (2) OVERPRESSURE

Affected BLEED Switch OFF

Wait 30 seconds.

Affected BLEED Switch AUTO

If the BLEED 1 (2) OVERPRES message persists:

Affected BLEED Switch OFF

XBLEED Knob OFF

ECS Knob UNAFFECTED
SIDE

Altitude 36000 ft
MAXIMUM

If below 30000 ft:

Icing Conditions EXIT/AVOID

CABIN DELTA PRESSURE FAILURE

If the cabin differential pressure indication is positive:

PRESN MODE Switch MAN

CABIN ALT Switch..... UP

If unable to modulate cabin altitude:

ECS Knob OFF VENT

If above 10000 ft:

Crew Oxygen Masks DON, 100%

Communication ESTABLISH

Altitude..... 10000 ft OR MEA,
WHICHEVER IS
HIGHER

At 10000 ft or MEA:

DUMP Button PUSH IN

If the indication is negative:

Airplane Descent Rate REDUCE



DUCT 1 (2) OVERTEMPERATURE

If both DUCT 1 OVERTEMP and DUCT 2 OVERTEMP messages are displayed:

DUAL DUCT OVERTEMPERATURE

Procedure ACCOMPLISH
AIR CONDITIONING MODE Switch MAN
AIR CONDITIONING TEMP Switch COLD FOR
5 SECONDS

Wait 1 minute.

If the DUCT 1 (2) OVERTEMP message extinguishes:

AIR CONDITIONING MODE Switch AUTO
Adjust the temperature as required.

If the DUCT 1 (2) OVERTEMP message persists:

ECS Knob UNAFFECTED
SIDE
Altitude 36000 ft
MAXIMUM

Wait 1 minute after the ECS Knob is positioned on UNAFFECTED SIDE.

If the DUCT 1 (2) OVERTEMP message still persists:

Associated BLEED Switch OFF
XBLEED Knob OFF

If below 30000 ft:

Icing Conditions EXIT/AVOID

Wait 1 minute.

If the DUCT 1 (2) OVERTEMP message persists:

Associated Thrust Lever IDLE

When applicable:

ONE ENGINE INOPERATIVE APPROACH
AND LANDING Procedure ACCOMPLISH

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ELECTRONIC BAY LEAKAGE

ECS Knob OFF VENT

If above 10000 ft:

Crew Oxygen Mask DON, 100%

Communication ESTABLISH

Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

At 10000 ft or MEA:

DUMP Button PUSH IN

If the EBAY LEAK message persists:

BLEED 1 & 2 Switches OFF

Leave and avoid icing conditions.

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ELECTRONIC BAY OVERHEAT

AIR CONDITIONING MODE Switch..... MAN
 AIR CONDITIONING TEMP Switch COLD FOR
 5 SECONDS

Wait 90 seconds.

If the EBAY OVHT message extinguishes:

AIR CONDITIONING MODE Switch AUTO
 CABIN TEMP Knob FULL COLD

If the EBAY OVHT message persists:

Non-Essential Equipment..... OFF

Wait 5 minutes and if the EBAY OVHT message persists:

Altitude 10000 ft OR MEA,
 WHICHEVER IS
 HIGHER

At 10000 ft or MEA:

DUMP Button PUSH IN
 ECS Knob OFF VENT
 LAND AS SOON AS PRACTICAL.



ENVIRONMENTAL CONTROL SYSTEM 1 (2) VALVE FAILURE

If the opposite side ECS is available:

ECS Knob UNAFFECTED
SIDE

Wait 1 minute.

ECS Knob BOTH

Temp Control Knob (Cockpit and Cabin) FULL HOT

Wait 2 minutes.

If the message persists:

ECS Knob UNAFFECTED
SIDE

Temp Control Knob (Cockpit and Cabin) OPERATE
NORMALLY

Altitude MAX 36000 FT

If the message disappears:

Temp Control Knob (Cockpit and Cabin) OPERATE
NORMALLY

If opposite side ECS is unavailable:

Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

ECS Knob OFF VENT



PRESSURIZATION AUTO FAILURE

PRESN MODE Switch MAN, THEN AUTO

If the PRESN AUTO FAIL message persists and the manual mode is operative:

In case of single pilot operation and above 10000 ft:

Crew Oxygen Mask	DON, NORMAL
Communication	ESTABLISH
PRESN MODE Switch.....	MAN
CABIN ALT Switch	AS REQUIRED
Altitude.....	25000 ft MAXIMUM
Cabin Pressurization Indications	MONITOR

If cabin altitude increases above 10000 ft:

CABIN ALTITUDE HIGH Procedure ACCOMPLISH

When appropriate before landing:

CABIN ALT Switch HOLD UP FOR
15 SECONDS

If the PRESN AUTO FAIL message persists and the manual mode is inoperative:

Crew Oxygen Masks	DON, 100%
Communication.....	ESTABLISH
ECS Knob.....	OFF VENT
Altitude.....	10000 ft OR MEA, WHICHEVER IS HIGHER

RAM AIR FAILURE

Report to maintenance after landing.

If associated with both ECS or bleed valves failure, LAND AS SOON AS PRACTICAL.



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AUTOMATIC FLIGHT

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AUTOPILOT FAILURE

Autopilot..... DISCONNECT

Fly the airplane manually and refer to the RVSM Operation Supplement, if in RVSM airspace.

If in single pilot operation, LAND AS SOON AS PRACTICAL.

AUTOPILOT PITCH MISTRIM

Control Wheel HOLD FIRMLY

QUICK DISCONNECT Button PRESS

Pitch Trim..... AS REQUIRED

If normal pitch trim is inoperative:

PITCH BKP MODE Switch BKP

AUTOPILOT ROLL MISTRIM

Roll Trim AS REQUIRED

AUTOMATIC PITCH TRIM FAILURE

Control Wheel HOLD FIRMLY

QUICK DISCONNECT Button PRESS

Pitch Trim..... AS REQUIRED

If normal pitch trim is inoperative:

PITCH BKP MODE Switch BKP

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YAW DAMPER FAILURE

If in icing conditions:

Airspeed 180 KIAS
MAXIMUM

If associated with VENTRAL RUD FAIL message:

If above 25000 ft:

Rudder Pedals FIXED
Thrust Levers IDLE
SPEED BRAKE Switch OPEN
Airspeed 210 KIAS
MAXIMUM
Control Wheel SEQUENCES OF
STEP INPUTS TO
REDUCE ROLL
OSCILLATIONS
Altitude 25000 ft
MAXIMUM

CAUTION: SEVERITY OF OSCILLATIONS DECREASES WITH
DECREASING ALTITUDE.

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If below 25000 ft:

Maintain the airspeed according to the table below:

ALTITUDE	AIRSPEED
BELOW 15000 FT	AS REQUIRED
FROM 15000 TO 20000 FT	200 KIAS MINIMUM
FROM 20000 TO 25000 FT	290 KIAS MINIMUM

NOTE: - Values above are also valid for icing conditions.

- The following failures will cause both the Ventral Rudder and Yaw Damper to be inoperative:
 - Electrical Emergency;
 - DC BUS 2 Off;
 - Dual AHRS Failure;
 - Dual ADS Failure;
 - ADS and AHRS Miscompare.

YAW DAMPER MISTRIM

Yaw Trim..... AS REQUIRED



VENTRAL RUDDER FAILURE

Yaw Damper ENGAGE

If associated with YD FAIL message:

If above 25000 ft:

Rudder Pedals FIXED
Thrust Levers IDLE
SPEED BRAKE Switch OPEN
Airspeed 210 KIAS
MAXIMUM
Control Wheel SEQUENCES OF
STEP INPUTS TO
REDUCE ROLL
OSCILLATIONS
Altitude 25000 ft
MAXIMUM

CAUTION: SEVERITY OF OSCILLATIONS DECREASES WITH
DECREASING ALTITUDE.

If below 25000 ft:

Maintain the airspeed according to the table below:

ALTITUDE	AIRSPEED
BELOW 15000 FT	AS REQUIRED
FROM 15000 TO 20000 FT	200 KIAS MINIMUM
FROM 20000 TO 25000 FT	290 KIAS MINIMUM

NOTE: - Values above are also valid for icing conditions.

- The following failures will cause both the Ventral Rudder and Yaw Damper to be inoperative:
 - Electrical Emergency;
 - DC BUS 2 Off;
 - Dual AHRS Failure;
 - Dual ADS Failure;
 - ADS and AHRS Miscompare.



ELECTRICAL

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ELECTRICAL EMERGENCY

- φ PRESN MODE Switch MAN
- φ CABIN ALT Switch HOLD DN FOR 10 SECONDS

If above 10000 ft:

- Rudder Pedal (above 25000 ft) FIXED
- CABIN ALTITUDE HIGH Procedure ACCOMPLISH

At 10000 ft or MEA:

- DUMP Button..... PUSH IN
- ECS Knob OFF VENT
- LDG GEAR Lever AS REQUIRED

Reset both generators.

If message persists:
LAND AS SOON AS POSSIBLE.

CAUTION: • WITH VENTRAL RUDDER AND YAW DAMPER FAILED SIMULTANEOUSLY, LATERAL/DIRECTIONAL OSCILLATIONS MAY OCCUR. IN THIS CASE, KEEPING A 20° TURN BANK HELP TO DAMP THE OSCILLATION.

- SEVERITY OF OSCILLATIONS DECREASES WITH DECREASING ALTITUDE.
- BATTERY DURATION IS 45 MINUTES MAXIMUM.

- EMER LT Switch OFF
- Airplane..... FLY MANUALLY
- Icing Conditions EXIT/AVOID

Confirm that IESI has reverted. If not, select ADSTBY on PFD.

Avoid side slipping the airplane and high angle of attack.

- Before landing:
- EMER LT Switch (if required)..... ON

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When landing maintain airspeed according to the following:

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	0 and 1	V _{REF FULL NO ICE} + 17 KIAS	1.35
	2 and 3		1.60
	FULL		1.70
IN ICING/ WITH ICE	0 and 1	V _{REF FULL NO ICE} + 28 KIAS	1.55
	2 and 3		1.90
	FULL		2.15

NOTE: - If flap stops between two positions, use the minimum airspeed associated with the next retracted position and the V_{FE} associated with the next extended position.

- Disregard green circle indication, as it may indicate slower speeds.

If a go-around is required, maintain the minimum airspeed presented in the applicable flap configuration from the table above, until the acceleration altitude is reached.

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The list below presents the relevant inoperative equipment/functions:

- | | |
|------------------------|--------------------------|
| - AHRS 2 | - Navigation/Taxi Lights |
| - Air Conditioning | - Passenger Signs |
| - AMS Controller | - PFD 2 |
| - Anti-Ice System | - Pitch Trim System |
| - Audio Panel 2 | - Pressurization Auto |
| - Autopilot | - Red Beacon |
| - Baggage Heater | - Roll Trim |
| - DME 1/2 | - Rudder Booster |
| - Engine 1/2 Flowmeter | - Stall Protection |
| - Flap System | - Stall Warning |
| - Flight Director 2 | - Spoiler System |
| - FMS Panel | - Strobe Lights |
| - Fuel Pump 2 | - Transponder 2 |
| - GIA 2 | - Ventral Rudder |
| - GPS 2/VOR 2/ILS 2 | - Windshield Heater 1/2 |
| - IASP 1/2 | - WX Radar |
| - Ice Detector | - Yaw Damper |
| - Landing Lights | - Yaw Trim |

ELECTRICAL EMERGENCY TRANSFER FAIL

⌀ ELEC EMER Button PUSH IN

LAND AS SOON AS POSSIBLE.

If transference is confirmed:

ELECTRICAL EMERGENCY Procedure ACCOMPLISH

(Continues on the next page)



(Continued from the previous page)

If transference is still failed:

Electrical Load REDUCE

CAUTION: IF EITHER PFD 2 OR MFD IS STILL ENERGIZED WITH ELEC XFR FAIL MESSAGE SHOWN, BATTERY DURATION IS LESS THAN 45 MINUTES.

BATTERY 1 (2) OFF BUS

Associated BATT Switch..... OFF, THEN ON

If message persists:

Associated BATT Switch ON

BATTERY DISCHARGE

LAND AS SOON AS PRACTICAL.

Check BATT 1 and BATT 2 voltage on MFD.

If BATT 1 or BATT 2 voltage is out of range (yellow color):

Associated GEN Switch..... OFF

If the message persists:

LAND AS SOON AS POSSIBLE.

ELEC EMER Button..... PUSH IN

CAUTION: BATTERY DURATION IS 45 MINUTES.

CAUTION: MONITOR THE ENGINE VIBRATION AND NOISE FOR REMAINDER OF THE FLIGHT. IF THERE IS ANY ENGINE ABNORMAL VIBRATION AND/OR NOISE, SHUTDOWN THE ASSOCIATED ENGINE.

If there is any engine abnormal vibration and/or noise, shutdown the associated engine:

ENGINE SHUTDOWN Procedure ACCOMPLISH



BATTERY EXCEEDANCE

LAND AS SOON AS PRACTICAL.

Check BATT 1 and BATT 2 voltage on MFD.

If BATT 1 or BATT 2 voltage is out of range (yellow color):

Associated GEN Switch..... OFF

If the message persists:

LAND AS SOON AS POSSIBLE.

ELEC EMER Button PUSH IN

CAUTION: BATTERY DURATION IS 45 MINUTES.

CAUTION: MONITOR THE ENGINE VIBRATION AND NOISE FOR REMAINDER OF THE FLIGHT. IF THERE IS ANY ENGINE ABNORMAL VIBRATION AND/OR NOISE, SHUTDOWN THE ASSOCIATED ENGINE.

If there is any engine abnormal vibration and/or noise, shutdown the associated engine:

ENGINE SHUTDOWN Procedure..... ACCOMPLISH

GENERATOR 1 (2) OFF BUS

Associated GEN Switch..... OFF, THEN AUTO

If the message persists:

LAND AS SOON AS PRACTICAL.

Associated GEN Switch..... OFF

CAUTION: MONITOR THE ENGINE VIBRATION AND NOISE FOR REMAINDER OF THE FLIGHT. IF THERE IS ANY ENGINE ABNORMAL VIBRATION AND/OR NOISE, SHUTDOWN THE ASSOCIATED ENGINE.

If there is any engine abnormal vibration and/or noise, shutdown the associated engine:

ENGINE SHUTDOWN Procedure..... ACCOMPLISH



GENERATOR OVERLOAD

AIR CONDITIONING MODE Switch OFF

If the message persists:

WSHLD 2 Switch OFF

If the message persists:

Electrical Load REDUCE

GENERATOR START FAULT

BATT 2 Switch OFF

ELEC EMER Button PUSH IN

ELECTRICAL EMERGENCY Procedure ACCOMPLISH

CAUTION: BATTERY DURATION IS 45 MINUTES.

DC BUS 1 OFF

Airplane FLY MANUALLY

Fly the airplane manually. If in RVSM airspace, notify the ATC and co-ordinate an appropriate action plan.

Flight Director SELECT FD2

Icing Conditions EXIT/AVOID

Altitude 36000 ft
MAXIMUM

Avoid side slipping the airplane and high angle of attack.

(Continues on the next page)



(Continued from the previous page)

When landing maintain airspeed according to the following:

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	0 and 1	V _{REF FULL NO ICE} + 17 KIAS	1.30
	2 and 3		1.35
	FULL		1.35
IN ICING/ WITH ICE	0 and 1	V _{REF FULL NO ICE} + 28 KIAS	1.50
	2 and 3		1.60
	FULL		1.60

NOTE: - If flap stops between two positions, use the V_{FE} associated with the next extended position.

- Disregard green circle indication, as it may indicate slower speeds.

The list below presents the relevant inoperative equipment/functions:

- AMS Controller Channel A
- Autopilot
- Baggage Heater
- Cockpit Evaporator
- Cockpit FCSOV
- DME 1
- Engine 1 Anti-Ice
- Engine 1 Flowmeter
- Flap System
- IASP 1
- Landing Lights
- Passenger Signs
- Pitch Trim (Back-Up)
- Roll Trim
- Speed Brake/Roll Spoiler
- Stall Protection
- Strobe Lights
- Windshield Heater 1
- WX Radar
- Yaw Trim



DC BUS 2 OFF

Airplane FLY MANUALLY
Fly the airplane manually. If in RVSM airspace, notify the ATC and co-ordinate an appropriate action plan.

Control Wheel HOLD FIRMLY

PITCH TRIM MODE Switch BKP

Use the Pitch Trim Backup Switch as required.

Icing Conditions EXIT/AVOID

Avoid side slipping the airplane and high angle of attack.

If in single pilot operation, LAND AS SOON AS PRACTICAL.

YAW DAMPER FAILURE Procedure ACCOMPLISH

Landing configuration:

Flap MAXIMUM
AVAILABLE

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 17 KIAS	1.80
IN ICING/ WITH ICE	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 28 KIAS	2.20

(Continues on the next page)



(Continued from the previous page)

The list below presents the relevant inoperative equipment/functions:

- | | |
|----------------------------|-----------------------------|
| - AHRS 2 | - IASP 2 |
| - AMS Controller Channel B | - Ice Detector |
| - Audio Panel 2 | - LH Navigation/Taxi Lights |
| - Autopilot | - PFD 2 |
| - Cabin Evaporator | - Pitch Trim (Normal) |
| - Cabin FCSOV | - Pressurization Auto |
| - DME 2 | - Red Beacon |
| - Engine 2 Anti-Ice | - Rudder Booster |
| - Engine 2 Flowmeter | - Spoiler System |
| - Flight Director 2 | - Stall Protection |
| - FMS Panel | - Transponder 2 |
| - Fuel Pump 2 | - Ventral Rudder |
| - GIA 2 | - Windshield Heater 2 |
| - GPS 2/VOR 2/ILS 2 | - Yaw Damper |



EMERGENCY BUS OFF

Airplane FLY MANUALLY
Fly the airplane manually. If in RVSM airspace, notify the ATC and co-ordinate an appropriate action plan.

Icing Conditions EXIT/AVOID

If in single pilot operation, LAND AS SOON AS PRACTICAL.

Before landing:

LDG GEAR Lever DN
Free Fall Handle PULL
UNTIL STOP

If gear does not lock down:

Airspeed 140 KIAS
MINIMUM

Slip the airplane to attempt to lock the main gear.

Landing configuration:

Flap MAXIMUM
AVAILABLE

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 2 KIAS	2.00
IN ICING/ WITH ICE	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 11 KIAS	2.30

(Continues on the next page)



(Continued from the previous page)

During landing run:

EMERGENCY BRAKING

TECHNIQUE Procedure..... ACCOMPLISH

The list below presents the relevant inoperative equipment/functions:

- | | |
|-----------------------------|--------------------------|
| - AHRS 1 | - Main Brake |
| - Audio Panel 1 | - Oxygen Transducer |
| - Autopilot | - Pax Mask Deploy (Auto) |
| - EFCU 1/2 | - PFD 1 |
| - Engines Fire Extinguisher | - Pressurization Manual |
| - Flight Director 1/2 | - Ram Air Valve |
| - Fuel Pump 1 | - Transponder 1 |
| - Fuel Shutoff Valves | - VHF 1 (Normal) |
| - GIA 1 | - WOW |
| - GPS 1/VOR 1/ILS 1 | - XFEED Valve |
| - Ground Spoiler | - Yaw Damper |
| - LDG Control Lever | |



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ENGINE

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ENGINE 1 (2) OIL LOW PRESSURE

If the affected oil pressure indication is abnormal:

ENGINE SHUTDOWN Procedure ACCOMPLISH

ENGINE 1 (2) CONTROL FAULT

LAND AS SOON AS PRACTICAL.

CAUTION: AVOID QUICK THRUST LEVER MOVEMENT AND HIGH ENGINE THRUST OPERATION.

If it is not possible to control engine:

Associated Thrust Lever IDLE

If engine parameters are still unstable:

ENGINE SHUTDOWN Procedure ACCOMPLISH

ENGINE 1 (2) FAILURE

LAND AS SOON AS POSSIBLE.

Associated Thrust Lever IDLE

If the affected engine was not successfully restarted via auto-relight:

Start/Stop Knob STOP

If not accomplishing the takeoff procedure:

Thrust Lever (operative engine) CON/CLB

If fuel leak is suspected:

FUEL LEAKAGE Procedure ACCOMPLISH

If fuel leak is not suspected and a restart is considered:

ENGINE IN-FLIGHT START Procedure ACCOMPLISH

If affected engine is not recovered and when appropriate:

ONE ENGINE INOPERATIVE APPROACH

AND LANDING Procedure ACCOMPLISH

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ENGINE 1 (2) FUEL IMPENDING BYPASS

Report to the maintenance personnel.

WARNING: IF BOTH E1 FUEL IMP BYP AND E2 FUEL IMP BYP MESSAGES ARE DISPLAYED, LAND AS SOON AS POSSIBLE.

ENGINE 1 (2) TLA FAILURE

The engine thrust will be set to idle automatically.

When appropriate:

ONE ENGINE INOPERATIVE APPROACH
AND LANDING Procedure.....ACCOMPLISH

ENGINE 1 (2) TT0 HEATER FAILURE

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

NOTE: Applicable to airplanes Post-Mod. SB 505-73-0001 or equipped with FADEC software version 4.3 and on. For airplanes Pre-Mod. SB 505-73-0001 refer to Supplement 3.

Exit or avoid icing conditions.

ENGINE 1 (2) TT0 PROBE ICE

If in icing conditions:

Icing ConditionsEXIT

After exit icing conditions:

ENG 1 & 2 SwitchesOFF

If not in icing conditions:

ENG 1 & 2 SwitchesOFF

CAUTION: BEFORE ENTER IN ICING CONDITIONS AT A LATER POINT, TURN ON ENG 1 & 2 SWITCHES AND MAKE SURE THE E1 (2) TT0 PROBE ICE MESSAGE IS NOT DISPLAYED.



ENGINE NO TO DATA

Enter the Engine Takeoff Data.

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FIRE PROTECTION

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BAGGAGE SMOKE

LAND AS SOON AS POSSIBLE.

ENGINE 1 (2) FIRE

- ϕ Thrust Lever (affected engine) IDLE
- ϕ Start/Stop Knob (affected engine) STOP
- ϕ Fire Shutoff Button (affected engine) PUSH IN

LAND AS SOON AS POSSIBLE.

Wait 30 seconds and if fire persists:

- BOTTLE Switch DISCH
- Altitude 36000 ft
MAXIMUM
- ECS Knob UNAFFECTED
SIDE
- Thrust Lever (operative engine) AS REQUIRED

If airplane high vibration exists:

- Airspeed REDUCE
(140 KIAS
MINIMUM)

Reduce airspeed as required to keep vibration within acceptable levels.

When appropriate:

- ONE ENGINE INOPERATIVE APPROACH
AND LANDING Procedure ACCOMPLISH

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AFT BAGGAGE COMPARTMENT SMOKE DETECTION FAILURE

LAND AS SOON AS PRACTICAL.

ENGINE 1 (2) FIRE DETECTION FAILURE

If engine fire or overheat is suspected or if E1 (2) FIRE DET FAIL appears simultaneously with engine failure:

ENGINE 1 (2) FIRE Procedure.....ACCOMPLISH

ENGINE 1 (2) FIRE EXTINGUISHER FAILURE

LAND AS SOON AS PRACTICAL.



FLIGHT CONTROLS

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FLAP FAILURE

If FLAP FAIL message is associated with PTRIM BKP FAIL, SWPS FAULT and SPOILER FAULT messages:

If FCE Reset is installed:

FCE Reset Knob SET TO FCE1
 FCE Reset Button PRESS FOR 1
 SECOND

NOTE: Only one actuation per flight is allowed for FCE 1.

FCE Reset Knob SET TO NONE
 Wait 5 seconds.

If FLAP FAIL message is NOT associated with PTRIM BKP FAIL, SWPS FAULT and SPOILER FAULT messages:

Altitude 18000 ft
 MAXIMUM

When below 18000ft:

Flap Lever CYCLE
 Up to 3 cycles may be attempted.

If FLAP FAIL message persists or FCE Reset is NOT installed:

If Flap indication is different than 0:

Altitude 18000 ft
 MAXIMUM

Leave and avoid icing conditions.

If it is not possible to avoid icing conditions:

LAND AS SOON AS POSSIBLE.

For landing:

- Check the available airports within the current range and choose the one that best matches the required runway length. Considerations for uphill slope and occurrence of headwind should be also made.
- Burn as much fuel as possible to reduce the landing weight.
- Maintain bank angle below 40°.

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- If applicable, press the TAWS FLAP OVRD Button.
- Maintain the airspeed according to the following:

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	0	V _{REF} FULL NO ICE + 27 KIAS	1.55
	1	V _{REF} FULL NO ICE + 19 KIAS	1.30
	2 and 3	V _{REF} FULL NO ICE + 6 KIAS	1.15
	FULL	V _{REF} FULL NO ICE	1.00
IN ICING/ WITH ICE	0	V _{REF} FULL NO ICE + 38 KIAS	1.70
	1	V _{REF} FULL NO ICE + 27 KIAS	1.45
	2 and 3	V _{REF} FULL NO ICE + 15 KIAS	1.30
	FULL	V _{REF} FULL NO ICE + 10 KIAS	1.20

NOTE: - If flap stops between two positions, use the minimum airspeed associated with the next retracted position and the V_{FE} associated with the next extended position.

- Disregard green circle indication, as it may indicate slower speeds.

If a go-around is required, maintain the minimum airspeed presented in the applicable flap configuration from the table above, until the acceleration altitude is reached.



GROUND SPOILER FAILURE

If associated with PTRIM NML FAIL and SPOILER FAULT messages and FCE Reset is installed:

FCE Reset Knob..... SET TO FCE2

FCE Reset Button..... PRESS FOR 1
SECOND

NOTE: Only one actuation per flight is allowed for FCE 2.

FCE Reset Knob..... SET TO NONE

Wait 5 seconds

If GND SPLR FAIL message is displayed or persists:

Landing configuration:

Flap 3

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	3	V _{REF} FULL NO ICE + 2 KIAS	1.20
IN ICING/ WITH ICE	3	V _{REF} FULL NO ICE + 11 KIAS	1.40



PITCH TRIM BACK-UP FAILURE

If associated with FLAP FAIL, SWPS FAULT and SPOILER FAULT messages and FCE Reset is installed:

FCE Reset Knob SET TO FCE1
FCE Reset Button PRESS FOR 1
SECOND

NOTE: Only one actuation per flight is allowed for FCE 1.

FCE Reset Knob SET TO NONE
Wait 5 seconds.

If PTRIM BKP FAIL message is displayed or persists:

PITCH BKP Switch OFF
Pitch Trim Switches USE AS
REQUIRED

PITCH TRIM NORMAL FAILURE

If associated with GND SPLR FAIL and SPOILER FAULT messages and FCE Reset is installed:

FCE Reset Knob SET TO FCE2
FCE Reset Button PRESS FOR 1
SECOND

NOTE: Only one actuation per flight is allowed for FCE 2.

FCE Reset Knob SET TO NONE
Wait 5 seconds

If PTRIM NML FAIL message is displayed or persists:

Control Wheel HOLD FIRMLY
QUICK DISCONNECT Button PRESS
PITCH BKP Switch BKP

Use the pitch trim backup mode as required.

NOTE: Autopilot engagement is inhibited.



RUDDER GUST LOCKED

On ground:

Gust Lock Pin CHECK
REMOVED

Rudder Pedals CHECK
UNLOCKED

If the RUDDER GUST LOCKED message persists:

Do not takeoff.

In flight:

LAND AS SOON AS POSSIBLE.

NOTE: Avoid landing with crosswinds higher than 15 kt.

RUDDER OVERBOOST

Airspeed (flap and gear up) 135 KIAS
MINIMUM

Avoid large rudder pedals deflection.

Landing configuration:

Flap MAXIMUM
AVAILABLE

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 17 KIAS	1.40
IN ICING/ WITH ICE	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 28 KIAS	1.60

NOTE: Avoid landing with crosswinds higher than 15 kt.



PITCH TRIM SWITCH 1 (2) FAILURE

For dual pilot operation:

Remaining PITCH TRIM SwitchUSE AS
REQUIRED

NOTE: The Pitch Backup Mode may also be used if available.

For single pilot operation and if the left pitch trim switch is failed:

Pitch Trim Backup Mode.....USE AS
REQUIRED

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FLIGHT INSTRUMENTS, COMMUNICATION AND NAVIGATION

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ADS 1 (2) FAILURE

Confirm the affected ADS automatic reversion. If necessary, manually revert to the ADS not affected.

If in RVSM airspace, notify the ATC and co-ordinate an appropriate action plan.

If both ADS are lost:

YAW DAMPER FAILURE Procedure ACCOMPLISH

NOTE: The airspeed and pressure altitude from IESI may be used.

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ADS 1 (2) HEATER FAILURE

Monitor the air data indications. If the information seems to be unreliable, revert to the not affected ADS.

If in RVSM airspace, notify the ATC and co-ordinate an appropriate action plan.

If more than one ADS heater is failed, set the ADS PROBES Knob to ON position and pull the PSTAT HTR CTRL circuit breaker (B6). If messages persist, leave and avoid icing conditions.

AHRS 1 (2) FAILURE

Confirm the affected AHRS automatic reversion. If necessary, manually revert to the other AHRS.

If both AHRS are lost:

YAW DAMPER FAILURE ProcedureACCOMPLISH

NOTE: The airspeed and pressure altitude from IESI may be used.



AUDIO PANEL 1 (2) FAILURE

Use headset as required.

NOTE: For affected side, the following items are operative:

- On-side VHF transmission/reception;
- Cross-side VHF reception.

BARO MISCOMPARE

Altimeters (pilots and IESI) SET & X-CHECK

If BARO MISCOMP persists or is displayed again:

BARO SYNC OFF

Altimeters (pilots and IESI) SET & X-CHECK

Use the PFD that displays the reliable altitude reading.

NOTE: Flight Director source must be according to the PFD in use.

CONFIGURATION MODULE FAILURE

Do not takeoff.



GIA 1 (2) FAILURE

Airplane FLY MANUALLY

If in RVSM airspace, notify the ATC and co-ordinate an appropriate action plan.

Icing Conditions EXIT/AVOID

If in single pilot operation, LAND AS SOON AS PRACTICAL.

Relevant inoperative items:

GIA 1 failed:

- Autopilot
- Crossfeed (Auto)
- DME 1
- Flight Director 1
- GPS 1
- TCAS
- VHF/NAV 1
- Yaw Damper

GIA 2 failed:

- Autopilot
- DME 2
- Flight Director 2
- GPS 2
- TCAS
- VHF/NAV 2
- Yaw Damper

GIA 1 (2) OVERHEAT

Avoid using radios from the affected side.

MFD CONFIGURATION

On ground, report to the maintenance personnel.

In flight:

DISPLAY BACKUP Button PRESS



MFD FAULT

On ground, report to the maintenance personnel.

In flight:

DISPLAY BACKUP Button PRESS

MFD OVERHEAT

If associated with smoke from the MFD:

Crew Oxygen Masks DON,
EMERGENCY
THEN 100%

EMERGENCY may be selected during 2 minutes maximum
then set to 100%.

Crew Oxygen Masks Auto Dilution Valve CLOSED

Smoke Goggles (if available) DON

Communication ESTABLISH

MFD Circuit Breakers (A9 and A24) PULL

SMOKE EVACUATION Procedure AS REQUIRED

PFD 1 (2) CONFIGURATION

On ground, report to the maintenance personnel.

In flight and associated with the flying pilot:

DISPLAY BACKUP Button PRESS

PFD 1 (2) FAULT

On ground, report to the maintenance personnel.

In flight and associated with the flying pilot:

DISPLAY BACKUP Button PRESS



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PFD 1 (2) OVERHEAT

If associated with smoke from the affected PFD:

- Crew Oxygen Masks.....DON,
EMERGENCY
THEN 100%
- EMERGENCY may be selected during 2 minutes maximum
then set to 100%.
- Crew Oxygen Masks Auto Dilution ValveCLOSED
- Smoke Goggles (if available).....DON
- Crew CommunicationESTABLISH
- PFD 1 (A7) or PFD 2 (B24)
- Circuit BreakerPULL
- DISPLAY BACKUP ButtonAS REQUIRED
- SMOKE EVACUATION Procedure.....AS REQUIRED
- If required during approach:
- Associated Circuit BreakerPUSH

STBY HEATER FAILURE

Monitor the air data indications. If the information seems to be unreliable, disregard IESI altitude and airspeed indication.

If more than one ADS heater is failed, set the ADS/AOA Knob to ON position and pull the PSTAT HTR CTRL circuit breaker (B6). If messages persist, leave and avoid icing conditions.

TRANSPONDER FAIL

Select and use the remaining Transponder, if available.

TRANSPONDER IN STANDBY

Transponder.....AS REQUIRED



FUEL

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FUEL 1 (2) LOW LEVEL

- Associated FUEL PUMP Switch..... ON
- Thrust Levers..... REDUCE TO LONG RANGE

Level the airplane and check both fuel quantities.

If fuel quantity is not sufficient:
LAND AS SOON AS POSSIBLE.

CAUTION: AVOID EXCESSIVE NOSE UP AND DOWN ATTITUDES, UNCOORDINATED MANEUVERS AND NEGATIVE G'S.

- If fuel leak is suspected:
 - FUEL LEAKAGE Procedure..... ACCOMPLISH
- If fuel leak is not suspected:
 - XFEED Knob AS REQUIRED

FUEL 1 (2) LOW PRESSURE

- Associated FUEL PUMP Switch..... ON
- If message still persists:
 - XFEED Knob OFF
 - Associated Thrust Lever..... AS REQUIRED
 - Engine Parameters..... MONITOR
- If unable to maintain desired thrust or engine parameters fluctuate:
 - Altitude..... DESCEND AS REQUIRED
- If condition persists:
 - LAND AS SOON AS PRACTICAL.
- If fuel leak is suspected:
 - FUEL LEAKAGE Procedure..... ACCOMPLISH

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FUEL 1 (2) SHUTOFF VALVE FAILURE

If associated with engine fire:

LAND AS SOON AS POSSIBLE.

Associated FUEL PUMP Switch OFF

XFEED Knob..... OFF

FUEL CROSSFEED FAILURE

If message is presented when opening the crossfeed valve for balancing procedure:

XFEED Knob..... OFF

Fuel Quantity..... MONITOR

Thrust Levers ADJUST

Use asymmetric thrust to improve or maintain fuel balance.

If message is presented after performing a fuel imbalance procedure:

FUEL PUMP Switches ON

XFEED Knob..... OFF

Fuel Quantity..... MONITOR

If further fuel balancing is required:

FUEL PUMP Switch (low level wing) OFF

FUEL PUMP Switch (high level wing)..... ON

After fuel balancing:

FUEL PUMP Switches ON

FUEL IMBALANCE

If fuel leak is suspected, accomplish the FUEL LEAKAGE Procedure.

If fuel leak is not suspected:

Attitude..... WINGS LEVEL

(Continues on the next page)

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(Continued from the previous page)

In case of right wing low level:

XFEED Knob LO2

If associated with FUEL PUMP 1 FAIL message:

FUEL PUMP 1 Switch ON

Fuel Quantity MONITOR

If imbalance does not decrease:

XFEED Knob OFF

Use the asymmetric thrust as required to achieve fuel balance.

If asymmetric thrust can not be used:

LAND AS SOON AS PRACTICAL.

When balance is achieved:

XFEED Knob OFF

FUEL PUMP Switches AUTO

Fuel Quantity MONITOR

In case of left wing low level:

XFEED Knob LO1

If associated with FUEL PUMP 2 FAIL message:

FUEL PUMP 2 Switch ON

Fuel Quantity MONITOR

If imbalance does not decrease:

XFEED Knob OFF

Use the asymmetric thrust as required to achieve fuel balance.

If asymmetric thrust can not be used:

LAND AS SOON AS PRACTICAL.

When balance is achieved:

XFEED Knob OFF

FUEL PUMP Switches AUTO

Fuel Quantity MONITOR

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FUEL PUMP 1 (2) FAILURE

On ground: do not takeoff.

During flight:

If FUEL PUMP 1 (2) FAIL message is displayed during FUEL IMBALANCE Procedure or during ENGINE IN-FLIGHT START Procedure:

Associated FUEL PUMP Switch..... ON

After procedure accomplished:

FUEL PUMP Switches.....AUTO



HYDRAULIC

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HYDRAULIC HIGH TEMPERATURE

HYD PUMP SOV 1 & 2 Switches CLOSED

Wait 7 minutes.

If HYD HI TEMP message does not extinguish and HYD SOV 1 (2) FAIL message is displayed:

LAND AS SOON AS POSSIBLE.

Thrust Lever (associated engine)..... IDLE

Start/Stop Knob (associated engine) STOP

If the HYD HI TEMP message extinguishes:

HYD PUMP SOV Switch
(non-associated engine)..... OPEN

When appropriate:

ONE ENGINE INOPERATIVE APPROACH
AND LANDING Procedure ACCOMPLISH

If HYD HI TEMP message extinguishes:

HYD PUMP SOV 2 Switch OPEN

Monitor system.

If HYD HI TEMP is displayed again:

HYD PUMP SOV 2 Switch CLOSED

Wait up to 7 minutes for HYD HI TEMP message extinguishing.

HYD PUMP SOV 1 Switch OPEN

Monitor system.

If HYD HI TEMP message is displayed:

HYD PUMP SOV 1 Switch CLOSED

HYDRAULIC LOW PRESSURE
Procedure..... ACCOMPLISH

Shutdown both engines as soon as possible after landing.

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HYDRAULIC LOW PRESSURE

NOTE: Expect lower roll rates and no speedbrake actuation.

Avoid side slipping the airplane and high angle of attack.

For landing:

ABNORMAL LANDING GEAR

EXTENSION Procedure ACCOMPLISH

Landing configuration:

Flap MAXIMUM
AVAILABLE

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 17 KIAS	2.70
IN ICING/ WITH ICE	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 28 KIAS	3.10

During landing run:

EMERGENCY BRAKING

TECHNIQUE Procedure ACCOMPLISH

Inoperative items list:

- Multifunction Spoiler
- Normal Brakes
- Normal Landing Gear Operation
- Pusher Actuator
- Spring Loaded Rudder Booster



HYDRAULIC SOV 1 (2) FAILURE

Associated Engine MONITOR

Hydraulic System..... MONITOR

If associated with fire or HYD HI TEMP message:

HYD PUMP SOV 1 (2) Switch CLOSED

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ICE AND RAIN PROTECTION

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ANTI-ICE ENGINE 1 (2) FAILURE

Affected ENG Switch OFF, THEN ON

If the A-I E1 (2) FAIL message persists:

Affected ENG Switch ON

Icing Conditions EXIT/AVOID

After exiting icing conditions:

Affected ENG Switch OFF

If high engine vibration develops:

Associated Thrust Lever IDLE

If the abnormal vibration persists:

ENGINE SHUTDOWN Procedure ACCOMPLISH

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ANTI-ICE LOW CAPACITY

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

Thrust Levers ADVANCE
Advance thrust levers until reach N1 anti-icing bug, if displayed.

If A-I LO CAPACITY message persists, after 15 seconds:

WINGSTAB Switch OFF
Icing Conditions EXIT/AVOID
Airspeed 165 KIAS
MINIMUM

Landing configuration:

WARNING: DO NOT SELECT FLAP 2 OR HIGHER.

Bank Angle BELOW 30°
Flap 1

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER THE FACTOR BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	1	V _{REF} FULL NO ICE + 30 KIAS	1.50
IN ICING/ WITH ICE			

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ANTI-ICE WING/STABILIZER FAILURE

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

Airspeed..... 165 KIAS
MINIMUM

Thrust Levers..... SYMMETRIC
POSITION

WINGSTAB Switch OFF, THEN ON

If an asymmetric thrust operation is required:

BLEED Switch (lower thrust engine) OFF

WINGSTAB Switch..... OFF, THEN ON

Altitude 15000 ft
MAXIMUM

In any case, if A-I WINGSTB FAIL message persists:

Icing Conditions EXIT/AVOID

Landing configuration:

WARNING: DO NOT SELECT FLAP 2 OR HIGHER.

Bank Angle BELOW 30°

Flap..... 1

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER THE FACTOR BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	1	V _{REF} FULL NO ICE + 30 KIAS	1.50
IN ICING/ WITH ICE			



ANTI-ICE WING/STABILIZER LEAKAGE

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

Airspeed 165 KIAS
MINIMUM

WINGSTAB Switch OFF

Exit and avoid icing conditions.

If A-I WINGSTB LEAK message persists:

BLEED 1 Switch..... OFF
Altitude 36000 ft
MAXIMUM
XBLEED Knob CLOSED

If A-I WINGSTB LEAK message persists:

BLEED 1 Switch AUTO
BLEED 2 Switch OFF

If A-I WINGSTB LEAK message persists:

BLEED 1 Switch OFF
Crew Oxygen Masks DON, 100%
Communication..... ESTABLISH
Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER

At 10000 ft or MEA:

DUMP Button..... PUSH IN
ECS Knob OFF VENT

(Continues on the next page)



(Continued from the previous page)

Landing configuration:

WARNING: DO NOT SELECT FLAP 2 OR HIGHER.

Bank Angle BELOW 30°

Flap 1

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER THE FACTOR BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	1	V _{REF} FULL NO ICE + 30 KIAS	1.50
IN ICING/ WITH ICE			



ANTI-ICE WING/STABILIZER INHIBIT

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

On ground:

Below 6°C:

Do not takeoff.

At or above 6°C:

WINGSTAB Switch OFF, THEN ICE
SPEED RESET

NOTE: If SAT is below 10°C, keep engine anti-ice system ON.

**WARNING: TAKEOFF MUST NOT BE PERFORMED WITH THE
MESSAGE PRESENTED.**

In flight:

Airspeed 165 KIAS
MINIMUM

Flap UP

WINGSTAB Switch KEEP ON

For Airplanes Pre-Mod. SB 505-00-0001 EXIT/AVOID
ICING
CONDITIONS

For Airplanes Post-Mod. SB 505-00-0001

Below 30000 ft EXIT/AVOID
ICING
CONDITIONS

After leaving icing conditions and if there is no ice accretion on the
airplane:

WINGSTAB Switch OFF, THEN ICE
SPEED RESET

**WARNING: THE WINGSTAB SWITCH MUST BE KEPT ON
UNTIL CREW IS CERTAIN THAT ALL ICE HAS
BEEN REMOVED.**

(Continues on the next page)



(Continued from the previous page)

During landing, if icing conditions are presented or if there is ice accretion:

WARNING: DO NOT SELECT FLAP 2 OR HIGHER.

Bank Angle BELOW 30°

Flap..... 1

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER THE FACTOR BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
IN ICING/ WITH ICE	1	V _{REF} FULL NO ICE + 30 KIAS	1.50

ICE CONDITION

ENG 1 & 2 Switches ON

WINGSTAB Switch ON

WSHLD 1 & 2 Switches ON

NOTE: Refer to the OPERATION IN ICING CONDITIONS Procedures in Section 3-04.



WINDSHIELD 1 (2) HEATER FAILURE

Both WSHLD Switches OFF, THEN ON

If the WSHLD 1 (2) HTR FAIL message persists:

Associated WSHLD Switch..... OFF

Icing Conditions (if required)..... EXIT/AVOID

ANTI-ICE WING/STABILIZER ARMED

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

In flight:

Airspeed 165 KIAS
MINIMUM

Flap ZERO

WINGSTAB Switch KEEP ON

After leaving icing conditions and if there is no ice accretion on the airplane:

WINGSTAB Switch OFF, THEN ICE
SPEED RESET

**WARNING: THE WINGSTAB SWITCH MUST BE KEPT ON
UNTIL CREW IS CERTAIN THAT ALL ICE HAS
BEEN REMOVED.**

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LANDING GEAR AND BRAKES

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LANDING GEAR LEVER DISAGREE

Airspeed..... 250 KIAS
MAXIMUM

LDG GEAR Lever CYCLE

If the LG LEVER DISAG message persists and it was displayed after a landing gear extension:

ABNORMAL LANDING GEAR
EXTENSION Procedure ACCOMPLISH

If the LG LEVER DISAG message persists and it was displayed after a landing gear retraction:

LDG GEAR Lever DN

Icing Conditions EXIT/AVOID

LAND AS SOON AS PRACTICAL.

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ANTISKID FAILURE

During landing run, apply the main brakes progressively until deceleration is felt and from this point, start to modulate the brakes to avoid tire blowout.

Landing configuration:

Flap MAXIMUM
AVAILABLE

CAUTION: • TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

- MAIN BRAKES AVAILABLE WITH POSSIBLE CHANGED PEDAL RESPONSE. IF NO DECELERATION IS FELT DURING INITIAL APPLICATION OF THE PEDAL, CONTINUE APPLYING THE MAIN BRAKES PROGRESSIVELY UNTIL DECELERATION IS FELT (APPROXIMATELY HALF PEDAL COURSE) AND FROM THIS POINT, START TO MODULATE THE BRAKES TO AVOID TIRE BLOWOUT.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 2 KIAS	1.65
IN ICING/ WITH ICE	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 11 KIAS	1.85



BRAKE FAILURE

Burn fuel to reduce the landing speed.

Landing configuration:

Flap..... MAXIMUM
AVAILABLE

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 2 KIAS	1.65
IN ICING/ WITH ICE	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 11 KIAS	1.85

During landing run:

EMERGENCY BRAKING
TECHNIQUE Procedure ACCOMPLISH

EMERGENCY BRAKE LOW PRESSURE

Emergency/parking brake is inoperative.

When parking the airplane, use wheel chocks.



LANDING GEAR WEIGHT-ON-WHEEL SYSTEM FAILURE

If climb performance is required to clear obstacles during takeoff and go-around, simultaneously proceed:

DN LCK REL Button PRESS
LDG GEAR Lever UP

The Weight on Wheel system failure may affect the functionality of some systems, as listed below. In this case, their associated failure procedures must be accomplished, at crew discretion.

Relevant affected systems:

- Environmental Control System
- Pressurization System
- Ice Protection System
- Stall Warning and Protection System

PARKING BRAKE NOT RELEASED

Before landing, release the emergency/parking brake.



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OXYGEN LOW PRESSURE

Oxygen Bottle Valve Handle..... **PRESS TO
RESTORE**

If message persists, limit altitude to 14000 ft or MEA, whichever is higher.

PASSENGER OXYGEN NO PRESSURE

Set the OXYGEN SUPPLY CONTROL Knob to PAX OVRD.

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WARNING

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NO TAKEOFF CONFIGURATION

Configure the airplane for takeoff.

AURAL WARNING FAILURE

Monitor visual indications.

NOTE: Aural warnings, including TCAS, TAWS, TIS and STALL callouts are lost.

PUSHER FAILURE

PUSHER CUTOUT Button PUSH IN

Avoid side slipping the airplane and high angle of attack.

NOTE: Low Speed Awareness cues (yellow and red band) and Stall aural warning will still be available.

PUSHER OFF

If the pusher was inadvertently cut off:

PUSHER CUTOUT Button PUSH OUT

If the pusher was intentionally cut off:

Avoid side slipping the airplane and high angle of attack.

NOTE: Low Speed Awareness cues (yellow and red band) and Stall aural warning will still be available.

STALL WARNING AND PROTECTION SYSTEM FAILURE

Avoid side slipping the airplane and high angle of attack.

NOTE: Low Speed Awareness cues (yellow and red band) and green circle will not be shown on airspeed tape. Stall aural message will not be activated.



STALL WARNING AND PROTECTION SYSTEM FAULT

If associated with FLAP FAIL, PTRIM BKP FAIL and SPOILER FAULT messages and FCE Reset is installed:

FCE Reset Knob SET TO FCE1

FCE Reset Button PRESS FOR 1
SECOND

NOTE: Only one actuation per flight is allowed for FCE 1.

FCE Reset Knob SET TO NONE

Wait 5 seconds.

If SWPS FAULT message is displayed or persists:

Avoid side slipping the airplane and high angle of attack.

For landing configuration:

Flap MAXIMUM
AVAILABLE

Maintain airspeed according to the following:

CAUTION: TO FIND THE REQUIRED LANDING DISTANCE, ENTER ONE OF THE FACTORS BELOW AND FLAP FULL FACTORED LANDING DISTANCE IN THE "LANDING DISTANCE CORRECTION" TABLE ON SECTION 5-25.

	FLAP POSITION	V _{REF}	LANDING DISTANCE CORRECTION FACTOR
NO ICING	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 6 KIAS	1.20
IN ICING/ WITH ICE	MAXIMUM AVAILABLE	V _{REF} FULL NO ICE + 15 KIAS	1.30

STALL WARNING AND PROTECTION SYSTEM UNTESTED

On ground, perform the pre-flight SWPS system test.



SMOKE PROCEDURES

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BAGGAGE SMOKE

LAND AS SOON AS POSSIBLE.

SMOKE/FIRE/FUME

- ☐ Crew Oxygen Masks DON,
EMERGENCY
- ☐ Crew Oxygen Masks Auto Dilution Valve..... CLOSED
- ☐ Smoke Goggles (if available) DON
- ☐ Communication..... ESTABLISH
- ☐ DUMP Button PUSH IN

LAND AS SOON AS POSSIBLE.

After 2 minutes at EMERGENCY:

Crew Oxygen Masks 100%

WARNING: ANY TIME SMOKE OR FUMES BECOME THE GREATEST THREAT, ACCOMPLISH THE SMOKE EVACUATION PROCEDURE.

NOTE: Smoke will initially decrease due to pressurization dumping, even if the correct source has not been eliminated.

If smoke origin is identified:

Smoke Source ELIMINATE

NOTE: Discharge of the entire contents of the fire extinguisher bottles into the occupied cabin area may exceed safe exposure limits. Use only the amount necessary to extinguish the fire.

If smoke source has been eliminated:

DUMP Button PUSH OUT

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If smoke origin is not identified:

EMERGENCY DESCENT Procedure.....ACCOMPLISH

If smoke persists:

GEN 2 Switch OFF

BUS TIE Knob 2 OPEN

Wait 1 minute and if smoke persists:

BUS TIE Knob AUTO

GEN 2 Switch AUTO

GEN 1 Switch OFF

BUS TIE Knob 1 OPEN

Wait 1 minute and if smoke persists:

GEN 1 SwitchAUTO

BATT 1 & 2 Switches..... OFF

Wait 1 minute and if smoke persists:

BUS TIE Knob AUTO

BATT 1 & 2 Switches..... ON

Non-Essential Equipment..... OFF

Passenger Oxygen SupplyAS REQUIRED

Depending on which bus is off, refer to DC BUS 1 OFF, DC BUS 2 OFF or EMERGENCY BUS OFF Procedure for additional guidance.

NOTE: Even after turning on the Emergency Bus, the main brake will be inoperative and the Emergency/Parking brake must be used for landing. If situation permits in flight, the main brake may be recovered by cycling the BRAKE circuit breaker (B32) with the landing gear extended.



SMOKE EVACUATION

- ☐ Crew Oxygen Masks DON,
EMERGENCY
- ☐ Crew Oxygen Masks Auto Dilution Valve..... CLOSED
- ☐ Smoke Goggles (if available) DON
- ☐ Oxygen SUPPLY CONTROL Knob..... CREW ONLY
- ☐ Communication..... ESTABLISH
- ☐ DUMP Button..... PUSH IN
- ☐ ECS Knob..... OFF VENT

LAND AS SOON AS POSSIBLE.

After 2 minutes at EMERGENCY:

- Crew Oxygen Masks 100%
- Fasten Seat Belts Signs ON
- Altitude 10000 ft OR MEA,
WHICHEVER IS
HIGHER
- Thrust Levers IDLE
- SPEED BRAKE Switch OPEN
- Airspeed..... 250 KIAS/M_{MO}
MAXIMUM
- LDG GEAR Lever DN
- Transponder 7700
- ATC..... NOTIFY

At 10000 ft or MEA:

- Oxygen SUPPLY CONTROL Knob..... AS REQUIRED

If there is no evidence of fire in the cabin, set the oxygen
SUPPLY CONTROL Knob to PAX AUTO.

- SPEED BRAKE Switch..... CLOSE



BAGGAGE SMOKE FAILURE

If baggage compartment is not empty:

LAND AS SOON AS PRACTICAL.



SECTION 5

PERFORMANCE

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INTRODUCTION

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

The performance information given in this Section is applicable to the PHENOM 300 airplane.

The performance data are based on the following conditions:

- Pertinent power less installation, air bleed, and accessory losses.
- Full temperature accountability within the operational limits for which the airplane is certified.
- The performance analysis meets the requirement CAT.POL.A.255(a)(2).

The performance information is not valid if:

- The airplane's gross weight exceeds the appropriate maximum allowable limits.
- Any of the limitations in Section 2 are not observed.
- A reading from any performance analysis is obtained by extrapolation (i.e. using values of parameters outside the range given on the graphs, tables or OPERA software).



PERFORMANCE CONFIGURATION

The configurations applicable to each performance phase are shown below.

	OPERATING ENGINES	TLA	FLAP	GEAR	AIRSPEED
TAKEOFF RUN	2 until V_{EF} , 1 after V_{EF}	TO/GA	1 or 2	DOWN	0 TO V_{LOF}
1ST SEGMENT	1	TO/GA	1 or 2	DOWN TO UP	V_{LOF} TO V_2
2ND SEGMENT	1	TO/GA	1 or 2	UP	V_2
3RD SEGMENT	1	TO/GA	TAKEOFF FLAP TO 0	UP	V_2 TO FINAL SEGMENT SPEED
FINAL SEGMENT	1	MAX CON	0	UP	FINAL SEGMENT SPEED
ENROUTE	1	MAX CON	0	UP	ENROUTE CLIMB SPEED
APPROACH CLIMB	1	TO/GA	1	UP	APPROACH CLIMB SPEED
LANDING CLIMB	2	TO/GA	3 or FULL	DOWN	LANDING CLIMB SPEED
LANDING	2	IDLE	3 or FULL	DOWN	V_{REF}



DEFINITIONS

OUTSIDE AIR TEMPERATURE – OAT OR SAT

Free static (ambient) temperature.

TOTAL AIR TEMPERATURE – TAT

Total temperature measured and corrected by the ADS system with airplane in motion.

INDICATED AIRSPEED – KIAS

It is the reading on the airspeed indicator (knots), as installed in the airplane. Zero instrument error is assumed by the OPERA.

CALIBRATED AIRSPEED – KCAS

It is the indicated airspeed corrected for the anemometric system position error.

TRUE AIRSPEED – TAS

It is the airplane speed in relation to the air and equal to the equivalent air speed corrected for air density. At sea level, ISA, the true airspeed is equal to the equivalent airspeed.

GROUND SPEED – GS

It is the airplane speed in relation to the ground. It is the true airspeed corrected for the wind speed.

TAKEOFF DECISION SPEED – V_1

It is the maximum speed in the takeoff at which the pilot must take the first action (e.g., apply brakes, reduce thrust) to stop the airplane within the accelerate-stop distance. V_1 also means the minimum speed in the takeoff, following a failure of the critical engine at V_{EF} , at which the pilot can continue the takeoff and achieve the required height above the takeoff surface within the takeoff distance.



STALL SPEED – V_S

It is the minimum steady flight speed at which the airplane is controllable.

ROTATION SPEED – V_R

It is the speed at which the pilot starts to rotate the airplane during the takeoff ground run.

LIFT-OFF SPEED – V_{LOF}

It is the speed at which the airplane first becomes airborne during takeoff.

FIRST SEGMENT

Starts at 35 ft above the takeoff surface, at the end of the one-engine-inoperative takeoff distance, and extends to the point where the landing gear is fully retracted using a constant V_2 speed and takeoff flap.

TAKEOFF SAFETY SPEED – V_2

It is the target speed to be attained at 35 ft height above runway surface, the critical engine having failed at V_{EF} .

SECOND SEGMENT

Extends from the end of the first segment up to the level-off height, with takeoff flap, gear up, and constant V_2 speed.

LEVEL-OFF HEIGHT

It is the height or altitude where the airplane is leveled for acceleration and flap retraction.

MINIMUM LEVEL-OFF HEIGHT

It is defined by regulations as 400 ft AGL. The user can define higher levels.

MAXIMUM LEVEL-OFF HEIGHT

It is the height where the takeoff thrust time limit occurs at the end of the final segment.

EXTENDED SECOND SEGMENT

It is an extension of the second takeoff climb segment, above the maximum level off height, to improve obstacle clearance performance.

THIRD (OR ACCELERATION) SEGMENT

Extends from the end of the second segment, at the level-off height, up to a point where the airplane is accelerated to final segment speed (V_{FS}) while retracting flap.

FINAL SEGMENT SPEED – V_{FS}

It is the speed to be achieved at the end of the acceleration segment and start of the final segment of the takeoff flight path, with one engine inoperative, landing gear retracted, and flap retracted.

FINAL SEGMENT

Extends from the end of the third segment to a gross height at least at 1500 ft AGL, with flap up and at final segment speed (V_{FS}).



NOISE LEVELS

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

The following Effective Perceived Noise Levels (EPNL's) comply with the Chapter 4 noise limits set forth in CS-36, equivalent to ICAO Annex 16, Volume 1, Appendix 2, Amendment 9 (*), effective July 20, 2008.

NOISE LEVEL IN EPNdb				
Engine Model	Noise Level	CONDITION		
		Flyover	Lateral	Approach
	Maximum Allowable	89.0	94.0	98.0
PW535E (1)	Actual	70.6	88.7	88.9
PW535E (2)	Actual	69.9	88.8	88.9
PW535E (3)	Actual	70.6	88.7	88.4
PW535E (4)	Actual	69.9	88.8	88.5

The noise levels for EMB-505 equipped with two Pratt & Whitney PW535E engines, were established as follows:

- Flyover: at maximum takeoff weight, flap setting 1, using thrust power cutback and $V_2 + 20$ KIAS;
- Lateral: at maximum takeoff weight, flap setting 1 and with all engines at maximum takeoff power setting;
- (1) and (2) Approach: at maximum landing weight, 3° glide slope, $V_{REF} + 10$ KIAS and flap setting FULL;
- (3) and (4) Approach: at maximum landing weight, 3° glide slope, $V_{REF} + 10$ KIAS and flap setting 3.

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NOTE: 1) For airplanes Post-Mod. SB 505-27-0011 or with an equivalent modification factory incorporated and Post-Mod. SB 505-00-0008 or with an equivalent modification factory incorporated.

NOTE: 2) For airplanes Post-Mod. SB 505-27-0011 or with an equivalent modification factory incorporated and Pre-Mod. SB 505-00-0008.

NOTE: 3) For airplanes Pre-Mod. SB 505-27-0011 and Post-Mod. SB 505-00-0008 or with an equivalent modification factory incorporated.

NOTE: 4) For airplanes Pre-Mod. SB 505-27-0011 and Pre-Mod. SB 505-00-0008. (*) Amendment 8, effective on July 11, 2005.

No determination has been made by the Airworthiness Authority that the noise levels of this airplane are or should be acceptable or unacceptable for operation at, into, or out of any airport.



SOFTWARE PRESENTATION

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

OPERA is the Optimized Performance Analyzer software certified to provide takeoff, landing and enroute approved performance data.

The OPERA replaces or supplements portions of the paper AFM, and is an EASA approved source for that AFM information. As paper AFM performance information is generally based on a simplified analysis, it may show performance data more conservative than OPERA. Any modification to the EASA approved OPERA software application, or subsequent alteration to the generated output, will cancel the airworthiness approval of the information, unless this change was approved by the appropriate airworthiness authority. This statement applies regardless of any approval notation printed on a generated output.

Performance wise, OPERA offers applicable aerodynamic and systems calculation options, to be selected according to the airplane specific configuration.

The OPERA installer incorporates the media identification, which will be displayed on the screen after installation and must be crosschecked with the AFM OPERA configuration.

The OPERA version is composed of two numbers. The second number identifies different databases. If there is an update only to the databases, only the second number moves on. When the OPERA is revised with changes on the Main Computer Interface or Calculation Modules the version moves on (e.g.: version 1.0 evolves to version 3.0).

Any time OPERA is revised, a new installer media is issued and the AFM is revised accordingly. After approval, both are provided to operators. Previous installer remains valid only if listed at Valid Versions table.

A dedicated User's Guide is also issued, as a separate document, providing detailed information on software usage aspects. In addition, the OPERA interface provides a Help feature with useful on line information.

If not mentioned, the latest Software Configuration Media incorporates all capabilities of the previous media.

For Operational Systems tested for running the OPERA software, please check in the Pilot's Operating Handbook Section 3.



SOFTWARE CONFIGURATION

WARNING: - OPERA VERSION MUST BE IN ACCORDANCE WITH AIRPLANE CONFIGURATION.

- OPERA UNITS MUST BE IN ACCORDANCE WITH AIRPLANE AVIONICS.

- VERSIONS NOT PRESENTED IN AFM ARE AUTOMATICALLY CANCELLED AND MUST NOT BE USED.

CAUTION: • AFTER USING THE AIRPORT ANALYSIS OPTION, CHECK THE MINIMUM REQUIRED RUNWAY LENGTH FOR TAKEOFF BY INPUTTING THE ACTUAL TAKEOFF WEIGHT IN THE GIVEN WEIGHT OPTION.

- THE TAKEOFF PERFORMANCE FLIGHT CARD DOES NOT PROVIDE RUNWAY SURFACE TYPE INFORMATION. BE AWARE OF THIS CONDITION WHEN SELECTING "GROOVED/PFC" FOR SURFACE TYPE ON OPERA SOFTWARE IN ORDER TO USE THE SUITABLE DATA.

NOTE: For operations in Region 2 of the Operational Envelope, driftdown must be calculated by setting the Minimum Operation Speed as the "given speed" and using SAT (OAT) as the temperature unit.

OP505EA535E version 1.0

The software OP505EA535E version 1.0 allows performance calculations for airplanes with the following configuration:

- PHENOM 300.
- PW535E engines.
- FADEC Software 3.04.
- Airplanes Pre-Mod. SB 505-00-0001.

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OP505EA535E version 17.0

The software OP505EA535E version 17.0 allows performance calculations for airplanes with the following configuration:

- PHENOM 300.
- PW535E engines.
- FADEC Software 3.5 and on.
- Airplanes Post-Mod. SB 505-00-0001 or with an equivalent modification factory incorporated.
- Airplanes Pre-Mod. SB 505-00-0008.

OP505EA535E version 17.1

The software OP505EA535E version 17.1 allows performance calculations for airplanes with the following configuration:

- PHENOM 300.
- PW535E engines.
- FADEC Software 3.5 and on.
- Airplanes Post-Mod. SB 505-00-0001 or with an equivalent modification factory incorporated.
- Airplanes Post-Mod. SB 505-00-0008 or with an equivalent modification factory incorporated.

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VALID VERSIONS

The table below presents valid OPERA versions.

AIRPLANE SERIAL NUMBER 505:	AIRPLANE CONFIGURATION	VALID OPERA VERSIONS	AFM LOCATION
00004 thru 00046	Pre-Mod. SB 505-00-0001	1.0 (1)	Section 5-02
	Post-Mod. SB 505-00-0001 and Pre-Mod. SB 505-00-0008	17.0	
	Post-Mod. SB 505-00-0001 and Post-Mod. SB 505-00-0008	17.1	
00047 and on	Pre-Mod. SB 505-00-0008	17.0	
	Post-Mod. SB 505-00-0008	17.1	

NOTE: 1) The OPERA version 1.0 requires WINDOWS® XP Operational System.

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POSITION ERROR CORRECTIONS

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

The corrections to be applied to indicated airspeed or altitude in order to eliminate the effect of location of the static port on instrument readings are shown in the Airspeed Position Error Correction chart and in the Altitude Position Error Correction chart. No position error correction is required for ADS 1 and ADS 2, as their indication is already corrected. The position correction charts are only applicable for ADS Stand-by for flap 0 (ZERO) and landing gear up position.

AIRSPEED POSITION ERROR CORRECTION CHART

USE

Enter the chart with indicated airspeed, considering the appropriate configuration, to read airspeed correction. The calibrated airspeed will be the sum of the indicated airspeed and airspeed correction.

EXAMPLE

Given:

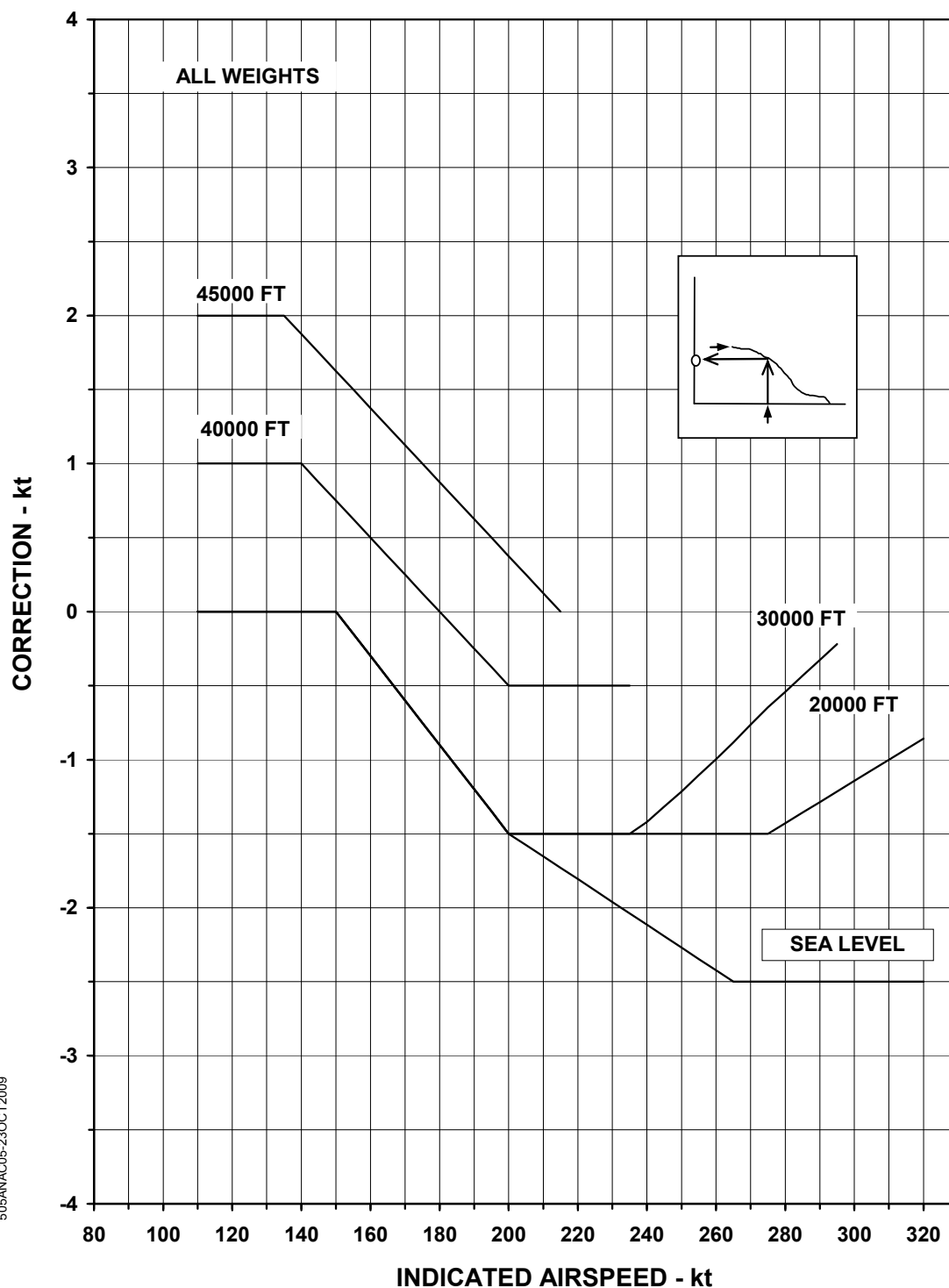
ADS Stand-By	
Indicated Airspeed	230 KIAS
Landing Gear	UP
Flap	ZERO
Altitude	SEA LEVEL

Determine:

Airspeed Correction	-2.0
Calibrated Airspeed	228 KCAS



**AIRSPEED POSITION ERROR CORRECTION
ADS STAND-BY
LANDING GEAR UP - FLAPS ZERO**



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ALTITUDE POSITION ERROR CORRECTION CHART

USE

Enter the chart with indicated airspeed, the indicated pressure altitude considering the appropriate airplane configuration to read the altimetric correction. The true pressure altitude will be the indicated pressure altitude plus the altimetric correction.

EXAMPLE

Given:

ADS Stand-By
Indicated Airspeed..... 280 KIAS
Landing Gear..... UP
Flap..... ZERO
Altitude..... 30000 ft

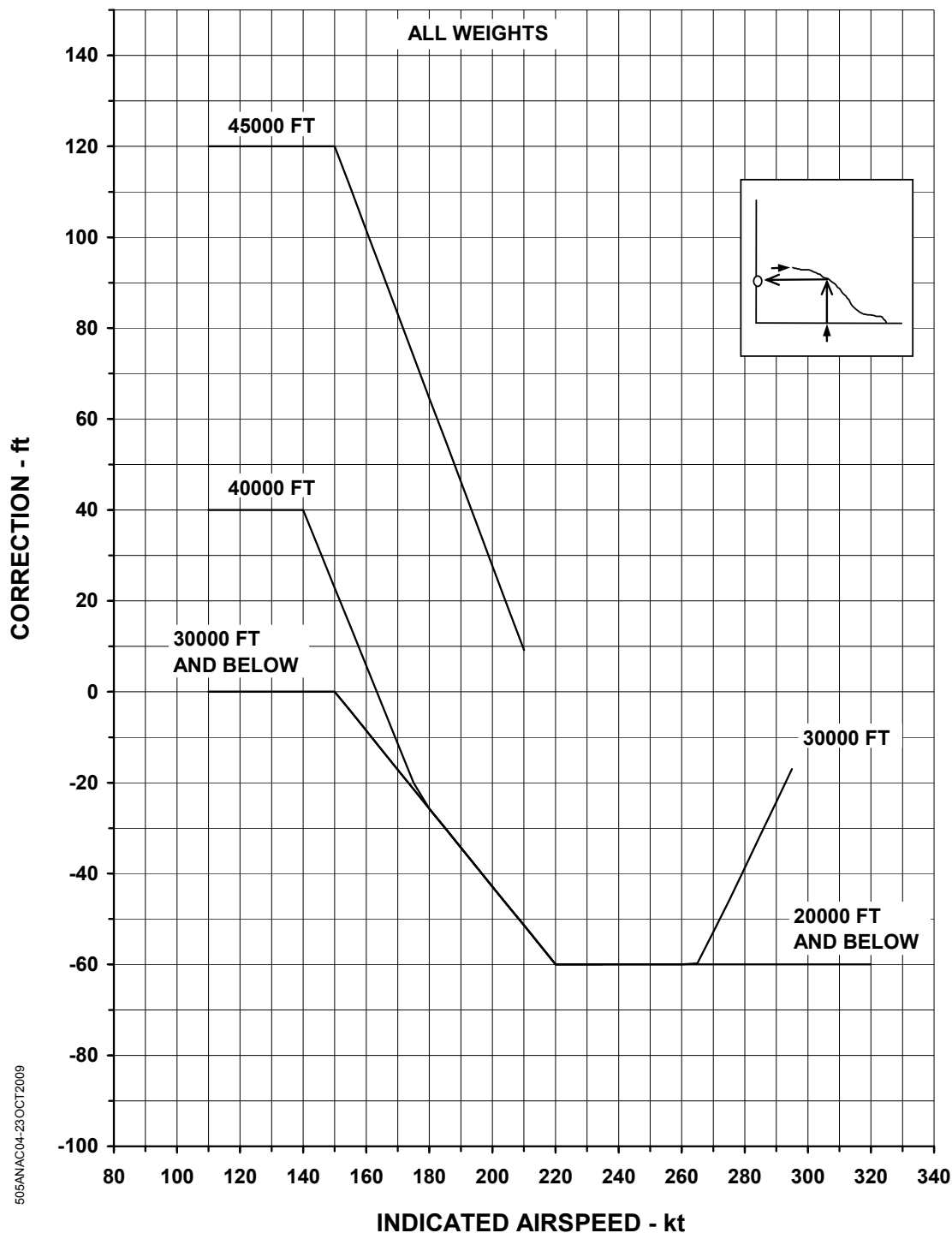
Determine:

Altimetric Correction -40 ft
True Pressure Altitude..... 29960 ft

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**ALTITUDE POSITION ERROR CORRECTION
ADS STAND-BY
LANDING GEAR UP - FLAPS ZERO**



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STALL SPEED

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF

NOTE: This table presents the stall speed value according to the airplane bank angle for the indicated weights with landing gear and flap up.

MTOW (lb)	BANK ANGLE				
	0°	15°	30°	45°	60°
	STALL SPEED (KIAS)				
17968	100	102	107	119	141
18387	101	103	108	120	143

NOTE: This table presents the stall speed value according to the airplane bank angle for the indicated weights with landing gear down and landing flap 3 or FULL.

MLW (lb)	FLAP	BANK ANGLE				
		0°	15°	30°	45°	60°
		STALL SPEED (KIAS)				
16865	3	89	90	95	106	125
	FULL	88	89	94	104	124
17042	3	89	91	96	106	126
	FULL	88	90	95	105	125



ENGINE ICE PROTECTION ON – WINGSTAB ON (SWPS ICE SPEED)

NOTE: This table presents the stall speed value according to the airplane bank angle for the indicated weights with landing gear and flap up.

MTOW (lb)	BANK ANGLE				
	0°	15°	30°	45°	60°
	STALL SPEED (KIAS)				
17968	111	113	120	132	157
18387	113	114	121	134	159

NOTE: This table presents the stall speed value according to the airplane bank angle for the indicated weights with landing gear down and landing flap 3 or FULL.

MLW (lb)	FLAP	BANK ANGLE				
		0°	15°	30°	45°	60°
		STALL SPEED (KIAS)				
16865	3	95	97	102	113	135
	FULL	93	95	100	110	131
17042	3	96	97	103	114	135
	FULL	93	95	100	111	132



THRUST SETTING TABLES

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

Thrust setting table is presented for various pressure altitudes and static air temperatures, with engine bleed open, and anti-ice on or off.

The following tables present Takeoff N1 setting for TO/GA mode on PW535E engines.

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**N1 FOR T/O MODE****ANTI-ICE OFF****AIRSPEED: 0 KIAS****PW535E ENGINES**

Pressure Altitude (ft)	Static Air Temperature (°C)								
	-40	-35	-30	-25	-20	-15	-10	-5	0
-1000	77.0	77.9	78.7	79.5	80.3	81.1	81.9	82.6	83.4
0	78.1	79.0	79.8	80.6	81.4	82.2	83.0	83.8	84.5
1000	79.2	80.1	80.9	81.8	82.6	83.4	84.2	85.0	85.7
2000	80.4	81.2	82.1	82.9	83.7	84.5	85.4	86.1	86.9
3000	81.5	82.3	83.2	84.0	84.9	85.7	86.5	87.3	88.1
4000	82.5	83.4	84.3	85.2	86.0	86.9	87.7	88.5	89.3
5000	83.7	84.6	85.5	86.3	87.2	88.0	88.9	89.7	90.5
6000	84.5	85.4	86.3	87.2	88.1	88.9	89.8	90.6	91.5
7000	84.7	85.7	86.6	87.5	88.4	89.3	90.1	91.0	91.6
8000	85.0	85.9	86.9	87.8	88.7	89.6	90.4	91.3	91.0

Pressure Altitude (ft)	Static Air Temperature (°C)								
	5	10	15	20	25	30	35	40	45
-1000	84.1	84.9	85.6	86.3	87.1	87.8	88.4	87.0	85.6
0	85.3	86.0	86.8	87.5	88.3	89.0	88.2	86.8	85.4
1000	86.5	87.3	88.0	88.8	89.5	89.3	87.9	86.4	85.0
2000	87.7	88.5	89.2	90.0	90.0	89.0	87.5	86.1	84.6
3000	88.9	89.7	90.4	90.6	89.8	88.6	87.2	85.7	-
4000	90.1	90.9	91.1	90.4	89.6	88.3	86.8	85.3	-
5000	91.3	91.6	91.0	90.4	89.5	88.1	86.6	85.1	-
6000	91.8	91.2	90.6	90.0	89.2	87.6	86.1	-	-
7000	91.2	90.7	90.1	89.5	88.4	86.9	85.4	-	-
8000	90.5	90.2	89.8	89.0	87.6	86.0	-	-	-

**N1 FOR T/O MODE****ANTI-ICE ON****AIRSPEED: 0 KIAS****PW535E ENGINES**

Pressure Altitude (ft)	Static Air Temperature (°C)				
	-35	-30	-25	-20	-15
-1000	78.0	78.8	79.6	80.4	81.2
0	79.1	79.9	80.7	81.5	82.3
1000	80.2	81.0	81.8	82.7	83.5
2000	81.3	82.2	83.0	83.8	84.6
3000	82.4	83.3	84.1	84.9	85.8
4000	83.6	84.4	85.3	86.1	86.9
5000	84.7	85.6	86.4	87.3	88.1
6000	85.5	86.4	87.3	88.2	89.0
7000	85.9	86.7	87.6	88.5	89.4
8000	86.2	87.1	87.9	88.8	89.4

Pressure Altitude (ft)	Static Air Temperature (°C)			
	-10	-5	0	5
-1000	82.0	82.7	83.5	84.2
0	83.1	83.8	84.6	85.4
1000	84.2	85.0	85.8	86.6
2000	85.4	86.2	87.0	87.8
3000	86.6	87.4	88.2	89.0
4000	87.8	88.6	89.4	88.9
5000	89.0	89.8	89.4	88.4
6000	89.7	89.7	88.8	87.9
7000	89.5	88.7	87.9	86.8
8000	88.6	87.7	86.8	85.5



TAKEOFF ANALYSIS

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

TAKEOFF TECHNIQUE

The takeoff performance data in this manual is based on the following conditions:

- Thrust lever set to TO/GA position with the airplane stopped and brakes applied;
- Brakes released after N1 stabilized;
- When V_R is attained the airplane is rotated to appropriate pitch angle according to flap selection until 35 ft or until reaching V_2 ;
- Pitch angle adjusted as required to maintain V_2 ;
- Landing gear retracted when a positive rate of climb is established.

SIMPLIFIED TAKEOFF ANALYSIS

Simplified Takeoff Analysis tables are presented for a set of pressure altitudes up to 8000 ft, temperatures and minimum required runway lengths for the conditions below:

- Takeoff flap 1 or 2;
- Ice Protection OFF or ON (WINGSTAB+ENG 1/2);
- ATR OFF;
- Given weight calculation type;
- Dry, wet or contaminated (standing water) runway;
- Zero slope;
- Zero wind;
- Drag index: 0;
- Level-off output: minimum only;
- Minimum level-off height (400 ft);
- Optimum V_1 ;
- No obstacles (analysis assumption).

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NOTE: - ATR OFF does not ensure a conservative performance when compared to ATR ON due to V_{MCG} characteristics.

- The Limited Weight columns represent the weight limited by climb or brake energy for the given temperature. Always check the structural weight limitation on Section 2.
- Refer to the OPERA software configuration to check if any special setting is required in addition to the choices above.
- To determine the required takeoff distance for slope (-2%, -1%, 1% and 2%) refer to Runway Slope Correction tables.

USE

Enter the outside air temperature and the airplane configuration (takeoff weight, ice protection system, runway condition and flap position) on the appropriated table according to the airport pressure altitude to obtain the required takeoff runway length, the associated speeds (V_1 , V_R , V_2 and V_{FS}) and the Climb Limited Weight.

TAKEOFF DISTANCE CORRECTION – RUNWAY SLOPE

In order to determine the minimum required runway length on runways which have a gradient, the Runway Slope Correction tables should be used. The runway lengths herein provided constitute a simplified analysis and therefore, do not represent the optimized takeoff performance for runway slopes.

For detailed data, OPERA software must be used.



USE

To use the Runway Slope Correction, follow the steps below:

- Determine the minimum required runway length and its respective speeds (V_1 , V_R , V_2) for zero slope runway and for the selected configuration (runway conditions, flap position, ice protection system, airport pressure altitude, temperature and takeoff weight) on the Simplified Takeoff Analysis tables;
- Determine the RUNWAY LENGTH FACTOR and SPEED CORRECTION in the Runway Length Factor and Speed (V_R) Correction columns, respectively, for the selected runway slope and configuration (ice protection system, runway conditions and flap position) in the Runway Slope Correction tables;
- Multiply the minimum required runway length obtained on the Simplified Takeoff Analysis tables for zero slope by the RUNWAY LENGTH FACTOR obtained on the last step. The result represents the minimum required runway length for the runway slope condition;
- Add the SPEED (V_R) CORRECTION value to the V_R obtained on the Simplified Takeoff Analysis tables for zero slope. The result represents the required V_R for the runway slope condition;
- The V_1 for the runway slope condition will be the same for the calculated runway slope V_R , thus $V_1 = V_R$;
- The V_2 for the desired runway slope condition will be the same for zero slope V_2 .

**EXAMPLE**

Given:

Airplane Takeoff Weight 15600 lb
 Airport Pressure Altitude 0 ft
 Outside Air Temperature 20°C
 Ice Protection OFF
 Flap 1
 Runway Condition DRY
 Slope 0% AND -1%

Determine:

For Zero Slope Runway:

Minimum Required Runway Length 2920 ft
 V_1 103 KIAS
 V_R 103 KIAS
 V_2 115 KIAS
 V_{FS} 127 KIAS
 Limited Weight 18386 lb

For -1% Slope Runway:

Runway Length Factor 1.03
 Speed (V_R) Correction ADD 1 KIAS
 Minimum Required Runway Length 3008 ft
 V_1 104 KIAS
 V_R 104 KIAS
 V_2 115 KIAS
 V_{FS} 127 KIAS
 Limited Weight 18386 lb


RUNWAY SLOPE CORRECTION – TABLE 1 OF 2

RUNWAY SLOPE (%)	ICE PROTECTION (WINGSTAB+ ENG)	RUNWAY CONDITION	FLAP POSITION	RUNWAY LENGTH FACTOR	SPEED (V _R) CORRECTION
-2	OFF	DRY	1	1.03	ADD 1 KIAS
			2	1.04	
		WET	1	1.35	
			2	1.40	
		STANDING WATER (0.125 in)	1	1.44	
			2	1.44	
		STANDING WATER (0.250 in)	1	1.24	
			2	1.31	
	ON	DRY	1	1.03	
			2	1.04	
		WET	1	1.34	
			2	1.39	
		STANDING WATER (0.125 in)	1	1.41	
			2	1.43	
-1	OFF	DRY	1	1.03	ADD 1 KIAS
			2	1.03	
		WET	1	1.34	
			2	1.38	
		STANDING WATER (0.125 in)	1	1.40	
			2	1.39	
		STANDING WATER (0.250 in)	1	1.21	
			2	1.28	
	ON	DRY	1	1.03	
			2	1.03	
		WET	1	1.32	
			2	1.38	
		STANDING WATER (0.125 in)	1	1.38	
			2	1.39	
		STANDING WATER (0.250 in)	1	1.23	
			2	1.28	

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RUNWAY SLOPE CORRECTION – TABLE 2 OF 2

RUNWAY SLOPE (%)	ICE PROTECTION (WINGSTAB+ ENG)	RUNWAY CONDITION	FLAP POSITION	RUNWAY LENGTH FACTOR	SPEED (V _R) CORRECTION
1	OFF	DRY	1	1.13	ADD 1 KIAS
			2	1.13	
		WET	1	1.32	
			2	1.36	
		STANDING WATER (0.125 in)	1	1.33	
			2	1.34	
	ON	STANDING WATER (0.250 in)	1	1.17	
			2	1.23	
		DRY	1	1.12	
			2	1.10	
		WET	1	1.31	
			2	1.35	
		STANDING WATER (0.125 in)	1	1.32	
			2	1.33	
		STANDING WATER (0.250 in)	1	1.18	
			2	1.23	
2	OFF	DRY	1	1.24	ADD 2 KIAS
			2	1.25	
		WET	1	1.38	
			2	1.37	
		STANDING WATER (0.125 in)	1	1.32	
			2	1.33	
	ON	STANDING WATER (0.250 in)	1	1.17	
			2	1.22	
		DRY	1	1.23	
			2	1.23	
		WET	1	1.32	
			2	1.36	
		STANDING WATER (0.125 in)	1	1.31	
			2	1.32	
		STANDING WATER (0.250 in)	1	1.18	
			2	1.23	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2133 104/104/118	2167 104/104/117	2294 104/104/116	2431 103/103/115	2636 105/105/116	2886 108/108/119	3333 113/113/123	18386
-35	2170 104/104/118	2200 104/104/117	2325 104/104/116	2465 103/103/115	2673 105/105/116	2927 108/108/119	3379 113/113/123	18386
-30	2207 104/104/118	2236 104/104/117	2357 104/104/116	2499 103/103/115	2710 105/105/116	2968 108/108/119	3426 113/113/123	18386
-25	2244 104/104/118	2274 104/104/117	2390 104/104/116	2534 103/103/115	2748 105/105/116	3010 108/108/119	3474 113/113/123	18386
-20	2280 104/104/118	2312 104/104/117	2422 104/104/116	2568 103/103/115	2786 105/105/116	3052 108/108/119	3522 113/113/123	18386
-15	2317 104/104/118	2349 104/104/117	2454 104/104/116	2603 103/103/115	2824 105/105/116	3093 108/108/119	3570 113/113/123	18386
-10	2353 104/104/118	2387 104/104/117	2487 104/104/116	2637 103/103/115	2862 105/105/116	3135 108/108/119	3619 113/113/123	18386
-5	2390 104/104/118	2424 104/104/117	2519 104/104/116	2671 103/103/115	2900 105/105/116	3178 108/108/119	3668 113/113/123	18386
0	2425 104/104/118	2461 104/104/117	2550 104/104/116	2705 103/103/115	2939 105/105/116	3221 108/108/119	3717 113/113/123	18386
5	2460 104/104/118	2497 104/104/117	2583 103/103/116	2740 103/103/115	2981 105/105/116	3266 108/108/119	3771 113/113/123	18386
10	2495 104/104/117	2532 104/104/117	2617 103/103/116	2775 103/103/115	3024 105/105/116	3314 108/108/119	3827 113/113/123	18386
15	2529 104/104/117	2568 104/104/116	2651 103/103/116	2811 103/103/115	3067 105/105/116	3361 108/108/119	3883 113/113/123	18386
20	2565 104/104/117	2604 103/103/116	2685 103/103/115	2847 103/103/115	3113 105/105/116	3411 108/108/119	3943 113/113/123	18386
25	2600 104/104/117	2641 103/103/116	2720 103/103/115	2884 103/103/114	3158 105/105/116	3461 108/108/119	4002 113/113/123	18386
30	2623 103/103/116	2666 103/103/115	2791 102/102/114	2971 103/103/114	3276 106/106/116	3614 109/109/119	4167 113/113/123	18386
35	2559 99/99/111	2722 99/99/111	2949 100/100/111	3268 103/103/114	3641 107/107/116	4031 110/110/119	4615 114/114/123	18386
40	2650 96/96/107	2907 98/98/108	3246 101/101/111	3651 104/104/114	4055 107/107/116	4465 110/110/119	5149 115/115/123	18386
45	2843 95/95/105	3201 99/99/108	3628 102/102/111	4053 105/105/114	4479 108/108/116	4978 111/111/119	5738 115/115/123	18386
50	3146 96/96/105	3587 99/99/108	4034 103/103/111	4502 106/106/114	5031 109/109/116	5566 112/112/119	6421 116/116/123	18386
52	3300 96/96/105	3749 100/100/108	4206 103/103/111	4728 106/106/114	5267 109/109/116	5815 112/112/119	- -	17996
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2206 105/105/118	2234 104/104/117	2346 104/104/116	2488 104/104/115	2690 105/105/116	2946 108/108/119	3396 113/113/123	18386
-35	2244 105/105/118	2274 104/104/117	2379 104/104/116	2523 104/104/115	2727 105/105/116	2987 108/108/119	3443 113/113/123	18386
-30	2283 105/105/118	2314 104/104/117	2412 104/104/116	2559 104/104/115	2766 105/105/116	3030 108/108/119	3491 113/113/123	18386
-25	2320 105/105/118	2352 104/104/117	2444 104/104/116	2593 104/104/115	2805 105/105/116	3072 108/108/119	3540 113/113/123	18386
-20	2357 105/105/118	2390 104/104/117	2478 104/104/116	2629 104/104/115	2845 105/105/116	3117 108/108/119	3591 113/113/123	18386
-15	2395 104/104/118	2429 104/104/117	2512 104/104/116	2665 103/103/115	2885 105/105/116	3161 108/108/119	3643 113/113/123	18386
-10	2432 104/104/118	2467 104/104/117	2545 104/104/116	2700 103/103/115	2926 105/105/116	3206 108/108/119	3694 113/113/123	18386
-5	2469 104/104/118	2506 104/104/117	2579 104/104/116	2736 103/103/115	2966 105/105/116	3250 108/108/119	3746 113/113/123	18386
0	2506 104/104/118	2544 104/104/117	2611 104/104/116	2771 103/103/115	3006 105/105/116	3294 108/108/119	3796 113/113/123	18386
5	2542 104/104/118	2580 104/104/117	2646 104/104/116	2808 103/103/115	3051 105/105/116	3343 108/108/119	3855 113/113/123	18386
10	2577 104/104/118	2617 104/104/117	2681 103/103/116	2845 103/103/115	3096 105/105/116	3393 108/108/119	3913 113/113/123	18386
15	2612 104/104/117	2653 104/104/116	2717 103/103/116	2882 103/103/115	3142 105/105/116	3443 108/108/119	3974 113/113/123	18386
20	2649 104/104/117	2691 103/103/116	2753 103/103/115	2920 103/103/115	3190 105/105/116	3496 108/108/119	4036 113/113/123	18386
25	2685 104/104/117	2729 103/103/116	2817 103/103/115	2985 103/103/114	3277 106/106/116	3600 109/109/119	4158 113/113/123	18386
30	2637 101/101/113	2730 100/100/112	2905 100/100/111	3213 103/103/114	3554 106/106/116	3945 109/109/119	4528 114/114/123	18386
35	2657 97/97/109	2856 97/97/108	3193 101/101/111	3572 104/104/114	3977 107/107/116	4387 110/110/119	5031 114/114/123	18386
40	2799 95/95/105	3146 98/98/108	3558 102/102/111	3983 105/105/114	4410 108/108/116	4871 111/111/119	5633 115/115/123	18386
45	3080 96/96/105	3512 99/99/108	3961 103/103/111	4408 106/106/114	4919 109/109/116	5455 112/112/119	6271 116/116/123	18386
50	3455 97/97/105	3919 100/100/108	4397 103/103/111	4955 106/106/114	5513 110/110/116	6098 112/112/119	- -	17658
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2273 105/105/118	2304 104/104/117	2405 104/104/116	2551 104/104/115	2759 105/105/116	3022 108/108/119	3480 113/113/123	18386
-35	2313 105/105/118	2345 104/104/117	2439 104/104/116	2587 104/104/115	2799 105/105/116	3065 108/108/119	3530 113/113/123	18386
-30	2352 105/105/118	2385 104/104/117	2473 104/104/116	2623 104/104/115	2839 105/105/116	3109 108/108/119	3580 113/113/123	18386
-25	2391 104/104/118	2425 104/104/117	2507 104/104/116	2660 103/103/115	2880 105/105/116	3154 108/108/119	3632 113/113/123	18386
-20	2430 104/104/118	2465 104/104/117	2541 104/104/116	2696 103/103/115	2921 105/105/116	3200 108/108/119	3684 113/113/123	18386
-15	2469 104/104/118	2505 104/104/117	2576 104/104/116	2733 103/103/115	2962 105/105/116	3245 108/108/119	3736 113/113/123	18386
-10	2507 104/104/118	2545 104/104/117	2610 104/104/116	2770 103/103/115	3003 105/105/116	3291 108/108/119	3789 113/113/123	18386
-5	2546 104/104/118	2585 104/104/117	2645 104/104/116	2806 103/103/115	3045 105/105/116	3336 108/108/119	3841 113/113/123	18386
0	2584 104/104/118	2624 104/104/117	2680 104/104/116	2843 103/103/115	3087 105/105/116	3383 108/108/119	3896 113/113/123	18386
5	2621 104/104/118	2662 104/104/117	2715 103/103/116	2881 103/103/115	3133 105/105/116	3433 108/108/119	3955 113/113/123	18386
10	2658 104/104/117	2700 104/104/117	2751 103/103/116	2918 103/103/115	3179 105/105/116	3484 108/108/119	4015 113/113/123	18386
15	2694 104/104/117	2738 104/104/116	2787 103/103/116	2956 103/103/115	3226 105/105/116	3535 108/108/119	4076 113/113/123	18386
20	2732 104/104/117	2777 103/103/116	2844 103/103/115	3014 103/103/114	3302 105/105/116	3621 109/109/119	4185 113/113/123	18386
25	2698 101/101/114	2756 101/101/113	2934 101/101/112	3220 103/103/114	3549 106/106/116	3942 109/109/119	4529 113/113/123	18386
30	2669 98/98/110	2852 98/98/109	3156 101/101/111	3509 104/104/114	3916 107/107/116	4328 110/110/119	4946 114/114/123	18386
35	2776 95/95/105	3116 98/98/108	3505 101/101/111	3933 105/105/114	4361 108/108/116	4795 111/111/119	5556 115/115/123	18386
40	3049 95/95/105	3458 99/99/108	3908 102/102/111	4358 105/105/114	4837 109/109/116	5375 112/112/119	6182 116/116/123	18386
45	3395 96/96/105	3861 100/100/108	4335 103/103/111	4866 106/106/114	5426 109/109/116	5994 112/112/119	-	17997
48	3637 97/97/105	4117 100/100/108	4645 104/104/111	5223 107/107/114	5801 110/110/116	6440 113/113/119	-	17281
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2340 104/104/118	2372 104/104/117	2469 104/104/116	2618 103/103/115	2842 105/105/117	3111 108/108/119	3583 113/113/123	18386
-35	2380 104/104/118	2414 104/104/117	2504 104/104/116	2656 103/103/115	2883 105/105/117	3157 108/108/119	3636 113/113/123	18386
-30	2421 104/104/118	2456 104/104/117	2539 104/104/116	2693 103/103/115	2925 105/105/117	3203 108/108/119	3689 113/113/123	18386
-25	2462 104/104/118	2498 104/104/117	2574 104/104/116	2731 103/103/115	2966 105/105/117	3249 108/108/119	3742 113/113/123	18386
-20	2501 104/104/118	2539 104/104/117	2610 104/104/116	2769 103/103/115	3009 105/105/117	3296 108/108/119	3796 113/113/123	18386
-15	2541 104/104/118	2580 104/104/117	2645 104/104/116	2806 103/103/115	3051 105/105/117	3343 108/108/119	3851 113/113/123	18386
-10	2581 104/104/118	2621 104/104/117	2680 103/103/116	2844 103/103/115	3094 105/105/117	3390 108/108/119	3905 113/113/123	18386
-5	2620 104/104/118	2661 104/104/117	2717 103/103/116	2882 103/103/115	3139 105/105/117	3439 108/108/119	3962 113/113/123	18386
0	2659 104/104/117	2701 104/104/117	2753 103/103/116	2920 103/103/115	3184 105/105/117	3489 108/108/119	4021 113/113/123	18386
5	2698 104/104/117	2741 104/104/116	2789 103/103/116	2959 103/103/115	3230 105/105/117	3539 108/108/119	4080 113/113/123	18386
10	2736 104/104/117	2780 103/103/116	2827 103/103/115	2998 103/103/115	3278 105/105/117	3592 108/108/119	4143 113/113/123	18386
15	2774 104/104/117	2819 103/103/116	2868 103/103/115	3041 103/103/114	3333 105/105/117	3652 108/108/119	4220 113/113/123	18386
20	2753 101/101/114	2801 101/101/113	2969 101/101/112	3243 103/103/114	3571 106/106/117	3965 109/109/119	4558 113/113/123	18386
25	2716 99/99/111	2879 98/98/110	3154 100/100/111	3494 104/104/114	3902 107/107/117	4315 110/110/119	4937 114/114/123	18386
30	2792 96/96/107	3079 98/98/108	3442 101/101/111	3872 104/104/114	4301 108/108/117	4737 111/111/119	5466 115/115/123	18386
35	3021 95/95/105	3408 99/99/108	3861 102/102/111	4313 105/105/114	4766 108/108/117	5303 111/111/119	6113 116/116/123	18386
40	3342 96/96/105	3810 100/100/108	4286 103/103/111	4786 106/106/114	5349 109/109/117	5919 112/112/119	-	18344
45	3746 97/97/105	4238 100/100/108	4788 104/104/111	5380 107/107/114	5974 110/110/117	-	-	17146
46	3830 97/97/105	4328 100/100/108	4907 104/104/111	5505 107/107/114	6115 110/110/117	-	-	16911
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2384 103/103/117	2419 103/103/116	2543 103/103/115	2696 102/102/114	2971 106/106/117	3257 109/109/119	3758 113/113/123	18386
-30	2467 103/103/117	2504 103/103/116	2616 103/103/115	2773 102/102/114	3058 106/106/117	3352 109/109/119	3869 113/113/123	18386
-20	2549 103/103/116	2588 103/103/116	2689 103/103/115	2853 102/102/114	3147 106/106/117	3451 109/109/119	3984 113/113/123	18386
-10	2631 103/103/116	2672 103/103/115	2763 102/102/115	2935 102/102/114	3238 106/106/117	3552 109/109/119	4101 113/113/123	18386
0	2711 103/103/116	2755 103/103/115	2836 102/102/114	3018 102/102/114	3330 106/106/117	3654 109/109/119	4219 113/113/123	18386
10	2790 103/103/116	2837 102/102/115	2913 102/102/114	3109 102/102/114	3430 106/106/117	3768 109/109/119	4349 113/113/123	18386
20	2771 99/99/111	2910 99/99/110	3173 100/100/111	3518 104/104/114	3921 107/107/117	4339 110/110/119	4967 114/114/123	18386
30	2987 95/95/105	3358 98/98/108	3805 102/102/111	4258 105/105/114	4713 108/108/117	5218 111/111/119	6031 115/115/123	18386
40	3693 97/97/105	4187 100/100/108	4706 103/103/111	5301 107/107/114	5897 110/110/117	6531 113/113/119	-	17465
44	4032 97/97/105	4557 101/101/108	5181 104/104/111	5802 107/107/114	6468 110/110/117	-	-	16543
V_{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2428 102/102/115	2465 102/102/114	2624 102/102/113	2829 103/103/114	3117 106/106/117	3440 109/109/119	3960 113/113/123	18386
-30	2513 102/102/115	2552 102/102/114	2700 102/102/113	2912 103/103/114	3209 106/106/117	3541 109/109/119	4078 113/113/123	18386
-20	2596 102/102/115	2638 102/102/114	2775 101/101/113	2997 103/103/114	3304 106/106/117	3646 109/109/119	4199 113/113/123	18386
-10	2680 102/102/115	2724 102/102/114	2851 101/101/113	3084 103/103/114	3399 106/106/117	3752 109/109/119	4322 113/113/123	18386
0	2763 102/102/115	2810 102/102/114	2928 101/101/113	3171 103/103/114	3496 106/106/117	3861 109/109/119	4447 113/113/123	18386
10	2849 102/102/114	2899 101/101/114	3031 101/101/113	3300 103/103/114	3636 106/106/117	4027 109/109/119	4636 113/113/123	18386
20	2844 97/97/108	3083 97/97/108	3446 101/101/111	3856 104/104/114	4292 107/107/117	4734 110/110/119	5428 115/115/123	18386
30	3257 96/96/105	3703 99/99/108	4183 102/102/111	4663 106/106/114	5187 109/109/117	5762 112/112/119	6625 116/116/123	18386
40	4067 97/97/105	4591 101/101/108	5222 104/104/111	5852 107/107/114	6516 110/110/117	-	-	16661
42	4246 98/98/105	4826 101/101/108	5472 104/104/111	6117 107/107/114	-	-	-	16207
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2473 101/101/114	2550 101/101/113	2717 100/100/112	2990 103/103/114	3290 106/106/117	3653 109/109/119	4195 113/113/123	18386
-30	2560 101/101/114	2623 101/101/113	2795 100/100/112	3077 103/103/114	3387 106/106/117	3760 109/109/119	4319 113/113/123	18386
-20	2645 101/101/113	2696 101/101/113	2873 100/100/112	3168 103/103/114	3487 106/106/117	3872 109/109/119	4448 113/113/123	18386
-10	2731 101/101/113	2778 100/100/112	2951 100/100/112	3259 103/103/114	3588 106/106/117	3984 109/109/119	4578 113/113/123	18386
0	2816 101/101/113	2866 100/100/112	3031 100/100/111	3352 103/103/114	3692 106/106/117	4101 109/109/119	4712 113/113/123	18386
10	2883 100/100/112	2981 99/99/111	3225 100/100/111	3578 103/103/114	3977 107/107/117	4405 110/110/119	5049 114/114/123	18386
20	2966 95/95/105	3339 98/98/108	3751 101/101/111	4210 105/105/114	4671 108/108/117	5138 111/111/119	5946 115/115/123	18386
30	3558 96/96/105	4056 100/100/108	4563 103/103/111	5102 106/106/114	5701 109/109/117	6309 112/112/119	-	18173
40	4440 98/98/105	5067 101/101/108	5735 104/104/111	6428 108/108/114	-	-	-	15878
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2519 100/100/112	2643 99/99/111	2851 100/100/111	3162 103/103/114	3502 106/106/117	3880 109/109/119	4445 114/114/123	18386
-30	2609 100/100/112	2719 99/99/111	2933 100/100/111	3254 103/103/114	3604 106/106/117	3994 109/109/119	4577 114/114/123	18386
-20	2696 100/100/112	2795 99/99/111	3020 100/100/111	3351 103/103/114	3712 106/106/117	4114 109/109/119	4715 114/114/123	18386
-10	2786 99/99/112	2872 99/99/111	3106 100/100/111	3448 103/103/114	3821 106/106/117	4235 109/109/119	4854 114/114/123	18386
0	2877 99/99/112	2949 99/99/111	3196 100/100/111	3547 103/103/114	3933 106/106/117	4359 110/110/119	4997 114/114/123	18386
10	2923 97/97/108	3168 97/97/108	3542 101/101/111	3963 104/104/114	4412 107/107/117	4868 110/110/119	5581 115/115/123	18386
20	3255 95/95/105	3676 99/99/108	4165 102/102/111	4653 105/105/114	5145 108/108/117	5725 111/111/119	6599 116/116/123	18386
30	3960 97/97/105	4488 100/100/108	5057 104/104/111	5692 107/107/114	6330 110/110/117	7021 113/113/119	-	17202
38	4705 98/98/105	5387 101/101/108	6083 105/105/111	-	-	-	-	15429
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2574 98/98/109	2752 97/97/109	3069 101/101/111	3410 104/104/114	3805 107/107/117	4202 110/110/119	4797 114/114/123	18386
-30	2647 98/98/109	2831 97/97/108	3159 101/101/111	3509 104/104/114	3916 107/107/117	4327 110/110/119	4943 114/114/123	18386
-20	2727 98/98/109	2911 97/97/108	3252 101/101/111	3614 104/104/114	4033 107/107/117	4456 110/110/119	5091 114/114/123	18386
-10	2820 98/98/109	2991 97/97/108	3347 101/101/111	3720 104/104/114	4152 107/107/117	4588 110/110/119	5242 114/114/123	18386
0	2914 97/97/109	3077 97/97/108	3446 101/101/111	3833 104/104/114	4278 107/107/117	4726 110/110/119	5401 114/114/123	18386
10	3113 95/95/105	3504 98/98/108	3952 102/102/111	4429 105/105/114	4908 108/108/117	5408 111/111/119	6263 115/115/123	18386
20	3584 96/96/105	4095 99/99/108	4614 103/103/111	5139 106/106/114	5753 109/109/117	6374 112/112/119	7347 116/116/123	18386
30	4421 97/97/105	5013 101/101/108	5693 104/104/111	6372 107/107/114	-	-	-	16176
36	5073 98/98/105	5786 102/102/108	6546 105/105/111	-	-	-	-	14890
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2683 96/96/107	2957 98/98/108	3302 101/101/111	3710 104/104/114	4125 107/107/117	4544 110/110/119	5222 115/115/123	18386
-30	2759 96/96/107	3043 98/98/108	3399 101/101/111	3818 104/104/114	4247 107/107/117	4679 110/110/119	5378 115/115/123	18386
-20	2837 96/96/107	3133 98/98/108	3500 101/101/111	3932 104/104/114	4373 107/107/117	4819 110/110/119	5540 115/115/123	18386
-10	2914 96/96/107	3223 98/98/108	3601 101/101/111	4046 104/104/114	4501 107/107/117	4960 110/110/119	5703 115/115/123	18386
0	3043 95/95/106	3394 98/98/108	3799 101/101/111	4270 105/105/114	4741 108/108/117	5221 111/111/119	6025 115/115/123	18386
10	3435 96/96/105	3899 99/99/108	4411 102/102/111	4923 106/106/114	5471 109/109/117	6079 112/112/119	6994 116/116/123	18386
20	3997 97/97/105	4536 100/100/108	5086 103/103/111	5730 107/107/114	6379 110/110/117	7052 113/113/119	-	17579
30	5017 98/98/105	5732 102/102/108	6469 105/105/111	-	-	-	-	15200
34	5534 99/99/105	6287 102/102/108	-	-	-	-	-	14416
V _{FS}	117	121	124	127	131	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3085 104/104/118	3099 104/104/117	3117 104/104/116	3133 103/103/115	3149 103/105/116	3168 103/108/119	3333 103/113/123	18386
-35	3144 104/104/118	3159 104/104/117	3179 104/104/116	3196 103/103/115	3213 103/105/116	3233 103/108/119	3379 103/113/123	18386
-30	3204 104/104/118	3220 104/104/117	3241 104/104/116	3259 103/103/115	3277 103/105/116	3298 103/108/119	3426 103/113/123	18386
-25	3264 104/104/118	3282 104/104/117	3304 104/104/116	3323 103/103/115	3342 103/105/116	3365 103/108/119	3474 103/113/123	18386
-20	3324 104/104/118	3343 104/104/117	3366 104/104/116	3387 103/103/115	3407 103/105/116	3430 103/108/119	3522 103/113/123	18386
-15	3383 104/104/118	3404 104/104/117	3428 104/104/116	3450 103/103/115	3472 103/105/116	3496 103/108/119	3570 103/113/123	18386
-10	3443 104/104/118	3465 104/104/117	3491 104/104/116	3514 103/103/115	3537 103/105/116	3562 103/108/119	3619 103/113/123	18386
-5	3503 104/104/118	3526 104/104/117	3554 104/104/116	3578 103/103/115	3601 103/105/116	3628 103/108/119	3670 103/113/123	18386
0	3563 104/104/118	3587 104/104/117	3616 104/104/116	3641 103/103/115	3666 103/105/116	3694 103/108/119	3737 102/113/123	18386
5	3621 104/104/118	3646 104/104/117	3676 103/103/116	3703 103/103/115	3728 103/105/116	3757 103/108/119	3803 102/113/123	18386
10	3677 104/104/117	3704 104/104/117	3735 103/103/116	3763 103/103/115	3789 103/105/116	3819 103/108/119	3867 102/113/123	18386
15	3734 104/104/117	3762 104/104/116	3795 103/103/116	3824 103/103/115	3852 103/105/116	3883 103/108/119	3932 102/113/123	18386
20	3798 104/104/117	3827 103/103/116	3860 103/103/115	3890 103/103/115	3918 103/105/116	3950 102/108/119	4000 102/113/123	18386
25	3861 104/104/117	3891 103/103/116	3925 103/103/115	3956 103/103/114	3986 103/105/116	4018 102/108/119	4069 102/113/123	18386
30	3887 103/103/116	3919 103/103/115	3956 102/102/114	3988 102/103/114	4020 102/106/116	4055 102/109/119	4167 101/113/123	18386
35	3722 99/99/111	3757 99/99/111	3796 99/100/111	3832 99/103/114	3867 98/107/116	4031 98/110/119	4615 103/114/123	18386
40	3573 96/96/107	3612 96/98/108	3654 96/101/111	3694 95/104/114	4055 97/107/116	4465 100/110/119	5149 105/115/123	18386
45	3434 93/95/105	3476 93/99/108	3628 92/102/111	4053 95/105/114	4479 99/108/116	4978 102/111/119	5738 108/115/123	18386
50	3300 90/96/105	3587 90/99/108	4034 94/103/111	4502 97/106/114	5031 101/109/116	5566 105/112/119	6421 110/116/123	18386
52	3300 88/96/105	3749 90/100/108	4206 94/103/111	4728 98/106/114	5267 102/109/116	5815 105/112/119	- -	17996
V _{FS}	117	120	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3207 105/105/118	3223 104/104/117	3243 104/104/116	3261 104/104/115	3278 103/105/116	3298 103/108/119	3396 103/113/123	18386
-35	3271 105/105/118	3288 104/104/117	3309 104/104/116	3328 104/104/115	3346 103/105/116	3367 103/108/119	3443 103/113/123	18386
-30	3334 105/105/118	3352 104/104/117	3375 104/104/116	3395 104/104/115	3414 103/105/116	3437 103/108/119	3491 103/113/123	18386
-25	3394 105/105/118	3414 104/104/117	3438 104/104/116	3459 104/104/115	3480 103/105/116	3504 103/108/119	3542 103/113/123	18386
-20	3455 105/105/118	3476 104/104/117	3501 104/104/116	3524 104/104/115	3545 103/105/116	3570 103/108/119	3610 103/113/123	18386
-15	3517 104/104/118	3539 104/104/117	3566 104/104/116	3590 103/103/115	3613 103/105/116	3639 103/108/119	3680 103/113/123	18386
-10	3579 104/104/118	3603 104/104/117	3631 104/104/116	3656 103/103/115	3680 103/105/116	3707 103/108/119	3750 103/113/123	18386
-5	3640 104/104/118	3665 104/104/117	3695 104/104/116	3721 103/103/115	3746 103/105/116	3775 103/108/119	3820 103/113/123	18386
0	3703 104/104/118	3730 104/104/117	3760 104/104/116	3788 103/103/115	3814 103/105/116	3844 103/108/119	3890 103/113/123	18386
5	3762 104/104/118	3790 104/104/117	3822 104/104/116	3850 103/103/115	3878 103/105/116	3909 103/108/119	3957 102/113/123	18386
10	3821 104/104/118	3850 104/104/117	3883 103/103/116	3913 103/103/115	3942 103/105/116	3974 103/108/119	4024 102/113/123	18386
15	3879 104/104/117	3910 104/104/116	3945 103/103/116	3976 103/103/115	4006 103/105/116	4039 103/108/119	4091 102/113/123	18386
20	3945 104/104/117	3976 103/103/116	4012 103/103/115	4044 103/103/115	4075 103/105/116	4109 102/108/119	4162 102/113/123	18386
25	4001 104/104/117	4035 103/103/116	4073 103/103/115	4106 103/103/114	4139 102/106/116	4175 102/109/119	4231 102/113/123	18386
30	3876 101/101/113	3912 100/100/112	3952 100/100/111	3988 100/103/114	4023 100/106/116	4062 99/109/119	4528 102/114/123	18386
35	3716 97/97/109	3755 97/97/108	3798 97/101/111	3838 96/104/114	3977 96/107/116	4387 99/110/119	5031 104/114/123	18386
40	3569 94/95/105	3612 94/98/108	3658 93/102/111	3983 94/105/114	4410 98/108/116	4871 101/111/119	5633 107/115/123	18386
45	3431 91/96/105	3512 90/99/108	3961 93/103/111	4408 96/106/114	4919 100/109/116	5455 104/112/119	6271 109/116/123	18386
50	3455 87/97/105	3919 91/100/108	4397 95/103/111	4955 99/106/114	5513 102/110/116	6098 106/112/119	- -	17658
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

**SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – WET RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES**

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3322 105/105/118	3340 104/104/117	3362 104/104/116	3381 104/104/115	3399 103/105/116	3421 103/108/119	3480 103/113/123	18386
-35	3387 105/105/118	3406 104/104/117	3430 104/104/116	3450 104/104/115	3470 103/105/116	3493 103/108/119	3530 103/113/123	18386
-30	3452 105/105/118	3472 104/104/117	3497 104/104/116	3519 104/104/115	3540 103/105/116	3564 103/108/119	3603 103/113/123	18386
-25	3515 104/104/118	3537 104/104/117	3563 104/104/116	3586 103/103/115	3608 103/105/116	3634 103/108/119	3675 103/113/123	18386
-20	3580 104/104/118	3603 104/104/117	3630 104/104/116	3654 103/103/115	3678 103/105/116	3705 103/108/119	3747 103/113/123	18386
-15	3645 104/104/118	3669 104/104/117	3698 104/104/116	3724 103/103/115	3748 103/105/116	3776 103/108/119	3821 103/113/123	18386
-10	3710 104/104/118	3735 104/104/117	3766 104/104/116	3793 103/103/115	3819 103/105/116	3848 103/108/119	3894 103/113/123	18386
-5	3775 104/104/118	3802 104/104/117	3834 104/104/116	3862 103/103/115	3889 103/105/116	3920 103/108/119	3968 103/113/123	18386
0	3839 104/104/118	3868 104/104/117	3901 104/104/116	3931 103/103/115	3959 103/105/116	3991 103/108/119	4040 103/113/123	18386
5	3901 104/104/118	3931 104/104/117	3966 103/103/116	3996 103/103/115	4026 103/105/116	4059 103/108/119	4110 102/113/123	18386
10	3963 104/104/117	3994 104/104/117	4030 103/103/116	4062 103/103/115	4093 103/105/116	4127 103/108/119	4180 102/113/123	18386
15	4025 104/104/117	4058 104/104/116	4095 103/103/116	4128 103/103/115	4160 103/105/116	4196 103/108/119	4250 102/113/123	18386
20	4088 104/104/117	4122 103/103/116	4161 103/103/115	4196 103/103/114	4230 103/105/116	4266 102/109/119	4324 102/113/123	18386
25	3987 101/101/114	4024 101/101/113	4065 101/101/112	4102 100/103/114	4139 100/106/116	4178 100/109/119	4529 101/113/123	18386
30	3857 98/98/110	3897 98/98/109	3941 98/101/111	3982 97/104/114	4021 97/107/116	4328 98/110/119	4946 103/114/123	18386
35	3700 95/95/105	3743 94/98/108	3790 94/101/111	3933 94/105/114	4361 97/108/116	4795 101/111/119	5556 106/115/123	18386
40	3560 92/95/105	3607 91/99/108	3908 92/102/111	4358 96/105/114	4837 99/109/116	5375 103/112/119	6182 108/116/123	18386
45	3423 88/96/105	3861 90/100/108	4335 94/103/111	4866 98/106/114	5426 102/109/116	5994 105/112/119	-	17997
48	3637 87/97/105	4117 91/100/108	4645 95/104/111	5223 99/107/114	5801 103/110/116	6440 106/113/119	-	17281
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3434 104/104/118	3453 104/104/117	3477 104/104/116	3498 103/103/115	3519 103/105/117	3543 103/108/119	3583 103/113/123	18386
-35	3501 104/104/118	3521 104/104/117	3547 104/104/116	3569 103/103/115	3591 103/105/117	3616 103/108/119	3656 103/113/123	18386
-30	3568 104/104/118	3591 104/104/117	3617 104/104/116	3641 103/103/115	3664 103/105/117	3690 103/108/119	3732 103/113/123	18386
-25	3636 104/104/118	3660 104/104/117	3688 104/104/116	3713 103/103/115	3737 103/105/117	3765 103/108/119	3809 102/113/123	18386
-20	3702 104/104/118	3727 104/104/117	3757 104/104/116	3784 103/103/115	3809 103/105/117	3838 103/108/119	3883 102/113/123	18386
-15	3769 104/104/118	3796 104/104/117	3827 104/104/116	3855 103/103/115	3882 103/105/117	3912 103/108/119	3959 102/113/123	18386
-10	3837 104/104/118	3865 104/104/117	3898 103/103/116	3927 103/103/115	3955 103/105/117	3986 103/108/119	4035 102/113/123	18386
-5	3903 104/104/118	3933 104/104/117	3967 103/103/116	3997 103/103/115	4026 103/105/117	4059 103/108/119	4110 102/113/123	18386
0	3969 104/104/117	4000 104/104/117	4035 103/103/116	4067 103/103/115	4098 103/105/117	4131 103/108/119	4184 102/113/123	18386
5	4035 104/104/117	4067 104/104/116	4104 103/103/116	4137 103/103/115	4169 103/105/117	4204 103/108/119	4259 102/113/123	18386
10	4099 104/104/117	4133 103/103/116	4172 103/103/115	4206 103/103/115	4239 103/105/117	4275 102/108/119	4332 102/113/123	18386
15	4164 104/104/117	4200 103/103/116	4240 103/103/115	4275 103/103/114	4310 103/105/117	4347 102/108/119	4406 102/113/123	18386
20	4087 101/101/114	4126 101/101/113	4168 101/101/112	4207 101/103/114	4245 100/106/117	4285 100/109/119	4558 101/113/123	18386
25	3977 99/99/111	4018 98/98/110	4063 98/100/111	4105 98/104/114	4145 98/107/117	4315 98/110/119	4937 103/114/123	18386
30	3842 96/96/107	3886 95/98/108	3934 95/101/111	3979 95/104/114	4301 96/108/117	4737 100/111/119	5466 105/115/123	18386
35	3689 92/95/105	3737 92/99/108	3861 92/102/111	4313 95/105/114	4766 99/108/117	5303 102/111/119	6113 107/116/123	18386
40	3550 89/96/105	3810 89/100/108	4286 93/103/111	4786 97/106/114	5349 101/109/117	5919 104/112/119	-	18344
45	3746 87/97/105	4238 91/100/108	4788 95/104/111	5380 99/107/114	5974 103/110/117	-	-	17146
46	3830 88/97/105	4328 92/100/108	4907 96/104/111	5505 100/107/114	6115 103/110/117	-	-	16911
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3500 103/103/117	3521 103/103/116	3548 103/103/115	3571 102/102/114	3593 102/106/117	3619 102/109/119	3758 102/113/123	18386
-30	3639 103/103/117	3663 103/103/116	3693 103/103/115	3718 102/102/114	3744 102/106/117	3772 102/109/119	3869 102/113/123	18386
-20	3774 103/103/116	3802 103/103/116	3834 103/103/115	3863 102/102/114	3890 102/106/117	3921 102/109/119	3984 102/113/123	18386
-10	3913 103/103/116	3944 103/103/115	3979 102/102/115	4010 102/102/114	4041 102/106/117	4074 102/109/119	4127 101/113/123	18386
0	4050 103/103/116	4084 103/103/115	4122 102/102/114	4156 102/102/114	4189 102/106/117	4225 102/109/119	4281 101/113/123	18386
10	4184 103/103/116	4221 102/102/115	4262 102/102/114	4299 102/102/114	4334 102/106/117	4373 101/109/119	4433 101/113/123	18386
20	4076 99/99/111	4119 99/99/110	4166 99/100/111	4208 98/104/114	4250 98/107/117	4339 98/110/119	4967 102/114/123	18386
30	3824 93/95/105	3873 93/98/108	3926 93/102/111	4258 94/105/114	4713 98/108/117	5218 101/111/119	6031 106/115/123	18386
40	3693 87/97/105	4187 91/100/108	4706 95/103/111	5301 98/107/114	5897 102/110/117	6531 106/113/119	-	17465
44	4032 88/97/105	4557 92/101/108	5181 96/104/111	5802 100/107/114	6468 104/110/117	-	-	16543
V_{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3564 102/102/115	3588 102/102/114	3617 102/102/113	3642 101/103/114	3667 101/106/117	3695 101/109/119	3960 101/113/123	18386
-30	3705 102/102/115	3732 102/102/114	3764 102/102/113	3792 101/103/114	3820 101/106/117	3851 101/109/119	4078 101/113/123	18386
-20	3845 102/102/115	3876 102/102/114	3911 101/101/113	3942 101/103/114	3972 101/106/117	4006 101/109/119	4199 101/113/123	18386
-10	3987 102/102/115	4020 102/102/114	4058 101/101/113	4092 101/103/114	4125 101/106/117	4161 101/109/119	4322 100/113/123	18386
0	4129 102/102/115	4166 102/102/114	4206 101/101/113	4243 101/103/114	4279 101/106/117	4318 101/109/119	4447 100/113/123	18386
10	4265 102/102/114	4305 101/101/114	4350 101/101/113	4391 101/103/114	4431 101/106/117	4473 100/109/119	4636 100/113/123	18386
20	4072 97/97/108	4119 96/97/108	4170 96/101/111	4217 96/104/114	4292 96/107/117	4734 99/110/119	5428 104/115/123	18386
30	3817 91/96/105	3870 91/99/108	4183 92/102/111	4663 95/106/114	5187 99/109/117	5762 103/112/119	6625 108/116/123	18386
40	4067 88/97/105	4591 92/101/108	5222 96/104/111	5852 100/107/114	6516 103/110/117	-	-	16661
42	4246 89/98/105	4826 93/101/108	5472 97/104/111	6117 101/107/114	-	-	-	16207
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3621 101/101/114	3648 101/101/113	3680 100/100/112	3709 100/103/114	3737 100/106/117	3768 100/109/119	4195 102/113/123	18386
-30	3767 101/101/114	3798 101/101/113	3833 100/100/112	3864 100/103/114	3895 100/106/117	3928 100/109/119	4319 102/113/123	18386
-20	3910 101/101/113	3943 101/101/113	3981 100/100/112	4016 100/103/114	4049 100/106/117	4086 100/109/119	4448 101/113/123	18386
-10	4056 101/101/113	4092 100/100/112	4134 100/100/112	4171 100/103/114	4207 100/106/117	4246 99/109/119	4578 101/113/123	18386
0	4201 101/101/113	4241 100/100/112	4285 100/100/111	4326 100/103/114	4364 100/106/117	4407 99/109/119	4712 101/113/123	18386
10	4281 100/100/112	4326 99/99/111	4376 99/100/111	4421 99/103/114	4465 99/107/117	4513 98/110/119	5049 102/114/123	18386
20	4082 95/95/105	4133 94/98/108	4188 94/101/111	4240 94/105/114	4671 96/108/117	5138 100/111/119	5946 105/115/123	18386
30	3825 89/96/105	4056 89/100/108	4563 93/103/111	5102 97/106/114	5701 100/109/117	6309 104/112/119	- -	18173
40	4440 89/98/105	5067 93/101/108	5735 97/104/111	6428 101/108/114	- -	- -	- -	15878
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3681 100/100/112	3712 99/99/111	3747 99/100/111	3779 99/103/114	3810 99/106/117	3880 98/109/119	4445 102/114/123	18386
-30	3831 100/100/112	3865 99/99/111	3903 99/100/111	3938 99/103/114	3971 99/106/117	4009 98/109/119	4577 102/114/123	18386
-20	3977 100/100/112	4014 99/99/111	4055 99/100/111	4093 99/103/114	4130 99/106/117	4170 98/109/119	4715 102/114/123	18386
-10	4134 99/99/112	4174 99/99/111	4218 99/100/111	4258 99/103/114	4298 98/106/117	4340 98/109/119	4854 102/114/123	18386
0	4301 99/99/112	4343 99/99/111	4390 99/100/111	4432 99/103/114	4473 98/106/117	4518 98/110/119	4997 101/114/123	18386
10	4284 97/97/108	4331 96/97/108	4384 96/101/111	4432 96/104/114	4479 96/107/117	4868 98/110/119	5581 103/115/123	18386
20	4098 92/95/105	4151 92/99/108	4209 92/102/111	4653 94/105/114	5145 98/108/117	5725 101/111/119	6599 107/116/123	18386
30	3960 86/97/105	4488 90/100/108	5057 94/104/111	5692 98/107/114	6330 102/110/117	7021 105/113/119	-	17202
38	4705 90/98/105	5387 94/101/108	6083 98/105/111	-	-	-	-	15429
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3695 98/98/109	3729 97/97/109	3768 97/101/111	3803 97/104/114	3838 97/107/117	4202 98/110/119	4797 104/114/123	18386
-30	3845 98/98/109	3882 97/97/108	3924 97/101/111	3963 97/104/114	4000 97/107/117	4327 98/110/119	4943 103/114/123	18386
-20	4001 98/98/109	4042 97/97/108	4087 97/101/111	4128 97/104/114	4168 97/107/117	4456 98/110/119	5091 103/114/123	18386
-10	4170 98/98/109	4212 97/97/108	4260 97/101/111	4303 97/104/114	4345 97/107/117	4588 98/110/119	5242 103/114/123	18386
0	4340 97/97/109	4385 97/97/108	4435 97/101/111	4480 97/104/114	4525 96/107/117	4726 97/110/119	5401 102/114/123	18386
10	4251 94/95/105	4303 94/98/108	4359 93/102/111	4429 93/105/114	4908 96/108/117	5408 100/111/119	6263 105/115/123	18386
20	4087 89/96/105	4145 89/99/108	4614 92/103/111	5139 96/106/114	5753 99/109/117	6374 103/112/119	7347 108/116/123	18386
30	4421 88/97/105	5013 92/101/108	5693 96/104/111	6372 100/107/114	-	-	-	16176
36	5073 90/98/105	5786 95/102/108	6546 98/105/111	-	-	-	-	14890
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3710 96/96/107	3748 95/98/108	3790 95/101/111	3830 95/104/114	4125 96/107/117	4544 100/110/119	5222 105/115/123	18386
-30	3865 96/96/107	3906 95/98/108	3952 95/101/111	3994 95/104/114	4247 96/107/117	4679 99/110/119	5378 104/115/123	18386
-20	4037 96/96/107	4080 95/98/108	4128 95/101/111	4173 95/104/114	4373 95/107/117	4819 99/110/119	5540 104/115/123	18386
-10	4209 96/96/107	4254 95/98/108	4305 95/101/111	4352 95/104/114	4501 95/107/117	4960 99/110/119	5703 104/115/123	18386
0	4372 95/95/106	4423 95/98/108	4478 95/101/111	4530 94/105/114	4741 95/108/117	5221 99/111/119	6025 104/115/123	18386
10	4241 91/96/105	4298 91/99/108	4411 91/102/111	4923 94/106/114	5471 98/109/117	6079 102/112/119	6994 107/116/123	18386
20	4073 87/97/105	4536 89/100/108	5086 93/103/111	5730 97/107/114	6379 101/110/117	7052 104/113/119	-	17579
30	5017 90/98/105	5732 94/102/108	6469 98/105/111	-	-	-	-	15200
34	5534 92/99/105	6287 96/102/108	-	-	-	-	-	14416
V _{FS}	117	121	124	127	131	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4153 104/104/117	4172 103/103/116	4195 103/103/115	4213 103/103/114	4229 102/105/116	4249 102/108/119	4280 102/113/123	18386
-35	4213 104/104/117	4235 103/103/116	4260 103/103/115	4280 103/103/114	4299 102/105/116	4321 102/108/119	4356 102/113/123	18386
-30	4273 104/104/117	4297 103/103/116	4325 103/103/115	4348 103/103/114	4370 102/105/116	4394 102/108/119	4432 102/113/123	18386
-25	4334 104/104/117	4361 103/103/116	4392 103/103/115	4417 103/103/114	4441 102/105/116	4467 102/108/119	4508 102/113/123	18386
-20	4395 104/104/117	4425 103/103/116	4458 103/103/115	4486 103/103/114	4512 102/105/116	4540 102/108/119	4584 102/113/123	18386
-15	4456 104/104/117	4488 103/103/116	4523 103/103/115	4554 103/103/114	4582 102/105/116	4613 102/108/119	4660 102/113/123	18386
-10	4517 104/104/117	4552 103/103/116	4590 103/103/115	4622 103/103/114	4653 102/105/116	4686 102/108/119	4736 102/113/123	18386
-5	4579 103/103/117	4616 103/103/116	4657 103/103/115	4692 103/103/114	4724 102/105/116	4759 102/108/119	4812 102/113/123	18386
0	4642 103/103/117	4681 103/103/116	4724 103/103/115	4761 103/103/114	4796 102/105/116	4833 102/108/119	4889 102/113/123	18386
5	4701 103/103/117	4743 103/103/116	4788 103/103/115	4827 102/102/114	4864 102/105/116	4903 102/108/119	4962 102/113/123	18386
10	4758 103/103/117	4803 103/103/116	4850 103/103/115	4892 102/102/114	4931 102/105/116	4972 102/108/119	5034 102/113/123	18386
15	4818 103/103/116	4864 103/103/116	4914 102/102/115	4958 102/102/114	4999 102/105/116	5042 102/108/119	5107 102/113/123	18386
20	4892 103/103/116	4940 103/103/115	4991 102/102/115	5036 102/102/114	5079 102/105/116	5123 102/108/119	5190 101/113/123	18386
25	4966 103/103/116	5015 103/103/115	5068 102/102/114	5115 102/102/114	5159 102/105/116	5205 102/108/119	5274 101/113/123	18386
30	4999 102/102/115	5051 102/102/114	5106 102/102/113	5155 101/103/114	5201 101/106/116	5250 101/109/119	5323 101/113/123	18386
35	4823 99/99/111	4875 98/98/110	4931 98/100/111	4981 98/103/114	5029 98/107/116	5079 98/110/119	5723 103/114/123	18386
40	4660 95/95/106	4713 95/98/108	4770 95/101/111	4821 95/104/114	4922 95/107/116	5491 100/110/119	6415 107/115/123	18386
45	4502 92/95/105	4556 92/99/108	4614 92/102/111	4893 94/105/114	5491 99/108/116	6143 104/111/119	7197 111/115/123	18386
50	4345 89/96/105	4401 89/99/108	4844 92/103/111	5479 98/106/114	6168 103/109/116	6914 108/112/119	8297 114/116/123	18386
52	4282 88/96/105	4452 89/100/108	5071 94/103/111	5745 99/106/114	6473 104/109/116	7264 109/112/119	- -	17996
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4290 104/104/117	4312 103/103/116	4338 103/103/115	4360 103/103/115	4380 103/105/116	4403 102/108/119	4439 102/113/123	18386
-35	4355 104/104/117	4380 103/103/116	4409 103/103/115	4433 103/103/115	4455 103/105/116	4480 102/108/119	4519 102/113/123	18386
-30	4419 104/104/117	4447 103/103/116	4479 103/103/115	4505 103/103/115	4530 103/105/116	4557 102/108/119	4600 102/113/123	18386
-25	4480 104/104/117	4511 103/103/116	4545 103/103/115	4574 103/103/115	4601 103/105/116	4631 102/108/119	4676 102/113/123	18386
-20	4540 104/104/117	4574 103/103/116	4611 103/103/115	4642 103/103/115	4672 103/105/116	4704 102/108/119	4753 102/113/123	18386
-15	4604 104/104/117	4640 103/103/116	4680 103/103/115	4714 103/103/114	4745 102/105/116	4780 102/108/119	4832 102/113/123	18386
-10	4668 104/104/117	4706 103/103/116	4748 103/103/115	4785 103/103/114	4819 102/105/116	4855 102/108/119	4910 102/113/123	18386
-5	4730 103/103/117	4771 103/103/116	4815 103/103/115	4854 103/103/114	4891 102/105/116	4929 102/108/119	4988 102/113/123	18386
0	4796 103/103/117	4839 103/103/116	4886 103/103/115	4927 103/103/114	4966 102/105/116	5006 102/108/119	5068 102/113/123	18386
5	4856 103/103/117	4902 103/103/116	4951 103/103/115	4995 102/102/114	5035 102/105/116	5078 102/108/119	5143 102/113/123	18386
10	4916 103/103/117	4965 103/103/116	5016 103/103/115	5062 102/102/114	5105 102/105/116	5150 102/108/119	5218 102/113/123	18386
15	4976 103/103/117	5027 103/103/116	5081 102/102/115	5129 102/102/114	5174 102/105/116	5221 102/108/119	5292 102/113/123	18386
20	5052 103/103/116	5105 103/103/115	5160 102/102/115	5210 102/102/114	5257 102/105/116	5305 102/108/119	5378 101/113/123	18386
25	5118 103/103/116	5174 102/102/115	5232 102/102/114	5284 102/102/114	5333 102/106/116	5384 101/109/119	5461 101/113/123	18386
30	4989 100/100/112	5044 100/100/111	5103 99/100/111	5155 99/103/114	5206 99/106/116	5258 99/109/119	5643 101/114/123	18386
35	4817 96/96/108	4873 96/97/108	4932 96/101/111	4987 96/104/114	5038 96/107/116	5424 99/110/119	6331 106/114/123	18386
40	4654 93/95/105	4711 93/98/108	4772 93/102/111	4842 93/105/114	5435 98/108/116	6077 103/111/119	7117 110/115/123	18386
45	4496 90/96/105	4555 90/99/108	4788 91/103/111	5415 96/106/114	6098 102/109/116	6832 106/112/119	8196 113/116/123	18386
50	4337 87/97/105	4707 90/100/108	5370 95/103/111	6095 100/106/114	6876 105/110/116	7759 110/112/119	- -	17658
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4418 104/104/117	4444 103/103/116	4474 103/103/115	4499 103/103/115	4522 103/105/116	4548 102/108/119	4589 102/113/123	18386
-35	4485 104/104/117	4514 103/103/116	4546 103/103/115	4574 103/103/115	4599 103/105/116	4628 102/108/119	4672 102/113/123	18386
-30	4550 104/104/117	4582 103/103/116	4617 103/103/115	4648 103/103/115	4676 103/105/116	4706 102/108/119	4753 102/113/123	18386
-25	4614 104/104/117	4648 103/103/116	4686 103/103/115	4719 103/103/115	4750 102/105/116	4783 102/108/119	4833 102/113/123	18386
-20	4678 104/104/117	4716 103/103/116	4757 103/103/115	4792 103/103/114	4825 102/105/116	4860 102/108/119	4914 102/113/123	18386
-15	4745 104/104/117	4785 103/103/116	4829 103/103/115	4867 103/103/114	4902 102/105/116	4940 102/108/119	4997 102/113/123	18386
-10	4811 103/103/117	4854 103/103/116	4900 103/103/115	4941 103/103/114	4978 102/105/116	5018 102/108/119	5079 102/113/123	18386
-5	4878 103/103/117	4924 103/103/116	4972 103/103/115	5015 103/103/114	5055 102/105/116	5097 102/108/119	5161 102/113/123	18386
0	4945 103/103/117	4993 103/103/116	5044 103/103/115	5089 103/103/114	5131 102/105/116	5176 102/108/119	5243 102/113/123	18386
5	5008 103/103/117	5059 103/103/116	5112 103/103/115	5160 102/102/114	5204 102/105/116	5251 102/108/119	5321 102/113/123	18386
10	5071 103/103/117	5124 103/103/116	5180 103/103/115	5230 102/102/114	5277 102/105/116	5326 102/108/119	5399 102/113/123	18386
15	5136 103/103/116	5192 103/103/116	5250 102/102/115	5302 102/102/114	5351 102/105/116	5402 102/108/119	5479 102/113/123	18386
20	5209 103/103/116	5267 103/103/115	5327 102/102/114	5382 102/102/114	5433 102/105/116	5487 102/109/119	5566 101/113/123	18386
25	5107 100/100/113	5165 100/100/112	5227 100/100/111	5283 100/103/114	5335 99/106/116	5390 99/109/119	5680 101/113/123	18386
30	4971 97/97/109	5030 97/97/108	5093 97/101/111	5150 97/104/114	5204 97/107/116	5383 98/110/119	6281 104/114/123	18386
35	4800 94/95/105	4860 94/98/108	4923 94/101/111	4982 93/105/114	5408 97/108/116	6045 102/111/119	7076 109/115/123	18386
40	4643 91/95/105	4704 91/99/108	4769 90/102/111	5384 96/105/114	6058 101/109/116	6791 106/112/119	8152 112/116/123	18386
45	4484 88/96/105	4673 89/100/108	5333 94/103/111	6051 99/106/114	6829 105/109/116	7669 109/112/119	-	17997
48	4388 86/97/105	5010 91/100/108	5731 96/104/111	6514 102/107/114	7363 107/110/116	8554 112/113/119	-	17281
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4542 104/104/117	4572 103/103/116	4606 103/103/115	4634 103/103/114	4661 102/105/117	4690 102/108/119	4735 102/113/123	18386
-35	4609 103/103/117	4642 103/103/116	4679 103/103/115	4710 103/103/114	4739 102/105/117	4771 102/108/119	4820 102/113/123	18386
-30	4678 103/103/117	4714 103/103/116	4753 103/103/115	4787 103/103/114	4819 102/105/117	4853 102/108/119	4905 102/113/123	18386
-25	4747 103/103/117	4786 103/103/116	4828 103/103/115	4864 103/103/114	4898 102/105/117	4935 102/108/119	4991 102/113/123	18386
-20	4813 103/103/117	4855 103/103/116	4900 103/103/115	4939 102/102/114	4976 102/105/117	5015 102/108/119	5074 102/113/123	18386
-15	4881 103/103/117	4926 103/103/116	4973 103/103/115	5015 102/102/114	5054 102/105/117	5096 102/108/119	5158 102/113/123	18386
-10	4951 103/103/117	4998 103/103/116	5048 103/103/115	5092 102/102/114	5134 102/105/117	5178 102/108/119	5244 102/113/123	18386
-5	5019 103/103/117	5068 103/103/116	5121 103/103/115	5168 102/102/114	5212 102/105/117	5258 102/108/119	5327 102/113/123	18386
0	5086 103/103/117	5139 103/103/116	5194 102/102/115	5243 102/102/114	5289 102/105/117	5338 102/108/119	5410 102/113/123	18386
5	5154 103/103/116	5209 103/103/116	5267 102/102/115	5318 102/102/114	5367 102/105/117	5418 102/108/119	5494 102/113/123	18386
10	5220 103/103/116	5277 103/103/115	5337 102/102/115	5392 102/102/114	5443 102/105/117	5496 102/108/119	5575 101/113/123	18386
15	5289 103/103/116	5349 102/102/115	5412 102/102/114	5468 102/102/114	5521 102/105/117	5576 102/108/119	5659 101/113/123	18386
20	5213 101/101/113	5275 100/100/112	5340 100/100/111	5398 100/103/114	5453 100/106/117	5511 99/109/119	5757 101/113/123	18386
25	5100 98/98/110	5162 98/98/109	5228 97/100/111	5287 97/104/114	5344 97/107/117	5405 97/110/119	6305 104/114/123	18386
30	4957 95/95/106	5020 95/98/108	5086 94/101/111	5148 94/104/114	5367 96/108/117	5999 101/111/119	7017 108/115/123	18386
35	4789 92/95/105	4853 91/99/108	4921 91/102/111	5363 95/105/114	6031 100/108/117	6761 105/111/119	8099 111/116/123	18386
40	4631 88/96/105	4697 88/100/108	5306 93/103/111	6017 98/106/114	6791 104/109/117	7620 108/112/119	-	18344
45	4537 86/97/105	5220 92/100/108	5978 97/104/111	6802 102/107/114	7695 108/110/117	-	-	17146
46	4645 87/97/105	5347 92/100/108	6130 98/104/111	6989 103/107/114	8060 108/110/117	-	-	16911
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4613 103/103/116	4647 102/102/115	4684 102/102/114	4716 102/102/114	4745 101/106/117	4778 101/109/119	4828 101/113/123	18386
-30	4756 102/102/116	4795 102/102/115	4838 102/102/114	4875 102/102/114	4910 101/106/117	4947 101/109/119	5004 101/113/123	18386
-20	4893 102/102/116	4938 102/102/115	4986 102/102/114	5029 102/102/114	5068 101/106/117	5111 101/109/119	5175 101/113/123	18386
-10	5035 102/102/115	5085 102/102/115	5139 102/102/114	5187 101/102/114	5232 101/106/117	5279 101/109/119	5350 101/113/123	18386
0	5176 102/102/115	5232 102/102/114	5291 102/102/114	5344 101/102/114	5394 101/106/117	5446 101/109/119	5523 101/113/123	18386
10	5314 102/102/115	5375 102/102/114	5439 101/101/113	5497 101/102/114	5552 101/106/117	5608 101/109/119	5693 100/113/123	18386
20	5207 98/98/110	5272 98/98/109	5340 98/100/111	5403 98/104/114	5462 97/107/117	5524 97/110/119	6382 104/114/123	18386
30	4939 93/95/105	5007 92/98/108	5078 92/102/111	5329 94/105/114	5994 99/108/117	6713 104/111/119	8039 110/115/123	18386
40	4618 86/97/105	5191 91/100/108	5943 96/103/111	6763 102/107/114	7643 107/110/117	8921 111/113/119	-	17465
44	4954 88/97/105	5714 94/101/108	6566 99/104/111	7490 105/107/114	8746 110/110/117	-	-	16543
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4685 101/101/114	4722 101/101/113	4763 101/101/113	4798 101/103/114	4831 100/106/117	4866 100/109/119	4921 100/113/123	18386
-30	4828 101/101/114	4871 101/101/113	4917 101/101/113	4958 101/103/114	4996 100/106/117	5037 100/109/119	5099 100/113/123	18386
-20	4972 101/101/114	5021 101/101/113	5073 101/101/112	5119 100/103/114	5162 100/106/117	5208 100/109/119	5277 100/113/123	18386
-10	5116 101/101/114	5171 101/101/113	5228 101/101/112	5280 100/103/114	5328 100/106/117	5379 100/109/119	5456 100/113/123	18386
0	5264 101/101/114	5324 101/101/113	5387 100/100/112	5444 100/103/114	5497 100/106/117	5553 100/109/119	5636 100/113/123	18386
10	5404 101/101/114	5470 100/100/113	5540 100/100/112	5603 100/103/114	5662 100/106/117	5724 100/109/119	5933 100/113/123	18386
20	5206 96/96/107	5276 96/97/108	5349 95/101/111	5416 95/104/114	5480 95/107/117	6071 100/110/119	7081 106/115/123	18386
30	4933 90/96/105	5005 90/99/108	5247 91/102/111	5950 97/106/114	6709 102/109/117	7594 106/112/119	9103 113/116/123	18386
40	5045 88/97/105	5823 94/101/108	6694 99/104/111	7634 105/107/114	9024 110/110/117	-	-	16661
42	5297 90/98/105	6126 95/101/108	7050 101/104/111	8259 106/107/114	-	-	-	16207
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4746 100/100/113	4787 100/100/112	4832 100/100/111	4871 99/103/114	4908 99/106/117	4948 99/109/119	5128 100/113/123	18386
-30	4898 100/100/113	4945 100/100/112	4995 100/100/111	5040 99/103/114	5082 99/106/117	5127 99/109/119	5349 100/113/123	18386
-20	5043 100/100/113	5096 100/100/112	5152 99/100/111	5202 99/103/114	5250 99/106/117	5300 99/109/119	5576 101/113/123	18386
-10	5194 100/100/113	5253 100/100/112	5314 99/100/111	5370 99/103/114	5423 99/106/117	5478 99/109/119	5811 101/113/123	18386
0	5345 100/100/112	5409 99/99/111	5476 99/100/111	5537 99/103/114	5595 99/106/117	5655 99/109/119	6049 101/113/123	18386
10	5426 99/99/111	5498 98/98/110	5572 98/100/111	5640 98/103/114	5705 98/107/117	5772 98/110/119	6567 103/114/123	18386
20	5221 94/95/105	5295 94/98/108	5373 93/101/111	5444 93/105/114	6013 97/108/117	6721 102/111/119	8037 108/115/123	18386
30	4943 88/96/105	5099 89/100/108	5837 94/103/111	6636 100/106/114	7539 104/109/117	8654 109/112/119	- -	18173
40	5629 91/98/105	6525 97/101/108	7525 102/104/111	8840 107/108/114	- -	- -	- -	15878
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4813 99/99/111	4858 99/99/110	4907 98/100/111	4950 98/103/114	4991 98/106/117	5035 98/109/119	5513 102/114/123	18386
-30	4968 99/99/111	5019 99/99/110	5074 98/100/111	5123 98/103/114	5170 98/106/117	5219 98/109/119	5752 102/114/123	18386
-20	5117 99/99/111	5175 98/98/110	5235 98/100/111	5290 98/103/114	5342 98/106/117	5396 98/109/119	6002 102/114/123	18386
-10	5287 99/99/111	5350 98/98/110	5416 98/100/111	5475 98/103/114	5532 98/106/117	5591 98/109/119	6261 102/114/123	18386
0	5479 98/98/111	5546 98/98/110	5616 98/100/111	5680 98/103/114	5740 98/106/117	5803 97/110/119	6534 103/114/123	18386
10	5475 96/96/107	5546 96/97/108	5622 95/101/111	5691 95/104/114	5757 95/107/117	6373 99/110/119	7515 106/115/123	18386
20	5284 91/95/105	5358 91/99/108	5437 91/102/111	6061 95/105/114	6828 100/108/117	7794 105/111/119	9245 111/116/123	18386
30	5028 86/97/105	5808 92/100/108	6673 97/104/111	7672 103/107/114	8868 107/110/117	10092 112/113/119	-	17202
38	6134 92/98/105	7134 98/101/108	8523 104/105/111	-	-	-	-	15429
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4828 97/97/109	4877 97/97/108	4929 96/101/111	4977 96/104/114	5022 96/107/117	5200 97/110/119	6059 104/114/123	18386
-30	4983 97/97/109	5038 97/97/108	5097 96/101/111	5150 96/104/114	5201 96/107/117	5425 97/110/119	6329 104/114/123	18386
-20	5152 97/97/108	5213 97/97/108	5277 96/101/111	5336 96/104/114	5391 96/107/117	5665 98/110/119	6609 105/114/123	18386
-10	5347 97/97/108	5412 96/97/108	5480 96/101/111	5543 96/104/114	5603 96/107/117	5923 98/110/119	6906 105/114/123	18386
0	5543 97/97/108	5612 96/97/108	5685 96/101/111	5752 96/104/114	5816 96/107/117	6189 98/110/119	7206 105/114/123	18386
10	5461 93/95/105	5536 93/98/108	5614 93/102/111	5766 93/105/114	6490 98/108/117	7255 103/111/119	8730 109/115/123	18386
20	5291 89/96/105	5369 88/99/108	6057 93/103/111	6885 99/106/114	7936 103/109/117	8999 108/112/119	10606 114/116/123	18386
30	5761 89/97/105	6682 95/101/108	7810 101/104/111	9147 106/107/114	-	-	-	16176
36	6749 94/98/105	8071 100/102/108	9299 105/105/111	-	-	-	-	14890
V_{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4843 95/95/106	4896 95/98/108	4953 94/101/111	5004 94/104/114	5120 95/107/117	5718 100/110/119	6682 107/115/123	18386
-30	5008 95/95/106	5067 95/98/108	5130 94/101/111	5187 94/104/114	5343 95/107/117	5972 100/110/119	6984 107/115/123	18386
-20	5209 95/95/106	5272 95/98/108	5340 94/101/111	5401 94/104/114	5592 95/107/117	6255 100/110/119	7301 107/115/123	18386
-10	5409 95/95/106	5477 95/98/108	5549 94/101/111	5615 94/104/114	5847 95/107/117	6540 100/110/119	7780 107/115/123	18386
0	5601 94/94/105	5675 94/98/108	5754 94/101/111	5827 94/105/114	6260 97/108/117	6993 101/111/119	8383 107/115/123	18386
10	5471 90/96/105	5550 90/99/108	5781 91/102/111	6571 97/106/114	7401 102/109/117	8540 106/112/119	10055 112/116/123	18386
20	5293 86/97/105	6000 91/100/108	6890 97/103/111	8004 102/107/114	9124 106/110/117	10271 110/113/119	-	17579
30	6758 94/98/105	8047 100/102/108	9403 105/105/111	-	-	-	-	15200
34	7614 97/99/105	8969 102/102/108	-	-	-	-	-	14416
V _{FS}	117	121	124	127	131	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3893 103/103/116	3928 103/103/115	3965 102/102/114	3998 102/102/114	4030 102/105/116	4064 102/108/119	4341 104/113/123	18386
-35	3948 103/103/116	3985 102/102/115	4025 102/102/114	4061 102/102/114	4095 102/105/116	4132 102/108/119	4444 105/113/123	18386
-30	4002 103/103/116	4042 102/102/115	4085 102/102/114	4124 102/102/114	4161 102/105/116	4200 102/108/119	4551 105/113/123	18386
-25	4057 103/103/116	4101 102/102/115	4147 102/102/114	4188 102/102/114	4227 102/105/116	4269 102/108/119	4664 105/113/123	18386
-20	4113 103/103/116	4159 102/102/115	4207 102/102/114	4251 102/102/114	4293 102/105/116	4337 101/108/119	4775 106/113/123	18386
-15	4168 103/103/116	4216 102/102/115	4268 102/102/114	4314 102/102/114	4359 102/105/116	4405 101/108/119	4890 106/113/123	18386
-10	4223 103/103/116	4275 102/102/115	4329 102/102/114	4378 102/102/114	4425 102/105/116	4474 101/108/119	5009 107/113/123	18386
-5	4279 103/103/116	4334 102/102/115	4390 102/102/114	4442 102/102/114	4492 102/105/116	4543 101/108/119	5133 107/113/123	18386
0	4335 102/102/116	4392 102/102/115	4452 102/102/114	4506 102/102/114	4558 101/105/116	4612 101/108/119	5268 107/113/123	18386
5	4389 102/102/116	4449 102/102/115	4511 102/102/114	4568 102/102/114	4622 101/105/116	4678 101/108/119	5397 108/113/123	18386
10	4441 102/102/116	4504 102/102/115	4568 102/102/114	4628 102/102/114	4685 101/105/116	4743 101/108/119	5540 108/113/123	18386
15	4495 102/102/116	4560 102/102/115	4627 102/102/114	4689 101/102/114	4748 101/105/116	4809 101/108/119	5686 109/113/123	18386
20	4560 102/102/115	4627 102/102/115	4696 101/101/114	4760 101/102/114	4822 101/105/116	4885 101/108/119	5852 110/113/123	18386
25	4626 102/102/115	4695 102/102/114	4766 101/101/114	4832 101/102/114	4895 101/105/116	4991 101/108/119	6010 110/113/123	18386
30	4660 101/101/114	4732 101/101/113	4806 101/101/113	4875 101/103/114	4942 100/106/116	5303 103/109/119	6425 112/113/123	18386
35	4520 98/98/110	4592 97/97/109	4667 97/100/111	4737 97/103/114	5365 103/107/116	6168 109/110/119	6946 114/114/123	18386
40	4391 94/94/105	4464 94/98/108	4634 95/101/111	5392 102/104/114	6089 107/107/116	6642 110/110/119	7489 115/115/123	18386
45	4265 91/95/105	4553 93/99/108	5412 100/102/111	6012 105/105/114	6557 108/108/116	7152 111/111/119	-	17425
50	4448 92/96/105	5285 99/99/108	5871 103/103/111	6459 106/106/114	-	-	-	16163
52	4722 94/96/105	5470 100/100/108	6054 103/103/111	6668 106/106/114	-	-	-	15678
V _{FS}	117	120	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4016 103/103/116	4054 103/103/116	4096 102/102/115	4134 102/102/114	4169 102/105/116	4207 102/108/119	4503 104/113/123	18386
-35	4074 103/103/116	4116 103/103/116	4161 102/102/115	4201 102/102/114	4239 102/105/116	4279 102/108/119	4613 105/113/123	18386
-30	4132 103/103/116	4177 103/103/116	4225 102/102/115	4268 102/102/114	4309 102/105/116	4351 102/108/119	4727 105/113/123	18386
-25	4188 103/103/116	4236 103/103/115	4286 102/102/115	4332 102/102/114	4375 102/105/116	4421 102/108/119	4844 106/113/123	18386
-20	4242 103/103/116	4293 102/102/115	4346 102/102/115	4395 102/102/114	4441 102/105/116	4489 102/108/119	4967 106/113/123	18386
-15	4300 103/103/116	4354 102/102/115	4410 102/102/114	4461 102/102/114	4510 102/105/116	4561 102/108/119	5095 107/113/123	18386
-10	4357 103/103/116	4414 102/102/115	4473 102/102/114	4527 102/102/114	4578 102/105/116	4632 101/108/119	5226 107/113/123	18386
-5	4414 103/103/116	4473 102/102/115	4535 102/102/114	4591 102/102/114	4645 102/105/116	4701 101/108/119	5359 107/113/123	18386
0	4473 102/102/116	4535 102/102/115	4599 102/102/114	4658 102/102/114	4715 102/105/116	4774 101/108/119	5507 108/113/123	18386
5	4528 102/102/116	4592 102/102/115	4659 102/102/114	4721 102/102/114	4781 101/105/116	4842 101/108/119	5645 109/113/123	18386
10	4582 102/102/116	4649 102/102/115	4719 102/102/114	4784 102/102/114	4846 101/105/116	4909 101/108/119	5798 109/113/123	18386
15	4636 102/102/116	4707 102/102/115	4779 102/102/114	4846 101/102/114	4911 101/105/116	4977 101/108/119	5955 110/113/123	18386
20	4704 102/102/115	4776 102/102/115	4851 101/101/114	4920 101/102/114	4987 101/105/116	5093 101/108/119	6155 110/113/123	18386
25	4766 102/102/115	4841 101/101/114	4919 101/101/113	4991 101/102/114	5061 101/106/116	5334 103/109/119	6461 112/113/123	18386
30	4663 99/99/111	4739 99/99/110	4817 98/100/111	4891 98/103/114	5270 101/106/116	6056 108/109/119	6945 114/114/123	18386
35	4527 96/96/107	4604 95/97/108	4683 95/101/111	5322 101/104/114	6086 107/107/116	6637 110/110/119	7604 114/114/123	18386
40	4399 92/95/105	4491 92/98/108	5307 99/102/111	6003 105/105/114	6565 108/108/116	7153 111/111/119	-	17715
45	4382 90/96/105	5251 98/99/108	5889 103/103/111	6460 106/106/114	7082 109/109/116	-	-	16438
50	5117 96/97/105	5727 100/100/108	6334 103/103/111	-	-	-	-	15241
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4130 103/103/116	4173 103/103/115	4219 102/102/115	4261 102/102/114	4300 102/105/116	4342 102/108/119	4710 105/113/123	18386
-35	4191 103/103/116	4237 103/103/115	4286 102/102/115	4330 102/102/114	4372 102/105/116	4417 102/108/119	4837 105/113/123	18386
-30	4250 103/103/116	4299 102/102/115	4351 102/102/115	4398 102/102/114	4443 102/105/116	4490 102/108/119	4956 106/113/123	18386
-25	4307 103/103/116	4360 102/102/115	4415 102/102/114	4465 102/102/114	4513 102/105/116	4562 102/108/119	5084 106/113/123	18386
-20	4366 103/103/116	4421 102/102/115	4479 102/102/114	4532 102/102/114	4583 102/105/116	4635 101/108/119	5218 107/113/123	18386
-15	4426 103/103/116	4484 102/102/115	4545 102/102/114	4601 102/102/114	4654 102/105/116	4710 101/108/119	5356 107/113/123	18386
-10	4486 103/103/116	4547 102/102/115	4611 102/102/114	4670 102/102/114	4726 102/105/116	4784 101/108/119	5497 108/113/123	18386
-5	4546 102/102/116	4610 102/102/115	4677 102/102/114	4738 102/102/114	4797 102/105/116	4858 101/108/119	5636 108/113/123	18386
0	4606 102/102/116	4673 102/102/115	4742 102/102/114	4807 102/102/114	4868 101/105/116	4931 101/108/119	5784 109/113/123	18386
5	4663 102/102/116	4733 102/102/115	4805 102/102/114	4872 102/102/114	4936 101/105/116	5002 101/108/119	5941 109/113/123	18386
10	4720 102/102/116	4793 102/102/115	4868 102/102/114	4938 101/102/114	5005 101/105/116	5076 101/108/119	6115 110/113/123	18386
15	4779 102/102/115	4854 102/102/115	4932 102/102/114	5004 101/102/114	5074 101/105/116	5209 102/108/119	6272 110/113/123	18386
20	4845 102/102/115	4924 102/102/114	5005 101/101/113	5080 101/102/114	5152 101/105/116	5432 103/109/119	6582 112/113/123	18386
25	4767 99/99/112	4847 99/99/111	4929 99/100/111	5006 99/103/114	5315 101/106/116	6092 107/109/119	7115 113/113/123	18386
30	4661 96/96/108	4742 96/97/108	4825 96/101/111	5250 99/104/114	6083 106/107/116	6671 110/110/119	7664 114/114/123	18386
35	4527 93/95/105	4608 93/98/108	5251 98/101/111	6046 105/105/114	6597 108/108/116	7182 111/111/119	-	17987
40	4404 90/95/105	5168 97/99/108	5900 102/102/111	6486 105/105/114	7105 109/109/116	-	-	16649
45	5060 95/96/105	5742 100/100/108	6353 103/103/111	-	-	-	-	15446
48	5413 97/97/105	6004 100/100/108	-	-	-	-	-	14759
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4242 103/103/116	4290 102/102/115	4340 102/102/114	4386 102/102/114	4429 102/105/117	4475 101/108/119	4961 106/113/123	18386
-35	4303 103/103/116	4353 102/102/115	4407 102/102/114	4456 102/102/114	4502 102/105/117	4551 101/108/119	5094 107/113/123	18386
-30	4365 103/103/116	4419 102/102/115	4475 102/102/114	4527 102/102/114	4576 102/105/117	4628 101/108/119	5232 107/113/123	18386
-25	4427 102/102/116	4484 102/102/115	4544 102/102/114	4598 102/102/114	4651 102/105/117	4705 101/108/119	5372 108/113/123	18386
-20	4487 102/102/116	4547 102/102/115	4610 102/102/114	4668 102/102/114	4723 101/105/117	4780 101/108/119	5516 108/113/123	18386
-15	4548 102/102/116	4611 102/102/115	4677 102/102/114	4738 102/102/114	4796 101/105/117	4856 101/108/119	5661 109/113/123	18386
-10	4611 102/102/116	4677 102/102/115	4746 102/102/114	4809 102/102/114	4870 101/105/117	4933 101/108/119	5813 109/113/123	18386
-5	4672 102/102/116	4741 102/102/115	4813 102/102/114	4879 101/102/114	4943 101/105/117	5008 101/108/119	5991 110/113/123	18386
0	4733 102/102/116	4805 102/102/115	4879 102/102/114	4949 101/102/114	5015 101/105/117	5098 101/108/119	6134 110/113/123	18386
5	4794 102/102/115	4869 102/102/115	4947 102/102/114	5019 101/102/114	5088 101/105/117	5229 102/108/119	6294 111/113/123	18386
10	4853 102/102/115	4932 102/102/114	5012 101/101/114	5087 101/102/114	5159 101/105/117	5376 102/108/119	6485 111/113/123	18386
15	4916 102/102/115	4997 101/101/114	5080 101/101/113	5157 101/102/114	5232 101/105/117	5531 103/108/119	6689 112/113/123	18386
20	4862 100/100/112	4945 99/99/111	5031 99/100/111	5111 99/103/114	5419 101/106/117	6255 107/109/119	7302 113/113/123	18386
25	4775 97/97/109	4859 97/97/108	4946 97/100/111	5281 99/104/114	6174 106/107/117	6726 110/110/119	7774 114/114/123	18386
30	4664 94/94/105	4749 94/98/108	5198 97/101/111	6041 104/104/114	6622 108/108/117	7203 111/111/119	8381 115/115/123	18386
35	4533 91/95/105	5144 96/99/108	5941 102/102/111	6522 105/105/114	7137 108/108/117	-	-	16845
40	5038 94/96/105	5780 100/100/108	6382 103/103/111	7036 106/106/114	-	-	-	15634
45	5590 97/97/105	6213 100/100/108	-	-	-	-	-	14503
46	5661 97/97/105	6313 100/100/108	-	-	-	-	-	14284
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4308 102/102/115	4360 101/101/114	4414 101/101/113	4464 101/102/114	4511 101/106/117	4561 100/109/119	5353 108/113/123	18386
-30	4437 102/102/115	4495 101/101/114	4555 101/101/113	4611 101/102/114	4664 101/106/117	4720 100/109/119	5659 109/113/123	18386
-20	4560 101/101/115	4625 101/101/114	4692 101/101/113	4753 101/102/114	4813 100/106/117	4969 101/109/119	5976 110/113/123	18386
-10	4689 101/101/115	4759 101/101/114	4832 101/101/113	4900 101/102/114	4965 100/106/117	5236 102/109/119	6377 111/113/123	18386
0	4816 101/101/114	4893 101/101/113	4972 101/101/113	5046 100/102/114	5117 100/106/117	5516 103/109/119	6662 112/113/123	18386
10	4941 101/101/114	5024 101/101/113	5109 100/100/112	5188 100/102/114	5265 100/106/117	5822 105/109/119	7084 113/113/123	18386
20	4870 97/97/109	4958 97/97/108	5048 97/100/111	5371 99/104/114	6247 106/107/117	6848 110/110/119	7920 114/114/123	18386
30	4665 92/95/105	5099 95/98/108	5983 102/102/111	6551 105/105/114	7162 108/108/117	-	-	17139
40	5593 96/97/105	6240 100/100/108	-	-	-	-	-	14675
44	5952 97/97/105	-	-	-	-	-	-	13818
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4375 101/101/113	4431 100/100/113	4489 100/100/112	4543 100/103/114	4595 100/106/117	4858 102/109/119	5839 111/113/123	18386
-30	4505 100/100/113	4567 100/100/113	4632 100/100/112	4692 100/103/114	4750 100/106/117	5125 103/109/119	6224 112/113/123	18386
-20	4635 100/100/113	4704 100/100/112	4775 100/100/112	4841 100/103/114	4905 99/106/117	5412 104/109/119	6499 113/113/123	18386
-10	4766 100/100/113	4841 100/100/112	4919 100/100/111	4991 100/103/114	5061 99/106/117	5710 105/109/119	6793 113/113/123	18386
0	4899 100/100/113	4981 100/100/112	5064 100/100/111	5143 99/103/114	5258 100/106/117	6023 106/109/119	7183 113/113/123	18386
10	5030 100/100/113	5119 99/99/112	5210 99/100/111	5296 99/103/114	5628 101/106/117	6521 108/109/119	7578 113/113/123	18386
20	4886 95/95/106	4979 95/97/108	5289 97/101/111	6151 103/104/114	6780 107/107/117	7548 110/110/119	8614 115/115/123	18386
30	4949 92/96/105	5802 99/99/108	6440 102/102/111	7087 106/106/114	-	-	-	16024
40	6066 97/97/105	-	-	-	-	-	-	13770
42	6266 98/98/105	-	-	-	-	-	-	13356
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4436 99/99/112	4496 99/99/111	4559 99/100/111	4618 99/103/114	4674 98/106/117	5322 105/109/119	6364 113/113/123	18386
-30	4573 99/99/112	4640 99/99/111	4710 99/100/111	4775 99/103/114	4922 99/106/117	5632 106/109/119	6670 113/113/123	18386
-20	4705 99/99/112	4779 99/99/111	4855 99/100/111	4926 98/103/114	5195 100/106/117	5985 107/109/119	6921 113/113/123	18386
-10	4842 99/99/112	4922 99/99/111	5005 98/100/111	5082 98/103/114	5484 101/106/117	6328 108/109/119	7340 113/113/123	18386
0	4979 99/99/111	5065 99/99/111	5155 98/100/111	5239 98/103/114	5784 102/106/117	6642 109/109/119	7704 113/113/123	18386
10	5066 98/98/110	5161 97/97/109	5259 97/100/111	5586 99/103/114	6456 106/107/117	7252 110/110/119	8238 114/114/123	18386
20	4915 93/95/105	5138 94/98/108	6053 101/101/111	6675 105/105/114	7446 108/108/117	8197 111/111/119	-	17775
30	5624 96/96/105	6272 100/100/108	6952 103/103/111	-	-	-	-	15094
38	6362 97/97/105	-	-	-	-	-	-	13369
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4502 98/98/110	4567 98/98/109	4635 98/100/111	4698 97/103/114	5121 101/106/117	5909 107/109/119	6779 114/114/123	18386
-30	4643 98/98/110	4715 98/98/109	4790 97/100/111	4860 97/103/114	5416 102/106/117	6256 109/109/119	7053 114/114/123	18386
-20	4779 98/98/110	4858 98/98/109	4939 97/100/111	5016 97/103/114	5747 103/106/117	6540 109/109/119	7490 114/114/123	18386
-10	4932 98/98/110	5017 97/97/109	5105 97/100/111	5251 98/103/114	6066 104/106/117	6787 109/109/119	7871 114/114/123	18386
0	5103 98/98/110	5194 97/97/109	5287 97/100/111	5558 99/103/114	6475 106/106/117	7279 110/110/119	8256 114/114/123	18386
10	5121 95/95/106	5219 95/97/108	5658 98/101/111	6516 104/104/114	7283 107/107/117	7989 110/110/119	9104 115/115/123	18386
20	5079 91/95/105	6042 99/99/108	6648 102/102/111	7455 105/105/114	-	-	-	16338
30	6198 97/97/105	6907 100/100/108	-	-	-	-	-	14094
34	6591 97/97/105	-	-	-	-	-	-	13263
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4525 96/96/108	4595 96/97/108	4668 96/101/111	5025 99/104/114	5815 105/107/117	6476 110/110/119	7260 114/114/123	18386
-30	4667 96/96/108	4744 96/97/108	4824 96/101/111	5318 100/104/114	6135 106/107/117	6732 110/110/119	7734 114/114/123	18386
-20	4820 96/96/108	4904 96/97/108	4990 95/101/111	5660 101/104/114	6426 107/107/117	7008 110/110/119	8120 114/114/123	18386
-10	4994 96/96/107	5084 95/97/108	5176 95/101/111	6054 102/104/114	6741 107/107/117	7485 110/110/119	8527 114/114/123	18386
0	5170 96/96/107	5265 95/97/108	5437 96/101/111	6365 103/104/114	7174 107/107/117	7874 110/110/119	8952 114/114/123	18386
10	5130 92/95/105	5715 96/98/108	6514 102/102/111	7275 105/105/114	8034 108/108/117	8837 111/111/119	-	17329
20	5920 96/96/105	6567 99/99/108	7369 103/103/111	-	-	-	-	15113
29	6706 97/97/105	-	-	-	-	-	-	13278
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4550 94/94/105	4624 94/98/108	4887 96/101/111	5746 103/104/114	6374 107/107/117	6926 110/110/119	7981 115/115/123	18386
-30	4701 94/94/105	4783 94/98/108	5176 97/101/111	6069 104/104/114	6626 107/107/117	7281 110/110/119	8387 115/115/123	18386
-20	4881 94/94/105	4969 94/98/108	5530 98/101/111	6321 104/104/114	6935 107/107/117	7733 110/110/119	8831 115/115/123	18386
-10	5061 94/94/105	5155 94/98/108	5914 99/101/111	6639 104/104/114	7409 107/107/117	8129 110/110/119	9276 115/115/123	18386
0	5244 93/94/105	5432 94/98/108	6384 101/101/111	7154 105/105/114	7889 108/108/117	8659 111/111/119	-	18242
10	5712 95/96/105	6411 99/99/108	7188 102/102/111	8037 106/106/114	-	-	-	15785
20	6450 97/97/105	7185 100/100/108	-	-	-	-	-	14091
24	6851 97/97/105	-	-	-	-	-	-	13261
V _{FS}	117	121	124	127	131	134	138	

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**SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – DRY RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES**

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2565 109/109/122	2455 106/106/118	2383 103/103/115	2353 101/101/112	2356 98/99/110	2528 102/102/112	2798 107/107/116	18386
-35	2608 109/109/122	2496 106/106/118	2419 103/103/115	2389 101/101/112	2392 98/99/110	2566 102/102/112	2842 107/107/116	18386
-30	2651 109/109/122	2538 106/106/118	2455 103/103/115	2425 101/101/112	2428 98/99/110	2605 102/102/112	2886 107/107/116	18386
-25	2694 109/109/122	2580 106/106/118	2491 103/103/115	2461 101/101/112	2464 98/99/110	2645 102/102/112	2930 107/107/116	18386
-20	2737 109/109/122	2622 106/106/118	2526 103/103/115	2496 101/101/112	2500 98/99/110	2684 102/102/112	2974 107/107/116	18386
-15	2779 109/109/122	2663 106/106/118	2562 103/103/115	2531 101/101/112	2536 98/99/110	2723 102/102/112	3018 107/107/116	18386
-10	2822 109/109/122	2705 106/106/118	2603 103/103/115	2566 101/101/112	2572 98/99/110	2762 102/102/112	3063 107/107/116	18386
-5	2864 109/109/122	2746 106/106/118	2643 103/103/115	2601 101/101/112	2608 98/99/110	2802 102/102/112	3108 107/107/116	18386
0	2906 109/109/122	2786 106/106/118	2682 103/103/115	2635 101/101/112	2644 98/99/110	2842 101/102/112	3153 107/107/116	18386
5	2947 109/109/121	2826 106/106/118	2721 103/103/115	2669 101/101/112	2682 98/99/110	2883 101/102/112	3199 107/107/116	18386
10	2986 109/109/121	2864 106/106/118	2758 103/103/115	2702 101/101/112	2721 98/99/110	2925 101/102/112	3247 107/107/116	18386
15	3026 109/109/121	2903 106/106/118	2796 103/103/115	2735 100/100/112	2760 98/99/110	2967 101/102/112	3295 107/107/116	18386
20	3068 109/109/121	2944 106/106/118	2835 103/103/115	2767 100/100/111	2799 98/99/110	3011 101/102/112	3344 107/107/116	18386
25	3109 109/109/121	2984 106/106/118	2874 103/103/114	2800 100/100/111	2838 98/99/110	3054 101/102/112	3393 107/107/116	18386
30	3131 108/108/120	3006 105/105/117	2897 102/102/113	2831 100/100/110	2913 98/100/110	3137 102/103/112	3490 107/107/116	18386
35	3032 104/104/115	2916 101/101/112	2842 99/99/109	2878 97/98/107	3111 100/101/110	3358 104/104/112	3863 108/108/116	18386
40	2946 101/101/111	2862 98/98/108	2834 95/95/104	3078 99/99/107	3369 102/102/110	3731 105/105/112	- -	17613
45	2881 97/97/107	2847 95/95/103	3027 96/96/104	3366 100/100/107	3758 103/103/110	- -	- -	16517
50	2866 94/94/102	2967 94/94/102	3350 97/97/104	- -	- -	- -	- -	15460
52	2860 93/93/101	3098 94/94/102	3509 98/98/104	- -	- -	- -	- -	15049
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2655 110/110/122	2540 107/107/119	2451 104/104/115	2420 101/101/112	2417 99/99/110	2593 102/102/112	2870 107/107/116	18386
-35	2701 110/110/122	2584 107/107/119	2488 104/104/115	2457 101/101/112	2454 98/99/110	2633 102/102/112	2915 107/107/116	18386
-30	2746 110/110/122	2628 107/107/119	2525 104/104/115	2494 101/101/112	2491 98/99/110	2673 102/102/112	2961 107/107/116	18386
-25	2790 110/110/122	2670 106/106/119	2566 104/104/115	2530 101/101/112	2528 98/99/110	2714 102/102/112	3006 107/107/116	18386
-20	2832 110/110/122	2712 106/106/119	2607 104/104/115	2566 101/101/112	2565 98/99/110	2755 101/102/112	3053 107/107/116	18386
-15	2876 110/110/122	2754 106/106/119	2649 103/103/115	2602 101/101/112	2603 98/99/110	2796 101/102/112	3099 107/107/116	18386
-10	2920 109/109/122	2797 106/106/119	2690 103/103/115	2638 101/101/112	2641 98/99/110	2837 101/102/112	3146 107/107/116	18386
-5	2963 109/109/122	2839 106/106/118	2731 103/103/115	2673 101/101/112	2679 98/99/110	2878 101/102/112	3192 107/107/116	18386
0	3006 109/109/122	2881 106/106/118	2772 103/103/115	2708 101/101/112	2716 98/99/110	2919 101/102/112	3238 106/107/116	18386
5	3047 109/109/122	2921 106/106/118	2811 103/103/115	2743 101/101/112	2756 98/99/110	2962 101/102/112	3288 106/107/116	18386
10	3088 109/109/122	2960 106/106/118	2850 103/103/115	2777 101/101/112	2795 98/99/110	3006 101/102/112	3337 106/107/116	18386
15	3128 109/109/121	3000 106/106/118	2888 103/103/115	2811 100/100/112	2836 98/99/110	3050 101/102/112	3387 106/107/116	18386
20	3170 109/109/121	3041 106/106/118	2928 103/103/115	2844 100/100/111	2877 98/99/110	3095 101/102/112	3438 106/107/116	18386
25	3210 109/109/121	3081 106/106/117	2968 103/103/114	2883 100/100/111	2935 98/100/110	3160 101/103/112	3514 107/107/116	18386
30	3132 106/106/117	3011 103/103/114	2905 100/100/110	2869 97/97/107	3094 99/100/110	3337 103/104/112	3784 108/108/116	18386
35	3039 102/102/112	2927 99/99/109	2886 96/96/106	3063 98/98/107	3317 101/101/110	3659 105/105/112	-	17993
40	2957 98/98/108	2906 96/96/105	3017 96/96/105	3303 99/99/107	3679 102/102/110	-	-	16856
45	2926 95/95/104	2954 94/94/102	3282 97/97/105	3686 100/100/107	-	-	-	15788
50	2910 92/92/99	3245 95/95/102	-	-	-	-	-	14786
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2735 110/110/122	2617 106/106/119	2515 104/104/115	2484 101/101/112	2483 98/99/110	2665 102/102/112	2952 107/107/116	18386
-35	2782 110/110/122	2662 106/106/119	2558 104/104/115	2522 101/101/112	2521 98/99/110	2706 102/102/112	2998 107/107/116	18386
-30	2828 110/110/122	2707 106/106/119	2602 104/104/115	2559 101/101/112	2559 98/99/110	2748 101/102/112	3045 107/107/116	18386
-25	2873 110/110/122	2751 106/106/119	2644 104/104/115	2596 101/101/112	2598 98/99/110	2790 101/102/112	3093 107/107/116	18386
-20	2918 110/110/122	2794 106/106/119	2687 103/103/115	2634 101/101/112	2636 98/99/110	2832 101/102/112	3140 107/107/116	18386
-15	2963 109/109/122	2839 106/106/119	2730 103/103/115	2671 101/101/112	2675 98/99/110	2874 101/102/112	3188 107/107/116	18386
-10	3008 109/109/122	2883 106/106/118	2773 103/103/115	2708 101/101/112	2714 98/99/110	2917 101/102/112	3236 106/107/116	18386
-5	3054 109/109/122	2927 106/106/118	2816 103/103/115	2745 101/101/112	2752 98/99/110	2959 101/102/112	3284 106/107/116	18386
0	3098 109/109/122	2970 106/106/118	2858 103/103/115	2781 101/101/112	2792 98/99/110	3002 101/102/112	3333 106/107/116	18386
5	3140 109/109/122	3011 106/106/118	2899 103/103/115	2816 101/101/112	2832 98/99/110	3046 101/102/112	3383 106/107/116	18386
10	3183 109/109/122	3052 106/106/118	2939 103/103/115	2851 100/100/112	2873 98/99/110	3091 101/102/112	3434 106/107/116	18386
15	3225 109/109/121	3094 106/106/118	2979 103/103/115	2886 100/100/112	2915 98/99/110	3136 101/102/112	3485 106/107/116	18386
20	3269 109/109/121	3136 106/106/118	3021 103/103/114	2923 100/100/111	2968 98/99/110	3195 101/103/112	3553 107/107/116	18386
25	3207 106/106/118	3082 103/103/114	2973 100/100/111	2920 98/98/108	3120 99/100/110	3365 103/103/112	3786 108/108/116	18386
30	3131 103/103/114	3014 100/100/111	2943 97/97/107	3057 97/98/107	3310 101/101/110	3604 104/104/112	-	18363
35	3042 99/99/109	2962 97/97/106	3015 95/96/105	3279 99/99/107	3633 102/102/110	-	-	17162
40	2986 96/96/105	2952 93/93/102	3234 97/97/105	3630 100/100/107	-	-	-	16087
45	2972 93/93/101	3192 94/94/102	3611 98/98/105	-	-	-	-	15057
48	3003 91/91/99	3417 95/95/102	-	-	-	-	-	14490
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2813 109/109/122	2692 106/106/118	2588 103/103/115	2548 101/101/112	2556 98/99/110	2745 102/102/112	3043 107/107/116	18386
-35	2860 109/109/122	2738 106/106/118	2632 103/103/115	2586 101/101/112	2596 98/99/110	2788 101/102/112	3091 107/107/116	18386
-30	2908 109/109/122	2784 106/106/118	2677 103/103/115	2625 101/101/112	2635 98/99/110	2831 101/102/112	3140 107/107/116	18386
-25	2955 109/109/122	2830 106/106/118	2722 103/103/115	2663 101/101/112	2675 98/99/110	2874 101/102/112	3189 107/107/116	18386
-20	3001 109/109/122	2876 106/106/118	2766 103/103/115	2701 101/101/112	2715 98/99/110	2918 101/102/112	3238 107/107/116	18386
-15	3048 109/109/122	2921 106/106/118	2810 103/103/115	2739 101/101/112	2755 98/99/110	2962 101/102/112	3287 106/107/116	18386
-10	3095 109/109/122	2966 106/106/118	2854 103/103/115	2777 101/101/112	2795 98/99/110	3005 101/102/112	3337 106/107/116	18386
-5	3140 109/109/122	3010 106/106/118	2898 103/103/115	2814 100/100/112	2835 98/99/110	3050 101/102/112	3387 106/107/116	18386
0	3185 109/109/121	3054 106/106/118	2940 103/103/115	2851 100/100/112	2876 98/99/110	3094 101/102/112	3438 106/107/116	18386
5	3230 109/109/121	3098 106/106/118	2983 103/103/115	2888 100/100/112	2918 98/99/110	3140 101/102/112	3489 106/107/116	18386
10	3273 109/109/121	3140 106/106/118	3025 103/103/115	2924 100/100/111	2960 98/99/110	3186 101/102/112	3542 106/107/116	18386
15	3317 109/109/121	3183 106/106/118	3066 103/103/114	2964 100/100/111	3004 98/99/110	3234 101/102/112	3598 106/107/116	18386
20	3274 106/106/118	3147 103/103/115	3035 101/101/112	2968 98/98/108	3152 99/100/110	3400 102/103/112	3811 108/108/116	18386
25	3210 104/104/115	3090 101/101/111	2998 98/98/108	3077 97/98/107	3330 100/101/110	3598 104/104/112	4145 108/108/116	18386
30	3134 100/100/111	3022 98/98/107	3009 95/95/105	3272 99/99/107	3578 102/102/110	3963 105/105/112	-	17500
35	3051 97/97/106	3009 94/94/103	3227 96/96/105	3588 100/100/107	-	-	-	16371
40	3032 94/94/102	3162 94/94/102	3564 97/97/105	-	-	-	-	15335
45	3089 91/91/99	3521 95/95/102	-	-	-	-	-	14383
46	3163 92/92/99	3606 95/95/102	-	-	-	-	-	14196
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2858 108/108/121	2737 105/105/117	2633 102/102/114	2597 100/100/111	2654 98/100/110	2852 102/103/112	3165 107/107/116	18386
-30	2955 108/108/121	2832 105/105/117	2725 102/102/114	2676 100/100/111	2736 98/100/110	2942 102/103/112	3267 107/107/116	18386
-20	3049 108/108/120	2924 105/105/117	2815 102/102/114	2753 100/100/111	2819 98/100/110	3033 102/103/112	3370 107/107/116	18386
-10	3144 108/108/120	3017 105/105/117	2905 102/102/114	2831 100/100/111	2903 98/100/110	3125 101/103/112	3475 107/107/116	18386
0	3238 108/108/120	3108 105/105/117	2994 102/102/114	2907 99/99/110	2988 98/100/110	3217 101/103/112	3579 107/107/116	18386
10	3329 108/108/120	3196 105/105/117	3081 102/102/113	2982 99/99/110	3076 98/100/110	3314 101/103/112	3690 107/107/116	18386
20	3278 104/104/115	3155 101/101/112	3047 98/98/108	3109 96/98/107	3365 100/101/110	3636 104/104/112	4170 108/108/116	18386
30	3139 98/98/107	3067 95/95/104	3223 96/96/105	3537 99/99/107	3939 103/103/110	-	-	16699
40	3084 91/91/99	3474 95/95/102	-	-	-	-	-	14638
44	3337 92/92/99	-	-	-	-	-	-	13905
V_{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2901 107/107/119	2781 104/104/116	2678 101/101/113	2647 99/99/109	2761 99/100/110	2972 102/103/112	3303 107/107/116	18386
-30	2999 107/107/119	2877 104/104/116	2771 101/101/113	2727 99/99/109	2848 99/100/110	3066 102/103/112	3410 107/107/116	18386
-20	3096 107/107/119	2972 104/104/116	2864 101/101/112	2807 99/99/109	2935 98/100/110	3161 102/103/112	3518 107/107/116	18386
-10	3194 107/107/119	3067 104/104/116	2956 101/101/112	2886 99/99/109	3023 98/100/110	3258 102/103/112	3628 107/107/116	18386
0	3290 107/107/119	3160 104/104/116	3047 101/101/112	2964 98/98/109	3111 98/100/110	3354 102/103/112	3737 107/107/116	18386
10	3388 107/107/118	3256 104/104/115	3142 101/101/112	3052 98/98/109	3227 98/100/110	3481 102/103/112	3882 108/108/116	18386
20	3289 101/101/112	3169 99/99/109	3097 96/96/105	3319 98/98/107	3599 102/102/110	3959 105/105/112	-	17897
30	3153 95/95/104	3156 93/93/102	3468 97/97/105	3891 100/100/107	-	-	-	15947
40	3366 92/92/99	3849 95/95/102	-	-	-	-	-	14000
42	3523 92/92/99	-	-	-	-	-	-	13637
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2945 106/106/118	2826 103/103/114	2734 100/100/111	2702 98/98/108	2892 99/100/110	3115 103/103/112	3503 108/108/116	18386
-30	3046 106/106/118	2925 103/103/114	2819 100/100/111	2784 98/98/108	2983 99/100/110	3215 103/103/112	3609 108/108/116	18386
-20	3145 106/106/117	3021 103/103/114	2913 100/100/111	2866 97/97/108	3075 99/100/110	3315 103/103/112	3719 108/108/116	18386
-10	3245 106/106/117	3118 103/103/114	3008 100/100/111	2947 97/97/108	3168 99/100/110	3417 103/103/112	3831 108/108/116	18386
0	3343 106/106/117	3214 103/103/114	3101 100/100/111	3027 97/97/107	3261 99/100/110	3519 102/103/112	3945 108/108/116	18386
10	3413 104/104/116	3285 101/101/112	3173 99/99/109	3181 96/98/107	3443 100/101/110	3721 103/104/112	4238 108/108/116	18386
20	3310 99/99/109	3194 96/96/106	3252 95/96/105	3541 99/99/107	3899 102/102/110	-	-	17098
30	3209 93/93/102	3374 94/94/102	3800 97/97/105	-	-	-	-	15204
40	3693 92/92/99	-	-	-	-	-	-	13377
V_{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2991 105/105/116	2873 102/102/113	2792 99/99/109	2808 96/97/107	3033 100/101/110	3271 104/104/112	3723 108/108/116	18386
-30	3095 105/105/116	2975 102/102/113	2877 99/99/109	2895 96/97/107	3129 100/101/110	3376 104/104/112	3835 108/108/116	18386
-20	3196 104/104/116	3073 101/101/113	2965 99/99/109	2984 96/97/107	3226 100/101/110	3483 103/104/112	3953 108/108/116	18386
-10	3301 104/104/116	3175 101/101/112	3064 99/99/109	3073 96/97/107	3325 99/101/110	3590 103/104/112	4072 108/108/116	18386
0	3407 104/104/116	3278 101/101/112	3165 99/99/109	3163 96/97/107	3423 99/101/110	3699 103/104/112	4195 108/108/116	18386
10	3418 101/101/112	3294 99/99/109	3191 96/96/105	3421 97/98/107	3712 101/102/110	4074 105/105/112	-	17779
20	3324 97/97/106	3234 94/94/103	3496 96/97/105	3878 100/100/107	-	-	-	16201
30	3305 91/91/99	3731 95/95/102	-	-	-	-	-	14436
36	3743 92/92/99	-	-	-	-	-	-	13369
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3008 102/102/113	2893 100/100/110	2837 97/97/107	2973 97/98/107	3217 101/101/110	3501 104/104/112	-	18082
-30	3112 102/102/113	2995 100/100/110	2925 97/97/107	3066 97/98/107	3320 101/101/110	3606 104/104/112	-	18122
-20	3218 102/102/113	3098 99/99/110	3011 97/97/107	3161 97/98/107	3424 101/101/110	3717 104/104/112	-	18152
-10	3328 102/102/113	3204 99/99/110	3097 97/97/107	3258 97/98/107	3531 101/101/110	3829 104/104/112	-	18169
0	3437 102/102/113	3310 99/99/110	3200 97/97/106	3357 97/98/107	3640 100/101/110	3950 104/104/112	-	18179
10	3413 98/98/108	3294 96/96/105	3399 95/96/105	3706 99/99/107	4106 102/102/110	-	-	16772
20	3336 94/94/102	3444 94/94/102	3840 97/97/105	-	-	-	-	15400
25	3332 91/91/99	3740 95/95/102	-	-	-	-	-	14545
30	3668 92/92/99	-	-	-	-	-	-	13619
32	3839 92/92/99	-	-	-	-	-	-	13261
V_{FS}	117	121	124	127	130	134	-	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3027 100/100/111	2920 98/98/108	2902 95/95/105	3153 98/99/107	3426 102/102/110	3798 105/105/112	-	17519
-30	3135 100/100/111	3021 98/98/108	2992 95/95/105	3253 98/99/107	3529 102/102/110	3913 105/105/112	-	17554
-20	3247 100/100/111	3130 98/98/107	3085 95/95/105	3355 98/99/107	3641 102/102/110	4033 105/105/112	-	17575
-10	3359 100/100/111	3238 97/97/107	3178 95/95/105	3458 98/99/107	3755 102/102/110	4154 105/105/112	-	17587
0	3480 100/100/110	3357 97/97/107	3329 94/96/105	3626 98/99/107	3958 102/102/110	4383 105/105/112	-	17340
10	3422 96/96/105	3352 93/93/102	3674 97/97/105	4109 100/100/107	-	-	-	15941
15	3403 94/94/102	3512 94/94/102	3911 97/97/105	-	-	-	-	15436
20	3378 91/91/99	3772 95/95/102	-	-	-	-	-	14706
25	3700 92/92/99	-	-	-	-	-	-	13774
27	3879 92/92/99	-	-	-	-	-	-	13391
V _{FS}	117	121	124	127	131	134	-	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3745 109/109/122	3558 106/106/118	3389 103/103/115	3230 101/101/112	3082 98/99/110	2950 96/102/112	2897 95/107/116	18386
-35	3813 109/109/122	3623 106/106/118	3453 103/103/115	3292 101/101/112	3142 98/99/110	3008 96/102/112	2955 95/107/116	18386
-30	3880 109/109/122	3689 106/106/118	3517 103/103/115	3354 101/101/112	3203 98/99/110	3067 96/102/112	3014 95/107/116	18386
-25	3949 109/109/122	3756 106/106/118	3582 103/103/115	3417 101/101/112	3263 98/99/110	3125 96/102/112	3072 95/107/116	18386
-20	4017 109/109/122	3822 106/106/118	3646 103/103/115	3479 101/101/112	3323 98/99/110	3183 96/102/112	3130 95/107/116	18386
-15	4084 109/109/122	3888 106/106/118	3710 103/103/115	3541 101/101/112	3383 98/99/110	3242 96/102/112	3189 95/107/116	18386
-10	4153 109/109/122	3954 106/106/118	3774 103/103/115	3603 101/101/112	3444 98/99/110	3300 96/102/112	3247 95/107/116	18386
-5	4222 109/109/122	4021 106/106/118	3839 103/103/115	3665 101/101/112	3504 98/99/110	3358 96/102/112	3306 95/107/116	18386
0	4291 109/109/122	4087 106/106/118	3903 103/103/115	3727 101/101/112	3564 98/99/110	3416 96/102/112	3363 95/107/116	18386
5	4357 109/109/121	4151 106/106/118	3964 103/103/115	3787 101/101/112	3621 98/99/110	3472 96/102/112	3419 95/107/116	18386
10	4420 109/109/121	4213 106/106/118	4024 103/103/115	3845 101/101/112	3677 98/99/110	3526 96/102/112	3473 95/107/116	18386
15	4487 109/109/121	4277 106/106/118	4086 103/103/115	3904 100/100/112	3734 98/99/110	3581 96/102/112	3528 95/107/116	18386
20	4561 109/109/121	4348 106/106/118	4154 103/103/115	3968 100/100/111	3796 98/99/110	3639 96/102/112	3586 94/107/116	18386
25	4636 109/109/121	4419 106/106/118	4222 103/103/114	4035 100/100/111	3860 98/99/110	3702 96/102/112	3648 94/107/116	18386
30	4662 108/108/120	4445 105/105/117	4248 102/102/113	4060 100/100/110	3886 97/100/110	3727 95/103/112	3712 94/107/116	18386
35	4449 104/104/115	4242 101/101/112	4055 99/99/109	3878 96/98/107	3714 94/101/110	3567 92/104/112	4076 97/108/116	18386
40	4254 101/101/111	4058 98/98/108	3883 95/95/104	3717 93/99/107	3564 91/102/110	3905 94/105/112	-	17613
45	4072 97/97/107	3887 95/95/103	3723 92/96/104	3569 90/100/107	3884 93/103/110	-	-	16517
50	3897 94/94/102	3724 91/94/102	3571 89/97/104	-	-	-	-	15460
52	3828 93/93/101	3660 90/94/102	3589 88/98/104	-	-	-	-	15049
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3895 110/110/122	3699 107/107/119	3522 104/104/115	3355 101/101/112	3201 99/99/110	3062 96/102/112	3007 95/107/116	18386
-35	3968 110/110/122	3769 107/107/119	3590 104/104/115	3421 101/101/112	3264 98/99/110	3123 96/102/112	3068 95/107/116	18386
-30	4040 110/110/122	3839 107/107/119	3658 104/104/115	3486 101/101/112	3327 98/99/110	3184 96/102/112	3129 95/107/116	18386
-25	4110 110/110/122	3906 106/106/119	3723 104/104/115	3549 101/101/112	3388 98/99/110	3243 96/102/112	3188 95/107/116	18386
-20	4178 110/110/122	3973 106/106/119	3787 104/104/115	3612 101/101/112	3449 98/99/110	3302 96/102/112	3247 95/107/116	18386
-15	4249 110/110/122	4042 106/106/119	3854 103/103/115	3676 101/101/112	3511 98/99/110	3362 96/102/112	3308 95/107/116	18386
-10	4320 109/109/122	4110 106/106/119	3921 103/103/115	3741 101/101/112	3573 98/99/110	3422 96/102/112	3368 95/107/116	18386
-5	4390 109/109/122	4178 106/106/118	3986 103/103/115	3804 101/101/112	3635 98/99/110	3482 96/102/112	3428 95/107/116	18386
0	4462 109/109/122	4248 106/106/118	4054 103/103/115	3870 101/101/112	3698 98/99/110	3543 96/102/112	3489 95/107/116	18386
5	4529 109/109/122	4312 106/106/118	4117 103/103/115	3931 101/101/112	3757 98/99/110	3600 96/102/112	3546 95/107/116	18386
10	4595 109/109/122	4377 106/106/118	4179 103/103/115	3991 101/101/112	3816 98/99/110	3657 96/102/112	3603 95/107/116	18386
15	4662 109/109/121	4442 106/106/118	4242 103/103/115	4052 100/100/112	3875 98/99/110	3714 96/102/112	3660 95/107/116	18386
20	4738 109/109/121	4515 106/106/118	4312 103/103/115	4119 100/100/111	3939 98/99/110	3776 96/102/112	3722 94/107/116	18386
25	4800 109/109/121	4576 106/106/117	4373 103/103/114	4179 100/100/111	3999 98/100/110	3835 95/103/112	3782 94/107/116	18386
30	4635 106/106/117	4420 103/103/114	4226 100/100/110	4041 97/97/107	3869 95/100/110	3714 93/104/112	4029 96/108/116	18386
35	4427 102/102/112	4223 99/99/109	4040 96/96/106	3867 94/98/107	3707 92/101/110	3864 93/105/112	-	17993
40	4236 98/98/108	4043 96/96/105	3871 93/96/105	3710 91/99/107	3848 92/102/110	-	-	16856
45	4056 95/95/104	3875 92/94/102	3715 90/97/105	3813 90/100/107	-	-	-	15788
50	3880 92/92/99	3712 89/95/102	-	-	-	-	-	14786
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4030 110/110/122	3828 106/106/119	3646 104/104/115	3474 101/101/112	3315 98/99/110	3171 96/102/112	3116 95/107/116	18386
-35	4105 110/110/122	3900 106/106/119	3716 104/104/115	3542 101/101/112	3380 98/99/110	3235 96/102/112	3179 95/107/116	18386
-30	4178 110/110/122	3972 106/106/119	3785 104/104/115	3609 101/101/112	3445 98/99/110	3298 96/102/112	3242 95/107/116	18386
-25	4250 110/110/122	4042 106/106/119	3853 104/104/115	3675 101/101/112	3509 98/99/110	3359 96/102/112	3304 95/107/116	18386
-20	4324 110/110/122	4113 106/106/119	3922 103/103/115	3741 101/101/112	3573 98/99/110	3422 96/102/112	3366 95/107/116	18386
-15	4398 109/109/122	4185 106/106/119	3992 103/103/115	3809 101/101/112	3638 98/99/110	3485 96/102/112	3429 95/107/116	18386
-10	4472 109/109/122	4256 106/106/118	4061 103/103/115	3876 101/101/112	3704 98/99/110	3548 96/102/112	3493 95/107/116	18386
-5	4546 109/109/122	4329 106/106/118	4131 103/103/115	3944 101/101/112	3770 98/99/110	3612 96/102/112	3557 95/107/116	18386
0	4620 109/109/122	4400 106/106/118	4201 103/103/115	4011 101/101/112	3834 98/99/110	3674 96/102/112	3619 95/107/116	18386
5	4690 109/109/122	4468 106/106/118	4266 103/103/115	4074 101/101/112	3895 98/99/110	3734 96/102/112	3678 95/107/116	18386
10	4761 109/109/122	4536 106/106/118	4332 103/103/115	4138 100/100/112	3957 98/99/110	3793 96/102/112	3738 95/107/116	18386
15	4833 109/109/121	4605 106/106/118	4399 103/103/115	4202 100/100/112	4019 98/99/110	3853 96/102/112	3798 94/107/116	18386
20	4905 109/109/121	4675 106/106/118	4466 103/103/114	4267 100/100/111	4081 98/99/110	3913 95/103/112	3858 94/107/116	18386
25	4769 106/106/118	4547 103/103/114	4347 100/100/111	4156 98/98/108	3979 95/100/110	3819 93/103/112	4056 95/108/116	18386
30	4598 103/103/114	4387 100/100/111	4196 97/97/107	4016 95/98/107	3848 93/101/110	3839 92/104/112	-	18363
35	4392 99/99/109	4193 97/97/106	4014 94/96/105	3846 92/99/107	3831 91/102/110	-	-	17162
40	4211 96/96/105	4022 93/93/102	3855 91/97/105	3793 90/100/107	-	-	-	16087
45	4033 93/93/101	3857 90/94/102	3737 88/98/105	-	-	-	-	15057
48	3927 91/91/99	3759 88/95/102	-	-	-	-	-	14490
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4160 109/109/122	3952 106/106/118	3765 103/103/115	3589 101/101/112	3425 98/99/110	3277 96/102/112	3221 95/107/116	18386
-35	4236 109/109/122	4026 106/106/118	3837 103/103/115	3658 101/101/112	3492 98/99/110	3342 96/102/112	3286 95/107/116	18386
-30	4313 109/109/122	4101 106/106/118	3910 103/103/115	3728 101/101/112	3560 98/99/110	3408 96/102/112	3352 95/107/116	18386
-25	4390 109/109/122	4176 106/106/118	3982 103/103/115	3799 101/101/112	3628 98/99/110	3474 96/102/112	3418 95/107/116	18386
-20	4466 109/109/122	4249 106/106/118	4053 103/103/115	3867 101/101/112	3694 98/99/110	3538 96/102/112	3482 95/107/116	18386
-15	4543 109/109/122	4323 106/106/118	4125 103/103/115	3936 101/101/112	3761 98/99/110	3603 96/102/112	3547 95/107/116	18386
-10	4620 109/109/122	4399 106/106/118	4198 103/103/115	4007 101/101/112	3829 98/99/110	3668 96/102/112	3612 95/107/116	18386
-5	4695 109/109/122	4471 106/106/118	4268 103/103/115	4075 100/100/112	3896 98/99/110	3733 96/102/112	3677 95/107/116	18386
0	4770 109/109/121	4544 106/106/118	4339 103/103/115	4144 100/100/112	3962 98/99/110	3797 96/102/112	3741 95/107/116	18386
5	4845 109/109/121	4617 106/106/118	4409 103/103/115	4212 100/100/112	4028 98/99/110	3861 96/102/112	3805 94/107/116	18386
10	4918 109/109/121	4687 106/106/118	4478 103/103/115	4278 100/100/111	4092 98/99/110	3923 96/102/112	3868 94/107/116	18386
15	4992 109/109/121	4759 106/106/118	4548 103/103/114	4346 100/100/111	4158 98/99/110	3987 95/102/112	3931 94/107/116	18386
20	4887 106/106/118	4661 103/103/115	4456 101/101/112	4261 98/98/108	4079 96/100/110	3915 93/103/112	4102 95/108/116	18386
25	4740 104/104/115	4523 101/101/111	4326 98/98/108	4140 95/98/107	3967 93/101/110	3855 92/104/112	4423 97/108/116	18386
30	4564 100/100/111	4357 98/98/107	4171 95/95/105	3995 93/99/107	3832 90/102/110	4201 94/105/112	-	17500
35	4366 97/97/106	4170 94/94/103	3996 92/96/105	3833 89/100/107	-	-	-	16371
40	4185 94/94/102	4002 91/94/102	3839 89/97/105	-	-	-	-	15335
45	4008 90/91/99	3837 88/95/102	-	-	-	-	-	14383
46	3973 90/92/99	3804 87/95/102	-	-	-	-	-	14196
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4228 108/108/121	4020 105/105/117	3831 102/102/114	3654 100/100/111	3489 97/100/110	3340 95/103/112	3308 94/107/116	18386
-30	4387 108/108/121	4174 105/105/117	3981 102/102/114	3798 100/100/111	3628 97/100/110	3475 95/103/112	3429 94/107/116	18386
-20	4542 108/108/120	4323 105/105/117	4126 102/102/114	3939 100/100/111	3764 97/100/110	3607 95/103/112	3553 94/107/116	18386
-10	4700 108/108/120	4477 105/105/117	4275 102/102/114	4082 100/100/111	3903 97/100/110	3741 95/103/112	3687 94/107/116	18386
0	4857 108/108/120	4629 105/105/117	4422 102/102/114	4224 99/99/110	4041 97/100/110	3874 95/103/112	3820 94/107/116	18386
10	5010 108/108/120	4777 105/105/117	4565 102/102/113	4363 99/99/110	4174 97/100/110	4003 95/103/112	3949 94/107/116	18386
20	4860 104/104/115	4637 101/101/112	4435 98/98/108	4243 96/98/107	4065 93/101/110	3905 91/104/112	4472 96/108/116	18386
30	4529 98/98/107	4326 95/95/104	4144 93/96/105	3973 90/99/107	4169 92/103/110	-	-	16699
40	4160 91/91/99	3982 89/95/102	-	-	-	-	-	14638
44	4021 88/92/99	-	-	-	-	-	-	13905
V_{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4294 107/107/119	4084 104/104/116	3896 101/101/113	3718 99/99/109	3552 96/100/110	3403 94/103/112	3492 95/107/116	18386
-30	4455 107/107/119	4241 104/104/116	4047 101/101/113	3863 99/99/109	3693 96/100/110	3539 94/103/112	3620 95/107/116	18386
-20	4615 107/107/119	4396 104/104/116	4198 101/101/112	4010 99/99/109	3834 96/100/110	3676 94/103/112	3752 94/107/116	18386
-10	4777 107/107/119	4552 104/104/116	4349 101/101/112	4156 99/99/109	3975 96/100/110	3813 94/103/112	3886 94/107/116	18386
0	4940 107/107/119	4710 104/104/116	4502 101/101/112	4304 98/98/109	4119 96/100/110	3951 94/103/112	4023 94/107/116	18386
10	5094 107/107/118	4860 104/104/115	4647 101/101/112	4444 98/98/109	4255 96/100/110	4084 94/103/112	4209 94/108/116	18386
20	4841 101/101/112	4620 99/99/109	4422 96/96/105	4234 94/98/107	4060 91/102/110	4249 93/105/112	-	17897
30	4506 95/95/104	4307 93/93/102	4130 90/97/105	4114 90/100/107	-	-	-	15947
40	4141 89/92/99	3977 86/95/102	-	-	-	-	-	14000
42	4071 87/92/99	-	-	-	-	-	-	13637
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4352 106/106/118	4142 103/103/114	3952 100/100/111	3772 98/98/108	3606 95/100/110	3457 93/103/112	3709 96/108/116	18386
-30	4521 106/106/118	4304 103/103/114	4109 100/100/111	3924 98/98/108	3752 95/100/110	3598 93/103/112	3846 95/108/116	18386
-20	4683 106/106/117	4461 103/103/114	4262 100/100/111	4072 97/97/108	3896 95/100/110	3737 93/103/112	3987 95/108/116	18386
-10	4850 106/106/117	4623 103/103/114	4418 100/100/111	4223 97/97/108	4042 95/100/110	3878 93/103/112	4130 95/108/116	18386
0	5017 106/106/117	4785 103/103/114	4574 100/100/111	4374 97/97/107	4187 95/100/110	4018 93/103/112	4276 95/108/116	18386
10	5100 104/104/116	4867 101/101/112	4656 99/99/109	4455 96/98/107	4268 94/101/110	4100 92/104/112	4587 96/108/116	18386
20	4839 99/99/109	4621 96/96/106	4425 94/96/105	4240 91/99/107	4185 90/102/110	-	-	17098
30	4501 93/93/102	4307 91/94/102	4134 88/97/105	-	-	-	-	15204
40	4143 87/92/99	-	-	-	-	-	-	13377
V_{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4416 105/105/116	4203 102/102/113	4012 99/99/109	3831 96/97/107	3664 94/101/110	3514 92/104/112	3944 96/108/116	18386
-30	4587 105/105/116	4369 102/102/113	4172 99/99/109	3986 96/97/107	3814 94/101/110	3659 92/104/112	4090 96/108/116	18386
-20	4754 104/104/116	4530 101/101/113	4328 99/99/109	4137 96/97/107	3960 94/101/110	3800 92/104/112	4241 96/108/116	18386
-10	4935 104/104/116	4705 101/101/112	4497 99/99/109	4300 96/97/107	4116 94/101/110	3951 92/104/112	4397 96/108/116	18386
0	5132 104/104/116	4893 101/101/112	4677 99/99/109	4471 96/97/107	4280 94/101/110	4108 92/104/112	4560 95/108/116	18386
10	5095 101/101/112	4861 99/99/109	4649 96/96/105	4449 94/98/107	4264 91/102/110	4416 92/105/112	-	17779
20	4845 97/97/106	4629 94/94/103	4434 91/97/105	4251 89/100/107	-	-	-	16201
30	4508 91/91/99	4314 88/95/102	-	-	-	-	-	14436
36	4296 87/92/99	-	-	-	-	-	-	13369
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4416 102/102/113	4206 100/100/110	4018 97/97/107	3841 94/98/107	3677 92/101/110	3720 92/104/112	-	18082
-30	4588 102/102/113	4373 100/100/110	4179 97/97/107	3996 94/98/107	3827 92/101/110	3857 92/104/112	-	18122
-20	4769 102/102/113	4547 99/99/110	4347 97/97/107	4158 94/98/107	3983 92/101/110	4000 92/104/112	-	18152
-10	4967 102/102/113	4736 99/99/110	4528 97/97/107	4331 94/98/107	4149 92/101/110	4150 92/104/112	-	18169
0	5166 102/102/113	4927 99/99/110	4711 97/97/106	4506 94/98/107	4316 92/101/110	4306 91/104/112	-	18179
10	5043 98/98/108	4811 96/96/105	4604 93/96/105	4409 91/99/107	4443 91/102/110	-	-	16772
20	4821 94/94/102	4607 91/94/102	4416 89/97/105	-	-	-	-	15400
25	4657 91/91/99	4455 88/95/102	-	-	-	-	-	14545
30	4474 88/92/99	-	-	-	-	-	-	13619
32	4402 86/92/99	-	-	-	-	-	-	13261
V_{FS}	117	121	124	127	130	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4419 100/100/111	4212 98/98/108	4027 95/95/105	3853 93/99/107	3693 90/102/110	4024 94/105/112	-	17519
-30	4597 100/100/111	4385 98/98/108	4194 95/95/105	4014 93/99/107	3848 90/102/110	4174 93/105/112	-	17554
-20	4800 100/100/111	4578 98/98/107	4378 95/95/105	4190 93/99/107	4017 90/102/110	4335 93/105/112	-	17575
-10	5003 100/100/111	4771 97/97/107	4563 95/95/105	4367 92/99/107	4185 90/102/110	4498 93/105/112	-	17587
0	5194 100/100/110	4955 97/97/107	4739 94/96/105	4536 92/99/107	4349 90/102/110	4765 93/105/112	-	17340
10	5012 96/96/105	4788 93/93/102	4587 90/97/105	4439 89/100/107	-	-	-	15941
15	4938 94/94/102	4717 91/94/102	4521 89/97/105	-	-	-	-	15436
20	4795 91/91/99	4583 89/95/102	-	-	-	-	-	14706
25	4610 88/92/99	-	-	-	-	-	-	13774
27	4529 87/92/99	-	-	-	-	-	-	13391
V _{FS}	117	121	124	127	131	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4804 109/109/121	4622 106/106/118	4450 103/103/115	4279 100/100/111	4114 98/99/110	3959 96/102/112	3891 94/107/116	18386
-35	4868 109/109/121	4686 106/106/118	4514 103/103/115	4343 100/100/111	4178 98/99/110	4023 96/102/112	3957 94/107/116	18386
-30	4931 109/109/121	4750 106/106/118	4578 103/103/115	4407 100/100/111	4242 98/99/110	4086 96/102/112	4022 94/107/116	18386
-25	4996 109/109/121	4815 106/106/118	4643 103/103/115	4472 100/100/111	4306 98/99/110	4150 96/102/112	4087 94/107/116	18386
-20	5061 109/109/121	4880 106/106/118	4708 103/103/115	4537 100/100/111	4371 98/99/110	4214 96/102/112	4153 94/107/116	18386
-15	5126 109/109/121	4945 106/106/118	4773 103/103/114	4601 100/100/111	4435 98/99/110	4278 96/102/112	4218 94/107/116	18386
-10	5192 109/109/121	5011 106/106/118	4839 103/103/114	4667 100/100/111	4499 98/99/110	4342 96/102/112	4284 94/107/116	18386
-5	5260 108/108/121	5078 105/105/118	4905 103/103/114	4733 100/100/111	4564 98/99/110	4406 95/102/112	4349 94/107/116	18386
0	5328 108/108/121	5145 105/105/118	4971 103/103/114	4798 100/100/111	4629 98/99/110	4470 95/102/112	4415 94/107/116	18386
5	5393 108/108/121	5209 105/105/117	5035 103/103/114	4861 100/100/111	4691 98/99/110	4531 95/102/112	4477 94/107/116	18386
10	5455 108/108/121	5271 105/105/117	5096 102/102/114	4921 100/100/111	4751 97/99/110	4590 95/102/112	4538 94/107/116	18386
15	5521 108/108/121	5335 105/105/117	5159 102/102/114	4984 100/100/111	4813 97/99/110	4651 95/102/112	4599 94/107/116	18386
20	5604 108/108/120	5416 105/105/117	5238 102/102/114	5060 100/100/111	4886 97/99/110	4722 95/102/112	4670 94/107/116	18386
25	5686 108/108/120	5496 105/105/117	5316 102/102/114	5137 100/100/111	4962 97/99/110	4797 95/102/112	4746 94/107/116	18386
30	5722 107/107/119	5531 104/104/116	5351 101/101/113	5171 99/99/110	4995 97/100/110	4829 94/103/112	4781 93/107/116	18386
35	5520 103/103/115	5331 101/101/111	5154 98/98/108	4977 96/98/107	4805 93/101/110	4645 91/104/112	5185 96/108/116	18386
40	5331 100/100/110	5146 97/97/107	4971 95/95/104	4799 92/99/107	4633 90/102/110	4964 93/105/112	-	17613
45	5148 97/97/106	4967 94/94/103	4797 91/96/104	4630 89/100/107	4964 92/103/110	-	-	16517
50	4967 93/93/102	4790 91/94/102	4625 88/97/104	-	-	-	-	15460
52	4894 92/92/100	4719 89/94/102	4571 87/98/104	-	-	-	-	15049
V _{FS}	117	120	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4960 109/109/121	4773 106/106/118	4595 103/103/115	4420 100/100/112	4250 98/99/110	4090 96/102/112	4023 95/107/116	18386
-35	5031 109/109/121	4843 106/106/118	4665 103/103/115	4489 100/100/112	4318 98/99/110	4158 96/102/112	4092 94/107/116	18386
-30	5099 109/109/121	4911 106/106/118	4734 103/103/115	4557 100/100/112	4385 98/99/110	4225 96/102/112	4160 94/107/116	18386
-25	5166 109/109/121	4977 106/106/118	4799 103/103/115	4622 100/100/112	4450 98/99/110	4289 96/102/112	4226 94/107/116	18386
-20	5230 109/109/121	5041 106/106/118	4863 103/103/115	4686 100/100/112	4514 98/99/110	4352 96/102/112	4291 94/107/116	18386
-15	5299 109/109/121	5110 106/106/118	4931 103/103/115	4753 100/100/111	4581 98/99/110	4418 96/102/112	4359 94/107/116	18386
-10	5367 109/109/121	5178 106/106/118	4999 103/103/115	4821 100/100/111	4647 98/99/110	4484 95/102/112	4426 94/107/116	18386
-5	5434 109/109/121	5245 106/106/118	5065 103/103/115	4887 100/100/111	4713 98/99/110	4549 95/102/112	4493 94/107/116	18386
0	5506 109/109/121	5316 105/105/118	5136 103/103/114	4956 100/100/111	4782 98/99/110	4617 95/102/112	4562 94/107/116	18386
5	5572 108/108/121	5381 105/105/118	5200 103/103/114	5020 100/100/111	4845 98/99/110	4680 95/102/112	4626 94/107/116	18386
10	5637 108/108/121	5446 105/105/117	5264 102/102/114	5084 100/100/111	4908 97/99/110	4742 95/102/112	4690 94/107/116	18386
15	5703 108/108/121	5510 105/105/117	5328 102/102/114	5147 100/100/111	4971 97/99/110	4804 95/102/112	4754 94/107/116	18386
20	5787 108/108/120	5593 105/105/117	5409 102/102/114	5226 100/100/111	5047 97/99/110	4879 95/102/112	4829 94/107/116	18386
25	5858 108/108/120	5664 105/105/117	5480 102/102/113	5297 99/99/110	5118 97/100/110	4949 95/103/112	4901 94/107/116	18386
30	5706 105/105/116	5515 102/102/113	5333 99/99/110	5154 97/97/107	4978 94/100/110	4814 92/104/112	5100 94/108/116	18386
35	5507 101/101/112	5319 98/98/108	5142 96/96/105	4966 93/98/107	4797 91/101/110	4891 92/105/112	-	17993
40	5319 98/98/107	5134 95/95/104	4961 92/96/105	4791 90/99/107	4902 91/102/110	-	-	16856
45	5136 94/94/103	4956 92/94/102	4788 89/97/105	4881 90/100/107	-	-	-	15788
50	4952 91/91/99	4776 88/95/102	-	-	-	-	-	14786
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5103 109/109/121	4912 106/106/118	4732 103/103/115	4554 100/100/112	4381 98/99/110	4219 96/102/112	4152 94/107/116	18386
-35	5174 109/109/121	4983 106/106/118	4803 103/103/115	4625 100/100/112	4451 98/99/110	4288 96/102/112	4224 94/107/116	18386
-30	5244 109/109/121	5053 106/106/118	4873 103/103/115	4694 100/100/112	4520 98/99/110	4357 96/102/112	4294 94/107/116	18386
-25	5313 109/109/121	5121 106/106/118	4941 103/103/115	4762 100/100/111	4587 98/99/110	4423 96/102/112	4362 94/107/116	18386
-20	5383 109/109/121	5191 106/106/118	5010 103/103/115	4830 100/100/111	4655 98/99/110	4491 95/102/112	4432 94/107/116	18386
-15	5455 109/109/121	5263 106/106/118	5082 103/103/115	4901 100/100/111	4726 98/99/110	4561 95/102/112	4503 94/107/116	18386
-10	5526 109/109/121	5334 106/106/118	5152 103/103/115	4972 100/100/111	4796 98/99/110	4630 95/102/112	4573 94/107/116	18386
-5	5599 109/109/121	5406 105/105/118	5224 103/103/115	5043 100/100/111	4866 98/99/110	4699 95/102/112	4645 94/107/116	18386
0	5671 108/108/121	5478 105/105/118	5295 103/103/114	5113 100/100/111	4935 98/99/110	4768 95/102/112	4715 94/107/116	18386
5	5741 108/108/121	5546 105/105/118	5363 102/102/114	5180 100/100/111	5001 97/99/110	4833 95/102/112	4781 94/107/116	18386
10	5810 108/108/121	5615 105/105/117	5430 102/102/114	5246 100/100/111	5067 97/99/110	4898 95/102/112	4848 94/107/116	18386
15	5882 108/108/121	5685 105/105/117	5499 102/102/114	5314 100/100/111	5134 97/99/110	4964 95/102/112	4915 94/107/116	18386
20	5964 108/108/120	5765 105/105/117	5577 102/102/114	5390 100/100/111	5208 97/99/110	5036 95/103/112	4988 94/107/116	18386
25	5840 105/105/117	5645 102/102/114	5461 100/100/110	5278 97/97/107	5100 95/100/110	4932 93/103/112	5132 94/108/116	18386
30	5681 102/102/113	5490 99/99/110	5309 97/97/107	5130 94/98/107	4957 92/101/110	4843 90/104/112	-	18363
35	5482 99/99/109	5294 96/96/105	5119 93/96/105	4946 91/99/107	4865 90/102/110	-	-	17162
40	5301 95/95/104	5117 93/93/102	4946 90/97/105	4841 89/100/107	-	-	-	16087
45	5118 92/92/100	4939 89/94/102	4792 87/98/105	-	-	-	-	15057
48	5007 90/91/99	4831 87/95/102	-	-	-	-	-	14490
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5239 109/109/121	5046 106/106/118	4864 103/103/114	4683 100/100/111	4507 98/99/110	4342 95/102/112	4277 94/107/116	18386
-35	5311 109/109/121	5118 106/106/118	4936 103/103/114	4754 100/100/111	4578 98/99/110	4413 95/102/112	4350 94/107/116	18386
-30	5386 109/109/121	5192 105/105/118	5009 103/103/114	4828 100/100/111	4651 98/99/110	4485 95/102/112	4423 94/107/116	18386
-25	5460 109/109/121	5266 105/105/118	5083 103/103/114	4900 100/100/111	4723 98/99/110	4557 95/102/112	4497 94/107/116	18386
-20	5532 108/108/121	5337 105/105/118	5154 103/103/114	4971 100/100/111	4793 98/99/110	4626 95/102/112	4568 94/107/116	18386
-15	5606 108/108/121	5411 105/105/118	5226 103/103/114	5043 100/100/111	4864 97/99/110	4696 95/102/112	4640 94/107/116	18386
-10	5682 108/108/121	5486 105/105/118	5301 102/102/114	5117 100/100/111	4937 97/99/110	4768 95/102/112	4713 94/107/116	18386
-5	5755 108/108/121	5559 105/105/117	5373 102/102/114	5188 100/100/111	5009 97/99/110	4839 95/102/112	4786 94/107/116	18386
0	5828 108/108/121	5632 105/105/117	5445 102/102/114	5260 100/100/111	5079 97/99/110	4909 95/102/112	4857 94/107/116	18386
5	5902 108/108/121	5705 105/105/117	5518 102/102/114	5331 100/100/111	5150 97/99/110	4979 95/102/112	4929 94/107/116	18386
10	5974 108/108/120	5775 105/105/117	5588 102/102/114	5401 100/100/111	5219 97/99/110	5046 95/102/112	4998 94/107/116	18386
15	6050 108/108/120	5850 105/105/117	5662 102/102/114	5474 99/99/111	5290 97/99/110	5118 95/102/112	5071 94/107/116	18386
20	5959 106/106/117	5762 103/103/114	5576 100/100/111	5391 97/97/108	5211 95/100/110	5041 93/103/112	5196 94/108/116	18386
25	5825 103/103/114	5631 100/100/111	5448 97/97/107	5267 95/98/107	5091 93/101/110	4925 90/104/112	5709 97/108/116	18386
30	5658 100/100/110	5468 97/97/107	5289 94/95/105	5112 92/99/107	4940 90/102/110	5411 94/105/112	-	17500
35	5464 96/96/105	5278 93/93/102	5103 91/96/105	4931 89/100/107	-	-	-	16371
40	5283 93/93/101	5100 90/94/102	4931 88/97/105	-	-	-	-	15335
45	5097 90/91/99	4920 87/95/102	-	-	-	-	-	14383
46	5060 89/92/99	4884 86/95/102	-	-	-	-	-	14196
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5313 108/108/120	5119 105/105/117	4936 102/102/113	4755 99/99/110	4579 97/100/110	4414 95/103/112	4352 93/107/116	18386
-30	5467 108/108/120	5272 104/104/116	5089 102/102/113	4906 99/99/110	4729 97/100/110	4562 94/103/112	4504 93/107/116	18386
-20	5615 107/107/120	5419 104/104/116	5235 102/102/113	5052 99/99/110	4873 97/100/110	4705 94/103/112	4650 93/107/116	18386
-10	5769 107/107/120	5573 104/104/116	5387 101/101/113	5202 99/99/110	5022 97/100/110	4852 94/103/112	4801 93/107/116	18386
0	5924 107/107/119	5726 104/104/116	5538 101/101/113	5351 99/99/110	5170 96/100/110	4998 94/103/112	4949 93/107/116	18386
10	6076 107/107/119	5874 104/104/116	5685 101/101/113	5496 99/99/109	5313 96/100/110	5139 94/103/112	5094 93/107/116	18386
20	5946 103/103/114	5749 100/100/111	5563 98/98/108	5378 95/98/107	5199 93/101/110	5031 91/104/112	5777 97/108/116	18386
30	5634 97/97/107	5444 94/94/103	5266 92/96/105	5091 90/99/107	5391 92/103/110	-	-	16699
40	5263 90/91/99	5083 88/95/102	-	-	-	-	-	14638
44	5117 88/92/99	-	-	-	-	-	-	13905
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5386 106/106/118	5192 103/103/115	5010 101/101/112	4828 98/98/109	4652 96/100/110	4487 94/103/112	4428 92/107/116	18386
-30	5541 106/106/118	5346 103/103/115	5163 101/101/112	4980 98/98/109	4802 96/100/110	4635 94/103/112	4580 92/107/116	18386
-20	5697 106/106/118	5501 103/103/115	5316 101/101/112	5133 98/98/109	4954 96/100/110	4786 93/103/112	4734 92/107/116	18386
-10	5855 106/106/118	5658 103/103/115	5471 100/100/112	5285 98/98/109	5105 96/100/110	4935 93/103/112	4886 92/107/116	18386
0	6016 106/106/118	5817 103/103/115	5629 100/100/112	5442 98/98/108	5260 95/100/110	5088 93/103/112	5075 93/107/116	18386
10	6171 106/106/118	5969 103/103/114	5779 100/100/111	5590 98/98/108	5406 95/100/110	5233 93/103/112	5348 93/108/116	18386
20	5941 101/101/111	5743 98/98/108	5558 95/95/105	5374 93/98/107	5196 91/102/110	5466 93/105/112	-	17897
30	5621 95/95/104	5432 92/93/102	5256 90/97/105	5337 90/100/107	-	-	-	15947
40	5249 88/92/99	5211 87/95/102	-	-	-	-	-	14000
42	5175 87/92/99	-	-	-	-	-	-	13637
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5450 105/105/117	5255 102/102/114	5071 99/99/110	4889 97/97/107	4712 95/100/110	4546 92/103/112	4647 93/108/116	18386
-30	5615 105/105/117	5419 102/102/114	5233 99/99/110	5049 97/97/107	4870 95/100/110	4703 92/103/112	4840 93/108/116	18386
-20	5772 105/105/117	5575 102/102/113	5388 99/99/110	5204 97/97/107	5024 94/100/110	4855 92/103/112	5040 94/108/116	18386
-10	5938 105/105/117	5739 102/102/113	5550 99/99/110	5363 97/97/107	5181 94/100/110	5010 92/103/112	5248 94/108/116	18386
0	6104 105/105/116	5902 102/102/113	5712 99/99/110	5522 97/97/107	5339 94/100/110	5165 92/103/112	5460 94/108/116	18386
10	6192 104/104/115	5990 101/101/112	5799 98/98/108	5609 96/98/107	5425 93/101/110	5253 91/104/112	5943 96/108/116	18386
20	5953 98/98/108	5755 96/96/105	5570 93/96/105	5387 91/99/107	5395 90/102/110	-	-	17098
30	5627 92/92/101	5440 90/94/102	5265 88/97/105	-	-	-	-	15204
40	5259 86/92/99	-	-	-	-	-	-	13377
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5522 104/104/115	5325 101/101/112	5139 98/98/109	4955 96/97/107	4777 93/101/110	4611 91/104/112	4999 95/108/116	18386
-30	5690 104/104/115	5492 101/101/112	5305 98/98/109	5120 96/97/107	4940 93/101/110	4772 91/104/112	5210 95/108/116	18386
-20	5853 104/104/115	5653 101/101/112	5465 98/98/109	5278 96/97/107	5097 93/101/110	4928 91/104/112	5430 95/108/116	18386
-10	6041 104/104/115	5838 101/101/112	5646 98/98/108	5457 95/97/107	5272 93/101/110	5099 91/104/112	5662 95/108/116	18386
0	6257 103/103/115	6048 100/100/112	5851 98/98/108	5655 95/97/107	5466 93/101/110	5288 91/104/112	5913 96/108/116	18386
10	6247 101/101/111	6038 98/98/108	5841 95/95/105	5647 93/98/107	5459 91/102/110	5736 92/105/112	-	17779
20	6020 96/96/105	5818 93/93/102	5628 91/97/105	5442 88/100/107	-	-	-	16201
30	5691 90/91/99	5496 87/95/102	-	-	-	-	-	14436
36	5472 86/92/99	-	-	-	-	-	-	13369
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5530 102/102/113	5334 99/99/109	5150 96/96/106	4968 94/98/107	4792 92/101/110	4689 90/104/112	-	18082
-30	5699 102/102/113	5502 99/99/109	5316 96/96/106	5133 94/98/107	4955 92/101/110	4884 90/104/112	-	18122
-20	5885 102/102/113	5686 99/99/109	5497 96/96/106	5311 94/98/107	5130 91/101/110	5094 91/104/112	-	18152
-10	6104 101/101/112	5898 99/99/109	5704 96/96/106	5512 94/98/107	5326 91/101/110	5320 91/104/112	-	18169
0	6324 101/101/112	6112 98/98/109	5912 96/96/106	5715 93/98/107	5524 91/101/110	5559 91/104/112	-	18179
10	6229 98/98/107	6017 95/95/104	5819 92/96/105	5625 90/99/107	5829 91/102/110	-	-	16772
20	6027 93/93/102	5822 91/94/102	5631 88/97/105	-	-	-	-	15400
25	5867 90/91/99	5667 88/95/102	-	-	-	-	-	14545
30	5682 87/92/99	-	-	-	-	-	-	13619
32	5607 86/92/99	-	-	-	-	-	-	13261
V _{FS}	117	121	124	127	130	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5539 100/100/110	5344 97/97/107	5161 94/95/105	4981 92/99/107	4807 90/102/110	5161 93/105/112	-	17519
-30	5720 100/100/110	5523 97/97/107	5338 94/95/105	5156 92/99/107	4980 90/102/110	5383 93/105/112	-	17554
-20	5946 100/100/110	5743 97/97/107	5552 94/95/105	5364 92/99/107	5183 90/102/110	5633 93/105/112	-	17575
-10	6171 99/99/110	5962 97/97/107	5765 94/95/105	5571 92/99/107	5384 90/102/110	5889 93/105/112	-	17587
0	6389 99/99/109	6173 96/96/106	5970 94/96/105	5771 91/99/107	5611 89/102/110	6320 95/105/112	-	17340
10	6231 95/95/104	6022 92/93/102	5826 90/97/105	5876 90/100/107	-	-	-	15941
15	6168 93/93/102	5957 90/94/102	5762 88/97/105	-	-	-	-	15436
20	6031 90/91/99	5823 88/95/102	-	-	-	-	-	14706
25	5846 87/92/99	-	-	-	-	-	-	13774
27	5763 86/92/99	-	-	-	-	-	-	13391
V _{FS}	117	121	124	127	131	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4468 108/108/120	4321 105/105/117	4181 102/102/114	4041 100/100/111	3903 97/99/110	3774 95/102/112	3854 96/107/116	18386
-35	4526 108/108/120	4379 105/105/117	4240 102/102/114	4100 100/100/111	3963 97/99/110	3833 95/102/112	3938 96/107/116	18386
-30	4583 108/108/120	4437 105/105/117	4299 102/102/114	4159 100/100/111	4022 97/99/110	3893 95/102/112	4024 96/107/116	18386
-25	4642 108/108/120	4497 105/105/117	4358 102/102/114	4219 100/100/111	4082 97/99/110	3952 95/102/112	4113 96/107/116	18386
-20	4701 108/108/120	4556 105/105/117	4418 102/102/114	4278 100/100/111	4141 97/99/110	4011 95/102/112	4206 97/107/116	18386
-15	4759 108/108/120	4615 105/105/117	4477 102/102/114	4338 99/99/111	4201 97/99/110	4071 95/102/112	4296 97/107/116	18386
-10	4819 108/108/120	4675 105/105/117	4537 102/102/114	4398 99/99/111	4260 97/99/110	4130 95/102/112	4391 97/107/116	18386
-5	4880 108/108/120	4735 105/105/117	4597 102/102/114	4458 99/99/111	4321 97/99/110	4190 95/102/112	4489 98/107/116	18386
0	4941 108/108/120	4796 105/105/117	4658 102/102/114	4518 99/99/111	4380 97/99/110	4250 95/102/112	4590 98/107/116	18386
5	4999 107/107/120	4854 104/104/117	4716 102/102/114	4576 99/99/110	4438 97/99/110	4307 95/102/112	4698 98/107/116	18386
10	5056 107/107/120	4910 104/104/117	4772 102/102/113	4632 99/99/110	4494 97/99/110	4362 95/102/112	4811 99/107/116	18386
15	5115 107/107/120	4969 104/104/116	4830 102/102/113	4689 99/99/110	4550 97/99/110	4418 94/102/112	4928 99/107/116	18386
20	5188 107/107/119	5041 104/104/116	4900 101/101/113	4757 99/99/110	4617 97/99/110	4483 94/102/112	5059 100/107/116	18386
25	5261 107/107/119	5112 104/104/116	4970 101/101/113	4827 99/99/110	4686 96/99/110	4552 94/102/112	5193 100/107/116	18386
30	5298 106/106/118	5149 103/103/115	5007 101/101/112	4863 98/98/109	4722 96/100/110	4587 94/103/112	5524 102/107/116	18386
35	5138 103/103/114	4989 100/100/111	4848 97/97/107	4707 95/98/107	4605 93/101/110	5281 99/104/112	6387 108/108/116	18386
40	4989 99/99/109	4843 96/96/106	4704 94/95/104	4618 92/99/107	5340 99/102/110	6082 104/105/112	-	17613
45	4846 96/96/105	4701 93/93/102	4581 91/96/104	5392 98/100/107	6041 103/103/110	-	-	16517
50	4704 92/92/101	4561 90/94/102	5350 97/97/104	-	-	-	-	15460
52	4647 91/91/99	4801 92/94/102	5577 98/98/104	-	-	-	-	15049
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4607 108/108/121	4456 105/105/117	4313 102/102/114	4169 100/100/111	4028 97/99/110	3895 95/102/112	3987 96/107/116	18386
-35	4670 108/108/121	4520 105/105/117	4376 102/102/114	4232 100/100/111	4091 97/99/110	3958 95/102/112	4077 96/107/116	18386
-30	4732 108/108/121	4582 105/105/117	4439 102/102/114	4295 100/100/111	4154 97/99/110	4020 95/102/112	4169 96/107/116	18386
-25	4792 108/108/121	4642 105/105/117	4498 102/102/114	4354 100/100/111	4213 97/99/110	4079 95/102/112	4263 97/107/116	18386
-20	4850 108/108/120	4700 105/105/117	4557 102/102/114	4413 100/100/111	4272 97/99/110	4138 95/102/112	4361 97/107/116	18386
-15	4912 108/108/120	4762 105/105/117	4619 102/102/114	4476 99/99/111	4334 97/99/110	4200 95/102/112	4463 97/107/116	18386
-10	4974 108/108/120	4824 105/105/117	4681 102/102/114	4537 99/99/111	4396 97/99/110	4261 95/102/112	4566 98/107/116	18386
-5	5034 108/108/120	4884 105/105/117	4742 102/102/114	4598 99/99/111	4457 97/99/110	4322 95/102/112	4671 98/107/116	18386
0	5099 108/108/120	4949 105/105/117	4806 102/102/114	4662 99/99/111	4520 97/99/110	4385 95/102/112	4779 98/107/116	18386
5	5158 107/107/120	5008 105/105/117	4865 102/102/114	4721 99/99/110	4579 97/99/110	4444 95/102/112	4900 99/107/116	18386
10	5217 107/107/120	5067 104/104/117	4923 102/102/113	4779 99/99/110	4637 97/99/110	4502 95/102/112	5023 99/107/116	18386
15	5277 107/107/120	5126 104/104/117	4982 102/102/113	4838 99/99/110	4695 97/99/110	4560 94/102/112	5152 100/107/116	18386
20	5351 107/107/120	5199 104/104/116	5054 101/101/113	4908 99/99/110	4765 97/99/110	4628 94/102/112	5295 100/107/116	18386
25	5418 107/107/119	5266 104/104/116	5121 101/101/113	4976 99/99/110	4832 96/100/110	4696 94/103/112	5561 102/107/116	18386
30	5297 104/104/115	5147 101/101/112	5004 98/98/109	4861 96/97/107	4720 94/100/110	5171 98/104/112	6326 107/108/116	18386
35	5141 100/100/111	4993 98/98/108	4852 95/95/105	4712 93/98/107	5237 97/101/110	6073 104/105/112	-	17993
40	4995 97/97/107	4848 94/94/103	4710 92/96/105	5261 96/99/107	6046 102/102/110	-	-	16856
45	4853 94/94/102	4708 91/94/102	5242 95/97/105	5941 100/100/107	-	-	-	15788
50	4709 90/91/99	5191 94/95/102	-	-	-	-	-	14786
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4733 108/108/121	4581 105/105/117	4436 102/102/114	4290 100/100/111	4148 97/99/110	4013 95/102/112	4159 96/107/116	18386
-35	4797 108/108/121	4646 105/105/117	4501 102/102/114	4355 100/100/111	4213 97/99/110	4078 95/102/112	4256 97/107/116	18386
-30	4861 108/108/120	4709 105/105/117	4565 102/102/114	4419 100/100/111	4277 97/99/110	4142 95/102/112	4356 97/107/116	18386
-25	4922 108/108/120	4771 105/105/117	4627 102/102/114	4482 99/99/111	4339 97/99/110	4204 95/102/112	4457 97/107/116	18386
-20	4986 108/108/120	4834 105/105/117	4690 102/102/114	4545 99/99/111	4402 97/99/110	4267 95/102/112	4562 98/107/116	18386
-15	5051 108/108/120	4899 105/105/117	4755 102/102/114	4610 99/99/111	4467 97/99/110	4331 95/102/112	4671 98/107/116	18386
-10	5115 108/108/120	4963 105/105/117	4819 102/102/114	4674 99/99/111	4531 97/99/110	4396 95/102/112	4782 98/107/116	18386
-5	5180 108/108/120	5029 105/105/117	4884 102/102/114	4739 99/99/111	4596 97/99/110	4460 95/102/112	4895 99/107/116	18386
0	5246 108/108/120	5094 105/105/117	4949 102/102/114	4804 99/99/111	4660 97/99/110	4524 95/102/112	5014 99/107/116	18386
5	5308 107/107/120	5156 104/104/117	5011 102/102/114	4865 99/99/110	4721 97/99/110	4584 95/102/112	5143 100/107/116	18386
10	5371 107/107/120	5218 104/104/117	5072 102/102/113	4926 99/99/110	4782 97/99/110	4645 94/102/112	5275 100/107/116	18386
15	5436 107/107/120	5282 104/104/116	5135 101/101/113	4989 99/99/110	4844 97/99/110	4706 94/102/112	5413 101/107/116	18386
20	5510 107/107/119	5355 104/104/116	5207 101/101/113	5059 99/99/110	4913 96/99/110	4774 94/103/112	5642 102/107/116	18386
25	5415 104/104/116	5263 101/101/113	5118 99/99/110	4973 96/97/107	4830 94/100/110	5213 97/103/112	6399 106/108/116	18386
30	5291 101/101/112	5141 99/99/109	4999 96/96/106	4856 94/98/107	5172 96/101/110	6014 103/104/112	-	18363
35	5136 98/98/108	4989 95/95/105	4849 93/96/105	5213 95/99/107	6029 102/102/110	-	-	17162
40	4997 94/94/104	4852 92/93/102	5201 94/97/105	5947 100/100/107	-	-	-	16087
45	4856 91/91/99	5152 92/94/102	5831 98/98/105	-	-	-	-	15057
48	4770 89/91/99	5504 95/95/102	-	-	-	-	-	14490
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4855 108/108/120	4701 105/105/117	4555 102/102/114	4408 99/99/111	4264 97/99/110	4128 95/102/112	4367 97/107/116	18386
-35	4920 108/108/120	4767 105/105/117	4621 102/102/114	4474 99/99/111	4330 97/99/110	4194 95/102/112	4471 97/107/116	18386
-30	4987 108/108/120	4834 105/105/117	4688 102/102/114	4541 99/99/111	4397 97/99/110	4260 95/102/112	4579 98/107/116	18386
-25	5054 108/108/120	4900 105/105/117	4755 102/102/114	4608 99/99/111	4464 97/99/110	4327 95/102/112	4688 98/107/116	18386
-20	5119 108/108/120	4965 105/105/117	4820 102/102/114	4673 99/99/110	4529 97/99/110	4392 95/102/112	4802 98/107/116	18386
-15	5185 107/107/120	5032 104/104/117	4886 102/102/114	4739 99/99/110	4594 97/99/110	4457 95/102/112	4919 99/107/116	18386
-10	5254 107/107/120	5100 104/104/117	4953 102/102/114	4806 99/99/110	4661 97/99/110	4523 95/102/112	5040 99/107/116	18386
-5	5320 107/107/120	5166 104/104/117	5019 102/102/113	4872 99/99/110	4727 97/99/110	4589 94/102/112	5165 100/107/116	18386
0	5386 107/107/120	5232 104/104/117	5085 102/102/113	4938 99/99/110	4793 97/99/110	4655 94/102/112	5299 100/107/116	18386
5	5453 107/107/120	5298 104/104/116	5151 101/101/113	5003 99/99/110	4858 97/99/110	4720 94/102/112	5435 101/107/116	18386
10	5518 107/107/120	5363 104/104/116	5215 101/101/113	5067 99/99/110	4922 96/99/110	4783 94/102/112	5578 101/107/116	18386
15	5586 107/107/119	5430 104/104/116	5282 101/101/113	5134 99/99/110	4988 96/99/110	4849 94/102/112	5746 102/107/116	18386
20	5521 105/105/116	5367 102/102/113	5221 99/99/110	5074 97/97/107	4930 94/100/110	5298 97/103/112	6480 107/108/116	18386
25	5417 102/102/113	5266 99/99/110	5122 96/96/107	4978 94/98/107	5200 95/101/110	6048 102/104/112	6958 108/108/116	18386
30	5289 99/99/109	5140 96/96/106	4998 94/95/105	5148 94/99/107	6018 101/102/110	6615 105/105/112	-	17500
35	5140 95/95/105	4993 93/93/102	5144 93/96/105	5986 100/100/107	-	-	-	16371
40	5002 92/92/100	5065 92/94/102	5839 97/97/105	-	-	-	-	15335
45	4964 90/91/99	5681 95/95/102	-	-	-	-	-	14383
46	5156 91/92/99	5772 95/95/102	-	-	-	-	-	14196
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4923 107/107/119	4770 104/104/116	4624 101/101/113	4477 98/98/109	4333 96/100/110	4197 94/103/112	4686 99/107/116	18386
-30	5062 107/107/119	4908 104/104/116	4763 101/101/112	4616 98/98/109	4472 96/100/110	4335 94/103/112	4931 100/107/116	18386
-20	5196 106/106/119	5042 104/104/116	4896 101/101/112	4750 98/98/109	4605 96/100/110	4468 94/103/112	5179 101/107/116	18386
-10	5335 106/106/119	5181 103/103/115	5035 101/101/112	4888 98/98/109	4743 96/100/110	4605 94/103/112	5451 102/107/116	18386
0	5475 106/106/119	5320 103/103/115	5173 101/101/112	5025 98/98/109	4879 96/100/110	4740 94/103/112	5738 103/107/116	18386
10	5612 106/106/118	5455 103/103/115	5307 100/100/112	5158 98/98/109	5012 96/100/110	4981 94/103/112	6059 104/107/116	18386
20	5526 102/102/113	5372 99/99/110	5225 97/97/107	5079 94/98/107	5276 95/101/110	6118 102/104/112	7070 108/108/116	18386
30	5288 96/96/106	5139 94/94/103	5087 92/96/105	5968 99/99/107	6560 103/103/110	-	-	16699
40	5008 90/91/99	5716 95/95/102	-	-	-	-	-	14638
44	5451 92/92/99	-	-	-	-	-	-	13905
V_{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4992 106/106/118	4839 103/103/114	4694 100/100/111	4547 97/97/108	4404 95/100/110	4268 93/103/112	5078 101/107/116	18386
-30	5132 105/105/118	4979 103/103/114	4834 100/100/111	4687 97/97/108	4543 95/100/110	4439 93/103/112	5349 102/107/116	18386
-20	5273 105/105/117	5120 102/102/114	4974 100/100/111	4828 97/97/108	4684 95/100/110	4660 94/103/112	5644 103/107/116	18386
-10	5416 105/105/117	5262 102/102/114	5115 100/100/111	4968 97/97/108	4824 95/100/110	4893 95/103/112	5948 104/107/116	18386
0	5562 105/105/117	5407 102/102/114	5260 99/99/111	5112 97/97/108	4966 95/100/110	5143 96/103/112	6279 105/107/116	18386
10	5707 105/105/117	5550 102/102/113	5402 99/99/110	5253 97/97/107	5107 94/100/110	5502 98/103/112	6721 107/108/116	18386
20	5539 100/100/110	5385 97/97/107	5239 94/95/105	5221 93/98/107	6105 100/102/110	6748 104/105/112	-	17897
30	5299 94/94/103	5150 91/93/102	5861 96/97/105	6474 100/100/107	-	-	-	15947
40	5545 92/92/99	6164 95/95/102	-	-	-	-	-	14000
42	5709 92/92/99	-	-	-	-	-	-	13637
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5056 104/104/116	4902 101/101/113	4756 99/99/110	4609 96/97/107	4466 94/100/110	4608 95/103/112	5568 104/108/116	18386
-30	5205 104/104/116	5051 101/101/113	4904 99/99/110	4757 96/97/107	4612 94/100/110	4846 96/103/112	5914 105/108/116	18386
-20	5348 104/104/116	5193 101/101/113	5047 98/98/109	4900 96/97/107	4755 94/100/110	5102 97/103/112	6236 106/108/116	18386
-10	5498 104/104/116	5343 101/101/112	5195 98/98/109	5047 96/97/107	4901 94/100/110	5380 98/103/112	6572 107/108/116	18386
0	5649 104/104/116	5492 101/101/112	5343 98/98/109	5194 96/97/107	5047 94/100/110	5655 99/103/112	6854 107/108/116	18386
10	5745 103/103/114	5587 100/100/111	5438 97/97/108	5288 95/98/107	5476 96/101/110	6327 102/104/112	7388 108/108/116	18386
20	5570 98/98/108	5415 95/95/104	5269 92/96/105	6032 98/99/107	6659 102/102/110	-	-	17098
30	5328 92/92/100	5699 94/94/102	6327 97/97/105	-	-	-	-	15204
40	5966 92/92/99	-	-	-	-	-	-	13377
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5126 103/103/114	4972 100/100/111	4825 97/97/108	4677 95/97/107	4533 93/101/110	5050 98/104/112	6180 107/108/116	18386
-30	5280 103/103/114	5125 100/100/111	4977 97/97/108	4829 95/97/107	4684 93/101/110	5323 99/104/112	6469 108/108/116	18386
-20	5428 103/103/114	5272 100/100/111	5124 97/97/108	4976 95/97/107	4884 93/101/110	5621 100/104/112	6761 108/108/116	18386
-10	5597 103/103/114	5439 100/100/111	5289 97/97/108	5139 95/97/107	5148 94/101/110	5963 101/104/112	7012 108/108/116	18386
0	5788 102/102/114	5626 100/100/111	5473 97/97/108	5319 95/97/107	5444 95/101/110	6291 102/104/112	7411 108/108/116	18386
10	5806 100/100/110	5644 97/97/107	5490 94/95/105	5573 94/98/107	6477 101/102/110	7079 104/105/112	-	17779
20	5644 95/95/104	5487 92/93/102	6043 96/97/105	6682 100/100/107	-	-	-	16201
30	5643 91/91/99	6290 95/95/102	-	-	-	-	-	14436
36	6176 92/92/99	-	-	-	-	-	-	13369
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5146 101/101/112	4992 98/98/109	4846 95/95/105	4700 93/98/107	4967 95/101/110	5724 102/104/112	-	18082
-30	5301 101/101/112	5147 98/98/109	5000 95/95/105	4854 93/98/107	5238 96/101/110	6071 103/104/112	-	18122
-20	5469 101/101/112	5313 98/98/108	5165 95/95/105	5018 93/98/107	5543 97/101/110	6403 104/104/112	-	18152
-10	5664 100/100/112	5504 98/98/108	5352 95/95/105	5201 93/98/107	5894 98/101/110	6717 104/104/112	-	18169
0	5860 100/100/111	5696 98/98/108	5541 95/95/105	5386 93/98/107	6285 99/101/110	6973 104/104/112	-	18179
10	5814 97/97/107	5650 94/94/103	5664 93/96/105	6496 99/99/107	7129 102/102/110	-	-	16772
20	5676 92/92/101	6000 94/94/102	6632 97/97/105	-	-	-	-	15400
25	5743 91/91/99	6400 95/95/102	-	-	-	-	-	14545
30	6192 92/92/99	-	-	-	-	-	-	13619
32	6383 92/92/99	-	-	-	-	-	-	13261
V_{FS}	117	121	124	127	130	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) OFF – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5168 99/99/109	5015 96/96/106	4870 94/95/105	4854 93/99/107	5658 99/102/110	6366 105/105/112	-	17519
-30	5333 99/99/109	5180 96/96/106	5034 94/95/105	5122 94/99/107	5993 101/102/110	6609 105/105/112	-	17554
-20	5535 99/99/109	5377 96/96/106	5227 93/95/105	5439 95/99/107	6357 102/102/110	6919 105/105/112	-	17575
-10	5736 98/98/109	5574 96/96/106	5421 93/95/105	5783 96/99/107	6605 101/102/110	7211 105/105/112	-	17587
0	5944 98/98/108	5776 95/95/105	5618 93/96/105	6359 98/99/107	7012 102/102/110	7749 105/105/112	-	17340
10	5842 94/94/103	5680 91/93/102	6467 97/97/105	7120 100/100/107	-	-	-	15941
15	5805 92/92/101	6139 94/94/102	6808 97/97/105	-	-	-	-	15436
20	5897 91/91/99	6539 95/95/102	-	-	-	-	-	14706
25	6327 92/92/99	-	-	-	-	-	-	13774
27	6530 92/92/99	-	-	-	-	-	-	13391
V _{FS}	117	121	124	127	131	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2139 105/105/118	2174 104/104/117	2303 104/104/116	2440 104/104/115	2641 105/105/116	2891 108/108/119	3340 113/113/123	18386
-35	2175 105/105/118	2205 104/104/117	2334 104/104/116	2473 104/104/115	2679 105/105/116	2933 108/108/119	3388 113/113/123	18386
-30	2211 105/105/118	2240 104/104/117	2366 104/104/116	2507 104/104/115	2717 105/105/116	2975 108/108/119	3436 113/113/123	18386
-25	2247 104/104/118	2278 104/104/117	2397 104/104/116	2541 103/103/115	2755 105/105/116	3017 108/108/119	3484 113/113/123	18386
-20	2283 104/104/118	2315 104/104/117	2429 104/104/116	2576 103/103/115	2793 105/105/116	3060 108/108/119	3533 113/113/123	18386
-15	2319 104/104/118	2351 104/104/117	2462 104/104/116	2610 103/103/115	2833 105/105/116	3103 108/108/119	3584 113/113/123	18386
-10	2354 104/104/118	2388 104/104/117	2494 104/104/116	2644 103/103/115	2872 105/105/116	3147 108/108/119	3634 113/113/123	18386
-5	2389 104/104/118	2424 104/104/117	2527 104/104/116	2679 103/103/115	2913 105/105/116	3191 108/108/119	3686 113/113/123	18386
0	2424 104/104/117	2459 104/104/117	2559 103/103/116	2713 103/103/115	2954 105/105/116	3237 108/108/119	3740 113/113/123	18386
5	2457 104/104/117	2494 104/104/116	2592 103/103/116	2748 103/103/115	2998 105/105/116	3284 108/108/119	3796 113/113/123	18386
V_{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 1 – DRY RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2215 105/105/118	2245 104/104/117	2358 104/104/116	2500 104/104/116	2695 105/105/116	2951 108/108/119	3403 113/113/123	18386
-35	2249 105/105/118	2279 104/104/117	2387 104/104/116	2531 104/104/115	2733 105/105/116	2993 108/108/119	3451 113/113/123	18386
-30	2286 105/105/118	2317 104/104/117	2419 104/104/116	2566 104/104/115	2773 105/105/116	3037 108/108/119	3501 113/113/123	18386
-25	2324 105/105/118	2356 104/104/117	2452 104/104/116	2601 104/104/115	2812 105/105/116	3081 108/108/119	3552 113/113/123	18386
-20	2359 105/105/118	2393 104/104/117	2486 104/104/116	2637 104/104/115	2854 105/105/116	3126 108/108/119	3605 113/113/123	18386
-15	2395 104/104/118	2430 104/104/117	2519 104/104/116	2672 103/103/115	2896 105/105/116	3172 108/108/119	3659 113/113/123	18386
-10	2432 104/104/118	2467 104/104/117	2553 104/104/116	2708 103/103/115	2938 105/105/116	3219 108/108/119	3713 113/113/123	18386
-5	2467 104/104/118	2504 104/104/117	2587 104/104/116	2744 103/103/115	2981 105/105/116	3266 108/108/119	3767 113/113/123	18386
0	2503 104/104/118	2541 104/104/117	2621 103/103/116	2780 103/103/115	3024 105/105/116	3313 108/108/119	3822 113/113/123	18386
5	2537 104/104/117	2576 104/104/116	2655 103/103/116	2816 103/103/115	3071 105/105/116	3364 108/108/119	3884 113/113/123	18386
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2281 105/105/118	2312 104/104/117	2415 104/104/116	2561 104/104/115	2765 105/105/116	3027 108/108/119	3488 113/113/123	18386
-35	2318 105/105/118	2350 104/104/117	2447 104/104/116	2595 104/104/115	2806 105/105/116	3072 108/108/119	3540 113/113/123	18386
-30	2356 105/105/118	2389 104/104/117	2481 104/104/116	2631 104/104/115	2847 105/105/116	3118 108/108/119	3592 113/113/123	18386
-25	2394 104/104/118	2428 104/104/117	2515 104/104/116	2668 104/104/115	2889 105/105/116	3164 108/108/119	3645 113/113/123	18386
-20	2432 104/104/118	2467 104/104/117	2550 104/104/116	2705 103/103/115	2932 105/105/116	3211 108/108/119	3700 113/113/123	18386
-15	2469 104/104/118	2506 104/104/117	2585 104/104/116	2742 103/103/115	2975 105/105/116	3258 108/108/119	3754 113/113/123	18386
-10	2506 104/104/118	2544 104/104/117	2619 104/104/116	2778 103/103/115	3018 105/105/116	3306 108/108/119	3810 113/113/123	18386
-5	2544 104/104/118	2582 104/104/117	2654 103/103/116	2815 103/103/115	3062 105/105/116	3354 108/108/119	3866 113/113/123	18386
0	2580 104/104/117	2620 104/104/117	2689 103/103/116	2852 103/103/115	3107 105/105/116	3404 108/108/119	3925 113/113/123	18386
5	2615 104/104/117	2656 104/104/116	2724 103/103/115	2889 103/103/115	3155 105/105/116	3456 108/108/119	3987 113/113/123	18386
V_{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 1 – DRY RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2351 105/105/118	2383 104/104/117	2482 104/104/116	2632 104/104/115	2848 105/105/117	3117 108/108/119	3592 113/113/123	18386
-35	2386 104/104/118	2420 104/104/117	2514 104/104/116	2666 103/103/115	2890 105/105/117	3165 108/108/119	3647 113/113/123	18386
-30	2424 104/104/118	2459 104/104/117	2548 104/104/116	2702 103/103/115	2933 105/105/117	3212 108/108/119	3701 113/113/123	18386
-25	2464 104/104/118	2500 104/104/117	2583 104/104/116	2739 103/103/115	2976 105/105/117	3259 108/108/119	3756 113/113/123	18386
-20	2503 104/104/118	2540 104/104/117	2618 104/104/116	2777 103/103/115	3020 105/105/117	3308 108/108/119	3813 113/113/123	18386
-15	2541 104/104/118	2580 104/104/117	2654 103/103/116	2815 103/103/115	3065 105/105/117	3357 108/108/119	3870 113/113/123	18386
-10	2580 104/104/117	2620 104/104/116	2690 103/103/116	2853 103/103/115	3111 105/105/117	3407 108/108/119	3929 113/113/123	18386
-5	2618 104/104/117	2659 104/104/116	2726 103/103/115	2891 103/103/115	3158 105/105/117	3459 108/108/119	3990 113/113/123	18386
0	2655 104/104/117	2697 103/103/116	2763 103/103/115	2930 103/103/114	3206 105/105/117	3511 108/108/119	4052 113/113/123	18386
5	2683 103/103/116	2726 103/103/116	2818 103/103/115	2986 102/102/114	3293 106/106/117	3617 109/109/119	4177 113/113/123	18386
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2395 104/104/117	2430 103/103/116	2556 103/103/115	2709 103/103/114	2977 106/106/117	3265 109/109/119	3767 113/113/123	18386
-30	2472 103/103/117	2509 103/103/116	2625 103/103/115	2783 102/102/114	3067 106/106/117	3365 109/109/119	3883 113/113/123	18386
-25	2512 103/103/116	2550 103/103/116	2662 103/103/115	2823 102/102/114	3113 106/106/117	3416 109/109/119	3942 113/113/123	18386
-20	2551 103/103/116	2591 103/103/115	2699 103/103/115	2866 102/102/114	3160 106/106/117	3469 109/109/119	4003 113/113/123	18386
-15	2590 103/103/116	2631 103/103/115	2736 102/102/114	2909 102/102/114	3208 106/106/117	3522 109/109/119	4064 113/113/123	18386
-10	2629 103/103/116	2671 103/103/115	2773 102/102/114	2952 102/102/114	3256 106/106/117	3576 109/109/119	4126 113/113/123	18386
-5	2668 103/103/116	2711 103/103/115	2810 102/102/114	2997 102/102/114	3305 106/106/117	3632 109/109/119	4190 113/113/123	18386
0	2706 103/103/116	2750 102/102/115	2847 102/102/114	3042 102/102/114	3355 106/106/117	3688 109/109/119	4255 113/113/123	18386
5	2687 101/101/113	2764 100/100/112	2943 100/100/111	3260 103/103/114	3599 106/106/117	3995 109/109/119	4586 114/114/123	18386
V_{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2437 102/102/115	2474 102/102/114	2637 102/102/114	2836 103/103/114	3124 106/106/117	3449 109/109/119	3970 113/113/123	18386
-30	2518 102/102/115	2557 102/102/114	2710 102/102/113	2922 103/103/114	3219 106/106/117	3554 109/109/119	4092 113/113/123	18386
-25	2559 102/102/115	2600 102/102/114	2748 102/102/113	2966 103/103/114	3268 106/106/117	3609 109/109/119	4155 113/113/123	18386
-20	2599 102/102/115	2641 102/102/114	2786 101/101/113	3011 103/103/114	3318 106/106/117	3665 109/109/119	4219 113/113/123	18386
-15	2639 102/102/115	2682 102/102/114	2824 101/101/113	3056 103/103/114	3368 106/106/117	3721 109/109/119	4284 113/113/123	18386
-10	2679 102/102/115	2723 102/102/114	2862 101/101/113	3102 103/103/114	3419 106/106/117	3779 109/109/119	4351 113/113/123	18386
-5	2719 102/102/115	2764 101/101/114	2901 101/101/113	3150 103/103/114	3471 106/106/117	3838 109/109/119	4418 113/113/123	18386
0	2746 101/101/114	2795 101/101/113	2977 100/100/112	3278 103/103/114	3612 106/106/117	4011 109/109/119	4609 114/114/123	18386
5	2709 98/98/110	2895 98/98/109	3215 101/101/111	3570 104/104/114	3983 107/107/117	4401 110/110/119	5030 114/114/123	18386
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2486 101/101/114	2565 101/101/113	2732 101/101/112	2996 103/103/114	3297 106/106/117	3662 109/109/119	4204 113/113/123	18386
-30	2565 101/101/114	2633 101/101/113	2805 100/100/112	3087 103/103/114	3398 106/106/117	3774 109/109/119	4334 113/113/123	18386
-25	2606 101/101/114	2670 101/101/113	2844 100/100/112	3134 103/103/114	3451 106/106/117	3832 109/109/119	4401 114/114/123	18386
-20	2648 101/101/113	2707 101/101/113	2883 100/100/112	3182 103/103/114	3504 106/106/117	3891 109/109/119	4469 114/114/123	18386
-15	2689 101/101/113	2745 100/100/112	2924 100/100/112	3231 103/103/114	3559 106/106/117	3952 109/109/119	4539 114/114/123	18386
-10	2730 101/101/113	2782 100/100/112	2963 100/100/111	3280 103/103/114	3615 106/106/117	4014 109/109/119	4609 114/114/123	18386
-5	2771 101/101/113	2820 100/100/112	3004 100/100/111	3330 103/103/114	3673 106/106/117	4078 109/109/119	4683 114/114/123	18386
0	2764 98/98/110	2933 98/98/109	3234 100/100/111	3585 104/104/114	4003 107/107/117	4426 110/110/119	5060 114/114/123	18386
5	2843 95/95/106	3153 98/98/108	3524 101/101/111	3961 104/104/114	4399 108/108/117	4843 111/111/119	5583 115/115/123	18386
V_{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2531 100/100/112	2656 100/100/111	2857 100/100/111	3168 103/103/114	3511 106/106/117	3889 110/110/119	4455 114/114/123	18386
-30	2614 100/100/112	2729 99/99/111	2943 100/100/111	3264 103/103/114	3618 106/106/117	4009 110/110/119	4593 114/114/123	18386
-25	2657 100/100/112	2768 99/99/111	2988 100/100/111	3315 103/103/114	3676 106/106/117	4072 110/110/119	4666 114/114/123	18386
-20	2699 100/100/112	2806 99/99/111	3034 100/100/111	3366 103/103/114	3733 106/106/117	4136 110/110/119	4739 114/114/123	18386
-15	2741 99/99/112	2844 99/99/111	3079 100/100/111	3417 103/103/114	3790 107/107/117	4199 110/110/119	4812 114/114/123	18386
-10	2789 99/99/111	2895 99/99/111	3142 100/100/111	3485 103/103/114	3871 107/107/117	4287 110/110/119	4911 114/114/123	18386
-5	2820 98/98/110	2991 98/98/109	3304 101/101/111	3668 104/104/114	4093 107/107/117	4523 110/110/119	5169 114/114/123	18386
0	2894 96/96/107	3202 98/98/108	3578 101/101/111	4021 104/104/114	4465 108/108/117	4917 111/111/119	5671 115/115/123	18386
5	3094 95/95/105	3475 99/99/108	3941 102/102/111	4405 105/105/114	4871 108/108/117	5397 111/111/119	6229 115/115/123	18386
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2590 98/98/110	2767 98/98/109	3076 101/101/111	3418 104/104/114	3814 107/107/117	4212 110/110/119	4809 114/114/123	18386
-30	2658 98/98/109	2841 97/97/109	3170 101/101/111	3523 104/104/114	3931 107/107/117	4342 110/110/119	4959 114/114/123	18386
-25	2695 98/98/109	2881 97/97/108	3218 101/101/111	3578 104/104/114	3992 107/107/117	4410 110/110/119	5036 114/114/123	18386
-20	2733 98/98/109	2921 97/97/108	3267 101/101/111	3634 104/104/114	4055 107/107/117	4479 110/110/119	5115 114/114/123	18386
-15	2775 97/97/109	2962 97/97/108	3317 101/101/111	3690 104/104/114	4117 107/107/117	4548 110/110/119	5194 114/114/123	18386
-10	2862 97/97/109	3082 97/97/108	3446 101/101/111	3852 104/104/114	4288 107/107/117	4731 110/110/119	5417 115/115/123	18386
-5	2963 95/95/105	3327 98/98/108	3740 102/102/111	4195 105/105/114	4651 108/108/117	5115 111/111/119	5918 115/115/123	18386
0	3210 95/95/105	3625 99/99/108	4102 102/102/111	4577 105/105/114	5056 108/108/117	5626 111/111/119	6481 116/116/123	18386
5	3509 96/96/105	4001 100/100/108	4502 103/103/111	5020 106/106/114	5611 109/109/117	6212 112/112/119	- -	17982
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2697 96/96/107	2964 98/98/108	3309 101/101/111	3719 104/104/114	4135 108/108/117	4554 111/111/119	5235 115/115/123	18386
-30	2769 96/96/107	3054 98/98/108	3411 101/101/111	3834 104/104/114	4263 108/108/117	4696 111/111/119	5400 115/115/123	18386
-25	2807 96/96/107	3100 98/98/108	3462 101/101/111	3892 104/104/114	4328 108/108/117	4768 111/111/119	5483 115/115/123	18386
-20	2847 96/96/107	3147 98/98/108	3515 101/101/111	3952 104/104/114	4395 108/108/117	4842 111/111/119	5571 115/115/123	18386
-15	2901 95/95/106	3218 98/98/108	3596 101/101/111	4046 104/104/114	4497 108/108/117	4952 111/111/119	5705 115/115/123	18386
-10	3048 95/95/105	3428 98/98/108	3867 102/102/111	4332 105/105/114	4798 108/108/117	5285 111/111/119	6117 115/115/123	18386
-5	3330 96/96/105	3778 99/99/108	4267 102/102/111	4756 106/106/114	5279 109/109/117	5862 112/112/119	6739 116/116/123	18386
0	3632 96/96/105	4136 100/100/108	4649 103/103/111	5202 106/106/114	5808 109/109/117	6424 112/112/119	-	17905
5	4071 97/97/105	4603 100/100/108	5210 104/104/111	5850 107/107/114	6493 110/110/117	-	-	16747
V _{FS}	117	121	124	127	131	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3094 105/105/118	3108 104/104/117	3127 104/104/116	3143 104/104/115	3159 103/105/116	3178 103/108/119	3340 103/113/123	18386
-35	3150 105/105/118	3166 104/104/117	3186 104/104/116	3203 104/104/115	3220 103/105/116	3241 103/108/119	3388 103/113/123	18386
-30	3208 105/105/118	3225 104/104/117	3247 104/104/116	3265 104/104/115	3283 103/105/116	3305 103/108/119	3436 103/113/123	18386
-25	3267 104/104/118	3286 104/104/117	3308 104/104/116	3328 103/103/115	3347 103/105/116	3370 103/108/119	3484 103/113/123	18386
-20	3326 104/104/118	3345 104/104/117	3369 104/104/116	3390 103/103/115	3410 103/105/116	3434 103/108/119	3533 103/113/123	18386
-15	3383 104/104/118	3404 104/104/117	3429 104/104/116	3451 103/103/115	3473 103/105/116	3498 103/108/119	3584 103/113/123	18386
-10	3441 104/104/118	3464 104/104/117	3490 104/104/116	3513 103/103/115	3536 103/105/116	3562 103/108/119	3634 102/113/123	18386
-5	3499 104/104/118	3523 104/104/117	3550 104/104/116	3575 103/103/115	3599 103/105/116	3626 103/108/119	3686 102/113/123	18386
0	3556 104/104/117	3581 104/104/117	3610 103/103/116	3635 103/103/115	3660 103/105/116	3688 103/108/119	3740 102/113/123	18386
5	3611 104/104/117	3637 104/104/116	3668 103/103/116	3694 103/103/115	3720 103/105/116	3750 103/108/119	3796 102/113/123	18386
V_{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 1 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3221 105/105/118	3237 104/104/117	3258 104/104/116	3276 104/104/116	3294 104/105/116	3315 103/108/119	3403 103/113/123	18386
-35	3277 105/105/118	3294 104/104/117	3316 104/104/116	3335 104/104/115	3353 103/105/116	3375 103/108/119	3451 103/113/123	18386
-30	3337 105/105/118	3356 104/104/117	3379 104/104/116	3399 104/104/115	3419 103/105/116	3442 103/108/119	3501 103/113/123	18386
-25	3398 105/105/118	3418 104/104/117	3442 104/104/116	3464 104/104/115	3485 103/105/116	3509 103/108/119	3552 103/113/123	18386
-20	3456 105/105/118	3477 104/104/117	3503 104/104/116	3526 104/104/115	3548 103/105/116	3573 103/108/119	3614 103/113/123	18386
-15	3515 104/104/118	3538 104/104/117	3565 104/104/116	3589 103/103/115	3612 103/105/116	3639 103/108/119	3681 103/113/123	18386
-10	3575 104/104/118	3599 104/104/117	3627 104/104/116	3652 103/103/115	3677 103/105/116	3704 103/108/119	3748 103/113/123	18386
-5	3633 104/104/118	3659 104/104/117	3689 104/104/116	3715 103/103/115	3741 103/105/116	3770 103/108/119	3815 102/113/123	18386
0	3694 104/104/118	3721 104/104/117	3752 103/103/116	3779 103/103/115	3806 103/105/116	3836 103/108/119	3883 102/113/123	18386
5	3749 104/104/117	3778 104/104/116	3810 103/103/116	3839 103/103/115	3867 103/105/116	3898 103/108/119	3947 102/113/123	18386
V _{FS}	117	120	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3334 105/105/118	3351 104/104/117	3374 104/104/116	3394 104/104/115	3413 103/105/116	3435 103/108/119	3488 103/113/123	18386
-35	3393 105/105/118	3412 104/104/117	3436 104/104/116	3457 104/104/115	3477 103/105/116	3500 103/108/119	3540 103/113/123	18386
-30	3455 105/105/118	3476 104/104/117	3501 104/104/116	3523 104/104/115	3545 103/105/116	3569 103/108/119	3609 103/113/123	18386
-25	3518 104/104/118	3540 104/104/117	3566 104/104/116	3590 104/104/115	3612 103/105/116	3638 103/108/119	3680 103/113/123	18386
-20	3580 104/104/118	3603 104/104/117	3631 104/104/116	3656 103/103/115	3680 103/105/116	3707 103/108/119	3750 103/113/123	18386
-15	3642 104/104/118	3667 104/104/117	3696 104/104/116	3722 103/103/115	3747 103/105/116	3775 103/108/119	3820 103/113/123	18386
-10	3704 104/104/118	3730 104/104/117	3761 104/104/116	3788 103/103/115	3814 103/105/116	3844 103/108/119	3890 102/113/123	18386
-5	3766 104/104/118	3794 104/104/117	3826 103/103/116	3854 103/103/115	3882 103/105/116	3913 103/108/119	3961 102/113/123	18386
0	3827 104/104/117	3856 104/104/117	3890 103/103/116	3920 103/103/115	3948 103/105/116	3980 103/108/119	4030 102/113/123	18386
5	3885 104/104/117	3916 104/104/116	3951 103/103/115	3982 103/103/115	4011 103/105/116	4045 102/108/119	4096 102/113/123	18386
V_{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3449 105/105/118	3469 104/104/117	3494 104/104/116	3515 104/104/115	3536 103/105/117	3561 103/108/119	3600 103/113/123	18386
-35	3508 104/104/118	3529 104/104/117	3555 104/104/116	3578 103/103/115	3600 103/105/117	3625 103/108/119	3666 103/113/123	18386
-30	3571 104/104/118	3594 104/104/117	3621 104/104/116	3645 103/103/115	3668 103/105/117	3695 103/108/119	3737 103/113/123	18386
-25	3636 104/104/118	3660 104/104/117	3689 104/104/116	3714 103/103/115	3739 103/105/117	3767 103/108/119	3811 102/113/123	18386
-20	3701 104/104/118	3726 104/104/117	3757 104/104/116	3783 103/103/115	3809 103/105/117	3838 103/108/119	3885 102/113/123	18386
-15	3765 104/104/118	3792 104/104/117	3824 103/103/116	3852 103/103/115	3879 103/105/117	3910 103/108/119	3958 102/113/123	18386
-10	3830 104/104/117	3859 104/104/116	3892 103/103/116	3921 103/103/115	3950 103/105/117	3981 103/108/119	4031 102/113/123	18386
-5	3893 104/104/117	3923 104/104/116	3958 103/103/115	3989 103/103/115	4018 103/105/117	4051 102/108/119	4103 102/113/123	18386
0	3956 104/104/117	3987 103/103/116	4023 103/103/115	4055 103/103/114	4086 103/105/117	4120 102/108/119	4174 102/113/123	18386
5	3994 103/103/116	4027 103/103/116	4065 103/103/115	4099 102/102/114	4132 102/106/117	4168 102/109/119	4224 102/113/123	18386
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3515 104/104/117	3538 103/103/116	3565 103/103/115	3589 103/103/114	3612 102/106/117	3638 102/109/119	3767 102/113/123	18386
-30	3643 103/103/117	3668 103/103/116	3698 103/103/115	3724 102/102/114	3749 102/106/117	3778 102/109/119	3883 102/113/123	18386
-25	3709 103/103/116	3736 103/103/116	3767 103/103/115	3795 102/102/114	3821 102/106/117	3852 102/109/119	3942 102/113/123	18386
-20	3775 103/103/116	3803 103/103/115	3836 103/103/115	3865 102/102/114	3893 102/106/117	3925 102/109/119	4003 102/113/123	18386
-15	3840 103/103/116	3870 103/103/115	3904 102/102/114	3935 102/102/114	3964 102/106/117	3997 102/109/119	4064 101/113/123	18386
-10	3906 103/103/116	3937 103/103/115	3972 102/102/114	4004 102/102/114	4035 102/106/117	4069 102/109/119	4126 101/113/123	18386
-5	3970 103/103/116	4003 103/103/115	4040 102/102/114	4073 102/102/114	4105 102/106/117	4141 102/109/119	4196 101/113/123	18386
0	4035 103/103/116	4069 102/102/115	4108 102/102/114	4142 102/102/114	4175 102/106/117	4212 101/109/119	4269 101/113/123	18386
5	3960 101/101/113	3997 100/100/112	4039 100/100/111	4076 100/103/114	4113 100/106/117	4153 99/109/119	4586 102/114/123	18386
V_{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 1 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3577 102/102/115	3601 102/102/114	3631 102/102/114	3657 102/103/114	3682 101/106/117	3711 101/109/119	3970 101/113/123	18386
-30	3710 102/102/115	3737 102/102/114	3770 102/102/113	3799 101/103/114	3827 101/106/117	3858 101/109/119	4092 101/113/123	18386
-25	3778 102/102/115	3808 102/102/114	3841 102/102/113	3872 101/103/114	3901 101/106/117	3934 101/109/119	4155 101/113/123	18386
-20	3846 102/102/115	3876 102/102/114	3912 101/101/113	3943 101/103/114	3974 101/106/117	4008 101/109/119	4219 101/113/123	18386
-15	3912 102/102/115	3944 102/102/114	3981 101/101/113	4014 101/103/114	4046 101/106/117	4082 101/109/119	4284 101/113/123	18386
-10	3979 102/102/115	4013 102/102/114	4052 101/101/113	4086 101/103/114	4119 101/106/117	4156 101/109/119	4351 101/113/123	18386
-5	4047 102/102/115	4082 101/101/114	4122 101/101/113	4158 101/103/114	4192 101/106/117	4230 100/109/119	4418 101/113/123	18386
0	4070 101/101/114	4109 101/101/113	4151 100/100/112	4190 100/103/114	4228 100/106/117	4269 100/109/119	4609 101/114/123	18386
5	3937 98/98/110	3979 98/98/109	4025 97/101/111	4067 97/104/114	4109 97/107/117	4401 98/110/119	5030 103/114/123	18386
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.



SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3640 101/101/114	3668 101/101/113	3701 101/101/112	3730 100/103/114	3759 100/106/117	3791 100/109/119	4204 102/113/123	18386
-30	3772 101/101/114	3803 101/101/113	3838 100/100/112	3870 100/103/114	3901 100/106/117	3935 100/109/119	4334 102/113/123	18386
-25	3841 101/101/114	3873 101/101/113	3910 100/100/112	3944 100/103/114	3976 100/106/117	4012 100/109/119	4401 102/114/123	18386
-20	3910 101/101/113	3944 101/101/113	3983 100/100/112	4017 100/103/114	4051 100/106/117	4088 100/109/119	4469 101/114/123	18386
-15	3980 101/101/113	4016 100/100/112	4056 100/100/112	4092 100/103/114	4127 100/106/117	4166 99/109/119	4539 101/114/123	18386
-10	4049 101/101/113	4086 100/100/112	4127 100/100/111	4165 100/103/114	4202 100/106/117	4241 99/109/119	4609 101/114/123	18386
-5	4117 101/101/113	4156 100/100/112	4199 100/100/111	4239 100/103/114	4277 99/106/117	4318 99/109/119	4683 101/114/123	18386
0	4051 98/98/110	4094 98/98/109	4142 98/100/111	4185 98/104/114	4228 97/107/117	4426 98/110/119	5060 103/114/123	18386
5	3918 95/95/106	3965 95/98/108	4016 95/101/111	4063 95/104/114	4399 96/108/117	4843 100/111/119	5583 105/115/123	18386
V_{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 1 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3697 100/100/112	3728 100/100/111	3764 99/100/111	3796 99/103/114	3828 99/106/117	3889 99/110/119	4455 102/114/123	18386
-30	3836 100/100/112	3870 99/99/111	3909 99/100/111	3944 99/103/114	3978 99/106/117	4016 98/110/119	4593 102/114/123	18386
-25	3907 100/100/112	3943 99/99/111	3983 99/100/111	4020 99/103/114	4055 99/106/117	4095 98/110/119	4666 102/114/123	18386
-20	3977 100/100/112	4014 99/99/111	4056 99/100/111	4094 99/103/114	4131 98/106/117	4172 98/110/119	4739 102/114/123	18386
-15	4049 99/99/112	4088 99/99/111	4131 99/100/111	4171 99/103/114	4209 98/107/117	4251 98/110/119	4812 102/114/123	18386
-10	4127 99/99/111	4168 99/99/111	4214 99/100/111	4255 99/103/114	4295 98/107/117	4339 98/110/119	4911 102/114/123	18386
-5	4152 98/98/110	4196 98/98/109	4245 98/101/111	4290 98/104/114	4333 97/107/117	4523 98/110/119	5169 103/114/123	18386
0	4042 96/96/107	4090 95/98/108	4141 95/101/111	4190 95/104/114	4465 96/108/117	4917 100/111/119	5671 105/115/123	18386
5	3929 93/95/105	3980 93/99/108	4035 92/102/111	4405 94/105/114	4871 98/108/117	5397 101/111/119	6229 106/115/123	18386
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3715 98/98/110	3750 98/98/109	3790 97/101/111	3826 97/104/114	3861 97/107/117	4212 98/110/119	4809 104/114/123	18386
-30	3850 98/98/109	3888 97/97/109	3931 97/101/111	3969 97/104/114	4008 97/107/117	4342 98/110/119	4959 103/114/123	18386
-25	3922 98/98/109	3961 97/97/108	4005 97/101/111	4046 97/104/114	4086 97/107/117	4410 98/110/119	5036 103/114/123	18386
-20	4001 98/98/109	4041 97/97/108	4087 97/101/111	4129 97/104/114	4169 97/107/117	4479 98/110/119	5115 103/114/123	18386
-15	4082 97/97/109	4124 97/97/108	4170 97/101/111	4213 97/104/114	4255 97/107/117	4548 98/110/119	5194 103/114/123	18386
-10	4160 97/97/109	4206 97/97/108	4256 97/101/111	4303 96/104/114	4348 96/107/117	4731 98/110/119	5417 104/115/123	18386
-5	4082 95/95/105	4131 95/98/108	4185 94/102/111	4235 94/105/114	4651 96/108/117	5115 100/111/119	5918 105/115/123	18386
0	3974 92/95/105	4027 92/99/108	4102 92/102/111	4577 95/105/114	5056 98/108/117	5626 102/111/119	6481 107/116/123	18386
5	3859 89/96/105	4001 89/100/108	4502 93/103/111	5020 96/106/114	5611 100/109/117	6212 104/112/119	- -	17982
V_{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 1 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3729 96/96/107	3768 96/98/108	3811 95/101/111	3851 95/104/114	4135 96/108/117	4554 100/111/119	5235 105/115/123	18386
-30	3869 96/96/107	3911 95/98/108	3957 95/101/111	4000 95/104/114	4263 96/108/117	4696 99/111/119	5400 105/115/123	18386
-25	3952 96/96/107	3995 95/98/108	4042 95/101/111	4086 95/104/114	4328 96/108/117	4768 99/111/119	5483 104/115/123	18386
-20	4036 96/96/107	4080 95/98/108	4128 95/101/111	4173 95/104/114	4395 96/108/117	4842 99/111/119	5571 104/115/123	18386
-15	4116 95/95/106	4162 95/98/108	4212 95/101/111	4259 95/104/114	4497 96/108/117	4952 99/111/119	5705 104/115/123	18386
-10	4148 94/95/105	4200 94/98/108	4256 94/102/111	4332 94/105/114	4798 97/108/117	5285 100/111/119	6117 105/115/123	18386
-5	4021 91/96/105	4076 91/99/108	4267 91/102/111	4756 95/106/114	5279 99/109/117	5862 102/112/119	6739 107/116/123	18386
0	3926 89/96/105	4136 89/100/108	4649 93/103/111	5202 97/106/114	5808 100/109/117	6424 104/112/119	-	17905
5	4071 87/97/105	4603 91/100/108	5210 95/104/111	5850 99/107/114	6493 103/110/117	-	-	16747
V _{FS}	117	121	124	127	131	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4163 104/104/117	4182 103/103/116	4206 103/103/115	4224 103/103/115	4241 103/105/116	4262 102/108/119	4294 102/113/123	18386
-35	4220 104/104/117	4242 103/103/116	4267 103/103/115	4289 103/103/114	4308 103/105/116	4330 102/108/119	4366 102/113/123	18386
-30	4278 104/104/117	4303 103/103/116	4331 103/103/115	4355 103/103/114	4376 103/105/116	4401 102/108/119	4439 102/113/123	18386
-25	4338 104/104/117	4365 103/103/116	4396 103/103/115	4422 103/103/114	4446 103/105/116	4473 102/108/119	4514 102/113/123	18386
-20	4397 104/104/117	4427 103/103/116	4460 103/103/115	4489 103/103/114	4515 102/105/116	4544 102/108/119	4589 102/113/123	18386
-15	4456 104/104/117	4488 103/103/116	4524 103/103/115	4555 103/103/114	4583 102/105/116	4614 102/108/119	4662 102/113/123	18386
-10	4515 103/103/117	4550 103/103/116	4588 103/103/115	4621 103/103/114	4652 102/105/116	4685 102/108/119	4736 102/113/123	18386
-5	4575 103/103/117	4612 103/103/116	4653 103/103/115	4688 103/103/114	4721 102/105/116	4756 102/108/119	4810 102/113/123	18386
0	4634 103/103/117	4674 103/103/116	4716 103/103/115	4754 102/102/114	4789 102/105/116	4826 102/108/119	4883 102/113/123	18386
5	4691 103/103/117	4733 103/103/116	4778 103/103/115	4818 102/102/114	4855 102/105/116	4894 102/108/119	4954 102/113/123	18386
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4305 104/104/118	4328 104/104/117	4355 103/103/116	4378 103/103/115	4398 103/105/116	4422 103/108/119	4459 102/113/123	18386
-35	4361 104/104/117	4387 104/104/116	4416 103/103/116	4441 103/103/115	4463 103/105/116	4489 102/108/119	4529 102/113/123	18386
-30	4423 104/104/117	4451 103/103/116	4483 103/103/116	4510 103/103/115	4535 103/105/116	4563 102/108/119	4606 102/113/123	18386
-25	4484 104/104/117	4515 103/103/116	4549 103/103/115	4579 103/103/115	4606 103/105/116	4636 102/108/119	4683 102/113/123	18386
-20	4541 104/104/117	4575 103/103/116	4612 103/103/115	4644 103/103/115	4674 103/105/116	4706 102/108/119	4756 102/113/123	18386
-15	4602 104/104/117	4639 103/103/116	4678 103/103/115	4713 103/103/114	4745 102/105/116	4779 102/108/119	4832 102/113/123	18386
-10	4663 103/103/117	4702 103/103/116	4744 103/103/115	4781 103/103/114	4815 102/105/116	4852 102/108/119	4907 102/113/123	18386
-5	4722 103/103/117	4764 103/103/116	4808 103/103/115	4848 103/103/114	4884 102/105/116	4923 102/108/119	4982 102/113/123	18386
0	4785 103/103/117	4829 103/103/116	4876 103/103/115	4918 102/102/114	4956 102/105/116	4997 102/108/119	5059 102/113/123	18386
5	4842 103/103/117	4889 103/103/116	4938 103/103/115	4982 102/102/114	5022 102/105/116	5066 102/108/119	5130 102/113/123	18386
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4431 104/104/117	4457 104/104/116	4488 103/103/116	4514 103/103/115	4537 103/105/116	4564 103/108/119	4605 102/113/123	18386
-35	4491 104/104/117	4520 103/103/116	4553 103/103/115	4581 103/103/115	4607 103/105/116	4636 102/108/119	4681 102/113/123	18386
-30	4554 104/104/117	4586 103/103/116	4622 103/103/115	4653 103/103/115	4681 103/105/116	4712 102/108/119	4760 102/113/123	18386
-25	4616 104/104/117	4651 103/103/116	4690 103/103/115	4723 103/103/114	4754 103/105/116	4787 102/108/119	4838 102/113/123	18386
-20	4679 104/104/117	4716 103/103/116	4758 103/103/115	4793 103/103/114	4827 102/105/116	4862 102/108/119	4917 102/113/123	18386
-15	4742 103/103/117	4782 103/103/116	4826 103/103/115	4864 103/103/114	4900 102/105/116	4938 102/108/119	4996 102/113/123	18386
-10	4805 103/103/117	4848 103/103/116	4894 103/103/115	4935 103/103/114	4973 102/105/116	5013 102/108/119	5074 102/113/123	18386
-5	4869 103/103/117	4914 103/103/116	4963 103/103/115	5006 102/102/114	5046 102/105/116	5089 102/108/119	5153 102/113/123	18386
0	4932 103/103/117	4980 103/103/116	5031 103/103/115	5076 102/102/114	5119 102/105/116	5164 102/108/119	5231 102/113/123	18386
5	4991 103/103/116	5042 103/103/116	5096 102/102/115	5143 102/102/114	5188 102/105/116	5235 102/108/119	5305 101/113/123	18386
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4558 104/104/117	4589 103/103/116	4624 103/103/115	4653 103/103/115	4681 103/105/117	4711 102/108/119	4757 102/113/123	18386
-35	4617 104/104/117	4650 103/103/116	4688 103/103/115	4719 103/103/114	4749 102/105/117	4781 102/108/119	4831 102/113/123	18386
-30	4681 103/103/117	4717 103/103/116	4757 103/103/115	4791 103/103/114	4823 102/105/117	4858 102/108/119	4911 102/113/123	18386
-25	4747 103/103/117	4786 103/103/116	4829 103/103/115	4865 103/103/114	4900 102/105/117	4937 102/108/119	4993 102/113/123	18386
-20	4812 103/103/117	4854 103/103/116	4899 103/103/115	4938 102/102/114	4975 102/105/117	5015 102/108/119	5075 102/113/123	18386
-15	4877 103/103/117	4922 103/103/116	4970 103/103/115	5012 102/102/114	5051 102/105/117	5093 102/108/119	5156 102/113/123	18386
-10	4944 103/103/117	4991 103/103/116	5042 103/103/115	5086 102/102/114	5128 102/105/117	5172 102/108/119	5238 102/113/123	18386
-5	5008 103/103/116	5058 103/103/116	5111 102/102/115	5158 102/102/114	5202 102/105/117	5249 102/108/119	5318 101/113/123	18386
0	5073 103/103/116	5125 103/103/115	5180 102/102/115	5230 102/102/114	5276 102/105/117	5325 102/108/119	5398 101/113/123	18386
5	5110 102/102/116	5166 102/102/115	5224 102/102/114	5276 102/102/114	5325 101/106/117	5376 101/109/119	5453 101/113/123	18386
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4631 103/103/116	4665 102/102/115	4703 102/102/114	4736 102/102/114	4766 102/106/117	4800 101/109/119	4851 101/113/123	18386
-30	4760 103/103/116	4800 102/102/115	4843 102/102/114	4881 102/102/114	4916 101/106/117	4954 101/109/119	5012 101/113/123	18386
-25	4827 102/102/116	4870 102/102/115	4916 102/102/114	4956 102/102/114	4994 101/106/117	5034 101/109/119	5096 101/113/123	18386
-20	4893 102/102/116	4939 102/102/115	4988 102/102/114	5031 102/102/114	5071 101/106/117	5114 101/109/119	5179 101/113/123	18386
-15	4960 102/102/115	5009 102/102/115	5060 102/102/114	5106 101/102/114	5149 101/106/117	5194 101/109/119	5262 101/113/123	18386
-10	5027 102/102/115	5078 102/102/114	5132 102/102/114	5180 101/102/114	5225 101/106/117	5273 101/109/119	5344 101/113/123	18386
-5	5093 102/102/115	5146 102/102/114	5203 101/101/113	5254 101/102/114	5301 101/106/117	5351 101/109/119	5426 101/113/123	18386
0	5160 102/102/115	5217 102/102/114	5276 101/101/113	5329 101/102/114	5379 101/106/117	5431 101/109/119	5509 100/113/123	18386
5	5077 100/100/112	5135 99/99/111	5197 99/100/111	5253 99/103/114	5305 99/106/117	5361 99/109/119	5781 102/114/123	18386
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4699 102/102/115	4737 101/101/114	4778 101/101/113	4814 101/103/114	4848 101/106/117	4884 100/109/119	4940 100/113/123	18386
-30	4834 101/101/114	4877 101/101/113	4924 101/101/113	4965 101/103/114	5004 100/106/117	5046 100/109/119	5108 100/113/123	18386
-25	4903 101/101/114	4950 101/101/113	5000 101/101/112	5044 101/103/114	5085 100/106/117	5129 100/109/119	5195 100/113/123	18386
-20	4972 101/101/114	5021 101/101/113	5074 101/101/112	5120 100/103/114	5164 100/106/117	5211 100/109/119	5281 100/113/123	18386
-15	5040 101/101/114	5092 101/101/113	5147 101/101/112	5196 100/103/114	5243 100/106/117	5292 100/109/119	5365 100/113/123	18386
-10	5109 101/101/114	5164 101/101/113	5221 100/100/112	5273 100/103/114	5322 100/106/117	5374 100/109/119	5453 100/113/123	18386
-5	5179 101/101/114	5236 101/101/113	5296 100/100/112	5351 100/103/114	5402 100/106/117	5456 100/109/119	5571 100/113/123	18386
0	5202 100/100/113	5263 100/100/112	5327 100/100/111	5385 99/103/114	5440 99/106/117	5497 99/109/119	5851 101/114/123	18386
5	5055 97/97/109	5118 97/97/108	5184 97/101/111	5244 96/104/114	5301 96/107/117	5541 98/110/119	6464 105/114/123	18386
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4767 100/100/113	4809 100/100/112	4855 100/100/111	4895 100/103/114	4933 99/106/117	4974 99/109/119	5135 100/113/123	18386
-30	4902 100/100/113	4950 100/100/112	5001 100/100/111	5046 99/103/114	5089 99/106/117	5135 99/109/119	5373 100/113/123	18386
-25	4973 100/100/113	5023 100/100/112	5077 100/100/111	5125 99/103/114	5171 99/106/117	5219 99/109/119	5491 101/114/123	18386
-20	5043 100/100/113	5096 100/100/112	5153 99/100/111	5204 99/103/114	5252 99/106/117	5303 99/109/119	5611 101/114/123	18386
-15	5115 100/100/112	5172 100/100/112	5231 99/100/111	5285 99/103/114	5335 99/106/117	5388 99/109/119	5737 101/114/123	18386
-10	5187 100/100/112	5246 99/99/111	5308 99/100/111	5364 99/103/114	5417 99/106/117	5472 99/109/119	5863 101/114/123	18386
-5	5257 100/100/112	5319 99/99/111	5384 99/100/111	5443 99/103/114	5499 99/106/117	5557 99/109/119	5991 101/114/123	18386
0	5184 98/98/109	5250 97/97/108	5318 97/100/111	5381 97/104/114	5441 97/107/117	5612 98/110/119	6546 104/114/123	18386
5	5037 95/95/106	5104 94/98/108	5175 94/101/111	5240 94/104/114	5558 96/108/117	6214 101/111/119	7257 108/115/123	18386
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4831 99/99/111	4877 99/99/110	4926 99/100/111	4970 98/103/114	5012 98/106/117	5057 98/110/119	5525 102/114/123	18386
-30	4974 99/99/111	5025 99/99/110	5081 98/100/111	5130 98/103/114	5177 98/106/117	5227 98/110/119	5780 102/114/123	18386
-25	5046 99/99/111	5101 99/99/110	5159 98/100/111	5211 98/103/114	5261 98/106/117	5314 98/110/119	5911 102/114/123	18386
-20	5117 99/99/111	5175 98/98/110	5236 98/100/111	5291 98/103/114	5344 98/106/117	5399 98/110/119	6042 102/114/123	18386
-15	5193 99/99/111	5253 98/98/110	5317 98/100/111	5375 98/103/114	5430 98/107/117	5487 98/110/119	6176 103/114/123	18386
-10	5280 98/98/111	5344 98/98/110	5411 98/100/111	5471 98/103/114	5529 98/107/117	5590 97/110/119	6351 103/114/123	18386
-5	5313 97/97/109	5381 97/97/108	5451 97/101/111	5515 97/104/114	5577 97/107/117	5795 98/110/119	6759 105/114/123	18386
0	5201 95/95/106	5269 95/98/108	5341 94/101/111	5407 94/104/114	5709 96/108/117	6384 101/111/119	7492 108/115/123	18386
5	5081 92/95/105	5150 92/99/108	5224 92/102/111	5618 94/105/114	6322 99/108/117	7078 104/111/119	8513 111/115/123	18386
V _{FS}	117	121	124	127	130	134	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4850 97/97/109	4900 97/97/108	4954 97/101/111	5003 96/104/114	5049 96/107/117	5208 97/110/119	6068 104/114/123	18386
-30	4989 97/97/109	5045 97/97/108	5104 96/101/111	5158 96/104/114	5209 96/107/117	5452 98/110/119	6360 105/114/123	18386
-25	5063 97/97/109	5122 97/97/108	5184 96/101/111	5241 96/104/114	5295 96/107/117	5575 98/110/119	6504 105/114/123	18386
-20	5151 97/97/108	5212 96/97/108	5277 96/101/111	5336 96/104/114	5393 96/107/117	5703 98/110/119	6656 105/114/123	18386
-15	5245 97/97/108	5309 96/97/108	5375 96/101/111	5436 96/104/114	5495 96/107/117	5837 98/110/119	6811 105/114/123	18386
-10	5337 96/96/108	5405 96/97/108	5477 96/101/111	5542 96/104/114	5605 95/107/117	6130 99/110/119	7149 106/115/123	18386
-5	5258 94/95/105	5328 94/98/108	5402 94/102/111	5471 93/105/114	6016 97/108/117	6729 102/111/119	8044 109/115/123	18386
0	5145 91/95/105	5217 91/99/108	5292 91/102/111	5913 95/105/114	6655 101/108/117	7449 105/111/119	9000 112/116/123	18386
5	5018 89/96/105	5092 88/100/108	5798 94/103/111	6588 99/106/114	7435 104/109/117	8582 109/112/119	- -	17982
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4864 95/95/106	4918 95/98/108	4976 95/101/111	5029 95/104/114	5129 95/108/117	5728 100/111/119	6695 107/115/123	18386
-30	5012 95/95/106	5072 95/98/108	5135 94/101/111	5194 94/104/114	5371 95/108/117	6004 100/111/119	7019 107/115/123	18386
-25	5110 95/95/106	5172 95/98/108	5237 94/101/111	5297 94/104/114	5499 95/108/117	6150 100/111/119	7186 107/115/123	18386
-20	5208 95/95/106	5272 95/98/108	5340 94/101/111	5402 94/104/114	5630 96/108/117	6299 100/111/119	7354 107/115/123	18386
-15	5302 95/95/106	5369 94/98/108	5440 94/101/111	5504 94/104/114	5810 96/108/117	6503 101/111/119	7718 107/115/123	18386
-10	5344 94/95/105	5417 93/98/108	5494 93/102/111	5579 93/105/114	6278 98/108/117	7023 103/111/119	8432 109/115/123	18386
-5	5209 91/96/105	5283 90/99/108	5483 91/102/111	6221 97/106/114	7011 102/109/117	8024 106/112/119	9511 112/116/123	18386
0	5106 88/96/105	5294 89/100/108	6065 95/103/111	6901 100/106/114	7944 105/109/117	9050 109/112/119	-	17905
5	5192 87/97/105	6002 93/100/108	6904 99/104/111	8040 104/107/114	9261 109/110/117	-	-	16747
V _{FS}	117	121	124	127	131	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3903 103/103/116	3938 103/103/115	3976 102/102/115	4010 102/102/114	4042 102/105/116	4076 102/108/119	4350 104/113/123	18386
-35	3954 103/103/116	3992 103/103/115	4033 102/102/115	4069 102/102/114	4104 102/105/116	4141 102/108/119	4459 105/113/123	18386
-30	4007 103/103/116	4048 103/103/115	4092 102/102/114	4131 102/102/114	4168 102/105/116	4207 102/108/119	4568 105/113/123	18386
-25	4061 103/103/116	4105 102/102/115	4151 102/102/114	4193 102/102/114	4233 102/105/116	4275 102/108/119	4681 106/113/123	18386
-20	4115 103/103/116	4162 102/102/115	4211 102/102/114	4255 102/102/114	4297 102/105/116	4342 102/108/119	4797 106/113/123	18386
-15	4168 103/103/116	4218 102/102/115	4269 102/102/114	4316 102/102/114	4361 102/105/116	4408 101/108/119	4918 106/113/123	18386
-10	4222 103/103/116	4274 102/102/115	4328 102/102/114	4378 102/102/114	4425 102/105/116	4475 101/108/119	5042 107/113/123	18386
-5	4276 102/102/116	4331 102/102/115	4388 102/102/114	4440 102/102/114	4490 101/105/116	4541 101/108/119	5173 107/113/123	18386
0	4330 102/102/116	4387 102/102/115	4446 102/102/114	4501 102/102/114	4553 101/105/116	4607 101/108/119	5314 108/113/123	18386
5	4381 102/102/116	4441 102/102/115	4503 102/102/114	4560 101/102/114	4615 101/105/116	4671 101/108/119	5453 108/113/123	18386
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4030 103/103/117	4070 103/103/116	4112 103/103/115	4151 102/102/114	4187 102/105/116	4226 102/108/119	4508 104/113/123	18386
-35	4081 103/103/116	4123 103/103/116	4168 102/102/115	4209 102/102/114	4248 102/105/116	4288 102/108/119	4629 105/113/123	18386
-30	4137 103/103/116	4182 103/103/116	4230 102/102/115	4273 102/102/114	4315 102/105/116	4358 102/108/119	4747 105/113/123	18386
-25	4192 103/103/116	4240 103/103/115	4291 102/102/115	4337 102/102/114	4381 102/105/116	4427 102/108/119	4868 106/113/123	18386
-20	4244 103/103/116	4295 102/102/115	4349 102/102/115	4398 102/102/114	4444 102/105/116	4493 102/108/119	4996 106/113/123	18386
-15	4299 103/103/116	4353 102/102/115	4410 102/102/114	4461 102/102/114	4510 102/105/116	4562 101/108/119	5130 107/113/123	18386
-10	4354 103/103/116	4411 102/102/115	4470 102/102/114	4524 102/102/114	4576 102/105/116	4630 101/108/119	5267 107/113/123	18386
-5	4408 102/102/116	4468 102/102/115	4530 102/102/114	4587 102/102/114	4641 101/105/116	4697 101/108/119	5408 108/113/123	18386
0	4465 102/102/116	4527 102/102/115	4592 102/102/114	4651 102/102/114	4708 101/105/116	4767 101/108/119	5563 108/113/123	18386
5	4517 102/102/116	4582 102/102/115	4649 102/102/114	4711 101/102/114	4770 101/105/116	4832 101/108/119	5710 109/113/123	18386
V _{FS}	117	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4142 103/103/117	4186 103/103/116	4233 102/102/115	4275 102/102/114	4315 102/105/116	4358 102/108/119	4721 105/113/123	18386
-35	4197 103/103/116	4243 103/103/115	4293 102/102/115	4338 102/102/114	4381 102/105/116	4425 102/108/119	4855 106/113/123	18386
-30	4254 103/103/116	4304 103/103/115	4356 102/102/115	4404 102/102/114	4449 102/105/116	4497 102/108/119	4980 106/113/123	18386
-25	4310 103/103/116	4363 102/102/115	4419 102/102/114	4469 102/102/114	4517 102/105/116	4568 102/108/119	5112 107/113/123	18386
-20	4367 103/103/116	4423 102/102/115	4481 102/102/114	4534 102/102/114	4586 102/105/116	4638 101/108/119	5251 107/113/123	18386
-15	4424 103/103/116	4483 102/102/115	4544 102/102/114	4600 102/102/114	4654 102/105/116	4710 101/108/119	5396 108/113/123	18386
-10	4481 102/102/116	4543 102/102/115	4607 102/102/114	4666 102/102/114	4722 101/105/116	4780 101/108/119	5543 108/113/123	18386
-5	4539 102/102/116	4603 102/102/115	4670 102/102/114	4732 102/102/114	4791 101/105/116	4852 101/108/119	5702 109/113/123	18386
0	4596 102/102/116	4663 102/102/115	4732 102/102/114	4797 101/102/114	4859 101/105/116	4922 101/108/119	5850 109/113/123	18386
5	4650 102/102/115	4720 102/102/115	4792 102/102/114	4859 101/102/114	4923 101/105/116	5007 101/108/119	6018 110/113/123	18386
V _{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4258 103/103/116	4306 103/103/115	4358 102/102/115	4404 102/102/114	4449 102/105/117	4496 102/108/119	4969 106/113/123	18386
-35	4310 103/103/116	4362 102/102/115	4416 102/102/114	4465 102/102/114	4512 102/105/117	4562 101/108/119	5114 107/113/123	18386
-30	4368 103/103/116	4423 102/102/115	4480 102/102/114	4532 102/102/114	4582 102/105/117	4634 101/108/119	5259 107/113/123	18386
-25	4428 102/102/116	4485 102/102/115	4545 102/102/114	4601 102/102/114	4653 102/105/117	4708 101/108/119	5405 108/113/123	18386
-20	4486 102/102/116	4547 102/102/115	4610 102/102/114	4668 102/102/114	4724 101/105/117	4781 101/108/119	5555 108/113/123	18386
-15	4546 102/102/116	4609 102/102/115	4675 102/102/114	4736 102/102/114	4795 101/105/117	4855 101/108/119	5710 109/113/123	18386
-10	4606 102/102/116	4672 102/102/115	4741 102/102/114	4805 101/102/114	4866 101/105/117	4929 101/108/119	5869 109/113/123	18386
-5	4664 102/102/115	4734 102/102/115	4805 102/102/114	4872 101/102/114	4936 101/105/117	5021 101/108/119	6066 110/113/123	18386
0	4722 102/102/115	4795 102/102/114	4869 101/101/114	4939 101/102/114	5005 101/105/117	5159 102/108/119	6237 111/113/123	18386
5	4760 101/101/115	4835 101/101/114	4913 101/101/113	4985 101/102/114	5055 101/106/117	5401 103/109/119	6516 112/113/123	18386
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4325 102/102/115	4377 102/102/114	4433 101/101/113	4483 101/102/114	4532 101/106/117	4582 101/109/119	5364 108/113/123	18386
-30	4442 102/102/115	4500 101/101/114	4562 101/101/113	4618 101/102/114	4672 101/106/117	4746 101/109/119	5694 110/113/123	18386
-25	4502 101/101/115	4564 101/101/114	4628 101/101/113	4688 101/102/114	4745 101/106/117	4874 101/109/119	5861 110/113/123	18386
-20	4562 101/101/115	4627 101/101/114	4695 101/101/113	4757 101/102/114	4817 100/106/117	5007 102/109/119	6023 111/113/123	18386
-15	4623 101/101/114	4691 101/101/114	4761 101/101/113	4827 101/102/114	4889 100/106/117	5151 102/109/119	6211 111/113/123	18386
-10	4683 101/101/114	4754 101/101/113	4828 101/101/113	4896 100/102/114	4961 100/106/117	5291 103/109/119	6386 112/113/123	18386
-5	4743 101/101/114	4817 101/101/113	4893 101/101/112	4964 100/102/114	5033 100/106/117	5438 103/109/119	6548 112/113/123	18386
0	4804 101/101/114	4881 101/101/113	4960 100/100/112	5034 100/102/114	5105 100/106/117	5590 104/109/119	6714 113/113/123	18386
5	4743 99/99/111	4823 99/99/110	4905 98/100/111	4982 98/103/114	5444 102/106/117	6245 108/109/119	7216 114/114/123	18386
V _{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4389 101/101/114	4446 100/100/113	4505 100/100/112	4559 100/103/114	4612 100/106/117	4874 102/109/119	5859 111/113/123	18386
-30	4511 100/100/113	4574 100/100/113	4639 100/100/112	4700 100/103/114	4758 100/106/117	5157 103/109/119	6229 112/113/123	18386
-25	4574 100/100/113	4640 100/100/112	4709 100/100/112	4773 100/103/114	4834 100/106/117	5304 104/109/119	6397 113/113/123	18386
-20	4637 100/100/113	4706 100/100/112	4778 100/100/112	4845 100/103/114	4909 99/106/117	5457 104/109/119	6510 113/113/123	18386
-15	4698 100/100/113	4770 100/100/112	4845 100/100/111	4915 100/103/114	4982 99/106/117	5618 105/109/119	6681 113/113/123	18386
-10	4761 100/100/113	4836 100/100/112	4914 100/100/111	4987 99/103/114	5057 99/106/117	5790 105/109/119	6795 113/113/123	18386
-5	4824 100/100/113	4902 100/100/112	4983 99/99/111	5059 99/103/114	5188 100/106/117	5942 106/109/119	7064 113/113/123	18386
0	4853 99/99/112	4936 99/99/111	5021 99/100/111	5101 99/103/114	5532 102/106/117	6394 109/109/119	7381 114/114/123	18386
5	4740 96/96/108	4825 96/97/108	4911 96/101/111	5479 100/104/114	6265 106/107/117	6832 110/110/119	7895 114/114/123	18386
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4456 100/100/112	4518 99/99/111	4582 99/100/111	4642 99/103/114	4699 99/106/117	5331 105/109/119	6400 113/113/123	18386
-30	4579 99/99/112	4646 99/99/111	4717 99/100/111	4782 99/103/114	4952 100/106/117	5669 106/109/119	6684 113/113/123	18386
-25	4643 99/99/112	4714 99/99/111	4787 99/100/111	4856 98/103/114	5095 100/106/117	5834 107/109/119	6810 113/114/123	18386
-20	4706 99/99/112	4781 99/99/111	4858 99/100/111	4929 98/103/114	5240 101/106/117	6062 107/109/119	6933 113/114/123	18386
-15	4772 99/99/112	4850 99/99/111	4930 98/100/111	5005 98/103/114	5404 101/106/117	6201 108/109/119	7137 113/114/123	18386
-10	4837 99/99/111	4918 99/99/111	5001 98/100/111	5079 98/103/114	5551 102/106/117	6366 108/109/119	7396 114/114/123	18386
-5	4902 99/99/111	4986 98/98/110	5072 98/100/111	5153 98/103/114	5713 103/106/117	6530 109/109/119	7574 114/114/123	18386
0	4855 97/97/108	4943 96/97/108	5033 96/100/111	5564 100/104/114	6361 106/107/117	6939 110/110/119	8053 114/114/123	18386
5	4742 94/94/105	4832 94/98/108	5463 99/101/111	6225 104/104/114	6814 108/108/117	7574 111/111/119	- -	18343
V_{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4519 98/98/110	4585 98/98/110	4654 98/100/111	4718 98/103/114	5135 101/106/117	5937 108/110/119	6788 114/114/123	18386
-30	4649 98/98/110	4722 98/98/109	4797 97/100/111	4868 97/103/114	5453 102/106/117	6270 109/110/119	7068 114/114/123	18386
-25	4715 98/98/110	4792 98/98/109	4870 97/100/111	4944 97/103/114	5618 103/106/117	6380 109/110/119	7312 114/114/123	18386
-20	4780 98/98/110	4860 98/98/109	4942 97/100/111	5019 97/103/114	5807 104/106/117	6550 109/110/119	7529 114/114/123	18386
-15	4849 98/98/110	4932 97/97/109	5017 97/100/111	5154 98/103/114	5973 104/107/117	6665 109/110/119	7712 114/114/123	18386
-10	4930 98/98/110	5016 97/97/109	5105 97/100/111	5347 99/103/114	6220 105/107/117	6857 110/110/119	7928 114/114/123	18386
-5	4970 97/97/108	5061 96/97/108	5155 96/101/111	5820 101/104/114	6586 107/107/117	7303 110/110/119	8319 114/114/123	18386
0	4887 94/94/105	4980 94/98/108	5675 99/101/111	6417 104/104/114	7006 108/108/117	7818 111/111/119	-	18303
5	4798 91/95/105	5518 97/99/108	6234 102/102/111	6846 105/105/114	7631 108/108/117	-	-	16893
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 1 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4547 96/96/108	4618 96/97/108	4692 96/101/111	5034 99/104/114	5867 105/107/117	6469 110/110/119	7272 114/114/123	18386
-30	4674 96/96/108	4752 96/97/108	4832 96/101/111	5356 100/104/114	6148 106/107/117	6747 110/110/119	7753 114/114/123	18386
-25	4742 96/96/108	4823 96/97/108	4907 95/101/111	5518 101/104/114	6314 107/107/117	6881 110/110/119	7944 114/114/123	18386
-20	4821 96/96/107	4905 96/97/108	4992 95/101/111	5721 101/104/114	6434 107/107/117	7022 110/110/119	8147 114/114/123	18386
-15	4905 96/96/107	4992 95/97/108	5082 95/101/111	5896 102/104/114	6622 107/107/117	7334 110/110/119	8358 114/114/123	18386
-10	4997 95/95/107	5090 95/97/108	5369 97/101/111	6279 104/104/114	6857 107/107/117	7644 110/110/119	8714 115/115/123	18386
-5	4945 93/95/105	5144 94/98/108	6078 101/102/111	6657 105/105/114	7420 108/108/117	8169 111/111/119	-	17736
0	4924 91/95/105	5832 98/99/108	6478 102/102/111	7116 105/105/114	7981 108/108/117	-	-	16430
5	5631 96/96/105	6265 100/100/108	6931 103/103/111	-	-	-	-	15277
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 1 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4571 94/94/105	4646 94/98/108	4898 96/101/111	5733 103/104/114	6392 108/108/117	6936 111/111/119	7995 115/115/123	18386
-30	4707 94/94/105	4789 94/98/108	5214 97/101/111	6081 104/104/114	6640 107/108/117	7300 111/111/119	8411 115/115/123	18386
-25	4794 94/94/105	4880 94/98/108	5388 98/101/111	6204 104/104/114	6803 108/108/117	7551 111/111/119	8636 115/115/123	18386
-20	4882 94/94/105	4971 94/98/108	5588 98/101/111	6389 104/104/114	6956 108/108/117	7758 111/111/119	8864 115/115/123	18386
-15	4970 94/94/105	5062 94/98/108	5855 100/101/111	6522 104/104/114	7265 108/108/117	7999 111/111/119	-	18379
-10	5028 93/95/105	5457 96/98/108	6258 101/102/111	6898 105/105/114	7714 108/108/117	8488 111/111/119	-	17363
-5	5318 93/96/105	6105 99/99/108	6727 102/102/111	7510 106/106/114	-	-	-	15977
0	5853 96/96/105	6482 100/100/108	7187 103/103/111	-	-	-	-	14958
5	6294 97/97/105	-	-	-	-	-	-	13842
V _{FS}	117	121	124	127	131	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2573 110/110/122	2463 106/106/118	2390 104/104/115	2360 101/101/112	2359 99/99/110	2531 102/102/112	2801 107/107/116	18386
-35	2614 110/110/122	2502 106/106/118	2424 104/104/115	2395 101/101/112	2395 99/99/110	2570 102/102/112	2846 107/107/116	18386
-30	2656 109/109/122	2543 106/106/118	2460 104/104/115	2430 101/101/112	2431 98/99/110	2610 102/102/112	2891 107/107/116	18386
-25	2698 109/109/122	2584 106/106/118	2495 104/104/115	2465 101/101/112	2467 98/99/110	2649 102/102/112	2935 107/107/116	18386
-20	2740 109/109/122	2625 106/106/118	2530 103/103/115	2500 101/101/112	2504 98/99/110	2689 102/102/112	2980 107/107/116	18386
-15	2781 109/109/122	2665 106/106/118	2565 103/103/115	2535 101/101/112	2541 98/99/110	2729 102/102/112	3026 107/107/116	18386
-10	2822 109/109/122	2705 106/106/118	2603 103/103/115	2569 101/101/112	2578 98/99/110	2769 102/102/112	3071 107/107/116	18386
-5	2863 109/109/121	2745 106/106/118	2642 103/103/115	2603 101/101/112	2615 98/99/110	2810 102/102/112	3117 107/107/116	18386
0	2903 109/109/121	2784 106/106/118	2680 103/103/115	2636 101/101/112	2653 98/99/110	2851 102/102/112	3163 107/107/116	18386
5	2942 109/109/121	2822 106/106/118	2717 103/103/115	2669 100/100/112	2691 98/99/110	2893 102/102/112	3212 107/107/116	18386
V_{FS}	121	120	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2668 110/110/122	2552 107/107/119	2461 104/104/116	2430 101/101/112	2420 99/99/110	2596 102/102/112	2874 107/107/116	18386
-35	2707 110/110/122	2590 107/107/119	2494 104/104/115	2463 101/101/112	2457 99/99/110	2637 102/102/112	2920 107/107/116	18386
-30	2750 110/110/122	2632 107/107/119	2530 104/104/115	2499 101/101/112	2495 99/99/110	2678 102/102/112	2966 107/107/116	18386
-25	2794 110/110/122	2674 107/107/119	2570 104/104/115	2535 101/101/112	2532 98/99/110	2719 102/102/112	3013 107/107/116	18386
-20	2835 110/110/122	2714 106/106/119	2610 104/104/115	2570 101/101/112	2571 98/99/110	2761 102/102/112	3060 107/107/116	18386
-15	2877 109/109/122	2755 106/106/118	2649 103/103/115	2605 101/101/112	2609 98/99/110	2803 101/102/112	3107 107/107/116	18386
-10	2918 109/109/122	2796 106/106/118	2689 103/103/115	2640 101/101/112	2648 98/99/110	2845 101/102/112	3155 107/107/116	18386
-5	2960 109/109/122	2836 106/106/118	2729 103/103/115	2674 101/101/112	2687 98/99/110	2887 101/102/112	3203 107/107/116	18386
0	3001 109/109/122	2877 106/106/118	2768 103/103/115	2708 101/101/112	2725 98/99/110	2929 101/102/112	3251 107/107/116	18386
5	3040 109/109/121	2914 106/106/118	2805 103/103/115	2741 100/100/112	2766 98/99/110	2974 101/102/112	3302 107/107/116	18386
V _{FS}	121	120	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2746 110/110/122	2627 107/107/119	2524 104/104/116	2493 101/101/112	2486 99/99/110	2669 102/102/112	2956 107/107/116	18386
-35	2787 110/110/122	2668 107/107/119	2564 104/104/115	2527 101/101/112	2525 99/99/110	2711 102/102/112	3003 107/107/116	18386
-30	2832 110/110/122	2711 106/106/119	2606 104/104/115	2564 101/101/112	2564 98/99/110	2753 102/102/112	3051 107/107/116	18386
-25	2876 110/110/122	2754 106/106/119	2648 104/104/115	2601 101/101/112	2603 98/99/110	2796 101/102/112	3100 107/107/116	18386
-20	2920 109/109/122	2797 106/106/119	2690 103/103/115	2637 101/101/112	2642 98/99/110	2839 101/102/112	3148 107/107/116	18386
-15	2963 109/109/122	2839 106/106/118	2731 103/103/115	2673 101/101/112	2682 98/99/110	2882 101/102/112	3197 107/107/116	18386
-10	3006 109/109/122	2881 106/106/118	2772 103/103/115	2709 101/101/112	2722 98/99/110	2926 101/102/112	3246 107/107/116	18386
-5	3049 109/109/122	2923 106/106/118	2813 103/103/115	2745 101/101/112	2762 98/99/110	2969 101/102/112	3296 107/107/116	18386
0	3091 109/109/121	2964 106/106/118	2853 103/103/115	2780 100/100/112	2802 98/99/110	3014 101/102/112	3346 107/107/116	18386
5	3131 109/109/121	3003 106/106/118	2891 103/103/115	2814 100/100/111	2844 98/99/110	3060 101/102/112	3399 107/107/116	18386
V_{FS}	121	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – DRY RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2827 110/110/122	2705 107/107/119	2600 104/104/115	2559 101/101/112	2560 98/99/110	2749 102/102/112	3047 107/107/116	18386
-35	2867 109/109/122	2745 106/106/118	2639 103/103/115	2593 101/101/112	2600 98/99/110	2793 102/102/112	3097 107/107/116	18386
-30	2911 109/109/122	2788 106/106/118	2681 103/103/115	2629 101/101/112	2640 98/99/110	2837 101/102/112	3146 107/107/116	18386
-25	2957 109/109/122	2833 106/106/118	2724 103/103/115	2667 101/101/112	2681 98/99/110	2881 101/102/112	3196 107/107/116	18386
-20	3002 109/109/122	2877 106/106/118	2768 103/103/115	2705 101/101/112	2722 98/99/110	2926 101/102/112	3247 107/107/116	18386
-15	3047 109/109/122	2920 106/106/118	2810 103/103/115	2742 101/101/112	2763 98/99/110	2970 101/102/112	3297 107/107/116	18386
-10	3092 109/109/121	2964 106/106/118	2853 103/103/115	2778 100/100/112	2804 98/99/110	3015 101/102/112	3349 107/107/116	18386
-5	3135 109/109/121	3006 106/106/118	2894 103/103/115	2815 100/100/111	2846 98/99/110	3061 101/102/112	3401 107/107/116	18386
0	3178 109/109/121	3048 106/106/118	2935 103/103/115	2850 100/100/111	2888 98/99/110	3108 101/102/112	3454 106/107/116	18386
5	3205 108/108/120	3076 105/105/117	2964 102/102/114	2882 100/100/111	2949 98/100/110	3175 101/103/112	3532 107/107/116	18386
V _{FS}	120	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2872 109/109/121	2751 106/106/117	2646 103/103/114	2608 100/100/111	2657 98/100/110	2856 102/103/112	3170 107/107/116	18386
-30	2960 108/108/121	2837 105/105/117	2730 102/102/114	2681 100/100/111	2741 98/100/110	2948 102/103/112	3275 107/107/116	18386
-25	3006 108/108/120	2882 105/105/117	2774 102/102/114	2720 100/100/111	2784 98/100/110	2995 102/103/112	3327 107/107/116	18386
-20	3052 108/108/120	2927 105/105/117	2818 102/102/114	2758 100/100/111	2827 98/100/110	3042 102/103/112	3381 107/107/116	18386
-15	3097 108/108/120	2971 105/105/117	2861 102/102/114	2795 100/100/110	2870 98/100/110	3089 102/103/112	3434 107/107/116	18386
-10	3142 108/108/120	3014 105/105/117	2903 102/102/114	2832 99/99/110	2914 98/100/110	3137 102/103/112	3488 107/107/116	18386
-5	3186 108/108/120	3058 105/105/117	2946 102/102/113	2869 99/99/110	2958 98/100/110	3185 102/103/112	3543 107/107/116	18386
0	3230 108/108/120	3100 105/105/117	2988 102/102/113	2906 99/99/110	3001 98/100/110	3233 102/103/112	3597 107/107/116	18386
5	3191 105/105/117	3068 102/102/114	2960 100/100/110	2917 97/97/107	3152 99/100/110	3400 103/103/112	3837 108/108/116	18386
V_{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – DRY RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2912 107/107/119	2793 104/104/116	2689 102/102/113	2656 99/99/110	2766 99/100/110	2976 102/103/112	3309 107/107/116	18386
-30	3005 107/107/119	2883 104/104/116	2777 101/101/113	2734 99/99/109	2854 99/100/110	3073 102/103/112	3418 107/107/116	18386
-25	3053 107/107/119	2929 104/104/116	2822 101/101/112	2773 99/99/109	2899 99/100/110	3122 102/103/112	3474 107/107/116	18386
-20	3099 107/107/119	2975 104/104/116	2867 101/101/112	2812 99/99/109	2944 98/100/110	3171 102/103/112	3530 107/107/116	18386
-15	3145 107/107/119	3020 104/104/116	2910 101/101/112	2850 99/99/109	2989 98/100/110	3221 102/103/112	3586 107/107/116	18386
-10	3191 107/107/119	3064 104/104/115	2954 101/101/112	2888 98/98/109	3035 98/100/110	3271 102/103/112	3643 107/107/116	18386
-5	3237 107/107/119	3109 104/104/115	2998 101/101/112	2926 98/98/109	3081 98/100/110	3322 102/103/112	3701 107/107/116	18386
0	3263 106/106/118	3136 103/103/114	3026 100/100/111	2966 98/98/108	3181 99/100/110	3432 103/103/112	3856 108/108/116	18386
5	3190 103/103/114	3071 100/100/110	2995 97/97/107	3124 97/98/107	3383 101/101/110	3671 104/104/112	- -	18231
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	2961 106/106/118	2842 103/103/114	2748 100/100/111	2715 98/98/108	2896 99/100/110	3120 103/103/112	3512 108/108/116	18386
-30	3051 106/106/118	2930 103/103/114	2825 100/100/111	2791 98/98/108	2989 99/100/110	3222 103/103/112	3623 108/108/116	18386
-25	3099 106/106/117	2977 103/103/114	2870 100/100/111	2831 98/98/108	3036 99/100/110	3273 103/103/112	3680 108/108/116	18386
-20	3147 106/106/117	3024 103/103/114	2916 100/100/111	2871 97/97/108	3084 99/100/110	3326 103/103/112	3738 108/108/116	18386
-15	3196 106/106/117	3071 103/103/114	2962 100/100/111	2910 97/97/108	3132 99/100/110	3379 103/103/112	3798 108/108/116	18386
-10	3242 106/106/117	3116 103/103/114	3006 100/100/111	2949 97/97/107	3181 99/100/110	3432 103/103/112	3858 108/108/116	18386
-5	3289 105/105/117	3162 102/102/114	3051 100/100/110	2989 97/97/107	3231 99/100/110	3486 103/103/112	3921 108/108/116	18386
0	3266 103/103/114	3144 100/100/111	3049 98/98/108	3155 97/98/107	3416 100/101/110	3692 104/104/112	-	18273
5	3195 100/100/110	3081 97/97/107	3087 95/95/105	3357 99/99/107	3666 102/102/110	4059 105/105/112	-	17287
V_{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3005 105/105/116	2887 102/102/113	2804 99/99/110	2812 97/97/107	3038 100/101/110	3277 104/104/112	3731 108/108/116	18386
-30	3101 105/105/116	2980 102/102/113	2884 99/99/109	2901 96/97/107	3136 100/101/110	3384 104/104/112	3849 108/108/116	18386
-25	3150 104/104/116	3028 102/102/113	2925 99/99/109	2947 96/97/107	3186 100/101/110	3439 104/104/112	3911 108/108/116	18386
-20	3198 104/104/116	3075 101/101/112	2968 99/99/109	2993 96/97/107	3237 100/101/110	3494 104/104/112	3974 108/108/116	18386
-15	3247 104/104/116	3123 101/101/112	3015 99/99/109	3039 96/97/107	3287 100/101/110	3550 103/104/112	4036 108/108/116	18386
-10	3302 104/104/115	3177 101/101/112	3067 99/99/109	3096 96/98/107	3350 100/101/110	3619 104/104/112	4122 108/108/116	18386
-5	3331 103/103/114	3207 100/100/111	3101 98/98/107	3217 97/98/107	3485 101/101/110	3775 104/104/112	-	18080
0	3275 100/100/111	3157 97/97/107	3138 95/96/105	3414 98/99/107	3724 102/102/110	-	-	17150
5	3217 97/97/107	3139 95/95/104	3335 96/96/105	3669 100/100/107	-	-	-	16251
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – DRY RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3025 103/103/114	2910 100/100/110	2852 97/97/107	2977 97/98/107	3222 101/101/110	3509 104/104/112	- -	18080
-30	3118 102/102/113	3001 100/100/110	2932 97/97/107	3073 97/98/107	3328 101/101/110	3620 104/104/112	- -	18034
-25	3168 102/102/113	3050 100/100/110	2974 97/97/107	3122 97/98/107	3381 101/101/110	3678 104/104/112	- -	18034
-20	3221 102/102/113	3101 99/99/110	3016 97/97/107	3171 97/98/107	3436 101/101/110	3736 104/104/112	- -	18033
-15	3273 102/102/113	3152 99/99/110	3058 97/97/107	3221 97/98/107	3490 101/101/110	3796 104/104/112	- -	18031
-10	3340 102/102/113	3218 99/99/109	3127 96/96/106	3327 97/98/107	3608 101/102/110	3957 105/105/112	- -	17686
-5	3306 99/99/109	3189 97/97/106	3236 95/96/105	3524 99/99/107	3885 102/102/110	- -	- -	16830
0	3253 97/97/106	3181 94/94/103	3439 97/97/105	3820 100/100/107	- -	- -	- -	15957
5	3214 94/94/102	3346 94/94/102	3751 97/97/105	- -	- -	- -	- -	15029
V_{FS}	117	121	124	127	130	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – DRY RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3044 101/101/111	2934 98/98/108	2907 95/95/105	3159 98/99/107	3435 102/102/110	3807 105/105/112	-	17508
-30	3140 100/100/111	3026 98/98/108	3000 95/95/105	3262 98/99/107	3543 102/102/110	3928 105/105/112	-	17468
-25	3195 100/100/111	3079 98/98/107	3047 95/95/105	3313 98/99/107	3598 102/102/110	3990 105/105/112	-	17468
-20	3250 100/100/111	3132 97/97/107	3095 95/95/105	3367 98/99/107	3656 102/102/110	4053 105/105/112	-	17466
-15	3306 100/100/110	3188 97/97/107	3159 95/95/105	3438 98/99/107	3744 102/102/110	4150 105/105/112	-	17369
-10	3354 99/99/109	3237 96/96/105	3322 95/96/105	3619 99/99/107	4014 102/102/110	-	-	16758
-5	3291 96/96/105	3239 93/93/102	3548 97/97/105	3976 100/100/107	-	-	-	15805
0	3267 93/93/102	3439 94/94/102	3878 98/98/105	-	-	-	-	14971
5	3368 92/92/99	3836 95/95/102	-	-	-	-	-	14055
V _{FS}	117	121	124	127	131	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3757 110/110/122	3569 106/106/118	3400 104/104/115	3241 101/101/112	3093 99/99/110	2960 96/102/112	2907 95/107/116	18386
-35	3820 110/110/122	3631 106/106/118	3460 104/104/115	3299 101/101/112	3150 99/99/110	3015 96/102/112	2962 95/107/116	18386
-30	3886 109/109/122	3695 106/106/118	3523 104/104/115	3360 101/101/112	3208 98/99/110	3072 96/102/112	3019 95/107/116	18386
-25	3953 109/109/122	3760 106/106/118	3586 104/104/115	3421 101/101/112	3268 98/99/110	3130 96/102/112	3077 95/107/116	18386
-20	4019 109/109/122	3824 106/106/118	3648 103/103/115	3482 101/101/112	3327 98/99/110	3187 96/102/112	3134 95/107/116	18386
-15	4084 109/109/122	3888 106/106/118	3710 103/103/115	3541 101/101/112	3385 98/99/110	3243 96/102/112	3191 95/107/116	18386
-10	4151 109/109/122	3952 106/106/118	3772 103/103/115	3602 101/101/112	3443 98/99/110	3300 96/102/112	3247 95/107/116	18386
-5	4217 109/109/121	4016 106/106/118	3834 103/103/115	3661 101/101/112	3501 98/99/110	3355 96/102/112	3303 95/107/116	18386
0	4282 109/109/121	4079 106/106/118	3895 103/103/115	3720 101/101/112	3558 98/99/110	3410 96/102/112	3358 95/107/116	18386
5	4345 109/109/121	4140 106/106/118	3954 103/103/115	3777 100/100/112	3612 98/99/110	3464 96/102/112	3411 95/107/116	18386
V_{FS}	121	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	3912 110/110/122	3716 107/107/119	3539 104/104/116	3371 101/101/112	3216 99/99/110	3077 96/102/112	3023 95/107/116	18386
-35	3975 110/110/122	3776 107/107/119	3597 104/104/115	3428 101/101/112	3271 99/99/110	3130 96/102/112	3076 95/107/116	18386
-30	4045 110/110/122	3844 107/107/119	3662 104/104/115	3491 101/101/112	3332 99/99/110	3189 96/102/112	3134 95/107/116	18386
-25	4113 110/110/122	3910 107/107/119	3727 104/104/115	3553 101/101/112	3392 98/99/110	3247 96/102/112	3192 95/107/116	18386
-20	4179 110/110/122	3974 106/106/119	3789 104/104/115	3614 101/101/112	3451 98/99/110	3304 96/102/112	3250 95/107/116	18386
-15	4247 109/109/122	4039 106/106/118	3852 103/103/115	3675 101/101/112	3511 98/99/110	3362 96/102/112	3308 95/107/116	18386
-10	4314 109/109/122	4105 106/106/118	3916 103/103/115	3737 101/101/112	3570 98/99/110	3420 96/102/112	3366 95/107/116	18386
-5	4381 109/109/122	4170 106/106/118	3979 103/103/115	3798 101/101/112	3629 98/99/110	3477 96/102/112	3423 95/107/116	18386
0	4449 109/109/122	4236 106/106/118	4043 103/103/115	3860 101/101/112	3689 98/99/110	3535 96/102/112	3481 95/107/116	18386
5	4512 109/109/121	4297 106/106/118	4102 103/103/115	3917 100/100/112	3745 98/99/110	3589 96/102/112	3535 95/107/116	18386
V _{FS}	121	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4044 110/110/122	3842 107/107/119	3659 104/104/116	3487 101/101/112	3328 99/99/110	3184 96/102/112	3129 95/107/116	18386
-35	4111 110/110/122	3907 107/107/119	3723 104/104/115	3548 101/101/112	3387 99/99/110	3241 96/102/112	3186 95/107/116	18386
-30	4182 110/110/122	3976 106/106/119	3789 104/104/115	3613 101/101/112	3449 98/99/110	3302 96/102/112	3247 95/107/116	18386
-25	4253 110/110/122	4044 106/106/119	3856 104/104/115	3678 101/101/112	3512 98/99/110	3362 96/102/112	3307 95/107/116	18386
-20	4323 109/109/122	4113 106/106/119	3922 103/103/115	3742 101/101/112	3574 98/99/110	3423 96/102/112	3368 95/107/116	18386
-15	4394 109/109/122	4181 106/106/118	3989 103/103/115	3806 101/101/112	3636 98/99/110	3483 96/102/112	3428 95/107/116	18386
-10	4464 109/109/122	4249 106/106/118	4055 103/103/115	3870 101/101/112	3698 98/99/110	3543 96/102/112	3489 95/107/116	18386
-5	4535 109/109/122	4318 106/106/118	4121 103/103/115	3935 101/101/112	3761 98/99/110	3604 96/102/112	3550 95/107/116	18386
0	4604 109/109/121	4385 106/106/118	4187 103/103/115	3998 100/100/112	3823 98/99/110	3663 96/102/112	3609 95/107/116	18386
5	4670 109/109/121	4449 106/106/118	4249 103/103/115	4058 100/100/111	3880 98/99/110	3719 96/102/112	3665 94/107/116	18386
V_{FS}	121	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4178 110/110/122	3970 107/107/119	3783 104/104/115	3606 101/101/112	3442 98/99/110	3294 96/102/112	3239 95/107/116	18386
-35	4244 109/109/122	4034 106/106/118	3845 103/103/115	3666 101/101/112	3500 98/99/110	3351 96/102/112	3295 95/107/116	18386
-30	4316 109/109/122	4104 106/106/118	3913 103/103/115	3732 101/101/112	3564 98/99/110	3412 96/102/112	3356 95/107/116	18386
-25	4390 109/109/122	4176 106/106/118	3983 103/103/115	3799 101/101/112	3629 98/99/110	3475 96/102/112	3420 95/107/116	18386
-20	4463 109/109/122	4247 106/106/118	4052 103/103/115	3866 101/101/112	3694 98/99/110	3538 96/102/112	3483 95/107/116	18386
-15	4537 109/109/122	4319 106/106/118	4121 103/103/115	3933 101/101/112	3758 98/99/110	3601 96/102/112	3545 95/107/116	18386
-10	4611 109/109/121	4390 106/106/118	4190 103/103/115	4000 100/100/112	3823 98/99/110	3664 96/102/112	3608 95/107/116	18386
-5	4682 109/109/121	4460 106/106/118	4257 103/103/115	4065 100/100/111	3886 98/99/110	3724 96/102/112	3670 94/107/116	18386
0	4753 109/109/121	4528 106/106/118	4324 103/103/115	4130 100/100/111	3949 98/99/110	3785 95/102/112	3730 94/107/116	18386
5	4791 108/108/120	4567 105/105/117	4363 102/102/114	4170 100/100/111	3989 97/100/110	3825 95/103/112	3772 94/107/116	18386
V _{FS}	120	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4248 109/109/121	4039 106/106/117	3850 103/103/114	3672 100/100/111	3506 98/100/110	3357 95/103/112	3313 94/107/116	18386
-30	4392 108/108/121	4179 105/105/117	3986 102/102/114	3803 100/100/111	3634 97/100/110	3481 95/103/112	3440 94/107/116	18386
-25	4467 108/108/120	4252 105/105/117	4057 102/102/114	3872 100/100/111	3700 97/100/110	3545 95/103/112	3504 94/107/116	18386
-20	4542 108/108/120	4324 105/105/117	4127 102/102/114	3940 100/100/111	3766 97/100/110	3609 95/103/112	3568 94/107/116	18386
-15	4617 108/108/120	4396 105/105/117	4197 102/102/114	4008 100/100/110	3831 97/100/110	3672 95/103/112	3635 94/107/116	18386
-10	4690 108/108/120	4468 105/105/117	4266 102/102/114	4075 99/99/110	3897 97/100/110	3735 95/103/112	3701 94/107/116	18386
-5	4764 108/108/120	4539 105/105/117	4336 102/102/113	4142 99/99/110	3961 97/100/110	3798 95/103/112	3769 94/107/116	18386
0	4838 108/108/120	4610 105/105/117	4405 102/102/113	4209 99/99/110	4026 97/100/110	3861 95/103/112	3837 94/107/116	18386
5	4734 105/105/117	4514 102/102/114	4314 100/100/110	4125 97/97/107	3949 95/100/110	3790 93/103/112	4112 96/108/116	18386
V_{FS}	117	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4310 107/107/119	4100 104/104/116	3911 102/102/113	3732 99/99/110	3567 97/100/110	3417 94/103/112	3499 95/107/116	18386
-30	4461 107/107/119	4246 104/104/116	4053 101/101/113	3870 99/99/109	3699 96/100/110	3546 94/103/112	3633 95/107/116	18386
-25	4539 107/107/119	4322 104/104/116	4126 101/101/112	3941 99/99/109	3768 96/100/110	3612 94/103/112	3701 95/107/116	18386
-20	4615 107/107/119	4396 104/104/116	4198 101/101/112	4010 99/99/109	3835 96/100/110	3678 94/103/112	3770 95/107/116	18386
-15	4691 107/107/119	4469 104/104/116	4269 101/101/112	4079 99/99/109	3902 96/100/110	3742 94/103/112	3840 95/107/116	18386
-10	4767 107/107/119	4544 104/104/115	4341 101/101/112	4149 98/98/109	3970 96/100/110	3807 94/103/112	3911 94/107/116	18386
-5	4844 107/107/119	4618 104/104/115	4413 101/101/112	4218 98/98/109	4037 96/100/110	3873 94/103/112	3984 94/107/116	18386
0	4866 106/106/118	4640 103/103/114	4435 100/100/111	4241 98/98/108	4060 95/100/110	3896 93/103/112	4154 95/108/116	18386
5	4691 103/103/114	4475 100/100/110	4280 97/97/107	4096 95/98/107	3925 92/101/110	3934 92/104/112	- -	18231
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4376 106/106/118	4165 103/103/114	3974 100/100/111	3795 98/98/108	3628 95/100/110	3478 93/103/112	3714 96/108/116	18386
-30	4525 106/106/118	4309 103/103/114	4114 100/100/111	3930 98/98/108	3759 95/100/110	3605 93/103/112	3859 95/108/116	18386
-25	4604 106/106/117	4385 103/103/114	4188 100/100/111	4002 98/98/108	3828 95/100/110	3672 93/103/112	3932 95/108/116	18386
-20	4682 106/106/117	4461 103/103/114	4262 100/100/111	4073 97/97/108	3897 95/100/110	3739 93/103/112	4005 95/108/116	18386
-15	4763 106/106/117	4539 103/103/114	4337 100/100/111	4145 97/97/108	3966 95/100/110	3805 93/103/112	4081 95/108/116	18386
-10	4841 106/106/117	4615 103/103/114	4410 100/100/111	4216 97/97/107	4035 95/100/110	3872 93/103/112	4158 95/108/116	18386
-5	4919 105/105/117	4690 102/102/114	4484 100/100/110	4287 97/97/107	4104 95/100/110	3939 93/103/112	4236 95/108/116	18386
0	4828 103/103/114	4606 100/100/111	4405 98/98/108	4215 95/98/107	4038 93/101/110	3975 92/104/112	-	18273
5	4654 100/100/110	4442 97/97/107	4252 95/95/105	4073 92/99/107	3919 90/102/110	4325 94/105/112	-	17287
V_{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4435 105/105/116	4222 102/102/113	4030 99/99/110	3849 97/97/107	3682 94/101/110	3531 92/104/112	3951 96/108/116	18386
-30	4592 105/105/116	4374 102/102/113	4178 99/99/109	3993 96/97/107	3820 94/101/110	3665 92/104/112	4104 96/108/116	18386
-25	4673 104/104/116	4453 102/102/113	4254 99/99/109	4066 96/97/107	3891 94/101/110	3734 92/104/112	4183 96/108/116	18386
-20	4752 104/104/116	4529 101/101/112	4328 99/99/109	4138 96/97/107	3961 94/101/110	3802 92/104/112	4263 96/108/116	18386
-15	4835 104/104/116	4609 101/101/112	4405 99/99/109	4212 96/97/107	4033 94/101/110	3871 92/104/112	4343 96/108/116	18386
-10	4926 104/104/115	4697 101/101/112	4490 99/99/109	4294 96/98/107	4112 94/101/110	3948 92/104/112	4446 96/108/116	18386
-5	4949 103/103/114	4720 100/100/111	4514 98/98/107	4319 95/98/107	4137 93/101/110	4080 92/104/112	-	18080
0	4807 100/100/111	4585 97/97/107	4386 95/96/105	4199 92/99/107	4027 90/102/110	-	-	17150
5	4657 97/97/107	4445 95/95/104	4256 92/96/105	4079 90/100/107	-	-	-	16251
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – WET RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4441 103/103/114	4231 100/100/110	4042 97/97/107	3864 95/98/107	3699 92/101/110	3726 92/104/112	- -	18080
-30	4594 102/102/113	4379 100/100/110	4185 97/97/107	4003 94/98/107	3834 92/101/110	3871 92/104/112	- -	18034
-25	4675 102/102/113	4458 100/100/110	4262 97/97/107	4077 94/98/107	3906 92/101/110	3944 92/104/112	- -	18034
-20	4767 102/102/113	4546 99/99/110	4347 97/97/107	4158 94/98/107	3984 92/101/110	4020 92/104/112	- -	18033
-15	4863 102/102/113	4637 99/99/110	4434 97/97/107	4241 94/98/107	4063 92/101/110	4098 92/104/112	- -	18031
-10	4954 102/102/113	4725 99/99/109	4519 96/96/106	4324 94/98/107	4143 92/102/110	4270 92/105/112	- -	17686
-5	4850 99/99/109	4627 97/97/106	4427 94/96/105	4240 92/99/107	4175 91/102/110	- -	- -	16830
0	4708 97/97/106	4494 94/94/103	4303 91/97/105	4125 89/100/107	- -	- -	- -	15957
5	4553 94/94/102	4351 91/94/102	4172 89/97/105	- -	- -	- -	- -	15029
V _{FS}	117	121	124	127	130	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS FLAP 2 – WET RUNWAY ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4442 101/101/111	4235 98/98/108	4049 95/95/105	3875 93/99/107	3714 91/102/110	4031 94/105/112	-	17508
-30	4602 100/100/111	4389 98/98/108	4199 95/95/105	4019 93/99/107	3853 90/102/110	4190 93/105/112	-	17468
-25	4700 100/100/111	4483 98/98/107	4288 95/95/105	4104 93/99/107	3935 90/102/110	4272 93/105/112	-	17468
-20	4799 100/100/111	4577 97/97/107	4377 95/95/105	4190 92/99/107	4017 90/102/110	4357 93/105/112	-	17466
-15	4892 100/100/110	4666 97/97/107	4464 95/95/105	4273 92/99/107	4096 90/102/110	4472 93/105/112	-	17369
-10	4926 99/99/109	4700 96/96/105	4498 94/96/105	4308 91/99/107	4320 91/102/110	-	-	16758
-5	4760 96/96/105	4544 93/93/102	4352 91/97/105	4251 89/100/107	-	-	-	15805
0	4632 93/93/102	4426 91/94/102	4244 88/98/105	-	-	-	-	14971
5	4441 90/92/99	4250 87/95/102	-	-	-	-	-	14055
V _{FS}	117	121	124	127	131	134	-	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4816 109/109/121	4633 106/106/118	4462 103/103/115	4291 100/100/112	4126 98/99/110	3971 96/102/112	3903 95/107/116	18386
-35	4875 109/109/121	4694 106/106/118	4522 103/103/115	4351 100/100/111	4186 98/99/110	4031 96/102/112	3965 95/107/116	18386
-30	4937 109/109/121	4756 106/106/118	4584 103/103/115	4413 100/100/111	4248 98/99/110	4092 96/102/112	4028 95/107/116	18386
-25	5001 109/109/121	4819 106/106/118	4648 103/103/115	4477 100/100/111	4311 98/99/110	4155 96/102/112	4093 94/107/116	18386
-20	5064 109/109/121	4883 106/106/118	4711 103/103/115	4540 100/100/111	4374 98/99/110	4217 96/102/112	4157 94/107/116	18386
-15	5126 109/109/121	4945 106/106/118	4773 103/103/114	4602 100/100/111	4435 98/99/110	4279 96/102/112	4220 94/107/116	18386
-10	5190 108/108/121	5009 105/105/118	4837 103/103/114	4665 100/100/111	4498 98/99/110	4341 95/102/112	4283 94/107/116	18386
-5	5255 108/108/121	5073 105/105/117	4900 103/103/114	4728 100/100/111	4560 98/99/110	4402 95/102/112	4346 94/107/116	18386
0	5319 108/108/121	5137 105/105/117	4963 102/102/114	4790 100/100/111	4622 98/99/110	4463 95/102/112	4408 94/107/116	18386
5	5381 108/108/121	5198 105/105/117	5024 102/102/114	4850 100/100/111	4681 97/99/110	4521 95/102/112	4468 94/107/116	18386
V _{FS}	121	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4978 109/109/122	4790 106/106/118	4613 103/103/115	4437 101/101/112	4267 98/99/110	4108 96/102/112	4040 95/107/116	18386
-35	5038 109/109/121	4850 106/106/118	4672 103/103/115	4496 100/100/112	4325 98/99/110	4166 96/102/112	4100 95/107/116	18386
-30	5104 109/109/121	4916 106/106/118	4738 103/103/115	4562 100/100/112	4390 98/99/110	4229 96/102/112	4165 95/107/116	18386
-25	5169 109/109/121	4981 106/106/118	4803 103/103/115	4626 100/100/112	4454 98/99/110	4292 96/102/112	4230 94/107/116	18386
-20	5230 109/109/121	5042 106/106/118	4864 103/103/115	4687 100/100/111	4515 98/99/110	4354 96/102/112	4293 94/107/116	18386
-15	5296 109/109/121	5108 106/106/118	4929 103/103/115	4752 100/100/111	4580 98/99/110	4418 95/102/112	4359 94/107/116	18386
-10	5362 109/109/121	5173 106/106/118	4994 103/103/114	4816 100/100/111	4644 98/99/110	4481 95/102/112	4424 94/107/116	18386
-5	5426 109/109/121	5236 105/105/118	5057 103/103/114	4879 100/100/111	4706 98/99/110	4543 95/102/112	4487 94/107/116	18386
0	5494 108/108/121	5304 105/105/118	5125 103/103/114	4946 100/100/111	4771 97/99/110	4607 95/102/112	4553 94/107/116	18386
5	5556 108/108/121	5366 105/105/117	5185 102/102/114	5006 100/100/111	4831 97/99/110	4666 95/102/112	4613 94/107/116	18386
V _{FS}	121	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5117 109/109/121	4927 106/106/118	4747 103/103/115	4568 100/100/112	4395 98/99/110	4233 96/102/112	4167 95/107/116	18386
-35	5181 109/109/121	4990 106/106/118	4810 103/103/115	4632 100/100/112	4458 98/99/110	4295 96/102/112	4231 95/107/116	18386
-30	5248 109/109/121	5058 106/106/118	4878 103/103/115	4699 100/100/112	4525 98/99/110	4361 96/102/112	4299 94/107/116	18386
-25	5315 109/109/121	5124 106/106/118	4944 103/103/115	4765 100/100/111	4590 98/99/110	4427 96/102/112	4366 94/107/116	18386
-20	5383 109/109/121	5192 106/106/118	5011 103/103/115	4831 100/100/111	4656 98/99/110	4492 95/102/112	4433 94/107/116	18386
-15	5451 109/109/121	5260 106/106/118	5078 103/103/114	4898 100/100/111	4723 98/99/110	4558 95/102/112	4501 94/107/116	18386
-10	5519 108/108/121	5327 105/105/118	5146 103/103/114	4965 100/100/111	4789 98/99/110	4624 95/102/112	4568 94/107/116	18386
-5	5588 108/108/121	5395 105/105/118	5213 103/103/114	5032 100/100/111	4856 98/99/110	4690 95/102/112	4636 94/107/116	18386
0	5656 108/108/121	5463 105/105/117	5281 102/102/114	5099 100/100/111	4922 97/99/110	4755 95/102/112	4703 94/107/116	18386
5	5721 108/108/121	5527 105/105/117	5344 102/102/114	5162 100/100/111	4984 97/99/110	4817 95/102/112	4765 94/107/116	18386
V _{FS}	121	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5258 109/109/121	5065 106/106/118	4883 103/103/115	4702 100/100/112	4526 98/99/110	4361 96/102/112	4297 94/107/116	18386
-35	5320 109/109/121	5127 106/106/118	4944 103/103/115	4763 100/100/111	4587 98/99/110	4422 95/102/112	4360 94/107/116	18386
-30	5389 109/109/121	5196 106/106/118	5013 103/103/114	4831 100/100/111	4655 98/99/110	4489 95/102/112	4428 94/107/116	18386
-25	5460 109/109/121	5266 105/105/118	5083 103/103/114	4901 100/100/111	4724 98/99/110	4558 95/102/112	4498 94/107/116	18386
-20	5530 108/108/121	5336 105/105/118	5152 103/103/114	4970 100/100/111	4792 98/99/110	4625 95/102/112	4568 94/107/116	18386
-15	5601 108/108/121	5406 105/105/117	5222 102/102/114	5039 100/100/111	4861 97/99/110	4694 95/102/112	4638 94/107/116	18386
-10	5673 108/108/121	5478 105/105/117	5293 102/102/114	5110 100/100/111	4931 97/99/110	4762 95/102/112	4708 94/107/116	18386
-5	5743 108/108/121	5547 105/105/117	5362 102/102/114	5178 100/100/111	4998 97/99/110	4829 95/102/112	4777 94/107/116	18386
0	5812 108/108/120	5616 105/105/117	5430 102/102/114	5245 100/100/111	5065 97/99/110	4895 95/102/112	4845 94/107/116	18386
5	5850 107/107/120	5655 104/104/116	5470 102/102/113	5286 99/99/110	5107 97/100/110	4938 95/103/112	4890 93/107/116	18386
V _{FS}	120	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5333 108/108/120	5139 105/105/117	4956 102/102/113	4775 99/99/110	4599 97/100/110	4433 95/103/112	4372 94/107/116	18386
-30	5472 108/108/120	5277 105/105/116	5094 102/102/113	4912 99/99/110	4735 97/100/110	4568 95/103/112	4510 93/107/116	18386
-25	5544 107/107/120	5349 104/104/116	5166 102/102/113	4983 99/99/110	4805 97/100/110	4638 94/103/112	4582 93/107/116	18386
-20	5616 107/107/120	5421 104/104/116	5236 102/102/113	5053 99/99/110	4875 97/100/110	4707 94/103/112	4653 93/107/116	18386
-15	5688 107/107/120	5493 104/104/116	5308 101/101/113	5124 99/99/110	4945 97/100/110	4776 94/103/112	4723 93/107/116	18386
-10	5760 107/107/119	5564 104/104/116	5379 101/101/113	5194 99/99/110	5014 96/100/110	4845 94/103/112	4794 93/107/116	18386
-5	5832 107/107/119	5635 104/104/116	5449 101/101/113	5264 99/99/110	5084 96/100/110	4914 94/103/112	4864 93/107/116	18386
0	5906 107/107/119	5708 104/104/116	5521 101/101/113	5335 99/99/109	5154 96/100/110	4983 94/103/112	4935 93/107/116	18386
5	5806 105/105/116	5611 102/102/113	5427 99/99/110	5244 97/97/107	5067 94/100/110	4899 92/103/112	5226 95/108/116	18386
V_{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5403 107/107/119	5209 104/104/115	5026 101/101/112	4844 98/98/109	4668 96/100/110	4503 94/103/112	4444 93/107/116	18386
-30	5547 106/106/118	5353 103/103/115	5169 101/101/112	4987 98/98/109	4809 96/100/110	4643 94/103/112	4588 92/107/116	18386
-25	5623 106/106/118	5428 103/103/115	5244 101/101/112	5061 98/98/109	4883 96/100/110	4715 94/103/112	4662 92/107/116	18386
-20	5697 106/106/118	5502 103/103/115	5317 101/101/112	5134 98/98/109	4955 96/100/110	4787 93/103/112	4736 92/107/116	18386
-15	5771 106/106/118	5574 103/103/115	5389 100/100/112	5205 98/98/108	5026 96/100/110	4857 93/103/112	4820 92/107/116	18386
-10	5846 106/106/118	5649 103/103/115	5463 100/100/111	5278 98/98/108	5098 95/100/110	4929 93/103/112	4922 93/107/116	18386
-5	5922 106/106/118	5724 103/103/115	5537 100/100/111	5352 98/98/108	5171 95/100/110	5001 93/103/112	5026 93/107/116	18386
0	5947 105/105/117	5749 102/102/113	5561 99/99/110	5375 97/97/107	5195 95/100/110	5024 92/103/112	5285 94/108/116	18386
5	5775 102/102/113	5581 99/99/110	5398 96/96/106	5217 94/98/107	5042 92/101/110	4987 91/104/112	- -	18231
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5475 105/105/117	5279 102/102/114	5095 100/100/111	4913 97/97/107	4736 95/100/110	4570 93/103/112	4653 93/108/116	18386
-30	5620 105/105/117	5424 102/102/114	5239 99/99/110	5056 97/97/107	4877 95/100/110	4710 92/103/112	4863 93/108/116	18386
-25	5697 105/105/117	5500 102/102/113	5314 99/99/110	5131 97/97/107	4952 95/100/110	4784 92/103/112	4967 94/108/116	18386
-20	5772 105/105/117	5575 102/102/113	5389 99/99/110	5205 97/97/107	5025 94/100/110	4856 92/103/112	5074 94/108/116	18386
-15	5852 105/105/117	5654 102/102/113	5467 99/99/110	5281 97/97/107	5100 94/100/110	4930 92/103/112	5186 94/108/116	18386
-10	5929 105/105/116	5731 102/102/113	5543 99/99/110	5356 97/97/107	5174 94/100/110	5004 92/103/112	5296 94/108/116	18386
-5	6007 105/105/116	5807 102/102/113	5618 99/99/110	5431 96/97/107	5249 94/100/110	5077 92/103/112	5411 94/108/116	18386
0	5922 102/102/113	5724 100/100/110	5537 97/97/107	5352 94/98/107	5173 92/101/110	5046 90/104/112	-	18273
5	5750 99/99/110	5556 97/97/106	5374 94/95/105	5195 92/99/107	5022 89/102/110	5608 94/105/112	-	17287
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5542 104/104/115	5345 101/101/112	5159 98/98/109	4975 96/97/107	4797 94/101/110	4631 91/104/112	5010 95/108/116	18386
-30	5696 104/104/115	5498 101/101/112	5312 98/98/109	5127 96/97/107	4948 93/101/110	4779 91/104/112	5237 95/108/116	18386
-25	5774 104/104/115	5576 101/101/112	5389 98/98/109	5204 96/97/107	5024 93/101/110	4855 91/104/112	5352 95/108/116	18386
-20	5852 104/104/115	5653 101/101/112	5465 98/98/108	5279 96/97/107	5098 93/101/110	4929 91/104/112	5469 95/108/116	18386
-15	5935 103/103/115	5735 101/101/112	5546 98/98/108	5359 95/97/107	5177 93/101/110	5006 91/104/112	5588 96/108/116	18386
-10	6033 103/103/115	5831 100/100/111	5640 98/98/108	5451 95/98/107	5267 93/101/110	5095 91/104/112	5748 96/108/116	18386
-5	6070 102/102/113	5866 99/99/110	5675 97/97/107	5485 94/98/107	5301 92/101/110	5211 91/104/112	-	18080
0	5941 100/100/110	5738 97/97/106	5548 94/96/105	5361 92/99/107	5181 90/102/110	-	-	17150
5	5800 97/97/106	5601 94/94/103	5415 91/96/105	5232 89/100/107	-	-	-	16251
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.125 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5556 102/102/113	5360 99/99/110	5176 96/96/106	4993 94/98/107	4817 92/101/110	4696 90/104/112	-	18080
-30	5706 102/102/113	5509 99/99/109	5323 96/96/106	5140 94/98/107	4963 92/101/110	4909 91/104/112	-	18034
-25	5786 102/102/113	5589 99/99/109	5403 96/96/106	5219 94/98/107	5040 92/101/110	5015 91/104/112	-	18034
-20	5885 102/102/112	5685 99/99/109	5497 96/96/106	5310 94/98/107	5130 91/101/110	5129 91/104/112	-	18033
-15	5990 101/101/112	5787 99/99/109	5596 96/96/106	5408 94/98/107	5225 91/101/110	5248 91/104/112	-	18031
-10	6096 101/101/112	5890 98/98/108	5696 96/96/105	5505 93/98/107	5320 91/102/110	5519 92/105/112	-	17686
-5	6004 99/99/109	5800 96/96/105	5608 93/96/105	5419 91/99/107	5406 90/102/110	-	-	16830
0	5873 96/96/105	5670 93/93/102	5482 91/97/105	5301 89/100/107	-	-	-	15957
5	5723 93/93/101	5526 90/94/102	5343 88/97/105	-	-	-	-	15029
V _{FS}	117	121	124	127	130	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.125 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5563 100/100/110	5369 97/97/107	5186 95/95/105	5005 92/99/107	4831 90/102/110	5170 93/105/112	-	17508
-30	5725 100/100/110	5528 97/97/107	5344 94/95/105	5162 92/99/107	4986 90/102/110	5413 93/105/112	-	17468
-25	5835 100/100/110	5635 97/97/107	5447 94/95/105	5263 92/99/107	5084 90/102/110	5541 93/105/112	-	17468
-20	5945 99/99/110	5742 97/97/107	5551 94/95/105	5364 92/99/107	5183 90/102/110	5674 93/105/112	-	17466
-15	6051 99/99/110	5845 97/97/106	5652 94/95/105	5462 92/99/107	5278 89/102/110	5859 94/105/112	-	17369
-10	6102 98/98/108	5894 95/95/105	5699 93/96/105	5508 91/99/107	5642 91/102/110	-	-	16758
-5	5946 95/95/104	5740 92/93/102	5550 90/97/105	5583 90/100/107	-	-	-	15805
0	5825 93/93/101	5624 90/94/102	5437 88/98/105	-	-	-	-	14971
5	5633 89/92/99	5438 87/95/102	-	-	-	-	-	14055
V _{FS}	117	121	124	127	131	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: -1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4479 108/108/120	4332 105/105/117	4192 102/102/114	4052 100/100/111	3915 97/99/110	3785 95/102/112	3862 96/107/116	18386
-35	4533 108/108/120	4387 105/105/117	4248 102/102/114	4108 100/100/111	3971 97/99/110	3841 95/102/112	3953 96/107/116	18386
-30	4589 108/108/120	4444 105/105/117	4305 102/102/114	4165 100/100/111	4028 97/99/110	3899 95/102/112	4040 96/107/116	18386
-25	4647 108/108/120	4502 105/105/117	4363 102/102/114	4224 100/100/111	4087 97/99/110	3957 95/102/112	4130 97/107/116	18386
-20	4704 108/108/120	4559 105/105/117	4421 102/102/114	4282 100/100/111	4145 97/99/110	4015 95/102/112	4224 97/107/116	18386
-15	4760 108/108/120	4616 105/105/117	4478 102/102/114	4339 99/99/111	4202 97/99/110	4073 95/102/112	4320 97/107/116	18386
-10	4818 108/108/120	4674 105/105/117	4536 102/102/114	4397 99/99/111	4260 97/99/110	4130 95/102/112	4419 98/107/116	18386
-5	4876 107/107/120	4732 105/105/117	4594 102/102/114	4455 99/99/110	4318 97/99/110	4187 95/102/112	4523 98/107/116	18386
0	4934 107/107/120	4790 104/104/117	4652 102/102/113	4512 99/99/110	4375 97/99/110	4244 95/102/112	4631 98/107/116	18386
5	4990 107/107/120	4845 104/104/116	4707 102/102/113	4567 99/99/110	4430 97/99/110	4299 95/102/112	4743 99/107/116	18386
V _{FS}	120	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 0 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4624 108/108/121	4473 105/105/117	4330 102/102/114	4186 100/100/111	4045 98/99/110	3912 95/102/112	3992 96/107/116	18386
-35	4677 108/108/121	4527 105/105/117	4384 102/102/114	4240 100/100/111	4099 97/99/110	3965 95/102/112	4091 96/107/116	18386
-30	4737 108/108/121	4587 105/105/117	4444 102/102/114	4300 100/100/111	4159 97/99/110	4025 95/102/112	4186 96/107/116	18386
-25	4796 108/108/121	4646 105/105/117	4503 102/102/114	4359 100/100/111	4218 97/99/110	4084 95/102/112	4283 97/107/116	18386
-20	4852 108/108/120	4702 105/105/117	4559 102/102/114	4416 100/100/111	4275 97/99/110	4141 95/102/112	4386 97/107/116	18386
-15	4911 108/108/120	4761 105/105/117	4619 102/102/114	4475 99/99/111	4335 97/99/110	4201 95/102/112	4492 98/107/116	18386
-10	4970 108/108/120	4820 105/105/117	4678 102/102/114	4535 99/99/111	4394 97/99/110	4260 95/102/112	4601 98/107/116	18386
-5	5028 108/108/120	4878 105/105/117	4736 102/102/114	4593 99/99/110	4451 97/99/110	4317 95/102/112	4713 98/107/116	18386
0	5090 107/107/120	4940 104/104/117	4797 102/102/113	4653 99/99/110	4512 97/99/110	4377 95/102/112	4828 99/107/116	18386
5	5146 107/107/120	4996 104/104/117	4853 102/102/113	4709 99/99/110	4567 97/99/110	4433 94/102/112	4956 99/107/116	18386
V _{FS}	120	120	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 1000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4747 108/108/121	4595 105/105/117	4450 102/102/114	4304 100/100/111	4162 97/99/110	4027 95/102/112	4168 96/107/116	18386
-35	4804 108/108/121	4653 105/105/117	4508 102/102/114	4363 100/100/111	4220 97/99/110	4085 95/102/112	4273 97/107/116	18386
-30	4865 108/108/120	4714 105/105/117	4570 102/102/114	4424 100/100/111	4282 97/99/110	4147 95/102/112	4376 97/107/116	18386
-25	4926 108/108/120	4775 105/105/117	4630 102/102/114	4485 100/100/111	4343 97/99/110	4208 95/102/112	4482 97/107/116	18386
-20	4987 108/108/120	4835 105/105/117	4691 102/102/114	4546 99/99/111	4404 97/99/110	4269 95/102/112	4592 98/107/116	18386
-15	5048 108/108/120	4897 105/105/117	4753 102/102/114	4608 99/99/111	4466 97/99/110	4330 95/102/112	4706 98/107/116	18386
-10	5110 108/108/120	4958 105/105/117	4814 102/102/114	4670 99/99/110	4527 97/99/110	4391 95/102/112	4823 99/107/116	18386
-5	5172 107/107/120	5020 104/104/117	4876 102/102/114	4732 99/99/110	4589 97/99/110	4453 95/102/112	4946 99/107/116	18386
0	5234 107/107/120	5082 104/104/117	4938 102/102/113	4793 99/99/110	4650 97/99/110	4514 95/102/112	5071 100/107/116	18386
5	5293 107/107/120	5140 104/104/116	4996 101/101/113	4851 99/99/110	4708 97/99/110	4571 94/102/112	5209 100/107/116	18386
V _{FS}	120	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 2000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4873 108/108/120	4720 105/105/117	4573 102/102/114	4427 100/100/111	4283 97/99/110	4147 95/102/112	4375 97/107/116	18386
-35	4929 108/108/120	4776 105/105/117	4630 102/102/114	4483 99/99/111	4339 97/99/110	4203 95/102/112	4487 98/107/116	18386
-30	4991 108/108/120	4838 105/105/117	4692 102/102/114	4546 99/99/111	4402 97/99/110	4265 95/102/112	4602 98/107/116	18386
-25	5055 108/108/120	4902 105/105/117	4756 102/102/114	4610 99/99/110	4466 97/99/110	4329 95/102/112	4717 98/107/116	18386
-20	5118 108/108/120	4965 105/105/117	4820 102/102/114	4673 99/99/110	4529 97/99/110	4392 95/102/112	4836 99/107/116	18386
-15	5182 107/107/120	5029 104/104/117	4883 102/102/113	4737 99/99/110	4593 97/99/110	4456 95/102/112	4958 99/107/116	18386
-10	5248 107/107/120	5094 104/104/117	4948 102/102/113	4802 99/99/110	4657 97/99/110	4520 94/102/112	5088 100/107/116	18386
-5	5311 107/107/120	5157 104/104/116	5011 101/101/113	4864 99/99/110	4720 97/99/110	4582 94/102/112	5224 100/107/116	18386
0	5373 107/107/120	5220 104/104/116	5073 101/101/113	4926 99/99/110	4782 96/99/110	4644 94/102/112	5366 101/107/116	18386
5	5412 107/107/119	5259 104/104/116	5114 101/101/112	4968 98/98/109	4824 96/100/110	4687 94/103/112	5621 102/107/116	18386
V _{FS}	119	121	124	127	130	133	138	

WARNING: THE VALUES ABOVE DO NOT TAKE INTO ACCOUNT OBSTACLES.

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 3000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	4942 107/107/119	4789 104/104/116	4643 101/101/113	4496 99/99/110	4352 96/100/110	4216 94/103/112	4695 99/107/116	18386
-30	5068 107/107/119	4914 104/104/116	4769 101/101/112	4622 98/98/109	4478 96/100/110	4341 94/103/112	4956 100/107/116	18386
-25	5133 107/107/119	4980 104/104/116	4834 101/101/112	4688 98/98/109	4543 96/100/110	4407 94/103/112	5084 100/107/116	18386
-20	5198 106/106/119	5045 104/104/116	4899 101/101/112	4752 98/98/109	4608 96/100/110	4471 94/103/112	5220 101/107/116	18386
-15	5264 106/106/119	5110 103/103/115	4964 101/101/112	4818 98/98/109	4673 96/100/110	4536 94/103/112	5363 101/107/116	18386
-10	5329 106/106/119	5175 103/103/115	5029 101/101/112	4882 98/98/109	4738 96/100/110	4600 94/103/112	5510 102/107/116	18386
-5	5394 106/106/118	5240 103/103/115	5093 100/100/112	4946 98/98/109	4802 96/100/110	4679 94/103/112	5662 103/107/116	18386
0	5461 106/106/118	5306 103/103/115	5159 100/100/112	5012 98/98/109	4867 96/100/110	4799 94/103/112	5819 103/107/116	18386
5	5388 104/104/115	5236 101/101/112	5091 98/98/109	4945 96/97/107	4802 94/100/110	5341 98/103/112	6507 107/108/116	18386
V_{FS}	117	121	124	127	130	133	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 4000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5008 106/106/118	4855 103/103/115	4710 100/100/111	4563 98/98/108	4420 95/100/110	4284 93/103/112	5095 101/107/116	18386
-30	5139 105/105/118	4986 103/103/114	4841 100/100/111	4695 97/97/108	4551 95/100/110	4464 94/103/112	5383 102/107/116	18386
-25	5208 105/105/117	5055 102/102/114	4909 100/100/111	4763 97/97/108	4619 95/100/110	4578 94/103/112	5534 103/107/116	18386
-20	5275 105/105/117	5122 102/102/114	4976 100/100/111	4830 97/97/108	4686 95/100/110	4697 94/103/112	5692 103/107/116	18386
-15	5342 105/105/117	5188 102/102/114	5042 100/100/111	4896 97/97/108	4752 95/100/110	4820 95/103/112	5860 104/107/116	18386
-10	5410 105/105/117	5256 102/102/114	5110 100/100/111	4964 97/97/108	4819 95/100/110	4947 95/103/112	6020 105/107/116	18386
-5	5479 105/105/117	5325 102/102/114	5178 99/99/111	5031 97/97/107	4887 95/100/110	5081 96/103/112	6231 105/107/116	18386
0	5511 104/104/116	5357 101/101/113	5210 99/99/109	5062 96/97/107	4918 94/100/110	5431 98/103/112	6597 107/108/116	18386
5	5380 101/101/112	5228 98/98/109	5083 96/96/106	4939 93/98/107	5381 97/101/110	6237 104/104/112	- -	18231
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS
FLAP 2 – STANDING WATER (0.250 in) RUNWAY
ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF
PW535E ENGINES

Airport Pressure Altitude: 5000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5079 104/104/116	4925 102/102/113	4779 99/99/110	4633 96/97/107	4489 94/100/110	4615 95/103/112	5577 104/108/116	18386
-30	5211 104/104/116	5057 101/101/113	4911 99/99/110	4764 96/97/107	4620 94/100/110	4876 96/103/112	5959 105/108/116	18386
-25	5280 104/104/116	5126 101/101/113	4980 99/99/109	4834 96/97/107	4689 94/100/110	5006 97/103/112	6129 106/108/116	18386
-20	5349 104/104/116	5195 101/101/113	5049 98/98/109	4902 96/97/107	4758 94/100/110	5144 97/103/112	6329 106/108/116	18386
-15	5422 104/104/116	5267 101/101/112	5120 98/98/109	4972 96/97/107	4827 94/100/110	5289 98/103/112	6467 107/108/116	18386
-10	5493 104/104/116	5337 101/101/112	5190 98/98/109	5042 96/97/107	4897 94/100/110	5439 98/103/112	6634 107/108/116	18386
-5	5563 104/104/115	5408 101/101/112	5260 98/98/109	5112 96/97/107	4966 93/100/110	5594 99/103/112	6742 107/108/116	18386
0	5509 101/101/113	5354 99/99/109	5207 96/96/106	5061 94/98/107	5468 97/101/110	6328 103/104/112	-	18273
5	5378 98/98/109	5226 96/96/105	5082 93/95/105	5404 95/99/107	6256 102/102/110	6802 105/105/112	-	17287
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 6000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5146 103/103/115	4991 100/100/111	4844 98/98/108	4697 95/97/107	4552 93/101/110	5064 98/104/112	6207 107/108/116	18386
-30	5286 103/103/114	5132 100/100/111	4985 97/97/108	4837 95/97/107	4692 93/101/110	5360 99/104/112	6483 108/108/116	18386
-25	5358 103/103/114	5203 100/100/111	5056 97/97/108	4909 95/97/107	4797 93/101/110	5522 99/104/112	6658 108/108/116	18386
-20	5429 103/103/114	5274 100/100/111	5126 97/97/108	4979 95/97/107	4927 94/101/110	5689 100/104/112	6771 108/108/116	18386
-15	5505 103/103/114	5349 100/100/111	5200 97/97/108	5052 95/97/107	5063 94/101/110	5841 101/104/112	6888 108/108/116	18386
-10	5595 102/102/114	5437 100/100/111	5288 97/97/107	5138 95/98/107	5249 95/101/110	6060 101/104/112	7078 108/108/116	18386
-5	5641 101/101/112	5482 99/99/109	5332 96/96/106	5182 94/98/107	5716 98/101/110	6549 104/104/112	-	18080
0	5546 99/99/109	5388 96/96/106	5238 93/96/105	5610 96/99/107	6443 102/102/110	-	-	17150
5	5443 96/96/105	5287 93/93/102	5504 94/96/105	6273 99/100/107	-	-	-	16251
V _{FS}	117	121	124	127	130	134	138	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 7000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5171 101/101/112	5017 98/98/109	4872 96/96/106	4725 93/98/107	4976 95/101/110	5735 102/104/112	-	18080
-30	5308 101/101/112	5154 98/98/109	5008 95/95/105	4862 93/98/107	5275 96/101/110	6131 103/104/112	-	18034
-25	5382 101/101/112	5228 98/98/108	5082 95/95/105	4935 93/98/107	5427 97/101/110	6293 104/104/112	-	18034
-20	5471 101/101/112	5315 98/98/108	5167 95/95/105	5020 93/98/107	5615 98/101/110	6410 104/104/112	-	18033
-15	5565 100/100/111	5407 98/98/108	5257 95/95/105	5108 93/98/107	5776 98/101/110	6595 104/104/112	-	18031
-10	5670 100/100/111	5510 97/97/108	5358 95/95/105	5296 93/98/107	6245 101/102/110	6816 104/105/112	-	17686
-5	5610 98/98/108	5451 95/95/105	5299 93/96/105	6059 98/99/107	6682 102/102/110	-	-	16830
0	5516 95/95/104	5357 92/93/102	5895 96/97/105	6512 100/100/107	-	-	-	15957
5	5407 92/92/101	5712 94/94/102	6328 97/97/105	-	-	-	-	15029
V _{FS}	117	121	124	127	130	134	-	

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SIMPLIFIED TAKEOFF ANALYSIS

FLAP 2 – STANDING WATER (0.250 in) RUNWAY

ICE PROTECTION (WINGSTAB+ENG) ON – ATR OFF

PW535E ENGINES

Airport Pressure Altitude: 8000 ft

TEMP (°C)	TAKEOFF WEIGHT (lb)							
	MINIMUM REQUIRED RUNWAY LENGTH (ft)							
	V ₁ /V _R /V ₂ (KIAS)							
	13200	14000	14800	15600	16400	17200	18387	LIMITED WEIGHT
-40	5192 99/99/109	5039 96/96/106	4894 94/95/105	4865 93/99/107	5674 99/102/110	6395 105/105/112	-	17508
-30	5340 99/99/109	5186 96/96/106	5041 94/95/105	5159 94/99/107	6060 101/102/110	6622 105/105/112	-	17468
-25	5438 99/99/109	5282 96/96/106	5135 93/95/105	5321 94/99/107	6240 102/102/110	6792 105/105/112	-	17468
-20	5537 99/99/109	5379 96/96/106	5229 93/95/105	5505 95/99/107	6363 102/102/110	6939 105/105/112	-	17466
-15	5635 98/98/109	5475 96/96/106	5324 93/95/105	5741 96/99/107	6554 102/102/110	7106 105/105/112	-	17369
-10	5703 97/97/107	5541 95/95/104	5438 92/96/105	6296 99/99/107	6885 102/102/110	-	-	16758
-5	5591 94/94/103	5429 92/93/102	6173 97/97/105	6760 100/100/107	-	-	-	15805
0	5507 92/92/100	5946 94/94/102	6553 98/98/105	-	-	-	-	14971
5	5729 91/92/99	6388 95/95/102	-	-	-	-	-	14055
V _{FS}	117	121	124	127	131	134	-	

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TAKEOFF CLIMB – ATR OFF

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

TAKEOFF 2ND SEGMENT NET GRADIENT (%)

EXAMPLE

Given:

Airplane Weight	17200 lb
Takeoff Flap.....	1
Anti-Ice	OFF
Altitude.....	2000 ft
OAT	10°C
Level-Off Height.....	400 ft

Determine:

Second Segment Net Gradient	8.4%
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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: -1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.3	12.9	11.6	10.4	9.4	8.4	7.2
-30	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-20	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-10	14.4	13.0	11.7	10.5	9.4	8.5	7.3
0	14.5	13.1	11.7	10.5	9.5	8.5	7.3
10	14.4	13.0	11.7	10.5	9.4	8.5	7.3
20	14.4	13.0	11.7	10.4	9.4	8.5	7.3
30	13.7	12.3	11.0	9.8	8.8	7.9	6.8
40	10.8	9.5	8.3	7.4	6.5	5.7	4.7
50	8.2	7.1	6.1	5.2	4.5	3.8	3.0
51	7.9	6.8	5.9	5.0	4.3	3.6	-

Altitude: -1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.5	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.5	13.1	11.8	10.5	9.5	8.6	7.4
-20	14.5	13.1	11.8	10.6	9.5	8.6	7.4
-10	14.5	13.1	11.8	10.6	9.5	8.6	7.4
0	14.6	13.1	11.8	10.6	9.5	8.6	7.4
10	14.5	13.1	11.8	10.6	9.5	8.6	7.4
20	14.5	13.1	11.7	10.5	9.5	8.5	7.3
30	13.2	11.8	10.6	9.4	8.4	7.6	6.4
40	10.4	9.1	8.0	7.0	6.2	5.4	4.5
50	7.8	6.7	5.8	4.9	4.2	3.6	2.7
51	7.6	6.5	5.6	4.8	4.0	3.4	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 0 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.5	13.1	11.8	10.5	9.5	8.6	7.3
-20	14.5	13.1	11.8	10.6	9.5	8.6	7.4
-10	14.5	13.1	11.8	10.6	9.5	8.6	7.4
0	14.6	13.1	11.8	10.6	9.5	8.6	7.4
10	14.5	13.1	11.8	10.5	9.5	8.6	7.4
20	14.5	13.1	11.7	10.5	9.4	8.5	7.3
30	12.7	11.3	10.0	9.0	8.0	7.2	6.1
40	9.8	8.6	7.5	6.6	5.8	5.1	4.1
50	7.3	6.3	5.4	4.5	3.8	3.2	-

Altitude: 0 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.5	13.1	11.7	10.5	9.5	8.5	7.4
-20	14.5	13.1	11.8	10.5	9.5	8.6	7.4
-10	14.5	13.1	11.8	10.6	9.5	8.6	7.4
0	14.5	13.1	11.8	10.6	9.5	8.6	7.4
10	14.5	13.1	11.8	10.5	9.5	8.6	7.4
20	14.3	12.9	11.6	10.4	9.3	8.4	7.2
30	12.2	10.8	9.6	8.5	7.6	6.8	5.7
40	9.4	8.2	7.2	6.3	5.5	4.7	3.8
50	7.0	6.0	5.0	4.2	3.6	2.9	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.5	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.5	13.1	11.8	10.5	9.5	8.6	7.4
-20	14.5	13.1	11.8	10.6	9.5	8.6	7.4
-10	14.5	13.1	11.8	10.6	9.5	8.6	7.4
0	14.5	13.1	11.8	10.6	9.5	8.6	7.4
10	14.5	13.1	11.8	10.5	9.5	8.6	7.4
20	14.0	12.6	11.3	10.1	9.1	8.2	7.0
30	11.7	10.3	9.1	8.1	7.2	6.4	5.3
40	8.9	7.8	6.7	5.8	5.1	4.4	3.5
48	7.0	5.9	5.0	4.2	3.6	2.9	-

Altitude: 1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.1	12.7	11.4	10.2	9.2	8.3	7.1
-30	14.2	12.8	11.5	10.3	9.2	8.3	7.1
-20	14.2	12.8	11.5	10.3	9.3	8.4	7.2
-10	14.3	12.9	11.5	10.3	9.3	8.4	7.2
0	14.3	12.9	11.6	10.3	9.3	8.4	7.2
10	14.3	12.9	11.5	10.3	9.3	8.4	7.2
20	13.4	12.1	10.8	9.6	8.6	7.7	6.6
30	11.2	9.8	8.7	7.7	6.8	6.0	5.0
40	8.5	7.4	6.4	5.5	4.8	4.1	3.2
48	6.6	5.6	4.7	3.9	3.3	2.7	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 2000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.3	12.9	11.5	10.3	9.3	8.4	7.2
-30	14.3	12.9	11.6	10.4	9.3	8.4	7.2
-20	14.3	12.9	11.6	10.4	9.3	8.4	7.3
-10	14.3	12.9	11.6	10.4	9.3	8.4	7.3
0	14.3	12.9	11.6	10.4	9.3	8.4	7.3
10	14.3	12.9	11.6	10.4	9.3	8.4	7.2
20	13.0	11.7	10.4	9.2	8.3	7.4	6.3
30	10.6	9.3	8.2	7.2	6.4	5.6	4.6
40	8.0	6.9	5.9	5.1	4.4	3.7	-
46	6.6	5.6	4.7	3.9	3.3	-	-

Altitude: 2000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.6	12.3	11.0	9.8	8.8	7.9	6.8
-30	13.7	12.4	11.1	9.9	8.9	8.0	6.8
-20	13.8	12.4	11.1	9.9	8.9	8.0	6.9
-10	13.8	12.4	11.1	10.0	8.9	8.0	6.9
0	13.9	12.5	11.2	10.0	9.0	8.1	6.9
10	13.8	12.5	11.2	10.0	9.0	8.1	6.9
20	12.4	11.1	9.8	8.7	7.8	7.0	5.9
30	10.2	8.9	7.8	6.9	6.1	5.3	4.3
40	7.6	6.6	5.6	4.8	4.1	3.4	-
46	6.3	5.3	4.4	3.6	3.0	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 3000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.8	12.4	11.1	9.9	8.9	8.1	6.9
-30	13.9	12.5	11.2	10.0	9.0	8.1	6.9
-20	13.9	12.5	11.2	10.0	9.0	8.1	7.0
-10	13.9	12.5	11.2	10.0	9.0	8.1	7.0
0	13.9	12.5	11.2	10.0	9.0	8.1	7.0
10	13.9	12.5	11.2	10.0	9.0	8.1	6.9
20	12.1	10.7	9.4	8.4	7.5	6.7	5.6
30	9.6	8.4	7.4	6.5	5.6	4.9	4.0
40	7.2	6.1	5.2	4.4	3.7	3.1	-
44	6.3	5.3	4.4	3.6	3.0	-	-

Altitude: 3000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.2	11.8	10.5	9.4	8.4	7.5	6.4
-30	13.2	11.9	10.6	9.4	8.5	7.6	6.5
-20	13.3	11.9	10.7	9.5	8.5	7.6	6.5
-10	13.3	12.0	10.7	9.5	8.5	7.7	6.6
0	13.4	12.0	10.7	9.5	8.6	7.7	6.6
10	13.4	12.0	10.7	9.5	8.6	7.7	6.6
20	11.5	10.2	8.9	7.9	7.1	6.3	5.2
30	9.3	8.1	7.0	6.1	5.3	4.6	3.7
40	6.8	5.8	4.9	4.1	3.4	2.8	-
44	5.9	5.0	4.1	3.4	2.7	-	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 4000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.4	12.0	10.7	9.5	8.6	7.7	6.6
-30	13.4	12.0	10.7	9.5	8.6	7.7	6.6
-20	13.4	12.0	10.7	9.6	8.6	7.7	6.6
-10	13.5	12.1	10.7	9.6	8.6	7.8	6.6
0	13.5	12.1	10.7	9.6	8.6	7.8	6.6
10	13.2	11.8	10.5	9.4	8.4	7.6	6.4
20	11.1	9.8	8.6	7.6	6.8	6.0	5.0
30	8.7	7.6	6.6	5.7	4.9	4.2	3.3
40	6.4	5.4	4.5	3.7	3.1	-	-
42	5.9	5.0	4.1	3.4	-	-	-

Altitude: 4000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	12.7	11.3	10.1	8.9	8.0	7.2	6.1
-30	12.7	11.4	10.1	9.0	8.1	7.2	6.1
-20	12.8	11.4	10.2	9.0	8.1	7.3	6.2
-10	12.8	11.5	10.2	9.1	8.1	7.3	6.2
0	12.9	11.5	10.2	9.1	8.2	7.3	6.2
10	12.5	11.2	9.9	8.8	7.9	7.0	5.9
20	10.5	9.2	8.1	7.1	6.3	5.5	4.6
30	8.4	7.2	6.2	5.4	4.6	4.0	3.1
40	6.0	5.1	4.2	3.5	2.8	-	-
42	5.6	4.7	3.8	3.1	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 5000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	12.9	11.5	10.2	9.1	8.1	7.3	6.2
-30	12.9	11.5	10.2	9.1	8.2	7.3	6.2
-20	12.9	11.5	10.2	9.1	8.2	7.4	6.3
-10	12.9	11.6	10.2	9.2	8.2	7.4	6.3
0	13.0	11.6	10.2	9.2	8.2	7.4	6.3
10	12.2	10.8	9.5	8.5	7.6	6.8	5.7
20	10.1	8.9	7.8	6.9	6.0	5.3	4.3
30	7.9	6.8	5.8	5.0	4.3	3.6	-
40	5.6	4.7	3.8	3.1	-	-	-

Altitude: 5000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	11.9	10.6	9.3	8.3	7.4	6.6	5.5
-30	12.0	10.7	9.4	8.4	7.5	6.7	5.6
-20	12.1	10.7	9.5	8.4	7.5	6.7	5.7
-10	12.1	10.8	9.5	8.5	7.6	6.8	5.7
0	12.2	10.8	9.6	8.5	7.6	6.8	5.7
10	11.3	10.0	8.8	7.8	6.9	6.1	5.1
20	9.4	8.2	7.2	6.3	5.5	4.7	3.8
30	7.4	6.3	5.4	4.6	3.9	3.2	-
40	5.2	4.3	3.4	2.7	-	-	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%) FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 6000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	12.2	10.8	9.5	8.5	7.6	6.8	5.7
-30	12.2	10.9	9.6	8.6	7.7	6.8	5.8
-20	12.3	10.9	9.6	8.6	7.7	6.9	5.8
-10	12.3	10.9	9.6	8.6	7.7	6.9	5.8
0	12.3	10.9	9.6	8.6	7.7	6.9	5.8
10	11.0	9.6	8.5	7.5	6.7	5.9	4.9
20	9.1	7.9	6.9	6.0	5.2	4.5	3.6
30	6.9	5.9	5.0	4.2	3.5	2.9	-
38	5.2	4.3	3.4	-	-	-	-

Altitude: 6000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	11.1	9.8	8.6	7.6	6.7	6.0	5.0
-30	11.2	9.9	8.7	7.7	6.8	6.0	5.0
-20	11.2	9.9	8.7	7.7	6.9	6.1	5.1
-10	11.3	10.0	8.8	7.8	6.9	6.1	5.1
0	11.4	10.0	8.8	7.8	7.0	6.2	5.2
10	10.1	8.8	7.7	6.8	6.0	5.2	4.3
20	8.4	7.3	6.3	5.5	4.7	4.0	3.2
30	6.4	5.4	4.5	3.7	3.1	2.5	-
38	4.6	3.8	3.0	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 7000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	11.3	10.0	8.8	7.8	6.9	6.2	5.1
-30	11.4	10.0	8.8	7.9	7.0	6.2	5.2
-20	11.4	10.0	8.9	7.9	7.0	6.2	5.2
-10	11.4	10.1	8.9	7.9	7.0	6.2	5.2
0	11.4	10.1	8.9	7.9	7.0	6.2	5.2
10	9.7	8.5	7.5	6.5	5.7	5.0	4.1
20	8.1	7.0	6.0	5.2	4.4	3.8	2.9
30	5.9	4.9	4.1	3.3	-	-	-
36	4.6	3.7	3.0	-	-	-	-

Altitude: 7000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.5	9.2	8.1	7.1	6.3	5.6	4.6
-30	10.6	9.3	8.2	7.2	6.4	5.6	4.6
-20	10.6	9.3	8.2	7.2	6.4	5.7	4.7
-10	10.7	9.4	8.2	7.3	6.4	5.7	4.7
0	10.7	9.4	8.3	7.3	6.5	5.7	4.7
10	9.0	7.8	6.8	5.9	5.2	4.5	3.6
20	7.6	6.6	5.6	4.8	4.1	3.4	2.6
30	5.4	4.5	3.6	2.9	-	-	-
36	4.1	3.3	2.5	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%) FLAP 1 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 8000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.6	9.3	8.2	7.3	6.4	5.7	4.7
-30	10.7	9.4	8.3	7.3	6.5	5.7	4.7
-20	10.7	9.4	8.3	7.3	6.5	5.7	4.7
-10	10.7	9.4	8.3	7.3	6.5	5.7	4.7
0	10.4	9.1	8.0	7.1	6.3	5.5	4.5
10	8.7	7.6	6.6	5.7	5.0	4.3	3.4
20	7.3	6.2	5.3	4.5	3.8	3.2	-
30	4.9	4.0	3.2	-	-	-	-
34	4.1	3.3	-	-	-	-	-

Altitude: 8000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.0	8.8	7.7	6.8	6.0	5.2	4.3
-30	10.1	8.8	7.7	6.8	6.0	5.3	4.3
-20	10.1	8.9	7.8	6.9	6.1	5.3	4.4
-10	10.2	8.9	7.8	6.9	6.1	5.3	4.4
0	9.9	8.7	7.6	6.7	5.9	5.2	4.2
10	8.4	7.2	6.3	5.4	4.7	4.0	3.1
20	6.9	5.9	5.0	4.2	3.5	2.9	-
30	4.7	3.8	3.0	-	-	-	-
34	3.9	3.1	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: -1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-30	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-20	10.5	9.9	9.1	8.3	7.4	6.5	5.3
-10	10.6	9.9	9.2	8.4	7.5	6.5	5.3
0	10.6	9.9	9.2	8.4	7.5	6.5	5.3
10	10.6	9.9	9.2	8.4	7.4	6.5	5.3
20	10.6	9.9	9.1	8.3	7.4	6.5	5.3
30	10.0	9.4	8.6	7.8	6.8	5.9	4.8
40	8.1	7.3	6.3	5.4	4.5	3.7	-
50	6.1	5.0	4.1	-	-	-	-
51	5.8	4.8	3.9	-	-	-	-

Altitude: -1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.6	9.9	9.2	8.3	7.4	6.5	5.3
-30	10.6	9.9	9.2	8.4	7.5	6.6	5.4
-20	10.7	10.0	9.2	8.4	7.5	6.6	5.4
-10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
0	10.7	10.0	9.3	8.4	7.5	6.6	5.4
10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
20	10.7	10.0	9.2	8.4	7.4	6.5	5.4
30	9.6	8.9	8.2	7.3	6.4	5.6	4.4
40	7.7	6.9	6.0	5.0	4.2	3.4	-
50	5.7	4.7	3.8	-	-	-	-
51	5.5	4.5	3.6	-	-	-	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 0 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.5	9.9	9.1	8.3	7.4	6.5	5.3
-30	10.6	9.9	9.2	8.4	7.5	6.6	5.4
-20	10.6	9.9	9.2	8.4	7.5	6.6	5.4
-10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
0	10.7	10.0	9.2	8.4	7.5	6.6	5.4
10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
20	10.7	10.0	9.2	8.4	7.4	6.5	5.3
30	9.4	8.7	7.9	7.0	6.0	5.2	4.1
40	7.4	6.5	5.5	4.6	3.8	-	-
50	5.3	4.3	-	-	-	-	-

Altitude: 0 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.6	9.9	9.1	8.3	7.4	6.5	5.3
-30	10.6	9.9	9.2	8.3	7.4	6.5	5.4
-20	10.6	9.9	9.2	8.4	7.5	6.5	5.4
-10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
0	10.7	10.0	9.2	8.4	7.5	6.6	5.4
10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
20	10.5	9.8	9.1	8.2	7.3	6.4	5.2
30	8.8	8.2	7.4	6.5	5.6	4.8	3.7
40	7.0	6.1	5.2	4.3	3.5	-	-
50	5.0	3.9	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.6	9.9	9.1	8.3	7.4	6.5	5.3
-30	10.6	9.9	9.2	8.4	7.5	6.5	5.4
-20	10.7	10.0	9.2	8.4	7.5	6.6	5.4
-10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
0	10.7	10.0	9.2	8.4	7.5	6.6	5.4
10	10.7	10.0	9.2	8.4	7.5	6.6	5.4
20	10.3	9.6	8.8	8.0	7.1	6.2	5.0
30	8.7	7.9	7.1	6.1	5.2	4.4	-
40	6.7	5.7	4.7	3.8	-	-	-
48	5.0	3.9	-	-	-	-	-

Altitude: 1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.2	9.6	8.8	8.0	7.1	6.2	5.1
-30	10.3	9.6	8.9	8.1	7.2	6.3	5.1
-20	10.4	9.7	8.9	8.1	7.2	6.3	5.2
-10	10.4	9.7	9.0	8.2	7.3	6.4	5.2
0	10.4	9.8	9.0	8.2	7.3	6.4	5.2
10	10.5	9.8	9.0	8.2	7.3	6.4	5.2
20	9.6	9.0	8.3	7.5	6.6	5.7	4.6
30	8.2	7.5	6.6	5.7	4.8	4.0	-
40	6.3	5.4	4.4	3.5	-	-	-
48	4.6	3.6	-	-	-	-	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%) FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 2000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.4	9.7	9.0	8.2	7.2	6.4	5.2
-30	10.5	9.8	9.0	8.2	7.3	6.4	5.2
-20	10.5	9.8	9.0	8.2	7.3	6.4	5.2
-10	10.5	9.8	9.1	8.2	7.3	6.4	5.3
0	10.5	9.8	9.1	8.2	7.3	6.4	5.3
10	10.5	9.8	9.1	8.2	7.3	6.4	5.2
20	9.6	8.9	8.1	7.2	6.3	5.4	4.3
30	8.0	7.2	6.2	5.2	4.4	3.6	-
40	5.9	4.9	3.9	-	-	-	-
46	4.6	3.6	-	-	-	-	-

Altitude: 2000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.8	9.2	8.5	7.7	6.8	5.9	4.8
-30	9.9	9.2	8.5	7.7	6.8	6.0	4.8
-20	10.0	9.3	8.6	7.8	6.9	6.0	4.9
-10	10.0	9.3	8.6	7.8	6.9	6.0	4.9
0	10.1	9.4	8.6	7.8	6.9	6.1	4.9
10	10.1	9.4	8.7	7.8	6.9	6.1	4.9
20	9.0	8.3	7.6	6.7	5.8	5.0	3.9
30	7.5	6.7	5.8	4.9	4.1	3.3	-
40	5.5	4.5	3.6	-	-	-	-
46	4.2	3.3	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 3000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.1	9.4	8.7	7.8	6.9	6.0	4.9
-30	10.2	9.5	8.7	7.9	6.9	6.1	4.9
-20	10.2	9.5	8.7	7.9	7.0	6.1	5.0
-10	10.2	9.5	8.8	7.9	7.0	6.1	5.0
0	10.3	9.6	8.8	7.9	7.0	6.1	5.0
10	10.3	9.5	8.8	7.9	7.0	6.1	4.9
20	8.9	8.2	7.3	6.4	5.5	4.7	3.6
30	7.3	6.4	5.4	4.5	3.7	-	-
40	5.1	4.1	-	-	-	-	-
44	4.2	-	-	-	-	-	-

Altitude: 3000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.5	8.8	8.1	7.3	6.4	5.5	4.4
-30	9.5	8.9	8.2	7.3	6.4	5.6	4.5
-20	9.6	9.0	8.2	7.4	6.5	5.6	4.5
-10	9.7	9.0	8.3	7.4	6.5	5.7	4.6
0	9.7	9.0	8.3	7.5	6.6	5.7	4.6
10	9.7	9.1	8.3	7.5	6.5	5.7	4.6
20	8.4	7.7	6.9	5.9	5.1	4.3	3.2
30	6.9	6.0	5.0	4.1	3.4	-	-
40	4.8	3.8	-	-	-	-	-
44	3.9	-	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 4000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.8	9.1	8.3	7.5	6.5	5.7	4.6
-30	9.9	9.1	8.4	7.5	6.6	5.7	4.6
-20	9.9	9.2	8.4	7.5	6.6	5.7	4.6
-10	9.9	9.2	8.4	7.5	6.6	5.7	4.6
0	9.9	9.2	8.4	7.5	6.6	5.7	4.6
10	9.7	9.0	8.2	7.4	6.4	5.6	4.5
20	8.3	7.5	6.6	5.6	4.8	4.0	-
30	6.6	5.6	4.6	3.7	-	-	-
40	4.3	3.4	-	-	-	-	-
42	3.9	-	-	-	-	-	-

Altitude: 4000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.1	8.5	7.7	6.9	6.0	5.2	4.1
-30	9.2	8.5	7.8	6.9	6.0	5.2	4.1
-20	9.3	8.6	7.8	7.0	6.1	5.3	4.2
-10	9.3	8.6	7.9	7.0	6.1	5.3	4.2
0	9.4	8.7	7.9	7.1	6.1	5.3	4.2
10	9.0	8.4	7.6	6.8	5.9	5.0	4.0
20	7.7	7.0	6.1	5.1	4.3	3.5	-
30	6.2	5.2	4.2	3.4	-	-	-
40	4.0	3.1	-	-	-	-	-
42	3.6	-	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 5000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.5	8.8	7.9	7.1	6.1	5.3	4.2
-30	9.5	8.8	8.0	7.1	6.2	5.3	4.2
-20	9.6	8.8	8.0	7.1	6.2	5.3	4.3
-10	9.6	8.9	8.0	7.1	6.2	5.4	4.3
0	9.6	8.9	8.0	7.1	6.2	5.4	4.3
10	9.0	8.3	7.4	6.5	5.6	4.8	3.7
20	7.6	6.8	5.8	4.9	4.0	-	-
30	5.8	4.8	3.8	-	-	-	-
40	3.6	-	-	-	-	-	-

Altitude: 5000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	8.5	7.9	7.1	6.3	5.4	4.6	3.5
-30	8.6	8.0	7.2	6.3	5.4	4.6	3.6
-20	8.7	8.0	7.3	6.4	5.5	4.7	3.7
-10	8.8	8.1	7.3	6.5	5.6	4.8	3.7
0	8.9	8.2	7.4	6.5	5.6	4.8	3.7
10	8.1	7.5	6.7	5.8	4.9	4.1	3.1
20	6.9	6.1	5.2	4.3	3.5	-	-
30	5.3	4.3	3.4	-	-	-	-
40	3.2	-	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%) FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 6000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.0	8.2	7.4	6.5	5.6	4.8	3.7
-30	9.0	8.3	7.5	6.5	5.6	4.8	3.8
-20	9.1	8.3	7.5	6.6	5.7	4.8	3.8
-10	9.1	8.4	7.5	6.6	5.7	4.9	3.8
0	9.1	8.4	7.5	6.6	5.7	4.9	3.8
10	8.2	7.4	6.5	5.5	4.7	3.9	-
20	6.8	5.9	4.9	4.0	-	-	-
30	4.9	3.9	-	-	-	-	-

Altitude: 6000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	7.9	7.2	6.5	5.6	4.7	4.0	3.0
-30	8.0	7.3	6.5	5.7	4.8	4.0	3.0
-20	8.1	7.4	6.6	5.7	4.9	4.1	3.1
-10	8.1	7.5	6.7	5.8	4.9	4.1	3.1
0	8.2	7.5	6.7	5.8	4.9	4.2	3.2
10	7.3	6.5	5.7	4.8	4.0	3.2	-
20	6.2	5.3	4.3	3.5	-	-	-
30	4.3	3.4	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 7000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	8.4	7.6	6.8	5.8	4.9	4.2	-
-30	8.4	7.7	6.8	5.8	5.0	4.2	-
-20	8.5	7.7	6.8	5.9	5.0	4.2	-
-10	8.5	7.7	6.9	5.9	5.0	4.2	-
0	8.5	7.8	6.9	5.9	5.0	4.2	-
10	7.3	6.4	5.4	4.5	3.7	-	-
20	6.0	5.0	4.0	-	-	-	-
30	3.9	-	-	-	-	-	-

Altitude: 7000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	7.6	6.9	6.1	5.1	4.3	3.6	-
-30	7.7	7.0	6.1	5.2	4.4	3.6	-
-20	7.7	7.0	6.2	5.2	4.4	3.6	-
-10	7.8	7.0	6.2	5.3	4.4	3.7	-
0	7.8	7.1	6.2	5.3	4.5	3.7	-
10	6.6	5.8	4.8	3.9	3.2	-	-
20	5.5	4.5	3.6	-	-	-	-
30	3.4	-	-	-	-	-	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE OFF

PW535E ENGINES

Altitude: 8000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	8.0	7.2	6.2	5.3	4.4	3.7	-
-30	8.0	7.2	6.3	5.3	4.5	3.7	-
-20	8.0	7.2	6.3	5.3	4.5	3.7	-
-10	8.1	7.2	6.3	5.3	4.5	3.7	-
0	7.8	7.0	6.0	5.1	4.3	3.5	-
10	6.5	5.6	4.6	3.7	-	-	-
20	5.2	4.2	-	-	-	-	-

Altitude: 8000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	7.4	6.6	5.7	4.8	3.9	3.2	-
-30	7.4	6.7	5.7	4.8	4.0	3.3	-
-20	7.5	6.7	5.8	4.9	4.0	3.3	-
-10	7.5	6.7	5.8	4.9	4.1	3.3	-
0	7.4	6.6	5.6	4.7	3.9	3.2	-
10	6.2	5.2	4.2	3.4	-	-	-
20	4.9	3.8	-	-	-	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: -1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.3	12.9	11.6	10.4	9.3	8.4	7.2
-30	14.3	12.9	11.6	10.4	9.3	8.4	7.2
-20	14.3	12.9	11.6	10.4	9.3	8.4	7.2
-10	14.3	12.9	11.6	10.4	9.3	8.4	7.2
0	14.3	12.9	11.6	10.4	9.3	8.4	7.2
5	14.3	12.9	11.6	10.3	9.3	8.4	7.2

Altitude: -1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-20	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-10	14.4	13.0	11.7	10.5	9.4	8.5	7.3
0	14.4	13.0	11.7	10.4	9.4	8.5	7.3
5	14.4	13.0	11.6	10.4	9.4	8.5	7.3



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 0 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-20	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-10	14.4	13.0	11.7	10.5	9.4	8.5	7.3
0	14.4	13.0	11.7	10.4	9.4	8.5	7.3
5	14.3	12.9	11.6	10.4	9.4	8.4	7.2

Altitude: 0 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.4	13.0	11.7	10.4	9.4	8.5	7.3
-20	14.4	13.0	11.7	10.4	9.4	8.5	7.3
-10	14.4	13.0	11.7	10.4	9.4	8.5	7.3
0	14.4	12.9	11.6	10.4	9.4	8.5	7.3
5	14.3	12.9	11.6	10.4	9.3	8.4	7.2

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-30	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-20	14.4	13.0	11.7	10.5	9.4	8.5	7.3
-10	14.4	13.0	11.7	10.4	9.4	8.5	7.3
0	14.4	12.9	11.6	10.4	9.4	8.4	7.3
5	14.3	12.9	11.6	10.4	9.3	8.4	7.2

Altitude: 1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.1	12.7	11.4	10.2	9.2	8.3	7.1
-30	14.1	12.7	11.4	10.2	9.1	8.2	7.1
-20	14.1	12.7	11.4	10.2	9.2	8.3	7.1
-10	14.1	12.7	11.4	10.2	9.2	8.3	7.1
0	14.1	12.7	11.4	10.2	9.1	8.2	7.1
5	14.0	12.6	11.3	10.1	9.1	8.2	7.0



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 2000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	14.2	12.9	11.5	10.3	9.3	8.4	7.2
-30	14.2	12.8	11.5	10.3	9.2	8.3	7.2
-20	14.2	12.8	11.5	10.3	9.2	8.3	7.2
-10	14.2	12.8	11.5	10.2	9.2	8.3	7.2
0	14.1	12.7	11.4	10.2	9.2	8.3	7.1
5	13.7	12.3	11.0	9.8	8.8	8.0	6.8

Altitude: 2000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.6	12.3	11.0	9.8	8.8	7.9	6.8
-30	13.6	12.3	11.0	9.8	8.8	7.9	6.8
-20	13.7	12.3	11.0	9.8	8.8	7.9	6.8
-10	13.7	12.3	11.0	9.8	8.8	7.9	6.8
0	13.6	12.3	11.0	9.8	8.8	7.9	6.8
5	13.0	11.7	10.4	9.2	8.3	7.4	6.3



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 3000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.8	12.4	11.1	9.9	8.9	8.0	6.9
-30	13.8	12.4	11.1	9.9	8.9	8.0	6.9
-20	13.8	12.4	11.1	9.9	8.9	8.0	6.9
-10	13.8	12.4	11.0	9.9	8.9	8.0	6.8
0	13.7	12.3	11.0	9.8	8.8	8.0	6.8
5	12.6	11.2	9.9	8.9	7.9	7.1	6.0

Altitude: 3000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.2	11.8	10.5	9.4	8.4	7.5	6.4
-30	13.1	11.8	10.5	9.3	8.4	7.5	6.4
-20	13.2	11.8	10.5	9.4	8.4	7.6	6.4
-10	13.2	11.8	10.5	9.4	8.4	7.6	6.4
0	13.1	11.7	10.4	9.3	8.3	7.5	6.4
5	11.9	10.6	9.3	8.3	7.4	6.6	5.5



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 4000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	13.3	12.0	10.7	9.5	8.5	7.7	6.5
-30	13.3	11.9	10.6	9.5	8.5	7.6	6.5
-20	13.3	11.9	10.6	9.5	8.5	7.6	6.5
-10	13.3	11.9	10.6	9.4	8.5	7.6	6.5
0	12.7	11.4	10.1	9.0	8.1	7.2	6.1
5	11.5	10.1	9.0	8.0	7.1	6.3	5.2

Altitude: 4000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	12.7	11.3	10.1	8.9	8.0	7.2	6.1
-30	12.7	11.3	10.0	8.9	8.0	7.2	6.1
-20	12.7	11.3	10.1	8.9	8.0	7.2	6.1
-10	12.7	11.3	10.0	8.9	8.0	7.2	6.1
0	11.9	10.6	9.3	8.3	7.4	6.6	5.5
5	10.8	9.5	8.3	7.4	6.5	5.7	4.7

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 5000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	12.9	11.5	10.2	9.1	8.1	7.3	6.2
-30	12.8	11.4	10.1	9.0	8.1	7.3	6.2
-20	12.8	11.4	10.1	9.0	8.1	7.3	6.2
-10	12.8	11.4	10.1	9.0	8.1	7.2	6.1
0	11.6	10.2	9.0	8.0	7.1	6.3	5.3
5	10.3	9.1	8.0	7.0	6.2	5.5	4.5

Altitude: 5000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	11.9	10.6	9.3	8.3	7.4	6.6	5.5
-30	11.9	10.6	9.3	8.3	7.4	6.6	5.5
-20	11.9	10.6	9.4	8.3	7.4	6.6	5.6
-10	12.0	10.6	9.4	8.3	7.5	6.6	5.6
0	10.5	9.2	8.1	7.1	6.3	5.5	4.6
5	9.4	8.2	7.2	6.3	5.5	4.8	3.9



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 6000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	12.2	10.8	9.5	8.5	7.6	6.8	5.7
-30	12.1	10.8	9.5	8.5	7.6	6.8	5.7
-20	12.1	10.8	9.5	8.5	7.6	6.8	5.7
-10	12.0	10.6	9.4	8.4	7.5	6.7	5.6
0	10.2	8.9	7.8	6.9	6.1	5.3	4.4
5	9.1	7.9	6.9	6.0	5.2	4.5	3.6

Altitude: 6000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	11.1	9.8	8.6	7.6	6.7	6.0	5.0
-30	11.1	9.8	8.6	7.6	6.7	6.0	5.0
-20	11.1	9.8	8.6	7.6	6.8	6.0	5.0
-10	10.9	9.6	8.5	7.5	6.6	5.9	4.9
0	9.1	7.9	6.9	6.0	5.2	4.5	3.6
5	8.0	7.0	6.0	5.1	4.4	3.7	2.9

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 7000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	11.3	10.0	8.8	7.8	6.9	6.2	5.1
-30	11.3	9.9	8.8	7.8	6.9	6.1	5.1
-20	11.3	9.9	8.8	7.8	6.9	6.1	5.1
-10	10.9	9.5	8.4	7.4	6.6	5.8	4.8
0	8.7	7.6	6.6	5.7	5.0	4.3	3.4
5	7.7	6.6	5.6	4.8	4.1	3.5	-

Altitude: 7000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.5	9.2	8.1	7.1	6.3	5.6	4.6
-30	10.5	9.2	8.1	7.1	6.3	5.5	4.6
-20	10.5	9.2	8.1	7.1	6.3	5.6	4.6
-10	10.0	8.7	7.7	6.7	5.9	5.2	4.2
0	7.9	6.9	5.9	5.1	4.3	3.7	2.8
5	6.9	5.8	4.9	4.2	3.5	2.9	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 1 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 8000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.6	9.3	8.2	7.3	6.4	5.7	4.7
-30	10.6	9.3	8.2	7.2	6.4	5.6	4.7
-20	10.6	9.3	8.2	7.2	6.4	5.6	4.6
-10	9.7	8.5	7.4	6.5	5.7	5.0	4.1
0	7.6	6.5	5.6	4.8	4.1	3.4	-
5	6.4	5.5	4.6	3.8	3.2	-	-

Altitude: 8000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.0	8.8	7.7	6.8	6.0	5.2	4.3
-30	10.0	8.7	7.7	6.7	5.9	5.2	4.3
-20	10.0	8.8	7.7	6.8	6.0	5.2	4.3
-10	9.2	8.0	7.0	6.1	5.3	4.6	3.7
0	7.1	6.1	5.2	4.4	3.7	3.1	-
5	6.1	5.1	4.3	3.5	2.9	-	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: -1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.4	9.8	9.0	8.2	7.4	6.4	5.2
-30	10.4	9.8	9.0	8.2	7.4	6.4	5.2
-20	10.5	9.8	9.1	8.3	7.4	6.4	5.3
-10	10.5	9.8	9.1	8.3	7.4	6.4	5.2
0	10.5	9.8	9.0	8.2	7.3	6.4	5.2
5	10.5	9.8	9.0	8.2	7.3	6.4	5.2

Altitude: -1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.6	9.9	9.1	8.3	7.4	6.5	5.3
-30	10.6	9.9	9.1	8.3	7.4	6.5	5.3
-20	10.6	9.9	9.1	8.3	7.4	6.5	5.3
-10	10.6	9.9	9.1	8.3	7.4	6.5	5.3
0	10.6	9.9	9.1	8.3	7.4	6.5	5.3
5	10.6	9.9	9.1	8.3	7.4	6.5	5.3



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 0 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-30	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-20	10.5	9.9	9.1	8.3	7.4	6.5	5.3
-10	10.5	9.9	9.1	8.3	7.4	6.5	5.3
0	10.5	9.9	9.1	8.3	7.4	6.5	5.3
5	10.5	9.8	9.1	8.3	7.3	6.4	5.3

Altitude: 0 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-30	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-20	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-10	10.5	9.8	9.1	8.3	7.4	6.5	5.3
0	10.5	9.8	9.1	8.3	7.3	6.4	5.3
5	10.5	9.8	9.1	8.2	7.3	6.4	5.2

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 1000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.5	9.9	9.1	8.3	7.4	6.5	5.3
-30	10.5	9.8	9.1	8.3	7.4	6.5	5.3
-20	10.5	9.9	9.1	8.3	7.4	6.5	5.3
-10	10.5	9.9	9.1	8.3	7.4	6.5	5.3
0	10.5	9.8	9.1	8.3	7.3	6.4	5.3
5	10.5	9.8	9.1	8.2	7.3	6.4	5.2

Altitude: 1000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.2	9.6	8.8	8.0	7.1	6.2	5.1
-30	10.2	9.5	8.8	8.0	7.1	6.2	5.1
-20	10.3	9.6	8.8	8.0	7.1	6.2	5.1
-10	10.3	9.6	8.9	8.0	7.1	6.2	5.1
0	10.3	9.6	8.8	8.0	7.1	6.2	5.1
5	10.2	9.6	8.8	8.0	7.1	6.2	5.0



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 2000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.4	9.7	9.0	8.1	7.2	6.4	5.2
-30	10.4	9.7	8.9	8.1	7.2	6.3	5.2
-20	10.4	9.7	8.9	8.1	7.2	6.3	5.2
-10	10.4	9.7	8.9	8.1	7.2	6.3	5.1
0	10.4	9.7	8.9	8.1	7.2	6.3	5.1
5	10.0	9.3	8.6	7.8	6.8	6.0	4.8

Altitude: 2000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.8	9.2	8.4	7.7	6.8	5.9	4.8
-30	9.8	9.2	8.4	7.6	6.8	5.9	4.8
-20	9.9	9.2	8.5	7.7	6.8	5.9	4.8
-10	9.9	9.2	8.5	7.7	6.8	5.9	4.8
0	9.9	9.2	8.5	7.7	6.8	5.9	4.8
5	9.3	8.7	8.0	7.2	6.3	5.4	4.3

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TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 3000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	10.1	9.4	8.7	7.8	6.9	6.0	4.9
-30	10.1	9.4	8.6	7.8	6.9	6.0	4.9
-20	10.1	9.4	8.6	7.8	6.9	6.0	4.9
-10	10.1	9.4	8.6	7.8	6.8	6.0	4.8
0	10.1	9.4	8.6	7.8	6.8	5.9	4.8
5	9.3	8.6	7.8	6.9	5.9	5.1	4.0

Altitude: 3000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.4	8.8	8.1	7.3	6.4	5.5	4.4
-30	9.5	8.8	8.1	7.3	6.4	5.5	4.4
-20	9.5	8.8	8.1	7.3	6.4	5.5	4.4
-10	9.5	8.9	8.1	7.3	6.4	5.5	4.4
0	9.5	8.8	8.1	7.2	6.3	5.5	4.4
5	8.6	7.9	7.2	6.3	5.4	4.6	3.5



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 4000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.8	9.1	8.3	7.4	6.5	5.7	4.5
-30	9.7	9.1	8.3	7.4	6.5	5.6	4.5
-20	9.8	9.1	8.3	7.4	6.5	5.6	4.5
-10	9.8	9.1	8.3	7.4	6.5	5.6	4.5
0	9.4	8.7	7.9	7.0	6.0	5.2	4.1
5	8.5	7.8	6.9	5.9	5.1	4.3	-

Altitude: 4000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.1	8.5	7.7	6.9	6.0	5.1	4.1
-30	9.1	8.5	7.7	6.9	6.0	5.1	4.1
-20	9.2	8.5	7.7	6.9	6.0	5.2	4.1
-10	9.2	8.5	7.7	6.9	6.0	5.2	4.1
0	8.5	7.9	7.1	6.3	5.4	4.6	3.5
5	7.8	7.1	6.3	5.4	4.5	3.7	-

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TAKEOFF 2ND SEGMENT NET GRADIENT (%) **FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON**

PW535E ENGINES

Altitude: 5000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	9.4	8.7	7.9	7.0	6.1	5.3	4.2
-30	9.4	8.7	7.9	7.0	6.1	5.2	4.2
-20	9.4	8.7	7.9	7.0	6.1	5.2	4.2
-10	9.4	8.7	7.9	7.0	6.1	5.2	4.1
0	8.5	7.8	6.9	6.0	5.1	4.3	-
5	7.7	6.9	6.0	5.0	4.2	3.5	-

Altitude: 5000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	8.5	7.9	7.1	6.3	5.4	4.6	3.5
-30	8.6	7.9	7.1	6.3	5.4	4.6	3.5
-20	8.6	7.9	7.2	6.3	5.4	4.6	3.6
-10	8.7	8.0	7.2	6.3	5.4	4.6	3.6
0	7.5	6.8	6.0	5.1	4.3	3.5	-
5	6.9	6.1	5.2	4.3	3.5	2.8	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 6000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	8.9	8.2	7.4	6.5	5.6	4.8	3.7
-30	8.9	8.2	7.4	6.5	5.6	4.7	3.7
-20	8.9	8.2	7.4	6.5	5.5	4.7	3.7
-10	8.8	8.1	7.3	6.3	5.4	4.6	3.6
0	7.6	6.8	5.8	4.9	4.1	-	-
5	6.8	5.9	4.9	4.0	-	-	-

Altitude: 6000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	7.9	7.2	6.4	5.6	4.7	4.0	3.0
-30	7.9	7.2	6.5	5.6	4.7	4.0	3.0
-20	8.0	7.3	6.5	5.6	4.8	4.0	3.0
-10	7.8	7.2	6.4	5.5	4.6	3.9	2.9
0	6.5	5.7	4.8	4.0	3.2	-	-
5	5.7	4.9	4.0	3.1	-	-	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 7000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	8.4	7.6	6.8	5.8	4.9	4.2	-
-30	8.3	7.6	6.7	5.8	4.9	4.1	-
-20	8.4	7.6	6.7	5.8	4.9	4.1	-
-10	8.0	7.3	6.4	5.4	4.6	3.8	-
0	6.5	5.6	4.6	3.7	-	-	-
5	5.6	4.6	3.6	-	-	-	-

Altitude: 7000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	7.6	6.9	6.1	5.1	4.3	3.6	-
-30	7.6	6.9	6.0	5.1	4.3	3.5	-
-20	7.6	6.9	6.1	5.1	4.3	3.6	-
-10	7.2	6.5	5.6	4.7	3.9	3.2	-
0	5.7	4.8	3.9	3.0	-	-	-
5	4.8	3.8	2.9	-	-	-	-



TAKEOFF 2ND SEGMENT NET GRADIENT (%)

FLAP 2 – V₂ – GEAR UP – ANTI-ICE ON

PW535E ENGINES

Altitude: 8000 ft

LEVEL-OFF: 400 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	7.9	7.1	6.2	5.2	4.4	3.7	-
-30	7.9	7.1	6.2	5.2	4.4	3.6	-
-20	7.9	7.1	6.2	5.2	4.4	3.6	-
-10	7.2	6.4	5.4	4.5	3.7	-	-
0	5.5	4.5	3.6	-	-	-	-
5	4.4	3.4	-	-	-	-	-

Altitude: 8000 ft

LEVEL-OFF: 1500 ft

TEMP (°C)	WEIGHT (lb)						
	13200	14000	14800	15600	16400	17200	18387
-40	7.3	6.6	5.7	4.7	3.9	3.2	-
-30	7.3	6.6	5.7	4.7	3.9	3.2	-
-20	7.4	6.6	5.7	4.8	3.9	3.2	-
-10	6.7	5.9	5.0	4.1	3.3	-	-
0	5.1	4.1	3.2	-	-	-	-
5	4.1	3.1	-	-	-	-	-

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OBSTACLE CLEARANCE

Tables below present the net takeoff flight path for two different level-off heights 400 ft and 1500 ft AGL. Close-in obstacles are normally cleared considering a 400 ft level-off while distant obstacles are normally cleared considering a 1500 ft level-off.

To use the data presented, one should enter the second segment gradient obtained from previous tables and the obstacle information from aerodrome obstacle charts.

Obstacle distance and height must be corrected for runway slope and takeoff distance. If obstacle data is referred to the beginning of takeoff run (brakes release), obstacle height increases for a downhill slope and decreases for an uphill slope. Considering that the simplified takeoff data uses fixed V_1 for slopes other than zero, thus making the takeoff distance not balanced, correct the takeoff data as follows:

- For negative slope:

Corrected obstacle height = Obstacle height + Slope x Minimum required runway length.

- For positive slope or slope zero:

Do not apply any correction.

EXAMPLE

Given:

Airplane Weight	16400 lb
Takeoff Flap	1
Anti-Ice	OFF
Altitude	2000 ft
OAT	10°C
Level-Off Height	400 ft
Runway	DRY, SLOPE ZERO
Obstacle (from brakes release)	250 ft HIGH AT 5000 ft



Determine:

Takeoff Distance: 3278 ft

Corrected Obstacle Distance: 1722 ft (5000 - 3278)

Second Segment Net Gradient 9.3%

Required Horizontal Distance..... 2293 ft

Since obstacle distance (1722 ft) is greater than minimum required (2293 ft), the obstacle will be cleared.

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NET TAKEOFF FLIGHT PATH (400 FT LEVEL-OFF) FLAP 1 – V₂ – GEAR DOWN TO UP

PW535E ENGINES

HEIGHT ABOVE REF. POINT (ft)	2 ND SEGMENT NET GRADIENT (%)						
	16	14	12	10	8	6	4
	HORIZONTAL DISTANCE FROM REFERENCE POINT (ft)						
50	98	121	148	174	253	525	797
100	426	525	636	748	1033	1364	1866
150	748	902	1128	1282	1627	2184	2916
200	1102	1305	1581	1784	2273	3021	4113
250	1420	1679	1991	2293	2883	3838	5360
300	1732	2011	2404	2788	3513	4664	6622
350	2063	2375	2798	3280	4116	5504	7872
400	2352	2736	3234	3795	4736	6271	8561

NOTE: Reference point: end of takeoff distance.



NET TAKEOFF FLIGHT PATH (1500 FT LEVEL-OFF) FLAP 1 – V₂ – GEAR DOWN TO UP

PW535E ENGINES

HEIGHT ABOVE REF. POINT (ft)	2 ND SEGMENT NET GRADIENT (%)						
	16	14	12	10	8	6	4
	HORIZONTAL DISTANCE FROM REFERENCE POINT (ft)						
50	95	112	138	161	220	380	771
100	407	482	604	705	945	1240	1866
150	712	850	1050	1260	1538	2001	2916
200	1036	1237	1486	1699	2119	2752	3949
250	1345	1588	1889	2145	2729	3487	5074
300	1676	1942	2286	2654	3296	4264	6173
350	1968	2296	2690	3100	3884	5028	7344
400	2286	2640	3077	3575	4612	5825	8561
450	2594	3021	3500	4123	5225	6649	9719
500	2913	3339	3936	4592	5825	7380	10847
550	3254	3703	4366	5107	6340	8259	12015
600	3477	4067	4723	5615	7085	8990	13166
650	3805	4428	5084	6101	7610	9804	14465
700	4100	4789	5576	6593	8233	10627	15678
750	4461	5117	6032	7150	8987	11414	16859
800	4789	5445	6330	7544	9479	12274	18030
850	5084	5838	6757	8036	10037	13153	19418
900	5412	6199	7282	8594	10726	14006	20566
950	5707	6527	7708	9086	11447	14694	21910
1000	6101	6954	8134	9610	12005	15547	22862
1050	6298	7183	8561	10037	12595	16269	24272
1100	6593	7577	8889	10568	13120	17056	25446
1150	6986	8003	9414	10988	13743	17876	26699
1200	7183	8266	9745	11578	14498	18657	27841
1250	7583	8725	10217	12143	15154	19546	29025
1300	7872	9063	10571	12677	15846	20467	30120
1350	8203	9417	11018	13120	16393	21110	31373
1400	8558	9781	11460	13566	17007	21963	32439
1450	8849	9994	11782	14048	17646	22842	33879
1500	9122	10453	12202	14511	18214	23544	35096

NOTE: Reference point: end of takeoff distance.



NET TAKEOFF FLIGHT PATH (400 FT LEVEL-OFF) FLAP 2 – V₂ – GEAR DOWN TO UP

PW535E ENGINES

HEIGHT ABOVE REF. POINT (ft)	2 ND SEGMENT NET GRADIENT (%)						
	12	10	8	7	6	5	4
	HORIZONTAL DISTANCE FROM REFERENCE POINT (ft)						
50	148	190	276	338	453	741	1168
100	640	813	1181	1374	1594	1968	2378
150	1125	1397	1788	2060	2453	2962	3601
200	1542	1879	2427	2785	3257	3969	4799
250	1955	2371	3050	3451	4074	4890	6029
300	2368	2857	3628	4175	4900	5937	7344
350	2785	3346	4280	4858	5733	6914	8485
400	3198	3864	4871	5583	6544	7659	9351

NOTE: Reference point: end of takeoff distance.



NET TAKEOFF FLIGHT PATH (1500 FT LEVEL-OFF) FLAP 2 – V₂ – GEAR DOWN TO UP

PW535E ENGINES

HEIGHT ABOVE REF. POINT (ft)	2 ND SEGMENT NET GRADIENT (%)						
	12	10	8	7	6	5	4
	HORIZONTAL DISTANCE FROM REFERENCE POINT (ft)						
50	144	167	239	295	394	551	932
100	610	722	1040	1253	1496	1752	2106
150	1073	1260	1640	1945	2296	2650	3178
200	1486	1715	2214	2627	3146	3575	4234
250	1896	2158	2818	3336	3972	4510	5392
300	2289	2673	3375	4080	4789	5438	6432
350	2732	3175	3962	4769	5733	6327	7616
400	3132	3651	4651	5484	6544	7291	8771
450	3500	4123	5225	6196	7442	8180	9929
500	3998	4664	5855	6947	8220	9223	11096
550	4385	5130	6458	7682	9050	10148	12284
600	4808	5661	7095	8341	9879	10945	13340
650	5271	6173	7724	9122	10886	11883	14619
700	5665	6717	8384	9781	11608	12769	15767
750	6104	7150	8987	10496	12487	13806	16938
800	6530	7701	9597	11270	13343	14734	18319
850	6911	8157	10217	11942	14143	15711	19418
900	7393	8672	10893	12671	14980	16630	20684
950	7783	9223	11480	13435	15885	17646	21874
1000	8193	9719	12100	14127	16721	18532	22862
1050	8561	10178	12667	14803	17558	19483	24272
1100	8997	10676	13310	15465	18368	20503	25292
1150	9414	11195	13973	16190	19073	21287	26516
1200	9856	11657	14498	16892	19949	22320	27768
1250	10260	12156	15154	17650	20825	23337	29025
1300	10686	12677	15846	18338	21632	24433	30120
1350	11136	13150	16393	19073	22580	25243	31373
1400	11487	13648	17112	19752	23242	26283	32439
1450	11942	14179	17646	20523	24151	27349	33879
1500	12421	14662	18338	21218	25030	28493	35424

NOTE: Reference point: end of takeoff distance.



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DRIFTDOWN TABLES (NET GRADIENT)

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

In the event of an engine failure during cruise the maximum airplane operating altitude is affected. The Driftdown tables present the one engine ceiling according to the airplane weight and temperature.

The airplane should level-off at the ceiling altitude.

NOTE: Static air temperatures were used to calculate the tables.

EXAMPLE

Given:

Airplane Weight 16800 lb
Temperature ISA + 10°C
Ice Protection..... OFF

Determine:

Ceiling..... 21000 ft
Driftdown Speed 134 KIAS



PW535E ENGINES

NET GRADIENT = 0%

WEIGHT (lb)	DRIFT DOWN SPEED (KIAS)	CEILING ALTITUDE (x 100 ft) ICE PROTECTION (WINGSTAB+ENG) OFF				
		ISA - 10	ISA	ISA + 10	ISA + 20	ISA + 35
12400	116	300	340	330	300	210
12800	118	300	330	320	280	200
13200	120	300	330	310	270	200
13600	122	300	320	300	250	190
14000	123	300	310	290	240	170
14400	125	300	300	280	230	160
14800	127	300	300	270	220	150
15200	128	290	290	250	210	130
15600	130	280	280	240	210	120
16000	131	270	270	230	200	110
16400	133	260	260	220	200	100
16800	134	250	250	210	190	90
17200	136	240	240	210	180	80
17600	137	240	240	200	170	60

CAUTION: SHADED AREAS REPRESENT THE CEILING ALTITUDES CALCULATED CONSIDERING THE MINIMUM MACH IN REGION 2 OF OPERATIONAL ENVELOPE.


PW535E ENGINES
NET GRADIENT = 0%

WEIGHT (lb)	DRIFT DOWN SPEED (KIAS)	CEILING ALTITUDE (x 100 ft) ICE PROTECTION (WINGSTAB+ENG) ON				
		ISA - 30	ISA - 20	ISA - 10	ISA	ISA + 10
12400	165	100	150	150	150	150
12800	165	100	150	150	150	150
13200	165	100	150	150	150	150
13600	165	100	150	150	150	150
14000	165	100	150	150	150	150
14400	165	100	150	150	150	140
14800	165	100	150	150	150	130
15200	165	100	150	150	150	120
15600	165	100	150	150	150	110
16000	165	100	150	150	150	110
16400	165	100	150	150	150	100
16800	165	100	150	150	150	100
17200	165	100	150	150	150	140

NOTE: The data above can be calculated only by OPERA versions 1.0, so it is applicable to airplanes Pre-Mod. SB 505-00-0001.

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PW535E ENGINES

NET GRADIENT = 0%

WEIGHT (lb)	DRIFT DOWN SPEED (KIAS)	CEILING ALTITUDE (x 100 ft) ICE PROTECTION (WINGSTAB+ENG) ON				
		ISA - 30	ISA - 20	ISA - 10	ISA	ISA + 10
12400	127	120	150	150	150	150
12800	129	120	150	150	150	150
13200	131	120	150	150	150	150
13600	133	120	150	150	150	150
14000	135	120	150	150	150	150
14400	137	120	150	150	150	150
14800	139	120	150	150	150	140
15200	141	120	150	150	150	130
15600	143	120	150	150	150	120
16000	145	120	150	150	150	120
16400	147	120	150	150	150	110
16800	149	120	150	150	150	100
17200	151	120	150	150	140	-
17600	153	120	150	150	130	-

NOTE: Applicable to airplanes Post-Mod. SB 505-00-0001 or with an equivalent modification factory incorporated.



ENROUTE GRADIENT

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

This table presents the corrected gradient value due to the wind effect on the enroute gradient of climb/descent with one engine inoperative.

The wing effect correction is applicable for all weights, temperature and altitudes, considering the landing gear and flap up, and must replace the gradient for zero wind.

EXAMPLE

Given:

Headwind Wind 40 kt

Gradient for Zero Wind 6.0%

Determine:

Gradient for 40 kt (headwind) 6.8%

CORRECTED GRADIENT %						
ZERO WIND	HEADWIND (kt)			TAILWIND (kt)		
	60	40	20	20	40	60
-6.0	-8.6	-7.3	-6.4	-6.0	-5.7	-5.6
-3.0	-4.8	-4.1	-3.6	-3.0	-3.0	-2.8
0.0	0.0	0.0	0.0	0.0	0.0	0.0
3.0	3.4	3.3	3.1	2.3	2.0	1.8
6.0	7.3	6.8	6.3	4.8	3.9	3.6
9.0	11.4	10.5	9.6	7.1	5.7	4.9
12.0	15.7	14.1	12.9	9.2	7.5	6.5
15.0	20.1	18.2	16.3	11.4	9.3	7.8
18.0	25.0	22.2	19.8	13.5	10.8	9.0



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APPROACH DATA

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

APPROACH CLIMB, LANDING CLIMB AND LANDING REFERENCE SPEEDS

- NOTE:** - For Approach Flap 1, Landing Flap 3 and Engine Ice Protection OFF/ON – Wingstab OFF, the Landing Climb Speed is equal to the Landing Reference Speed.
- For Approach Flap 1, Landing Flap 3 and Engine Ice Protection ON – Wingstab ON (SWPS Ice Speed), the Approach Climb Speed, Landing Climb Speed and Landing Reference Speed have the same value.
 - For Approach Flap 2, Landing Flap FULL and Engine Ice Protection OFF/ON – Wingstab OFF, the Approach Climb Speed, Landing Climb Speed and Landing Reference Speed have the same value.
 - For Approach Flap 2, Landing Flap FULL and Engine Ice Protection ON – Wingstab ON (SWPS Ice Speed), the Approach Climb Speed, Landing Climb Speed and Landing Reference Speed have the same value.

EXAMPLE

Given:

Airplane Landing Weight	15600 lb
Approach Flap	1
Landing Flap	3
Engine Ice Protection	OFF
Wingstab	OFF

Determine:

Approach Climb Speed	116 KIAS
Landing Climb Speed	113 KIAS
V _{REF}	113 KIAS
V _{FS}	127 KIAS



APPROACH FLAP 1 AND LANDING FLAP 3 ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF

PW535E ENGINES

WEIGHT (lb)	APPROACH	LANDING (CLIMB/REFERENCE)	V _{FS} – KIAS
	FLAP 1	FLAP 3	
	V _{AC} – KIAS	V _{REF} – KIAS	
12400	104	101	113
12800	105	102	115
13200	107	104	117
13600	109	105	119
14000	110	107	121
14400	111	108	122
14800	113	110	124
15200	114	111	126
15600	116	113	127
16000	117	114	129
16400	119	115	130
16800	120	117	132
17200	121	118	133
17600	122	119	135
18000	124	121	136
18387	125	122	138

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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APPROACH FLAP 1 AND LANDING FLAP 3 ENGINE ICE PROTECTION ON – WINGSTAB ON (SWPS ICE SPEED)

PW535E ENGINES

WEIGHT (lb)	APPROACH CLIMB/LANDING (CLIMB/REFERENCE)	V _{FS} – KIAS
	SPEED – KIAS	
12400	108	126
12800	110	128
13200	111	130
13600	113	132
14000	115	134
14400	116	136
14800	118	138
15200	119	140
15600	121	142
16000	122	143
16400	124	145
16800	125	147
17200	127	149
17600	128	150
18000	130	152
18387	131	154

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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APPROACH FLAP 2 AND LANDING FLAP FULL ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF

PW535E ENGINES

WEIGHT (lb)	APPROACH	LANDING (CLIMB/REFERENCE)	V _{FS} – KIAS
	FLAP 2	FLAP FULL	
	V _{AC} – KIAS	V _{REF} – KIAS	
12400	100	100	113
12800	101	101	115
13200	103	103	117
13600	104	104	119
14000	106	106	121
14400	107	107	122
14800	109	109	124
15200	110	110	126
15600	111	111	127
16000	113	113	129
16400	114	114	130
16800	115	115	132
17200	117	117	133
17600	118	118	135
18000	119	119	136
18387	121	121	138

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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APPROACH FLAP 2 AND LANDING FLAP FULL ENGINE ICE PROTECTION ON – WINGSTAB ON (SWPS ICE SPEED)

PW535E ENGINES

WEIGHT (lb)	APPROACH CLIMB/LANDING (CLIMB/REFERENCE)	V _{FS} – KIAS
	SPEED – KIAS	
12400	106	126
12800	108	128
13200	109	130
13600	111	132
14000	112	134
14400	114	136
14800	115	138
15200	117	140
15600	118	142
16000	119	143
16400	121	145
16800	122	147
17200	124	149
17600	125	150
18000	126	152
18387	128	154

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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APPROACH CLIMB GRADIENT

EXAMPLE

Given:

Airplane Landing Weight..... 15600 lb
Approach Flap 1
Engine Ice Protection..... OFF
Wingstab OFF
Altitude 2000 ft
OAT..... 10°C

Determine:

Approach Climb Gradient 11.2%


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: -1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	16.9	16.0	15.2	14.4	13.7	13.0	12.4	11.8
-35	16.9	16.0	15.2	14.4	13.7	13.0	12.4	11.8
-30	16.9	16.0	15.2	14.5	13.7	13.1	12.4	11.8
-25	16.9	16.1	15.2	14.5	13.8	13.1	12.4	11.8
-20	17.0	16.1	15.3	14.5	13.8	13.1	12.4	11.8
-15	17.0	16.1	15.3	14.5	13.8	13.1	12.4	11.8
-10	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.8
-5	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
0	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
5	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.8
10	16.9	16.1	15.3	14.5	13.8	13.1	12.4	11.8
15	16.9	16.1	15.2	14.5	13.7	13.1	12.4	11.8
20	16.9	16.0	15.2	14.4	13.7	13.0	12.4	11.8
25	16.9	16.0	15.2	14.4	13.7	13.0	12.4	11.8
30	16.1	15.3	14.5	13.7	13.0	12.4	11.8	11.2
35	14.5	13.7	13.0	12.3	11.6	11.0	10.4	9.9
40	13.0	12.2	11.5	10.9	10.3	9.7	9.2	8.7
45	11.6	10.9	10.2	9.6	9.1	8.5	8.0	7.5
50	10.2	9.6	9.0	8.4	7.9	7.4	6.9	6.5
52	9.7	9.1	8.5	8.0	7.4	7.0	6.5	6.1

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: -1000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	11.2	10.7	10.2	9.7	9.3	8.8	8.4	8.1
-35	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
-30	11.2	10.7	10.2	9.8	9.3	8.9	8.5	8.1
-25	11.3	10.7	10.2	9.8	9.3	8.9	8.5	8.1
-20	11.3	10.7	10.3	9.8	9.3	8.9	8.5	8.1
-15	11.3	10.8	10.3	9.8	9.3	8.9	8.5	8.1
-10	11.3	10.8	10.3	9.8	9.4	8.9	8.5	8.2
-5	11.3	10.8	10.3	9.8	9.4	8.9	8.5	8.2
0	11.3	10.8	10.3	9.8	9.4	8.9	8.5	8.2
5	11.3	10.8	10.3	9.8	9.4	8.9	8.5	8.2
10	11.3	10.8	10.3	9.8	9.3	8.9	8.5	8.1
15	11.3	10.7	10.2	9.8	9.3	8.9	8.5	8.1
20	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
25	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
30	10.6	10.1	9.7	9.2	8.8	8.4	8.0	7.6
35	9.4	8.9	8.5	8.0	7.6	7.3	6.9	6.6
40	8.2	7.8	7.3	7.0	6.6	6.2	5.9	5.6
45	7.1	6.7	6.3	6.0	5.6	5.3	5.0	4.7
50	6.1	5.7	5.3	5.0	4.7	4.4	4.1	3.8
52	5.7	5.3	5.0	4.6	4.3	4.0	3.7	3.5

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 1**
PW535E ENGINES

Altitude: 0 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	16.9	16.1	15.3	14.5	13.8	13.1	12.4	11.8
-35	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
-30	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
-25	17.0	16.1	15.3	14.6	13.8	13.2	12.5	11.9
-20	17.0	16.1	15.3	14.6	13.8	13.2	12.5	11.9
-15	17.0	16.1	15.3	14.6	13.8	13.2	12.5	11.9
-10	17.0	16.2	15.3	14.6	13.9	13.2	12.5	11.9
-5	17.0	16.2	15.3	14.6	13.9	13.2	12.5	11.9
0	17.0	16.2	15.3	14.6	13.9	13.2	12.5	11.9
5	17.0	16.1	15.3	14.6	13.8	13.2	12.5	11.9
10	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
15	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
20	16.9	16.1	15.2	14.5	13.8	13.1	12.4	11.8
25	16.3	15.5	14.7	14.0	13.2	12.6	12.0	11.4
30	15.0	14.2	13.5	12.8	12.1	11.5	10.9	10.3
35	13.5	12.7	12.0	11.3	10.7	10.1	9.6	9.0
40	12.0	11.3	10.6	10.0	9.4	8.9	8.4	7.9
45	10.6	10.0	9.4	8.8	8.3	7.8	7.3	6.8
50	9.4	8.8	8.2	7.6	7.1	6.7	6.2	5.8

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 0 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	11.3	10.8	10.3	9.8	9.4	8.9	8.5	8.2
-35	11.3	10.8	10.3	9.8	9.4	8.9	8.5	8.2
-30	11.3	10.8	10.3	9.8	9.4	9.0	8.6	8.2
-25	11.3	10.8	10.3	9.9	9.4	9.0	8.6	8.2
-20	11.4	10.8	10.3	9.9	9.4	9.0	8.6	8.2
-15	11.4	10.8	10.3	9.9	9.4	9.0	8.6	8.2
-10	11.4	10.8	10.3	9.9	9.4	9.0	8.6	8.2
-5	11.4	10.8	10.4	9.9	9.4	9.0	8.6	8.2
0	11.4	10.9	10.4	9.9	9.4	9.0	8.6	8.2
5	11.4	10.8	10.3	9.9	9.4	9.0	8.6	8.2
10	11.3	10.8	10.3	9.9	9.4	9.0	8.6	8.2
15	11.3	10.8	10.3	9.8	9.4	9.0	8.6	8.2
20	11.3	10.8	10.3	9.8	9.4	8.9	8.5	8.2
25	10.8	10.3	9.8	9.4	9.0	8.5	8.1	7.8
30	9.8	9.3	8.9	8.4	8.0	7.6	7.2	6.9
35	8.6	8.1	7.7	7.3	6.9	6.5	6.2	5.9
40	7.5	7.0	6.6	6.3	5.9	5.6	5.2	5.0
45	6.4	6.0	5.6	5.3	5.0	4.7	4.3	4.1
50	5.4	5.0	4.7	4.4	4.1	3.8	3.5	3.3

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	16.9	16.0	15.2	14.5	13.7	13.1	12.4	11.8
-35	16.9	16.1	15.2	14.5	13.8	13.1	12.4	11.9
-30	16.9	16.1	15.3	14.5	13.8	13.1	12.5	11.9
-25	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
-20	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
-15	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
-10	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
-5	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
0	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
5	17.0	16.1	15.3	14.5	13.8	13.1	12.5	11.9
10	16.9	16.1	15.3	14.5	13.8	13.1	12.5	11.9
15	16.9	16.1	15.2	14.5	13.8	13.1	12.5	11.9
20	16.5	15.6	14.8	14.1	13.4	12.7	12.1	11.5
25	15.2	14.4	13.7	13.0	12.3	11.7	11.1	10.5
30	13.9	13.1	12.4	11.7	11.1	10.5	9.9	9.4
35	12.4	11.7	11.0	10.4	9.8	9.2	8.7	8.2
40	11.0	10.4	9.7	9.2	8.6	8.1	7.6	7.1
45	9.7	9.1	8.5	8.0	7.5	7.0	6.5	6.1
48	9.0	8.4	7.8	7.3	6.8	6.3	5.9	5.5

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 1000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	11.3	10.8	10.3	9.8	9.4	9.0	8.6	8.2
-35	11.3	10.8	10.3	9.8	9.4	9.0	8.6	8.2
-30	11.3	10.8	10.3	9.8	9.4	9.0	8.6	8.2
-25	11.3	10.8	10.3	9.9	9.4	9.0	8.6	8.2
-20	11.3	10.8	10.3	9.9	9.4	9.0	8.6	8.2
-15	11.4	10.8	10.3	9.9	9.4	9.0	8.6	8.3
-10	11.4	10.8	10.4	9.9	9.4	9.0	8.6	8.3
-5	11.4	10.8	10.4	9.9	9.4	9.0	8.6	8.3
0	11.4	10.8	10.4	9.9	9.4	9.0	8.6	8.3
5	11.3	10.8	10.3	9.9	9.4	9.0	8.6	8.3
10	11.3	10.8	10.3	9.9	9.4	9.0	8.6	8.2
15	11.3	10.8	10.3	9.8	9.4	9.0	8.6	8.2
20	10.9	10.4	9.9	9.5	9.0	8.6	8.2	7.9
25	10.0	9.5	9.0	8.6	8.2	7.8	7.4	7.1
30	8.9	8.5	8.0	7.6	7.2	6.9	6.5	6.2
35	7.8	7.3	6.9	6.5	6.2	5.8	5.5	5.2
40	6.7	6.3	5.9	5.6	5.2	4.9	4.6	4.3
45	5.7	5.3	5.0	4.6	4.3	4.0	3.7	3.5
48	5.1	4.7	4.4	4.1	3.8	3.5	3.2	3.0

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 1**
PW535E ENGINES
Altitude: 2000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	16.7	15.8	15.0	14.3	13.5	12.9	12.2	11.7
-35	16.7	15.8	15.0	14.3	13.6	12.9	12.3	11.7
-30	16.7	15.8	15.0	14.3	13.6	12.9	12.3	11.7
-25	16.7	15.9	15.1	14.3	13.6	12.9	12.3	11.7
-20	16.7	15.9	15.1	14.3	13.6	12.9	12.3	11.7
-15	16.7	15.9	15.1	14.3	13.6	12.9	12.3	11.7
-10	16.8	15.9	15.1	14.3	13.6	13.0	12.3	11.7
-5	16.8	15.9	15.1	14.3	13.6	12.9	12.3	11.7
0	16.8	15.9	15.1	14.3	13.6	12.9	12.3	11.7
5	16.7	15.9	15.1	14.3	13.6	12.9	12.3	11.7
10	16.7	15.9	15.1	14.3	13.6	12.9	12.3	11.7
15	16.6	15.7	14.9	14.2	13.4	12.8	12.1	11.6
20	15.4	14.5	13.8	13.1	12.4	11.8	11.1	10.6
25	14.2	13.4	12.7	12.0	11.3	10.7	10.2	9.6
30	12.8	12.1	11.4	10.8	10.2	9.6	9.0	8.5
35	11.4	10.7	10.1	9.5	8.9	8.4	7.9	7.4
40	10.1	9.4	8.9	8.3	7.8	7.3	6.8	6.4
45	8.8	8.2	7.7	7.2	6.7	6.2	5.8	5.4
46	8.6	8.0	7.5	6.9	6.5	6.0	5.6	5.2

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1

PW535E ENGINES

Altitude: 2000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	11.1	10.6	10.1	9.7	9.2	8.8	8.4	8.1
-35	11.1	10.6	10.1	9.7	9.2	8.8	8.4	8.1
-30	11.1	10.6	10.2	9.7	9.3	8.8	8.4	8.1
-25	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
-20	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
-15	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
-10	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
-5	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
0	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
5	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
10	11.2	10.7	10.2	9.7	9.3	8.9	8.5	8.1
15	11.0	10.5	10.0	9.6	9.1	8.7	8.3	8.0
20	10.1	9.6	9.1	8.7	8.3	7.9	7.5	7.2
25	9.1	8.7	8.3	7.8	7.4	7.1	6.7	6.4
30	8.1	7.7	7.2	6.9	6.5	6.1	5.8	5.5
35	7.0	6.6	6.2	5.8	5.5	5.2	4.8	4.6
40	6.0	5.6	5.2	4.9	4.6	4.3	4.0	3.7
45	5.0	4.6	4.3	4.0	3.7	3.4	3.1	2.9
46	4.8	4.5	4.1	3.8	3.5	3.2	3.0	2.7

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 3000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	16.2	15.4	14.6	13.8	13.1	12.5	11.9	11.3
-35	16.2	15.4	14.6	13.9	13.2	12.5	11.9	11.3
-30	16.3	15.4	14.6	13.9	13.2	12.5	11.9	11.3
-25	16.3	15.4	14.6	13.9	13.2	12.5	11.9	11.3
-20	16.3	15.4	14.7	13.9	13.2	12.5	11.9	11.3
-15	16.3	15.5	14.7	13.9	13.2	12.6	11.9	11.4
-10	16.3	15.5	14.7	13.9	13.2	12.6	11.9	11.4
-5	16.3	15.5	14.7	13.9	13.2	12.6	11.9	11.4
0	16.3	15.5	14.7	13.9	13.2	12.6	11.9	11.4
5	16.3	15.4	14.7	13.9	13.2	12.6	11.9	11.3
10	16.3	15.4	14.6	13.9	13.2	12.5	11.9	11.3
15	15.5	14.6	13.9	13.2	12.5	11.8	11.2	10.7
20	14.3	13.5	12.8	12.1	11.5	10.8	10.3	9.7
25	13.1	12.4	11.7	11.1	10.5	9.9	9.3	8.8
30	11.8	11.1	10.5	9.9	9.3	8.7	8.2	7.7
35	10.4	9.8	9.2	8.6	8.1	7.6	7.1	6.6
40	9.2	8.6	8.0	7.5	7.0	6.5	6.0	5.6
44	8.2	7.6	7.1	6.6	6.1	5.7	5.2	4.9

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 3000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	10.8	10.3	9.8	9.3	8.9	8.5	8.1	7.8
-35	10.8	10.3	9.8	9.4	8.9	8.5	8.1	7.8
-30	10.8	10.3	9.8	9.4	8.9	8.5	8.1	7.8
-25	10.8	10.3	9.8	9.4	9.0	8.5	8.2	7.8
-20	10.8	10.3	9.8	9.4	9.0	8.6	8.2	7.8
-15	10.8	10.3	9.8	9.4	9.0	8.6	8.2	7.8
-10	10.8	10.3	9.9	9.4	9.0	8.6	8.2	7.8
-5	10.8	10.3	9.9	9.4	9.0	8.6	8.2	7.8
0	10.8	10.3	9.9	9.4	9.0	8.6	8.2	7.8
5	10.8	10.3	9.8	9.4	9.0	8.6	8.2	7.8
10	10.8	10.3	9.8	9.4	9.0	8.5	8.2	7.8
15	10.2	9.7	9.2	8.8	8.4	8.0	7.6	7.2
20	9.3	8.8	8.4	7.9	7.5	7.2	6.8	6.5
25	8.4	7.9	7.5	7.1	6.7	6.4	6.0	5.7
30	7.3	6.9	6.5	6.1	5.8	5.4	5.1	4.8
35	6.2	5.9	5.5	5.1	4.8	4.5	4.2	3.9
40	5.3	4.9	4.6	4.2	3.9	3.6	3.4	3.1
44	4.5	4.2	3.8	3.5	3.3	3.0	2.7	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 4000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	15.7	14.9	14.1	13.4	12.7	12.0	11.4	10.9
-35	15.7	14.9	14.1	13.4	12.7	12.1	11.5	10.9
-30	15.7	14.9	14.1	13.4	12.7	12.1	11.5	10.9
-25	15.8	14.9	14.2	13.4	12.7	12.1	11.5	10.9
-20	15.8	14.9	14.2	13.4	12.7	12.1	11.5	10.9
-15	15.8	14.9	14.2	13.4	12.8	12.1	11.5	10.9
-10	15.8	15.0	14.2	13.4	12.8	12.1	11.5	10.9
-5	15.8	15.0	14.2	13.5	12.8	12.1	11.5	10.9
0	15.8	15.0	14.2	13.5	12.8	12.1	11.5	10.9
5	15.8	14.9	14.2	13.4	12.8	12.1	11.5	10.9
10	15.5	14.7	14.0	13.2	12.6	11.9	11.3	10.7
15	14.4	13.6	12.9	12.2	11.6	10.9	10.4	9.8
20	13.3	12.6	11.9	11.2	10.6	10.0	9.5	8.9
25	12.2	11.5	10.8	10.2	9.6	9.1	8.5	8.1
30	10.9	10.2	9.6	9.0	8.4	7.9	7.4	7.0
35	9.5	8.9	8.3	7.8	7.3	6.8	6.4	5.9
40	8.3	7.7	7.2	6.7	6.2	5.8	5.4	5.0
42	7.8	7.3	6.8	6.3	5.8	5.4	5.0	4.6

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 4000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	10.4	9.9	9.4	9.0	8.5	8.1	7.8	7.4
-35	10.4	9.9	9.4	9.0	8.6	8.2	7.8	7.4
-30	10.4	9.9	9.4	9.0	8.6	8.2	7.8	7.5
-25	10.4	9.9	9.4	9.0	8.6	8.2	7.8	7.5
-20	10.4	9.9	9.5	9.0	8.6	8.2	7.8	7.5
-15	10.4	9.9	9.5	9.0	8.6	8.2	7.8	7.5
-10	10.4	9.9	9.5	9.0	8.6	8.2	7.8	7.5
-5	10.4	9.9	9.5	9.0	8.6	8.2	7.8	7.5
0	10.4	9.9	9.5	9.0	8.6	8.2	7.8	7.5
5	10.4	9.9	9.5	9.0	8.6	8.2	7.8	7.5
10	10.2	9.7	9.3	8.8	8.4	8.0	7.6	7.3
15	9.3	8.9	8.4	8.0	7.6	7.2	6.9	6.6
20	8.5	8.0	7.6	7.2	6.9	6.5	6.1	5.8
25	7.6	7.2	6.8	6.4	6.1	5.7	5.4	5.1
30	6.6	6.2	5.8	5.4	5.1	4.8	4.5	4.2
35	5.5	5.2	4.8	4.5	4.2	3.9	3.6	3.4
40	4.6	4.3	3.9	3.6	3.3	3.1	2.8	2.6
42	4.2	3.9	3.6	3.3	3.0	2.7	2.5	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 1**
PW535E ENGINES
Altitude: 5000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	15.1	14.3	13.6	12.9	12.2	11.6	11.0	10.4
-35	15.2	14.3	13.6	12.9	12.2	11.6	11.0	10.4
-30	15.2	14.4	13.6	12.9	12.2	11.6	11.0	10.5
-25	15.2	14.4	13.6	12.9	12.3	11.6	11.0	10.5
-20	15.2	14.4	13.6	12.9	12.3	11.6	11.0	10.5
-15	15.2	14.4	13.6	12.9	12.3	11.6	11.0	10.5
-10	15.2	14.4	13.7	12.9	12.3	11.6	11.1	10.5
-5	15.2	14.4	13.7	12.9	12.3	11.6	11.1	10.5
0	15.2	14.4	13.7	13.0	12.3	11.7	11.1	10.5
5	15.2	14.4	13.6	12.9	12.3	11.6	11.0	10.5
10	14.4	13.6	12.9	12.2	11.6	11.0	10.4	9.8
15	13.3	12.6	11.9	11.2	10.6	10.0	9.5	9.0
20	12.3	11.6	10.9	10.3	9.7	9.2	8.6	8.1
25	11.2	10.5	9.9	9.3	8.8	8.2	7.7	7.3
30	9.9	9.3	8.7	8.1	7.6	7.1	6.7	6.2
35	8.7	8.1	7.5	7.0	6.5	6.1	5.6	5.2
40	7.5	7.0	6.4	6.0	5.5	5.1	4.7	4.3

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 5000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.9	9.4	9.0	8.6	8.2	7.8	7.4	7.1
-35	9.9	9.5	9.0	8.6	8.2	7.8	7.4	7.1
-30	10.0	9.5	9.0	8.6	8.2	7.8	7.4	7.1
-25	10.0	9.5	9.0	8.6	8.2	7.8	7.4	7.1
-20	10.0	9.5	9.0	8.6	8.2	7.8	7.4	7.1
-15	10.0	9.5	9.1	8.6	8.2	7.8	7.5	7.1
-10	10.0	9.5	9.1	8.6	8.2	7.8	7.5	7.1
-5	10.0	9.5	9.1	8.6	8.2	7.8	7.5	7.1
0	10.0	9.5	9.1	8.6	8.2	7.8	7.5	7.1
5	10.0	9.5	9.1	8.6	8.2	7.8	7.5	7.1
10	9.3	8.9	8.4	8.0	7.6	7.2	6.9	6.6
15	8.5	8.1	7.6	7.2	6.9	6.5	6.2	5.9
20	7.7	7.3	6.9	6.5	6.1	5.8	5.5	5.2
25	6.8	6.4	6.1	5.7	5.4	5.0	4.7	4.5
30	5.8	5.5	5.1	4.8	4.5	4.2	3.9	3.6
35	4.9	4.5	4.2	3.9	3.6	3.3	3.0	2.8
40	4.0	3.6	3.3	3.0	2.8	2.5	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 1**
PW535E ENGINES
Altitude: 6000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.4	13.6	12.9	12.2	11.6	11.0	10.4	9.8
-35	14.4	13.7	12.9	12.2	11.6	11.0	10.4	9.9
-30	14.5	13.7	12.9	12.3	11.6	11.0	10.4	9.9
-25	14.5	13.7	13.0	12.3	11.6	11.0	10.4	9.9
-20	14.5	13.7	13.0	12.3	11.6	11.0	10.5	9.9
-15	14.5	13.7	13.0	12.3	11.6	11.0	10.5	9.9
-10	14.5	13.7	13.0	12.3	11.7	11.0	10.5	9.9
-5	14.5	13.7	13.0	12.3	11.7	11.1	10.5	9.9
0	14.5	13.7	13.0	12.3	11.7	11.1	10.5	9.9
5	14.2	13.4	12.7	12.0	11.4	10.8	10.2	9.7
10	13.1	12.4	11.7	11.1	10.5	9.9	9.3	8.8
15	12.2	11.5	10.8	10.2	9.6	9.0	8.5	8.0
20	11.2	10.5	9.9	9.3	8.7	8.2	7.7	7.2
25	10.1	9.5	8.9	8.3	7.8	7.3	6.8	6.4
30	8.9	8.3	7.7	7.2	6.7	6.3	5.8	5.4
35	7.7	7.2	6.6	6.2	5.7	5.3	4.9	4.5
38	7.0	6.5	6.0	5.5	5.1	4.7	4.3	3.9

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1

PW535E ENGINES

Altitude: 6000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.4	8.9	8.5	8.0	7.7	7.3	6.9	6.6
-35	9.4	8.9	8.5	8.1	7.7	7.3	6.9	6.6
-30	9.4	8.9	8.5	8.1	7.7	7.3	7.0	6.6
-25	9.4	9.0	8.5	8.1	7.7	7.3	7.0	6.6
-20	9.4	9.0	8.5	8.1	7.7	7.3	7.0	6.7
-15	9.4	9.0	8.5	8.1	7.7	7.3	7.0	6.7
-10	9.4	9.0	8.5	8.1	7.7	7.4	7.0	6.7
-5	9.5	9.0	8.6	8.1	7.7	7.4	7.0	6.7
0	9.5	9.0	8.6	8.1	7.7	7.4	7.0	6.7
5	9.2	8.7	8.3	7.9	7.5	7.1	6.7	6.4
10	8.4	7.9	7.5	7.1	6.7	6.4	6.0	5.7
15	7.6	7.2	6.8	6.4	6.0	5.7	5.4	5.1
20	6.8	6.4	6.0	5.7	5.4	5.0	4.7	4.4
25	6.0	5.6	5.3	4.9	4.6	4.3	4.0	3.7
30	5.0	4.7	4.4	4.0	3.7	3.5	3.2	2.9
35	4.1	3.8	3.5	3.2	2.9	2.7	2.4	-
38	3.6	3.3	3.0	2.7	2.4	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 1**
PW535E ENGINES
Altitude: 7000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	13.5	12.8	12.1	11.4	10.8	10.2	9.6	9.1
-35	13.5	12.8	12.1	11.4	10.8	10.2	9.7	9.1
-30	13.6	12.8	12.1	11.4	10.8	10.2	9.7	9.2
-25	13.6	12.8	12.1	11.5	10.8	10.3	9.7	9.2
-20	13.6	12.8	12.1	11.5	10.8	10.3	9.7	9.2
-15	13.6	12.9	12.2	11.5	10.9	10.3	9.7	9.2
-10	13.6	12.9	12.2	11.5	10.9	10.3	9.7	9.2
-5	13.6	12.9	12.2	11.5	10.9	10.3	9.7	9.2
0	13.6	12.9	12.2	11.5	10.9	10.3	9.7	9.2
5	12.8	12.1	11.4	10.8	10.2	9.6	9.1	8.6
10	11.9	11.2	10.5	9.9	9.4	8.8	8.3	7.8
15	11.0	10.4	9.7	9.2	8.6	8.1	7.6	7.1
20	10.2	9.5	8.9	8.4	7.8	7.3	6.9	6.4
25	9.0	8.4	7.9	7.4	6.9	6.4	6.0	5.5
30	7.8	7.3	6.7	6.3	5.8	5.4	4.9	4.6
35	6.7	6.2	5.7	5.2	4.8	4.4	4.0	3.6
36	6.5	5.9	5.5	5.0	4.6	4.2	3.8	3.5

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 7000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.7	8.2	7.8	7.4	7.0	6.7	6.3	6.0
-35	8.7	8.2	7.8	7.4	7.0	6.7	6.3	6.0
-30	8.7	8.3	7.8	7.4	7.1	6.7	6.4	6.0
-25	8.7	8.3	7.9	7.5	7.1	6.7	6.4	6.1
-20	8.7	8.3	7.9	7.5	7.1	6.7	6.4	6.1
-15	8.7	8.3	7.9	7.5	7.1	6.7	6.4	6.1
-10	8.7	8.3	7.9	7.5	7.1	6.7	6.4	6.1
-5	8.8	8.3	7.9	7.5	7.1	6.7	6.4	6.1
0	8.8	8.3	7.9	7.5	7.1	6.7	6.4	6.1
5	8.1	7.7	7.3	6.9	6.5	6.2	5.8	5.5
10	7.4	7.0	6.6	6.2	5.9	5.5	5.2	4.9
15	6.7	6.3	6.0	5.6	5.3	4.9	4.6	4.4
20	6.0	5.7	5.3	5.0	4.6	4.3	4.0	3.8
25	5.2	4.8	4.5	4.2	3.9	3.6	3.3	3.0
30	4.2	3.9	3.6	3.3	3.0	2.7	2.5	2.2
35	3.3	3.0	2.7	2.5	2.2	-	-	-
36	3.2	2.9	2.6	2.3	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 1**
PW535E ENGINES
Altitude: 8000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	12.8	12.1	11.4	10.8	10.2	9.6	9.1	8.6
-35	12.8	12.1	11.4	10.8	10.2	9.6	9.1	8.6
-30	12.9	12.1	11.5	10.8	10.2	9.6	9.1	8.6
-25	12.9	12.2	11.5	10.8	10.2	9.7	9.1	8.6
-20	12.9	12.2	11.5	10.8	10.2	9.7	9.1	8.6
-15	12.9	12.2	11.5	10.8	10.2	9.7	9.1	8.6
-10	12.9	12.2	11.5	10.8	10.2	9.7	9.1	8.6
-5	12.9	12.2	11.5	10.9	10.2	9.7	9.1	8.6
0	12.6	11.9	11.2	10.6	10.0	9.4	8.9	8.4
5	11.7	11.0	10.3	9.7	9.2	8.6	8.1	7.7
10	10.8	10.2	9.6	9.0	8.4	7.9	7.4	7.0
15	10.2	9.6	9.0	8.4	7.9	7.4	6.9	6.5
20	9.3	8.7	8.1	7.6	7.1	6.6	6.2	5.7
25	8.0	7.5	6.9	6.4	6.0	5.5	5.1	4.7
30	6.8	6.3	5.8	5.3	4.9	4.5	4.1	3.7
34	5.9	5.4	5.0	4.5	4.1	3.7	3.4	3.0



OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 8000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.1	7.7	7.3	6.9	6.5	6.2	5.9	5.6
-35	8.1	7.7	7.3	6.9	6.6	6.2	5.9	5.6
-30	8.2	7.7	7.3	6.9	6.6	6.2	5.9	5.6
-25	8.2	7.7	7.3	6.9	6.6	6.2	5.9	5.6
-20	8.2	7.7	7.3	7.0	6.6	6.2	5.9	5.6
-15	8.2	7.8	7.3	7.0	6.6	6.2	5.9	5.6
-10	8.2	7.8	7.3	7.0	6.6	6.2	5.9	5.6
-5	8.2	7.8	7.4	7.0	6.6	6.2	5.9	5.6
0	7.9	7.5	7.1	6.7	6.4	6.0	5.7	5.4
5	7.2	6.8	6.4	6.1	5.7	5.4	5.1	4.8
10	6.6	6.2	5.8	5.5	5.1	4.8	4.5	4.3
15	6.1	5.7	5.4	5.0	4.7	4.4	4.1	3.8
20	5.4	5.0	4.7	4.3	4.0	3.7	3.5	3.2
25	4.4	4.0	3.7	3.4	3.1	2.9	2.6	2.4
30	3.4	3.1	2.8	2.5	2.3	-	-	-
34	2.7	2.5	2.2	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 8300 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	12.7	11.9	11.3	10.6	10.0	9.5	8.9	8.5
-35	12.7	12.0	11.3	10.7	10.1	9.5	9.0	8.5
-30	12.7	12.0	11.3	10.7	10.1	9.5	9.0	8.5
-25	12.7	12.0	11.3	10.7	10.1	9.5	9.0	8.5
-20	12.7	12.0	11.3	10.7	10.1	9.5	9.0	8.5
-15	12.7	12.0	11.3	10.7	10.1	9.5	9.0	8.5
-10	12.8	12.0	11.3	10.7	10.1	9.5	9.0	8.5
-5	12.8	12.0	11.4	10.7	10.1	9.6	9.0	8.5
0	12.3	11.6	11.0	10.3	9.7	9.2	8.7	8.2
5	11.4	10.7	10.1	9.5	8.9	8.4	7.9	7.5
10	10.6	10.0	9.4	8.8	8.3	7.8	7.3	6.8
15	10.0	9.4	8.8	8.2	7.7	7.2	6.7	6.3
20	9.0	8.4	7.9	7.4	6.9	6.4	6.0	5.5
25	7.8	7.2	6.7	6.2	5.7	5.3	4.9	4.5
30	6.6	6.1	5.6	5.1	4.7	4.3	3.9	3.6
33	5.9	5.4	5.0	4.5	4.1	3.7	3.4	3.0



OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 1**

PW535E ENGINES

Altitude: 8300 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.0	7.6	7.2	6.8	6.4	6.1	5.8	5.5
-35	8.0	7.6	7.2	6.8	6.4	6.1	5.8	5.5
-30	8.0	7.6	7.2	6.8	6.5	6.1	5.8	5.5
-25	8.1	7.6	7.2	6.8	6.5	6.1	5.8	5.5
-20	8.1	7.6	7.2	6.8	6.5	6.1	5.8	5.5
-15	8.1	7.6	7.2	6.9	6.5	6.1	5.8	5.5
-10	8.1	7.6	7.2	6.9	6.5	6.1	5.8	5.5
-5	8.1	7.6	7.2	6.9	6.5	6.1	5.8	5.5
0	7.7	7.3	6.9	6.5	6.2	5.8	5.5	5.2
5	7.0	6.6	6.3	5.9	5.5	5.2	4.9	4.6
10	6.4	6.0	5.7	5.3	5.0	4.7	4.4	4.1
15	5.9	5.5	5.2	4.9	4.5	4.2	3.9	3.7
20	5.2	4.8	4.5	4.2	3.9	3.6	3.3	3.0
25	4.2	3.8	3.5	3.2	3.0	2.7	2.4	2.2
30	3.2	2.9	2.7	2.4	2.1	-	-	-
33	2.7	2.5	2.2	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 1**
PW535E ENGINES

Altitude: -1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.6	13.7	12.9	12.2	11.5	10.8	10.2	9.6
-35	14.6	13.8	13.0	12.2	11.5	10.8	10.2	9.6
-30	14.6	13.8	13.0	12.2	11.5	10.8	10.2	9.6
-25	14.6	13.8	13.0	12.2	11.5	10.9	10.2	9.6
-20	14.6	13.8	13.0	12.2	11.5	10.9	10.2	9.6
-15	14.6	13.8	13.0	12.2	11.5	10.9	10.2	9.6
-10	14.6	13.8	13.0	12.2	11.5	10.9	10.2	9.6
-5	14.6	13.8	13.0	12.2	11.5	10.8	10.2	9.6
0	14.6	13.8	13.0	12.2	11.5	10.8	10.2	9.6
5	14.6	13.7	12.9	12.2	11.5	10.8	10.2	9.6

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.1	8.5	8.1	7.6	7.2	6.8	6.4	6.0
-35	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-30	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-25	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-20	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-15	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-10	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-5	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
0	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
5	9.1	8.5	8.1	7.6	7.2	6.7	6.4	6.0

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 1

PW535E ENGINES

Altitude: 0 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-35	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-30	14.7	13.8	13.1	12.3	11.6	10.9	10.3	9.7
-25	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
-20	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
-15	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-10	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-5	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
0	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
5	14.6	13.8	13.0	12.2	11.5	10.9	10.2	9.7

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.2	8.6	8.2	7.7	7.3	6.8	6.5	6.1
-35	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-30	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-25	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-20	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-15	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-10	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-5	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
0	9.2	8.6	8.2	7.7	7.3	6.9	6.5	6.1
5	9.1	8.6	8.1	7.7	7.2	6.8	6.4	6.1

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 1**
PW535E ENGINES

Altitude: 1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-35	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-30	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-25	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-20	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-15	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-10	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-5	14.6	13.8	13.0	12.2	11.5	10.9	10.3	9.7
0	14.6	13.7	13.0	12.2	11.5	10.9	10.3	9.7
5	14.5	13.7	12.9	12.2	11.5	10.8	10.2	9.6

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.2
-35	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.2
-30	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.2
-25	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.2
-20	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.2
-15	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.2
-10	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.2
-5	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.2
0	9.1	8.6	8.2	7.7	7.3	6.9	6.5	6.1
5	9.1	8.6	8.1	7.7	7.3	6.9	6.5	6.1

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 1

PW535E ENGINES

Altitude: 2000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-35	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-30	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-25	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.6
-20	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-15	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-10	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-5	14.4	13.5	12.8	12.0	11.3	10.7	10.1	9.5
0	14.3	13.5	12.7	12.0	11.3	10.7	10.1	9.5
5	14.3	13.5	12.7	12.0	11.3	10.6	10.0	9.5

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.0	8.5	8.0	7.6	7.2	6.8	6.4	6.0
-35	9.0	8.5	8.0	7.6	7.2	6.8	6.4	6.0
-30	9.0	8.5	8.1	7.6	7.2	6.8	6.4	6.0
-25	9.0	8.5	8.1	7.6	7.2	6.8	6.4	6.0
-20	9.0	8.5	8.1	7.6	7.2	6.8	6.4	6.0
-15	9.0	8.5	8.0	7.6	7.2	6.8	6.4	6.0
-10	9.0	8.5	8.0	7.6	7.2	6.8	6.4	6.0
-5	9.0	8.5	8.0	7.6	7.1	6.7	6.4	6.0
0	9.0	8.5	8.0	7.6	7.1	6.7	6.3	6.0
5	8.9	8.4	8.0	7.5	7.1	6.7	6.3	6.0

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 1**
PW535E ENGINES

Altitude: 3000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	13.9	13.1	12.4	11.6	11.0	10.3	9.7	9.2
-35	14.0	13.1	12.4	11.7	11.0	10.3	9.7	9.2
-30	14.0	13.1	12.4	11.7	11.0	10.3	9.7	9.2
-25	14.0	13.2	12.4	11.7	11.0	10.4	9.8	9.2
-20	14.0	13.1	12.4	11.7	11.0	10.3	9.7	9.2
-15	13.9	13.1	12.4	11.6	11.0	10.3	9.7	9.2
-10	13.9	13.1	12.4	11.6	11.0	10.3	9.7	9.2
-5	13.9	13.1	12.3	11.6	10.9	10.3	9.7	9.1
0	13.9	13.1	12.3	11.6	10.9	10.3	9.7	9.1
5	13.8	13.0	12.3	11.6	10.9	10.3	9.7	9.1

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7
-35	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7
-30	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7
-25	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7
-20	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7
-15	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7
-10	8.6	8.2	7.7	7.3	6.8	6.4	6.1	5.7
-5	8.6	8.1	7.7	7.2	6.8	6.4	6.1	5.7
0	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
5	8.6	8.1	7.6	7.2	6.8	6.4	6.0	5.7

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 1

PW535E ENGINES

Altitude: 4000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.8
-35	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.8
-30	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.8
-25	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.8
-20	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.8
-15	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.8
-10	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.7
-5	13.4	12.6	11.8	11.1	10.5	9.9	9.3	8.7
0	13.4	12.6	11.8	11.1	10.5	9.8	9.3	8.7
5	13.2	12.4	11.7	11.0	10.3	9.7	9.1	8.5

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-35	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-30	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-25	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-20	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-15	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-10	8.2	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-5	8.2	7.8	7.3	6.9	6.5	6.1	5.7	5.4
0	8.2	7.7	7.3	6.9	6.4	6.1	5.7	5.4
5	8.0	7.6	7.1	6.7	6.3	5.9	5.5	5.2

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 1**
PW535E ENGINES

Altitude: 5000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	12.9	12.1	11.4	10.7	10.0	9.4	8.9	8.3
-35	12.9	12.1	11.4	10.7	10.1	9.4	8.9	8.3
-30	12.9	12.1	11.4	10.7	10.1	9.5	8.9	8.3
-25	12.9	12.1	11.4	10.7	10.1	9.5	8.9	8.3
-20	12.9	12.1	11.4	10.7	10.0	9.4	8.9	8.3
-15	12.9	12.1	11.4	10.7	10.0	9.4	8.9	8.3
-10	12.9	12.1	11.4	10.7	10.0	9.4	8.8	8.3
-5	12.8	12.1	11.3	10.7	10.0	9.4	8.8	8.3
0	12.8	12.0	11.3	10.6	10.0	9.4	8.8	8.3
5	12.2	11.5	10.8	10.1	9.5	8.9	8.3	7.8

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.1
-35	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.1
-30	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.1
-25	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.1
-20	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.1
-15	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.1
-10	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.0
-5	7.8	7.3	6.9	6.5	6.1	5.7	5.4	5.0
0	7.8	7.3	6.9	6.5	6.1	5.7	5.3	5.0
5	7.3	6.9	6.4	6.0	5.6	5.3	4.9	4.6

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 1

PW535E ENGINES

Altitude: 6000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	12.2	11.4	10.7	10.0	9.4	8.8	8.3	7.8
-35	12.2	11.4	10.7	10.1	9.4	8.8	8.3	7.8
-30	12.2	11.4	10.7	10.1	9.4	8.9	8.3	7.8
-25	12.2	11.4	10.7	10.1	9.4	8.9	8.3	7.8
-20	12.2	11.4	10.7	10.1	9.4	8.8	8.3	7.8
-15	12.2	11.4	10.7	10.1	9.4	8.8	8.3	7.8
-10	12.2	11.4	10.7	10.0	9.4	8.8	8.3	7.8
-5	12.2	11.4	10.7	10.0	9.4	8.8	8.3	7.7
0	12.1	11.4	10.7	10.0	9.4	8.8	8.2	7.7
5	11.2	10.5	9.8	9.1	8.5	8.0	7.4	6.9

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	7.3	6.8	6.4	6.0	5.6	5.3	4.9	4.6
-35	7.3	6.8	6.4	6.0	5.6	5.3	4.9	4.6
-30	7.3	6.8	6.4	6.0	5.6	5.3	4.9	4.6
-25	7.3	6.8	6.4	6.0	5.6	5.3	4.9	4.6
-20	7.3	6.8	6.4	6.0	5.6	5.3	4.9	4.6
-15	7.3	6.8	6.4	6.0	5.6	5.3	4.9	4.6
-10	7.3	6.8	6.4	6.0	5.6	5.3	4.9	4.6
-5	7.3	6.8	6.4	6.0	5.6	5.2	4.9	4.6
0	7.2	6.8	6.4	6.0	5.6	5.2	4.9	4.6
5	6.5	6.0	5.6	5.2	4.9	4.5	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 1**
PW535E ENGINES

Altitude: 7000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	11.3	10.6	9.9	9.2	8.6	8.1	7.5	7.0
-35	11.3	10.6	9.9	9.3	8.7	8.1	7.6	7.1
-30	11.3	10.6	9.9	9.3	8.7	8.1	7.6	7.1
-25	11.3	10.6	9.9	9.3	8.7	8.1	7.6	7.1
-20	11.3	10.6	9.9	9.3	8.7	8.1	7.6	7.1
-15	11.3	10.6	9.9	9.3	8.7	8.1	7.6	7.1
-10	11.3	10.6	9.9	9.2	8.6	8.1	7.5	7.0
-5	11.3	10.5	9.9	9.2	8.6	8.1	7.5	7.0
0	10.8	10.1	9.4	8.8	8.2	7.6	7.1	6.6
5	9.5	8.8	8.2	7.6	7.1	6.5	6.0	5.6

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	6.6	6.2	5.8	5.4	5.0	4.6	4.3	4.0
-35	6.6	6.2	5.8	5.4	5.0	4.7	4.3	4.0
-30	6.6	6.2	5.8	5.4	5.0	4.7	4.3	4.0
-25	6.6	6.2	5.8	5.4	5.0	4.7	4.3	4.0
-20	6.6	6.2	5.8	5.4	5.0	4.7	4.3	4.0
-15	6.6	6.2	5.8	5.4	5.0	4.7	4.3	4.0
-10	6.6	6.2	5.8	5.4	5.0	4.6	4.3	4.0
-5	6.6	6.1	5.7	5.4	5.0	4.6	4.3	4.0
0	6.2	5.7	5.3	5.0	4.6	4.2	3.9	-
5	5.1	4.7	4.4	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 1

PW535E ENGINES

Altitude: 8000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	10.6	9.9	9.2	8.6	8.0	7.5	7.0	6.5
-35	10.6	9.9	9.2	8.6	8.1	7.5	7.0	6.5
-30	10.6	9.9	9.3	8.6	8.1	7.5	7.0	6.5
-25	10.6	9.9	9.3	8.6	8.1	7.5	7.0	6.5
-20	10.6	9.9	9.2	8.6	8.1	7.5	7.0	6.5
-15	10.6	9.9	9.2	8.6	8.0	7.5	7.0	6.5
-10	10.6	9.9	9.2	8.6	8.0	7.5	7.0	6.5
-5	10.6	9.9	9.2	8.6	8.0	7.5	6.9	6.5
0	9.4	8.7	8.1	7.5	7.0	6.5	6.0	5.5
5	8.1	7.5	6.9	6.3	5.8	5.3	4.9	4.4

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	6.1	5.6	5.2	4.9	4.5	4.2	3.9	3.6
-35	6.1	5.6	5.3	4.9	4.5	4.2	3.9	3.6
-30	6.1	5.7	5.3	4.9	4.5	4.2	3.9	3.6
-25	6.1	5.7	5.3	4.9	4.5	4.2	3.9	3.6
-20	6.1	5.6	5.3	4.9	4.5	4.2	3.9	3.6
-15	6.1	5.6	5.2	4.9	4.5	4.2	3.8	3.6
-10	6.0	5.6	5.2	4.9	4.5	4.2	3.8	3.6
-5	6.0	5.6	5.2	4.8	4.5	4.1	3.8	-
0	5.1	4.7	4.3	4.0	-	-	-	-
5	4.0	-	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 1**
PW535E ENGINES

Altitude: 8300 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	10.5	9.8	9.1	8.5	7.9	7.4	6.9	6.4
-35	10.5	9.8	9.1	8.5	7.9	7.4	6.9	6.4
-30	10.5	9.8	9.1	8.5	7.9	7.4	6.9	6.4
-25	10.5	9.8	9.1	8.5	7.9	7.4	6.9	6.4
-20	10.5	9.8	9.1	8.5	7.9	7.4	6.9	6.4
-15	10.5	9.8	9.1	8.5	7.9	7.4	6.9	6.4
-10	10.4	9.7	9.1	8.5	7.9	7.4	6.8	6.4
-5	10.2	9.6	8.9	8.3	7.7	7.2	6.7	6.2
0	9.1	8.4	7.8	7.2	6.7	6.2	5.7	5.2
5	7.8	7.2	6.6	6.0	5.5	5.1	4.6	4.2

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	5.9	5.5	5.1	4.8	4.4	4.1	3.8	3.5
-35	6.0	5.5	5.1	4.8	4.4	4.1	3.8	3.5
-30	6.0	5.5	5.2	4.8	4.4	4.1	3.8	3.5
-25	6.0	5.5	5.2	4.8	4.4	4.1	3.8	3.5
-20	6.0	5.5	5.1	4.8	4.4	4.1	3.8	3.5
-15	5.9	5.5	5.1	4.8	4.4	4.1	3.8	3.5
-10	5.9	5.5	5.1	4.8	4.4	4.1	3.7	3.5
-5	5.8	5.4	5.0	4.6	4.3	3.9	-	-
0	4.8	4.4	4.1	3.7	-	-	-	-
5	3.8	-	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: -1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.6	13.8	13.0	12.2	11.5	10.8	10.2	9.6
-35	14.6	13.8	13.0	12.2	11.5	10.9	10.2	9.6
-30	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-25	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-20	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-15	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
-10	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
-5	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
0	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
5	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
10	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
15	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
20	14.6	13.8	13.0	12.2	11.5	10.9	10.2	9.6
25	14.6	13.8	13.0	12.2	11.5	10.8	10.2	9.6
30	13.9	13.1	12.3	11.6	10.9	10.2	9.6	9.0
35	12.3	11.5	10.8	10.1	9.5	8.8	8.3	7.7
40	10.8	10.0	9.4	8.7	8.1	7.6	7.0	6.5
45	9.4	8.7	8.1	7.5	6.9	6.4	5.9	5.4
50	8.0	7.4	6.8	6.3	5.8	5.3	4.8	4.3
52	7.5	6.9	6.3	5.8	5.3	4.8	4.4	3.9


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: -1000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.1	8.6	8.1	7.6	7.2	6.7	6.3	6.0
-35	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-30	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
-25	9.1	8.6	8.1	7.7	7.2	6.8	6.4	6.0
-20	9.1	8.6	8.1	7.7	7.2	6.8	6.4	6.0
-15	9.2	8.6	8.1	7.7	7.2	6.8	6.4	6.0
-10	9.2	8.6	8.2	7.7	7.2	6.8	6.4	6.1
-5	9.2	8.7	8.2	7.7	7.3	6.8	6.4	6.1
0	9.2	8.7	8.2	7.7	7.3	6.8	6.4	6.1
5	9.2	8.6	8.2	7.7	7.3	6.8	6.4	6.1
10	9.1	8.6	8.1	7.7	7.2	6.8	6.4	6.0
15	9.1	8.6	8.1	7.7	7.2	6.8	6.4	6.0
20	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
25	9.1	8.6	8.1	7.6	7.2	6.8	6.4	6.0
30	8.5	8.0	7.5	7.1	6.7	6.3	5.9	5.5
35	7.2	6.8	6.3	5.9	5.5	5.1	4.8	4.5
40	6.1	5.6	5.2	4.8	4.5	4.1	3.8	3.5
45	5.0	4.6	4.2	3.8	3.5	3.2	2.9	2.6
50	4.0	3.6	3.2	2.9	2.6	2.3	-	-
52	3.6	3.2	2.8	2.5	2.2	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 0 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-35	14.7	13.8	13.1	12.3	11.6	10.9	10.3	9.7
-30	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.7
-25	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.8
-20	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.8
-15	14.8	13.9	13.1	12.4	11.7	11.0	10.4	9.8
-10	14.8	13.9	13.1	12.4	11.7	11.0	10.4	9.8
-5	14.8	13.9	13.1	12.4	11.7	11.0	10.4	9.8
0	14.8	13.9	13.1	12.4	11.7	11.0	10.4	9.8
5	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.8
10	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.7
15	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
20	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
25	14.1	13.3	12.5	11.8	11.1	10.4	9.8	9.2
30	12.8	12.0	11.3	10.6	9.9	9.3	8.7	8.2
35	11.2	10.5	9.8	9.2	8.6	8.0	7.4	6.9
40	9.8	9.1	8.5	7.9	7.3	6.8	6.2	5.8
45	8.5	7.8	7.2	6.6	6.1	5.6	5.1	4.7
50	7.2	6.6	6.0	5.5	5.0	4.5	4.1	3.7

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OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 0 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.2	8.6	8.2	7.7	7.2	6.8	6.4	6.1
-35	9.2	8.7	8.2	7.7	7.3	6.8	6.4	6.1
-30	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-25	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-20	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-15	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
-10	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
-5	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
0	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
5	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
10	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
15	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
20	9.2	8.6	8.2	7.7	7.3	6.8	6.4	6.1
25	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
30	7.7	7.2	6.7	6.3	5.9	5.5	5.2	4.8
35	6.4	6.0	5.6	5.2	4.8	4.5	4.1	3.8
40	5.3	4.9	4.5	4.2	3.8	3.5	3.1	2.9
45	4.3	3.9	3.5	3.2	2.9	2.5	2.2	-
50	3.3	2.9	2.6	2.3	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.6	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-35	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-30	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
-25	14.7	13.8	13.1	12.3	11.6	10.9	10.3	9.7
-20	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
-15	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.7
-10	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.7
-5	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.7
0	14.7	13.9	13.1	12.3	11.6	11.0	10.3	9.7
5	14.7	13.9	13.1	12.3	11.6	10.9	10.3	9.7
10	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
15	14.7	13.8	13.0	12.3	11.6	10.9	10.3	9.7
20	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
25	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.3
30	11.7	10.9	10.2	9.6	9.0	8.4	7.8	7.3
35	10.2	9.5	8.8	8.2	7.6	7.1	6.6	6.1
40	8.8	8.2	7.6	7.0	6.4	5.9	5.4	5.0
45	7.5	6.9	6.4	5.8	5.3	4.8	4.4	3.9
48	6.8	6.2	5.6	5.1	4.6	4.2	3.7	3.3

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 1000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.1	8.6	8.1	7.7	7.3	6.8	6.4	6.1
-35	9.2	8.6	8.2	7.7	7.3	6.9	6.5	6.1
-30	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-25	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-20	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-15	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
-10	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
-5	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
0	9.2	8.7	8.2	7.8	7.3	6.9	6.5	6.1
5	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
10	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
15	9.2	8.7	8.2	7.7	7.3	6.9	6.5	6.1
20	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
25	7.8	7.4	6.9	6.5	6.1	5.7	5.3	5.0
30	6.8	6.4	5.9	5.5	5.1	4.8	4.4	4.1
35	5.6	5.2	4.8	4.4	4.1	3.7	3.4	3.1
40	4.6	4.2	3.8	3.5	3.1	2.8	2.5	2.2
45	3.6	3.2	2.9	2.5	2.2	-	-	-
48	3.0	2.6	2.3	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: 2000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.4	13.6	12.8	12.0	11.3	10.7	10.1	9.5
-35	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-30	14.4	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-25	14.5	13.6	12.8	12.1	11.4	10.7	10.1	9.5
-20	14.5	13.6	12.9	12.1	11.4	10.8	10.1	9.5
-15	14.5	13.7	12.9	12.1	11.4	10.8	10.1	9.6
-10	14.5	13.7	12.9	12.1	11.4	10.8	10.1	9.6
-5	14.5	13.7	12.9	12.1	11.4	10.8	10.1	9.6
0	14.5	13.7	12.9	12.1	11.4	10.8	10.1	9.6
5	14.5	13.6	12.9	12.1	11.4	10.8	10.1	9.6
10	14.5	13.6	12.8	12.1	11.4	10.7	10.1	9.5
15	14.3	13.5	12.7	12.0	11.3	10.6	10.0	9.4
20	13.1	12.3	11.6	10.9	10.2	9.6	9.0	8.4
25	11.9	11.2	10.5	9.8	9.2	8.6	8.0	7.5
30	10.6	9.9	9.2	8.6	8.0	7.4	6.9	6.4
35	9.2	8.5	7.9	7.3	6.8	6.2	5.7	5.3
40	7.9	7.3	6.7	6.1	5.6	5.1	4.7	4.2
45	6.6	6.1	5.5	5.0	4.5	4.1	3.6	3.2
46	6.4	5.8	5.3	4.8	4.3	3.9	3.4	3.0

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OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 2000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	9.0	8.5	8.0	7.5	7.1	6.7	6.3	5.9
-35	9.0	8.5	8.0	7.5	7.1	6.7	6.3	6.0
-30	9.0	8.5	8.0	7.6	7.1	6.7	6.3	6.0
-25	9.0	8.5	8.0	7.6	7.1	6.7	6.3	6.0
-20	9.0	8.5	8.0	7.6	7.2	6.7	6.4	6.0
-15	9.0	8.5	8.0	7.6	7.2	6.7	6.4	6.0
-10	9.0	8.5	8.1	7.6	7.2	6.8	6.4	6.0
-5	9.0	8.5	8.1	7.6	7.2	6.8	6.4	6.0
0	9.0	8.5	8.0	7.6	7.2	6.8	6.4	6.0
5	9.0	8.5	8.0	7.6	7.2	6.7	6.4	6.0
10	9.0	8.5	8.0	7.6	7.1	6.7	6.3	6.0
15	8.9	8.4	7.9	7.4	7.0	6.6	6.2	5.9
20	7.9	7.5	7.0	6.6	6.2	5.8	5.4	5.1
25	7.0	6.6	6.1	5.7	5.3	5.0	4.6	4.3
30	6.0	5.5	5.1	4.8	4.4	4.0	3.7	3.4
35	4.9	4.5	4.1	3.7	3.4	3.1	2.7	2.5
40	3.8	3.5	3.1	2.8	2.5	2.2	-	-
45	2.9	2.5	2.2	-	-	-	-	-
46	2.7	2.3	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 3000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.0	13.1	12.4	11.6	11.0	10.3	9.7	9.1
-35	14.0	13.2	12.4	11.7	11.0	10.3	9.7	9.1
-30	14.0	13.2	12.4	11.7	11.0	10.4	9.7	9.2
-25	14.0	13.2	12.4	11.7	11.0	10.4	9.8	9.2
-20	14.0	13.2	12.4	11.7	11.0	10.4	9.8	9.2
-15	14.1	13.2	12.5	11.7	11.0	10.4	9.8	9.2
-10	14.1	13.2	12.5	11.7	11.0	10.4	9.8	9.2
-5	14.1	13.2	12.5	11.7	11.0	10.4	9.8	9.2
0	14.1	13.2	12.5	11.7	11.0	10.4	9.8	9.2
5	14.0	13.2	12.4	11.7	11.0	10.4	9.8	9.2
10	14.0	13.2	12.4	11.7	11.0	10.4	9.8	9.2
15	13.2	12.4	11.7	11.0	10.3	9.7	9.1	8.5
20	12.1	11.3	10.6	9.9	9.3	8.7	8.1	7.6
25	10.9	10.2	9.5	8.9	8.3	7.7	7.2	6.7
30	9.6	8.9	8.3	7.7	7.1	6.6	6.1	5.6
35	8.2	7.6	7.0	6.5	5.9	5.4	5.0	4.5
40	7.0	6.4	5.8	5.3	4.8	4.4	3.9	3.5
44	6.0	5.5	4.9	4.4	4.0	3.5	3.1	2.7

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OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 3000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.6	8.1	7.6	7.2	6.8	6.4	6.0	5.7
-35	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
-30	8.6	8.2	7.7	7.2	6.8	6.4	6.0	5.7
-25	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
-20	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
-15	8.7	8.2	7.7	7.3	6.8	6.4	6.1	5.7
-10	8.7	8.2	7.7	7.3	6.8	6.4	6.1	5.7
-5	8.7	8.2	7.7	7.3	6.9	6.4	6.1	5.7
0	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7
5	8.7	8.2	7.7	7.3	6.8	6.4	6.1	5.7
10	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
15	8.0	7.5	7.1	6.7	6.2	5.9	5.5	5.1
20	7.1	6.7	6.2	5.8	5.4	5.1	4.7	4.4
25	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
30	5.2	4.8	4.4	4.0	3.7	3.3	3.0	2.7
35	4.1	3.7	3.4	3.0	2.7	2.4	2.1	-
40	3.1	2.8	2.4	2.1	-	-	-	-
44	2.4	-	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 4000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	13.5	12.7	11.9	11.2	10.5	9.9	9.3	8.7
-35	13.5	12.7	11.9	11.2	10.5	9.9	9.3	8.7
-30	13.5	12.7	11.9	11.2	10.6	9.9	9.3	8.7
-25	13.5	12.7	12.0	11.2	10.6	9.9	9.3	8.8
-20	13.5	12.7	12.0	11.3	10.6	9.9	9.3	8.8
-15	13.5	12.7	12.0	11.3	10.6	9.9	9.3	8.8
-10	13.6	12.7	12.0	11.3	10.6	10.0	9.4	8.8
-5	13.6	12.7	12.0	11.3	10.6	10.0	9.4	8.8
0	13.6	12.8	12.0	11.3	10.6	10.0	9.4	8.8
5	13.5	12.7	12.0	11.3	10.6	9.9	9.3	8.8
10	13.3	12.5	11.8	11.1	10.4	9.8	9.2	8.6
15	12.2	11.4	10.7	10.0	9.4	8.8	8.2	7.7
20	11.1	10.4	9.7	9.0	8.4	7.9	7.3	6.8
25	10.0	9.3	8.7	8.1	7.5	6.9	6.4	5.9
30	8.7	8.0	7.4	6.8	6.3	5.8	5.3	4.9
35	7.4	6.8	6.2	5.7	5.2	4.7	4.2	3.8
40	6.1	5.6	5.0	4.6	4.1	3.6	3.2	2.8
42	5.7	5.1	4.6	4.1	3.7	3.2	2.8	2.4


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 4000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.2	7.7	7.3	6.8	6.4	6.0	5.7	5.3
-35	8.2	7.7	7.3	6.9	6.4	6.0	5.7	5.3
-30	8.2	7.8	7.3	6.9	6.5	6.1	5.7	5.3
-25	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-20	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-15	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-10	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-5	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
0	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
5	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
10	8.1	7.6	7.2	6.7	6.3	5.9	5.5	5.2
15	7.2	6.8	6.3	5.9	5.5	5.1	4.8	4.5
20	6.4	5.9	5.5	5.1	4.7	4.4	4.1	3.7
25	5.5	5.1	4.7	4.3	4.0	3.6	3.3	3.0
30	4.4	4.1	3.7	3.3	3.0	2.7	2.4	2.1
35	3.4	3.1	2.7	2.4	-	-	-	-
40	2.5	2.1	-	-	-	-	-	-
42	2.1	-	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 5000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	12.9	12.1	11.4	10.7	10.0	9.4	8.8	8.3
-35	12.9	12.1	11.4	10.7	10.0	9.4	8.8	8.3
-30	13.0	12.2	11.4	10.7	10.1	9.4	8.9	8.3
-25	13.0	12.2	11.4	10.7	10.1	9.5	8.9	8.3
-20	13.0	12.2	11.4	10.7	10.1	9.5	8.9	8.3
-15	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.3
-10	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.3
-5	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.4
0	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.4
5	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.3
10	12.2	11.4	10.7	10.0	9.4	8.8	8.2	7.7
15	11.1	10.4	9.7	9.1	8.5	7.9	7.4	6.8
20	10.1	9.4	8.8	8.1	7.6	7.0	6.5	6.0
25	9.0	8.4	7.8	7.2	6.6	6.1	5.6	5.1
30	7.7	7.1	6.5	6.0	5.5	5.0	4.5	4.1
35	6.5	5.9	5.4	4.9	4.4	3.9	3.5	3.1
40	5.3	4.8	4.3	3.8	3.4	2.9	2.5	2.2


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 5000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	7.8	7.3	6.9	6.4	6.0	5.7	5.3	5.0
-35	7.8	7.3	6.9	6.5	6.1	5.7	5.3	5.0
-30	7.8	7.3	6.9	6.5	6.1	5.7	5.3	5.0
-25	7.8	7.4	6.9	6.5	6.1	5.7	5.3	5.0
-20	7.8	7.4	6.9	6.5	6.1	5.7	5.3	5.0
-15	7.8	7.4	6.9	6.5	6.1	5.7	5.3	5.0
-10	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.0
-5	7.9	7.4	6.9	6.5	6.1	5.7	5.4	5.0
0	7.9	7.4	6.9	6.5	6.1	5.7	5.4	5.0
5	7.8	7.4	6.9	6.5	6.1	5.7	5.4	5.0
10	7.2	6.8	6.3	5.9	5.5	5.1	4.8	4.5
15	6.4	5.9	5.5	5.1	4.8	4.4	4.1	3.8
20	5.6	5.2	4.8	4.4	4.0	3.7	3.4	3.1
25	4.7	4.3	4.0	3.6	3.3	2.9	2.6	2.4
30	3.7	3.4	3.0	2.7	2.4	-	-	-
35	2.8	2.4	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 6000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	12.2	11.4	10.7	10.0	9.4	8.8	8.2	7.7
-35	12.2	11.5	10.7	10.1	9.4	8.8	8.3	7.7
-30	12.2	11.5	10.8	10.1	9.4	8.8	8.3	7.7
-25	12.3	11.5	10.8	10.1	9.5	8.9	8.3	7.8
-20	12.3	11.5	10.8	10.1	9.5	8.9	8.3	7.8
-15	12.3	11.5	10.8	10.1	9.5	8.9	8.3	7.8
-10	12.3	11.5	10.8	10.1	9.5	8.9	8.3	7.8
-5	12.3	11.5	10.8	10.1	9.5	8.9	8.3	7.8
0	12.3	11.5	10.8	10.1	9.5	8.9	8.3	7.8
5	12.0	11.2	10.5	9.8	9.2	8.6	8.0	7.5
10	10.9	10.2	9.5	8.9	8.3	7.7	7.2	6.7
15	10.0	9.3	8.6	8.0	7.4	6.9	6.4	5.9
20	9.0	8.3	7.7	7.1	6.6	6.1	5.6	5.1
25	7.9	7.3	6.7	6.2	5.7	5.2	4.7	4.3
30	6.7	6.1	5.6	5.1	4.6	4.1	3.7	3.3
35	5.5	5.0	4.5	4.0	3.6	3.1	2.7	2.3
38	4.9	4.3	3.9	3.4	3.0	2.6	2.2	-


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 6000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	7.2	6.8	6.3	5.9	5.5	5.2	4.8	4.5
-35	7.2	6.8	6.4	6.0	5.6	5.2	4.8	4.5
-30	7.3	6.8	6.4	6.0	5.6	5.2	4.9	4.5
-25	7.3	6.8	6.4	6.0	5.6	5.2	4.9	4.5
-20	7.3	6.8	6.4	6.0	5.6	5.2	4.9	4.6
-15	7.3	6.8	6.4	6.0	5.6	5.2	4.9	4.6
-10	7.3	6.9	6.4	6.0	5.6	5.2	4.9	4.6
-5	7.3	6.9	6.4	6.0	5.6	5.3	4.9	4.6
0	7.3	6.9	6.4	6.0	5.6	5.3	4.9	4.6
5	7.0	6.6	6.2	5.8	5.4	5.0	4.6	4.3
10	6.2	5.8	5.4	5.0	4.6	4.3	4.0	3.7
15	5.5	5.1	4.7	4.3	4.0	3.6	3.3	3.0
20	4.7	4.3	3.9	3.6	3.3	2.9	2.6	2.4
25	3.9	3.5	3.2	2.8	2.5	2.2	-	-
30	2.9	2.6	2.3	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.



OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 7000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	11.3	10.6	9.9	9.2	8.6	8.1	7.5	7.0
-35	11.3	10.6	9.9	9.3	8.7	8.1	7.5	7.0
-30	11.4	10.6	9.9	9.3	8.7	8.1	7.5	7.0
-25	11.4	10.7	10.0	9.3	8.7	8.1	7.6	7.0
-20	11.4	10.7	10.0	9.3	8.7	8.1	7.6	7.1
-15	11.4	10.7	10.0	9.3	8.7	8.1	7.6	7.1
-10	11.4	10.7	10.0	9.3	8.7	8.1	7.6	7.1
-5	11.4	10.7	10.0	9.3	8.7	8.1	7.6	7.1
0	11.4	10.7	10.0	9.4	8.7	8.2	7.6	7.1
5	10.6	9.9	9.2	8.6	8.0	7.5	6.9	6.4
10	9.7	9.0	8.4	7.8	7.2	6.7	6.2	5.7
15	8.8	8.2	7.6	7.0	6.5	6.0	5.5	5.0
20	8.0	7.4	6.8	6.2	5.7	5.2	4.8	4.3
25	6.9	6.3	5.7	5.2	4.7	4.3	3.8	3.4
30	5.6	5.1	4.6	4.1	3.7	3.2	2.8	2.4
35	4.5	4.0	3.5	3.1	2.7	2.2	-	-
36	4.3	3.8	3.3	2.9	2.5	-	-	-


OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 7000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	6.5	6.1	5.7	5.3	4.9	4.6	4.2	3.9
-35	6.5	6.1	5.7	5.3	4.9	4.6	4.2	3.9
-30	6.6	6.1	5.7	5.3	4.9	4.6	4.3	3.9
-25	6.6	6.2	5.7	5.3	5.0	4.6	4.3	4.0
-20	6.6	6.2	5.7	5.4	5.0	4.6	4.3	4.0
-15	6.6	6.2	5.8	5.4	5.0	4.6	4.3	4.0
-10	6.6	6.2	5.8	5.4	5.0	4.6	4.3	4.0
-5	6.6	6.2	5.8	5.4	5.0	4.6	4.3	4.0
0	6.6	6.2	5.8	5.4	5.0	4.6	4.3	4.0
5	6.0	5.6	5.2	4.8	4.4	4.1	3.7	3.4
10	5.3	4.9	4.5	4.1	3.8	3.4	3.1	2.8
15	4.6	4.2	3.8	3.5	3.2	2.9	2.5	2.3
20	3.9	3.6	3.2	2.9	2.5	2.2	-	-
25	3.1	2.7	2.4	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.



OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: 8000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	10.6	9.9	9.2	8.6	8.0	7.5	6.9	6.4
-35	10.7	9.9	9.3	8.6	8.0	7.5	7.0	6.5
-30	10.7	10.0	9.3	8.7	8.1	7.5	7.0	6.5
-25	10.7	10.0	9.3	8.7	8.1	7.5	7.0	6.5
-20	10.7	10.0	9.3	8.7	8.1	7.5	7.0	6.5
-15	10.7	10.0	9.3	8.7	8.1	7.5	7.0	6.5
-10	10.7	10.0	9.3	8.7	8.1	7.5	7.0	6.5
-5	10.7	10.0	9.3	8.7	8.1	7.5	7.0	6.5
0	10.4	9.7	9.0	8.4	7.8	7.3	6.8	6.3
5	9.5	8.8	8.2	7.6	7.0	6.5	6.0	5.5
10	8.7	8.0	7.4	6.8	6.3	5.8	5.3	4.9
15	8.0	7.4	6.8	6.3	5.7	5.3	4.8	4.4
20	7.1	6.5	5.9	5.4	4.9	4.5	4.0	3.6
25	5.8	5.3	4.8	4.3	3.8	3.4	3.0	2.6
30	4.6	4.1	3.6	3.2	2.8	2.4	-	-
34	3.8	3.3	2.8	2.4	-	-	-	-


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 8000 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	6.0	5.6	5.2	4.8	4.4	4.1	3.8	3.5
-35	6.0	5.6	5.2	4.8	4.4	4.1	3.8	3.5
-30	6.0	5.6	5.2	4.8	4.5	4.1	3.8	3.5
-25	6.0	5.6	5.2	4.8	4.5	4.1	3.8	3.5
-20	6.0	5.6	5.2	4.8	4.5	4.1	3.8	3.5
-15	6.1	5.6	5.2	4.8	4.5	4.1	3.8	3.5
-10	6.1	5.6	5.2	4.9	4.5	4.1	3.8	3.5
-5	6.1	5.6	5.2	4.9	4.5	4.1	3.8	3.5
0	5.8	5.4	5.0	4.6	4.3	3.9	3.6	3.3
5	5.1	4.7	4.3	4.0	3.6	3.3	3.0	2.7
10	4.5	4.1	3.7	3.4	3.0	2.7	2.4	2.2
15	4.0	3.6	3.3	2.9	2.6	2.3	-	-
20	3.2	2.9	2.6	2.2	-	-	-	-
25	2.3	-	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.



OEI – APPROACH CLIMB GRADIENT (%)

**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
GEAR UP – FLAP 2**

PW535E ENGINES

Altitude: 8300 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	10.5	9.8	9.1	8.5	7.9	7.3	6.8	6.3
-35	10.5	9.8	9.1	8.5	7.9	7.4	6.8	6.3
-30	10.5	9.8	9.2	8.5	7.9	7.4	6.8	6.4
-25	10.5	9.8	9.2	8.5	7.9	7.4	6.9	6.4
-20	10.5	9.8	9.2	8.5	8.0	7.4	6.9	6.4
-15	10.6	9.9	9.2	8.6	8.0	7.4	6.9	6.4
-10	10.6	9.9	9.2	8.6	8.0	7.4	6.9	6.4
-5	10.6	9.9	9.2	8.6	8.0	7.4	6.9	6.4
0	10.1	9.4	8.8	8.2	7.6	7.1	6.5	6.1
5	9.2	8.6	7.9	7.4	6.8	6.3	5.8	5.3
10	8.4	7.8	7.2	6.7	6.1	5.6	5.2	4.7
15	7.8	7.2	6.6	6.1	5.6	5.1	4.6	4.2
20	6.8	6.3	5.7	5.2	4.7	4.3	3.8	3.4
25	5.6	5.0	4.5	4.1	3.6	3.2	2.8	2.4
30	4.4	3.9	3.4	3.0	2.6	2.2	-	-
33	3.8	3.3	2.8	2.4	-	-	-	-


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION OFF – WINGSTAB OFF
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 8300 ft

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	5.9	5.5	5.1	4.7	4.3	4.0	3.7	3.4
-35	5.9	5.5	5.1	4.7	4.3	4.0	3.7	3.4
-30	5.9	5.5	5.1	4.7	4.4	4.0	3.7	3.4
-25	5.9	5.5	5.1	4.7	4.4	4.0	3.7	3.4
-20	5.9	5.5	5.1	4.7	4.4	4.0	3.7	3.4
-15	5.9	5.5	5.1	4.7	4.4	4.0	3.7	3.4
-10	5.9	5.5	5.1	4.7	4.4	4.0	3.7	3.4
-5	5.9	5.5	5.1	4.8	4.4	4.0	3.7	3.4
0	5.6	5.2	4.8	4.4	4.1	3.7	3.4	3.1
5	4.9	4.5	4.1	3.8	3.5	3.1	2.8	2.5
10	4.3	3.9	3.6	3.2	2.9	2.6	2.3	-
15	3.8	3.4	3.1	2.8	2.4	2.1	-	-
20	3.1	2.7	2.4	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.



OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: -1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.1	13.3	12.5	11.8	11.1	10.4	9.8	9.2
-35	14.1	13.3	12.5	11.8	11.1	10.4	9.8	9.2
-30	14.2	13.3	12.5	11.8	11.1	10.5	9.8	9.2
-25	14.2	13.3	12.6	11.8	11.1	10.5	9.8	9.2
-20	14.2	13.3	12.6	11.8	11.1	10.5	9.8	9.3
-15	14.2	13.3	12.5	11.8	11.1	10.5	9.8	9.3
-10	14.2	13.3	12.5	11.8	11.1	10.5	9.8	9.2
-5	14.1	13.3	12.5	11.8	11.1	10.4	9.8	9.2
0	14.1	13.3	12.5	11.8	11.1	10.4	9.8	9.2
5	14.1	13.3	12.5	11.8	11.1	10.4	9.8	9.2

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.7	8.2	7.7	7.2	6.8	6.4	6.0	5.6
-35	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.6
-30	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
-25	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
-20	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
-15	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
-10	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
-5	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
0	8.7	8.2	7.7	7.3	6.8	6.4	6.0	5.7
5	8.7	8.2	7.7	7.2	6.8	6.4	6.0	5.6

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 0 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
-35	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
-30	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
-25	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
-20	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
-15	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
-10	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
-5	14.2	13.4	12.6	11.9	11.2	10.5	9.9	9.3
0	14.2	13.3	12.6	11.8	11.1	10.5	9.9	9.3
5	14.1	13.3	12.5	11.8	11.1	10.5	9.8	9.3

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.8	8.3	7.8	7.3	6.9	6.5	6.1	5.7
-35	8.8	8.3	7.8	7.3	6.9	6.5	6.1	5.7
-30	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-25	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-20	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-15	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-10	8.8	8.3	7.8	7.3	6.9	6.5	6.1	5.8
-5	8.8	8.3	7.8	7.3	6.9	6.5	6.1	5.7
0	8.8	8.3	7.8	7.3	6.9	6.5	6.1	5.7
5	8.7	8.2	7.8	7.3	6.9	6.5	6.1	5.7

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: 1000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	14.2	13.3	12.6	11.8	11.1	10.5	9.9	9.3
-35	14.2	13.3	12.6	11.8	11.1	10.5	9.9	9.3
-30	14.2	13.3	12.6	11.8	11.2	10.5	9.9	9.3
-25	14.2	13.4	12.6	11.8	11.2	10.5	9.9	9.3
-20	14.2	13.4	12.6	11.8	11.2	10.5	9.9	9.3
-15	14.2	13.3	12.6	11.8	11.2	10.5	9.9	9.3
-10	14.2	13.3	12.6	11.8	11.1	10.5	9.9	9.3
-5	14.1	13.3	12.5	11.8	11.1	10.5	9.9	9.3
0	14.1	13.3	12.5	11.8	11.1	10.5	9.8	9.3
5	14.1	13.2	12.5	11.8	11.1	10.4	9.8	9.2

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-35	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-30	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-25	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-20	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-15	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-10	8.8	8.3	7.8	7.4	6.9	6.5	6.1	5.8
-5	8.8	8.3	7.8	7.3	6.9	6.5	6.1	5.8
0	8.7	8.3	7.8	7.3	6.9	6.5	6.1	5.8
5	8.7	8.2	7.7	7.3	6.9	6.5	6.1	5.7

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 2000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	13.9	13.1	12.3	11.6	10.9	10.3	9.7	9.1
-35	13.9	13.1	12.3	11.6	10.9	10.3	9.7	9.1
-30	13.9	13.1	12.4	11.6	11.0	10.3	9.7	9.1
-25	14.0	13.1	12.4	11.6	11.0	10.3	9.7	9.1
-20	13.9	13.1	12.4	11.6	11.0	10.3	9.7	9.1
-15	13.9	13.1	12.4	11.6	11.0	10.3	9.7	9.1
-10	13.9	13.1	12.3	11.6	10.9	10.3	9.7	9.1
-5	13.9	13.1	12.3	11.6	10.9	10.3	9.7	9.1
0	13.9	13.1	12.3	11.6	10.9	10.3	9.7	9.1
5	13.8	13.0	12.3	11.5	10.9	10.2	9.6	9.1

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
-35	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
-30	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
-25	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
-20	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
-15	8.6	8.1	7.7	7.2	6.8	6.4	6.0	5.7
-10	8.6	8.1	7.6	7.2	6.8	6.4	6.0	5.6
-5	8.6	8.1	7.6	7.2	6.8	6.4	6.0	5.6
0	8.6	8.1	7.6	7.2	6.7	6.3	6.0	5.6
5	8.5	8.1	7.6	7.1	6.7	6.3	5.9	5.6

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.



OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: 3000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	13.5	12.7	11.9	11.2	10.5	9.9	9.3	8.8
-35	13.5	12.7	11.9	11.2	10.6	9.9	9.3	8.8
-30	13.5	12.7	11.9	11.2	10.6	9.9	9.3	8.8
-25	13.5	12.7	11.9	11.2	10.6	9.9	9.3	8.8
-20	13.5	12.7	11.9	11.2	10.6	9.9	9.3	8.8
-15	13.5	12.7	11.9	11.2	10.5	9.9	9.3	8.8
-10	13.5	12.7	11.9	11.2	10.5	9.9	9.3	8.8
-5	13.5	12.6	11.9	11.2	10.5	9.9	9.3	8.7
0	13.4	12.6	11.9	11.2	10.5	9.9	9.3	8.7
5	13.4	12.6	11.8	11.1	10.5	9.8	9.2	8.7

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-35	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-30	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-25	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-20	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-15	8.3	7.8	7.3	6.9	6.5	6.1	5.7	5.4
-10	8.2	7.8	7.3	6.9	6.5	6.1	5.7	5.3
-5	8.2	7.8	7.3	6.9	6.4	6.1	5.7	5.3
0	8.2	7.7	7.3	6.8	6.4	6.0	5.7	5.3
5	8.2	7.7	7.2	6.8	6.4	6.0	5.6	5.3

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 4000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	13.0	12.2	11.4	10.7	10.1	9.5	8.9	8.4
-35	13.0	12.2	11.4	10.8	10.1	9.5	8.9	8.4
-30	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.4
-25	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.4
-20	13.0	12.2	11.5	10.8	10.1	9.5	8.9	8.4
-15	13.0	12.2	11.4	10.7	10.1	9.5	8.9	8.4
-10	13.0	12.2	11.4	10.7	10.1	9.5	8.9	8.3
-5	12.9	12.1	11.4	10.7	10.1	9.5	8.9	8.3
0	12.9	12.1	11.4	10.7	10.0	9.4	8.9	8.3
5	12.7	12.0	11.2	10.5	9.9	9.3	8.7	8.2

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	7.9	7.4	6.9	6.5	6.1	5.7	5.3	5.0
-35	7.9	7.4	6.9	6.5	6.1	5.7	5.4	5.0
-30	7.9	7.4	6.9	6.5	6.1	5.7	5.4	5.0
-25	7.9	7.4	6.9	6.5	6.1	5.7	5.4	5.0
-20	7.9	7.4	6.9	6.5	6.1	5.7	5.4	5.0
-15	7.9	7.4	6.9	6.5	6.1	5.7	5.3	5.0
-10	7.8	7.4	6.9	6.5	6.1	5.7	5.3	5.0
-5	7.8	7.4	6.9	6.5	6.1	5.7	5.3	5.0
0	7.8	7.3	6.9	6.5	6.1	5.7	5.3	5.0
5	7.7	7.2	6.7	6.3	5.9	5.5	5.2	4.8

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: 5000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	12.4	11.7	10.9	10.3	9.6	9.0	8.5	7.9
-35	12.4	11.7	10.9	10.3	9.6	9.0	8.5	7.9
-30	12.4	11.7	11.0	10.3	9.6	9.0	8.5	7.9
-25	12.4	11.7	11.0	10.3	9.6	9.0	8.5	7.9
-20	12.4	11.7	10.9	10.3	9.6	9.0	8.5	7.9
-15	12.4	11.6	10.9	10.2	9.6	9.0	8.5	7.9
-10	12.4	11.6	10.9	10.2	9.6	9.0	8.4	7.9
-5	12.4	11.6	10.9	10.2	9.6	9.0	8.4	7.9
0	12.4	11.6	10.9	10.2	9.6	9.0	8.4	7.9
5	11.8	11.0	10.3	9.7	9.1	8.5	7.9	7.4

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	7.4	7.0	6.5	6.1	5.7	5.4	5.0	4.7
-35	7.4	7.0	6.5	6.1	5.7	5.4	5.0	4.7
-30	7.5	7.0	6.6	6.1	5.7	5.4	5.0	4.7
-25	7.5	7.0	6.6	6.1	5.7	5.4	5.0	4.7
-20	7.4	7.0	6.5	6.1	5.7	5.4	5.0	4.7
-15	7.4	7.0	6.5	6.1	5.7	5.4	5.0	4.7
-10	7.4	7.0	6.5	6.1	5.7	5.3	5.0	4.7
-5	7.4	7.0	6.5	6.1	5.7	5.3	5.0	4.6
0	7.4	6.9	6.5	6.1	5.7	5.3	4.9	4.6
5	6.9	6.5	6.0	5.6	5.3	4.9	4.5	4.2

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 6000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	11.7	11.0	10.3	9.6	9.0	8.4	7.9	7.4
-35	11.7	11.0	10.3	9.6	9.0	8.4	7.9	7.4
-30	11.7	11.0	10.3	9.6	9.0	8.4	7.9	7.4
-25	11.7	11.0	10.3	9.6	9.0	8.4	7.9	7.4
-20	11.7	11.0	10.3	9.6	9.0	8.4	7.9	7.4
-15	11.7	11.0	10.3	9.6	9.0	8.4	7.9	7.4
-10	11.7	11.0	10.3	9.6	9.0	8.4	7.9	7.4
-5	11.7	10.9	10.3	9.6	9.0	8.4	7.9	7.3
0	11.7	10.9	10.2	9.6	9.0	8.4	7.8	7.3
5	10.7	10.0	9.4	8.7	8.1	7.6	7.0	6.5

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	6.9	6.4	6.0	5.6	5.2	4.9	4.5	4.2
-35	6.9	6.5	6.0	5.6	5.2	4.9	4.5	4.2
-30	6.9	6.5	6.0	5.6	5.3	4.9	4.5	4.2
-25	6.9	6.5	6.0	5.6	5.3	4.9	4.5	4.2
-20	6.9	6.5	6.0	5.6	5.2	4.9	4.5	4.2
-15	6.9	6.4	6.0	5.6	5.2	4.9	4.5	4.2
-10	6.9	6.4	6.0	5.6	5.2	4.9	4.5	4.2
-5	6.9	6.4	6.0	5.6	5.2	4.9	4.5	4.2
0	6.8	6.4	6.0	5.6	5.2	4.8	4.5	4.2
5	6.1	5.7	5.2	4.9	4.5	4.1	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: 7000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	10.8	10.1	9.5	8.8	8.2	7.7	7.1	6.6
-35	10.9	10.1	9.5	8.8	8.2	7.7	7.2	6.7
-30	10.9	10.1	9.5	8.8	8.2	7.7	7.2	6.7
-25	10.9	10.1	9.5	8.8	8.3	7.7	7.2	6.7
-20	10.9	10.1	9.5	8.8	8.2	7.7	7.2	6.7
-15	10.9	10.1	9.5	8.8	8.2	7.7	7.2	6.7
-10	10.8	10.1	9.5	8.8	8.2	7.7	7.1	6.6
-5	10.8	10.1	9.4	8.8	8.2	7.7	7.1	6.6
0	10.3	9.6	9.0	8.4	7.8	7.2	6.7	6.2
5	9.0	8.4	7.8	7.2	6.6	6.1	5.6	5.2

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
-35	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
-30	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
-25	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
-20	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
-15	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
-10	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
-5	6.2	5.8	5.4	5.0	4.6	4.3	3.9	3.6
0	5.8	5.4	5.0	4.6	4.2	3.9	3.5	-
5	4.8	4.4	4.0	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.


OEI – APPROACH CLIMB GRADIENT (%)
**V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
 GEAR UP – FLAP 2**
PW535E ENGINES

Altitude: 8000 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	10.2	9.5	8.8	8.2	7.6	7.1	6.6	6.1
-35	10.2	9.5	8.8	8.2	7.6	7.1	6.6	6.1
-30	10.2	9.5	8.8	8.2	7.6	7.1	6.6	6.1
-25	10.2	9.5	8.8	8.2	7.6	7.1	6.6	6.1
-20	10.2	9.5	8.8	8.2	7.6	7.1	6.6	6.1
-15	10.2	9.5	8.8	8.2	7.6	7.1	6.6	6.1
-10	10.1	9.4	8.8	8.2	7.6	7.1	6.6	6.1
-5	10.1	9.4	8.8	8.2	7.6	7.1	6.5	6.1
0	9.0	8.3	7.7	7.1	6.6	6.1	5.6	5.1
5	7.6	7.0	6.4	5.9	5.4	4.9	4.5	4.0

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	5.7	5.3	4.9	4.5	4.1	3.8	3.5	3.2
-35	5.7	5.3	4.9	4.5	4.1	3.8	3.5	3.2
-30	5.7	5.3	4.9	4.5	4.1	3.8	3.5	3.2
-25	5.7	5.3	4.9	4.5	4.1	3.8	3.5	3.2
-20	5.7	5.3	4.9	4.5	4.1	3.8	3.5	3.2
-15	5.7	5.2	4.9	4.5	4.1	3.8	3.5	3.2
-10	5.6	5.2	4.8	4.5	4.1	3.8	3.5	3.2
-5	5.6	5.2	4.8	4.4	4.1	3.8	3.4	-
0	4.7	4.3	3.9	3.6	-	-	-	-
5	3.6	-	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.

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OEI – APPROACH CLIMB GRADIENT (%)

V_{AC} – ENGINE ICE PROTECTION ON – WINGSTAB ON
GEAR UP – FLAP 2

PW535E ENGINES

Altitude: 8300 ft

TEMP (°C)	WEIGHT (lb)							
	12400	12800	13200	13600	14000	14400	14800	15200
-40	10.0	9.3	8.7	8.1	7.5	7.0	6.5	6.0
-35	10.0	9.3	8.7	8.1	7.5	7.0	6.5	6.0
-30	10.0	9.3	8.7	8.1	7.5	7.0	6.5	6.0
-25	10.0	9.3	8.7	8.1	7.5	7.0	6.5	6.0
-20	10.0	9.3	8.7	8.1	7.5	7.0	6.5	6.0
-15	10.0	9.3	8.7	8.1	7.5	7.0	6.5	6.0
-10	10.0	9.3	8.7	8.1	7.5	7.0	6.4	6.0
-5	9.8	9.1	8.5	7.9	7.3	6.8	6.3	5.8
0	8.6	8.0	7.4	6.8	6.3	5.8	5.3	4.8
5	7.3	6.7	6.1	5.6	5.1	4.6	4.2	3.8

TEMP (°C)	WEIGHT (lb)							
	15600	16000	16400	16800	17200	17600	18000	18387
-40	5.6	5.1	4.8	4.4	4.0	3.7	3.4	3.1
-35	5.6	5.1	4.8	4.4	4.0	3.7	3.4	3.1
-30	5.6	5.2	4.8	4.4	4.0	3.7	3.4	3.1
-25	5.6	5.2	4.8	4.4	4.0	3.7	3.4	3.1
-20	5.6	5.1	4.8	4.4	4.0	3.7	3.4	3.1
-15	5.5	5.1	4.7	4.4	4.0	3.7	3.4	3.1
-10	5.5	5.1	4.7	4.4	4.0	3.7	3.4	3.1
-5	5.4	5.0	4.6	4.2	3.9	3.5	-	-
0	4.4	4.1	3.7	3.3	-	-	-	-
5	3.4	-	-	-	-	-	-	-

CAUTION: SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT WAS EXCEEDED.



LANDING DATA

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

LANDING TECHNIQUE

The landing performance data in this manual is based on the following conditions:

- Steady three degree angle approach at V_{REF} in landing configuration;
- V_{REF} airspeed maintained at runway threshold;
- Idle thrust established at runway threshold;
- Attitude maintained until MLG touchdown;
- Maximum brake applied immediately after MLG touchdown;
- Antiskid system operative.

If these performance techniques are not strictly used for a typical landing made during normal operations, the distances may be longer.

MAXIMUM LANDING WEIGHT – CLIMB LIMITED

The Maximum Landing Weight – Climb Limited tables present the maximum landing climb weight in compliance with the airworthiness climb requirements in function of the temperature and altitude and according to the ice protection system configuration.

The landing climb weights shown may be above the Maximum Landing Weight and are codified as follows:

- (A) Approach Climb Limited;
- (L) Landing Climb Limited;
- (E) Enroute Climb Limited;
- (O) Overweight Landing Limited.

NOTE: The overweight landing notation (O) is used to indicate that the airplane is not climb limited. Always check this structural limitation on Section 2.



EXAMPLE

Given:

- Airport Pressure Altitude 2000 ft
- Approach Flap 1
- Landing Flap 3
- Engine Ice Protection..... OFF
- Wingstab OFF
- Temperature 20°C

Determine:

- Maximum Landing Weight –
Overweight Landing Limited 18386 lb

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MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 1 – LANDING FLAP 3
ENGINE ICE PROTECTION OFF – WINGSTAB OFF
PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)					
	ALTITUDE					
	-1000 ft	0 ft	1000 ft	2000 ft	3000 ft	4000 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
0	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
45	18386 (O)	18386 (O)	18386 (O)	18386 (O)	-	-
50	18386 (O)	18386 (O)	-	-	-	-

NOTE: The values above Maximum Landing Weight must not be used for dispatch.

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**MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 1 – LANDING FLAP 3
ENGINE ICE PROTECTION OFF – WINGSTAB OFF
PW535E ENGINES**

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)				
	ALTITUDE				
	5000 ft	6000 ft	7000 ft	8000 ft	8300 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
0	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
30	18386 (O)	18386 (O)	18386 (O)	17500 (A)	17251 (A)
35	18386 (O)	18110 (E)	17251 (E)	-	-
40	17670 (E)	-	-	-	-

NOTE: The values above Maximum Landing Weight must not be used for dispatch.



MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 1 – LANDING FLAP 3
ENGINE ICE PROTECTION ON – WINGSTAB OFF
PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)					
	ALTITUDE					
	-1000 ft	0 ft	1000 ft	2000 ft	3000 ft	4000 ft
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)				
	ALTITUDE				
	5000 ft	6000 ft	7000 ft	8000 ft	8300 ft
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)

NOTE: The values above Maximum Landing Weight must not be used for dispatch.

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**MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 1 – LANDING FLAP 3
ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED)**

PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)					
	ALTITUDE					
	-1000 ft	0 ft	1000 ft	2000 ft	3000 ft	4000 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
0	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
8	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)

NOTE: The shaded area and values above Maximum Landing Weight must not be used for dispatch.

**MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 1 – LANDING FLAP 3
ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED)**

PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)				
	ALTITUDE				
	5000 ft	6000 ft	7000 ft	8000 ft	8300 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18236 (E)	17965 (E)
0	18386 (O)	18386 (O)	18013 (E)	17151 (E)	16898 (E)
5	18386 (O)	17654 (E)	16755 (E)	15901 (E)	15657 (E)
8	17824 (E)	16865 (E)	15954 (E)	15101 (E)	14870 (E)

NOTE: The shaded area and values above Maximum Landing Weight must not be used for dispatch.



**MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 2 – LANDING FLAP FULL
ENGINE ICE PROTECTION OFF – WINGSTAB OFF
PW535E ENGINES**

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)					
	ALTITUDE					
	-1000 ft	0 ft	1000 ft	2000 ft	3000 ft	4000 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
0	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18018 (A)	17198 (A)
40	18386 (O)	18386 (O)	18386 (O)	17687 (A)	16843 (A)	16056 (A)
45	18386 (O)	18212 (A)	17356 (A)	16521 (A)	-	-
50	17822 (A)	17017 (A)	-	-	-	-

NOTE: The values above Maximum Landing Weight must not be used for dispatch.

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MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 2 – LANDING FLAP FULL
ENGINE ICE PROTECTION OFF – WINGSTAB OFF
PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)				
	ALTITUDE				
	5000 ft	6000 ft	7000 ft	8000 ft	8300 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
0	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18300 (A)
15	18386 (O)	18386 (O)	18386 (O)	17877 (A)	17665 (A)
20	18386 (O)	18386 (O)	17797 (A)	16986 (A)	16759 (A)
25	18386 (O)	17729 (A)	16748 (A)	15796 (A)	15549 (A)
30	17546 (A)	16600 (A)	15593 (A)	14667 (A)	14468 (A)
35	16386 (A)	15494 (A)	14557 (A)	-	-
40	15286 (A)	-	-	-	-

NOTE: The values above Maximum Landing Weight must not be used for dispatch.

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**MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 2 – LANDING FLAP FULL
ENGINE ICE PROTECTION ON – WINGSTAB OFF
PW535E ENGINES**

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)					
	ALTITUDE					
	-1000 ft	0 ft	1000 ft	2000 ft	3000 ft	4000 ft
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)				
	ALTITUDE				
	5000 ft	6000 ft	7000 ft	8000 ft	8300 ft
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
10	18386 (O)	18386 (O)	18386 (O)	17974 (A)	17786 (A)

NOTE: The values above Maximum Landing Weight must not be used for dispatch.



MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 2 – LANDING FLAP FULL
ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED)

PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)					
	ALTITUDE					
	-1000 ft	0 ft	1000 ft	2000 ft	3000 ft	4000 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
0	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
5	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
8	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)

NOTE: The shaded area and values above Maximum Landing Weight must not be used for dispatch.



**MAXIMUM LANDING WEIGHT – CLIMB LIMITED
APPROACH FLAP 2 – LANDING FLAP FULL
ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED)**

PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)				
	ALTITUDE				
	5000 ft	6000 ft	7000 ft	8000 ft	8300 ft
-40	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-35	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-30	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-25	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-20	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-15	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-10	18386 (O)	18386 (O)	18386 (O)	18386 (O)	18386 (O)
-5	18386 (O)	18386 (O)	18386 (O)	18236 (E)	17965 (E)
0	18386 (O)	18386 (O)	18013 (E)	17151 (E)	16898 (E)
5	18386 (O)	17654 (E)	16755 (E)	15901 (E)	15657 (E)
8	17824 (E)	16865 (E)	15954 (E)	15101 (E)	14870 (E)

NOTE: The shaded area and values above Maximum Landing Weight must not be used for dispatch.



FACTORED LANDING DISTANCES

Factored landing distance is the actual distance to land the airplane from a point 50 ft above runway threshold to complete stop, factored according operational rules, using the landing technique described in the beginning of this section.

Factored Landing Distance tables are presented for a set of pressure altitudes and winds for the conditions below:

- Dry runway;
- Zero slope;
- Ice Protection System:
For Wingstab OFF, Engine Ice Protection OFF or ON,
For Wingstab ON, Engine Ice Protection ON;
- Temperature: ISA (1);
- No SAT effect;
- No V_{REF} Overspeed;
- No Drag Index.

NOTE: (1) Wing and Stabilizer Ice-Protection system is auto-inhibited outside its operational envelope. Refer to its operational envelope in Section 2 for further information.

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



EXAMPLE

Given:

Airplane Landing Weight..... 16000 lb
Airport Pressure Altitude..... 2000 ft
Engine Ice Protection..... OFF
Wingstab OFF
Flap 3
Wind..... 10 kt

Determine:

Factored Landing Distance..... 3925 ft
V_{REF} 114 KIAS

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FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP 3**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	-1000 ft				0 ft				
	WIND								
-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt		
12400	4009	3358	3151	2950	4070	3416	3208	3006	101
12800	4074	3419	3211	3009	4137	3479	3270	3066	102
13200	4140	3482	3273	3069	4204	3543	3333	3128	104
13600	4207	3545	3335	3130	4273	3608	3397	3191	105
14000	4275	3609	3398	3192	4342	3674	3461	3254	107
14400	4338	3670	3458	3250	4408	3736	3522	3314	108
14800	4403	3731	3518	3309	4474	3799	3584	3374	110
15200	4468	3793	3578	3369	4541	3862	3646	3435	111
15600	4534	3856	3640	3429	4609	3926	3709	3497	113
16000	4601	3919	3702	3490	4677	3991	3773	3560	114
16400	4664	3978	3760	3547	4741	4052	3833	3618	115
16800	4727	4038	3819	3605	4806	4114	3893	3677	117
17200	4795	4099	3878	3663	4895	4176	3954	3737	118
17600	4891	4160	3939	3722	4996	4239	4016	3798	119
18000	4989	4221	3999	3781	5098	4301	4077	3858	121
18387	5086	4279	4056	3837	5199	4361	4136	3916	122

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.

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FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	1000 ft				2000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4136	3477	3269	3065	4203	3541	3331	3126	101
12800	4203	3542	3332	3127	4272	3607	3396	3190	102
13200	4273	3608	3396	3190	4343	3674	3462	3254	104
13600	4343	3674	3462	3254	4416	3743	3529	3320	105
14000	4414	3742	3528	3320	4489	3812	3597	3387	107
14400	4481	3806	3591	3381	4558	3878	3661	3450	108
14800	4549	3870	3654	3443	4627	3944	3726	3514	110
15200	4618	3935	3718	3506	4698	4011	3792	3578	111
15600	4688	4001	3782	3569	4769	4078	3858	3643	113
16000	4758	4068	3848	3633	4841	4146	3925	3709	114
16400	4824	4130	3909	3693	4909	4210	3988	3770	115
16800	4901	4193	3971	3754	5009	4275	4051	3833	117
17200	5002	4257	4033	3815	5113	4340	4116	3896	118
17600	5107	4322	4097	3878	5224	4407	4181	3960	119
18000	5214	4386	4160	3939	5336	4473	4246	4023	121
18387	5320	4451	4220	3998	5446	4560	4307	4084	122

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP 3**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	3000 ft				4000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4272	3607	3395	3189	4344	3675	3462	3255	101
12800	4344	3675	3462	3254	4417	3744	3530	3321	102
13200	4417	3744	3530	3321	4492	3815	3600	3390	104
13600	4491	3814	3599	3389	4568	3887	3671	3459	105
14000	4566	3885	3668	3457	4645	3960	3742	3529	107
14400	4636	3952	3734	3522	4718	4029	3810	3596	108
14800	4708	4020	3801	3587	4791	4099	3878	3663	110
15200	4780	4088	3868	3653	4866	4169	3947	3731	111
15600	4853	4158	3936	3720	4941	4240	4017	3799	113
16000	4927	4228	4005	3787	5022	4312	4088	3868	114
16400	5015	4294	4070	3851	5128	4380	4154	3933	115
16800	5121	4360	4135	3915	5239	4448	4221	3999	117
17200	5231	4427	4201	3979	5354	4517	4289	4065	118
17600	5346	4496	4268	4045	5475	4595	4358	4133	119
18000	5464	4580	4334	4110	5599	4695	4426	4200	121
18387	5580	4674	4401	4172	5721	4794	4515	4264	122

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	5000 ft				6000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4418	3745	3531	3322	4514	3832	3616	3405	101
12800	4493	3816	3601	3390	4591	3906	3688	3475	102
13200	4570	3889	3672	3461	4669	3980	3761	3547	104
13600	4648	3963	3745	3532	4749	4056	3835	3620	105
14000	4727	4038	3818	3604	4830	4133	3911	3694	107
14400	4802	4109	3888	3672	4907	4206	3982	3764	108
14800	4877	4180	3958	3741	4984	4279	4054	3835	110
15200	4954	4252	4029	3811	5062	4353	4127	3906	111
15600	5031	4326	4101	3881	5153	4428	4201	3978	113
16000	5135	4400	4173	3952	5272	4504	4275	4052	114
16400	5247	4469	4241	4019	5389	4576	4345	4120	115
16800	5363	4539	4310	4086	5511	4648	4416	4190	117
17200	5483	4610	4380	4155	5639	4738	4488	4260	118
17600	5610	4711	4451	4225	5773	4847	4568	4332	119
18000	5740	4816	4537	4293	5910	4958	4672	4402	121
18387	5869	4919	4634	4362	6047	5067	4774	4495	122

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP 3**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	7000 ft				8000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4602	3915	3697	3483	4694	4000	3780	3564	101
12800	4681	3990	3770	3555	4775	4077	3855	3639	102
13200	4762	4067	3845	3629	4857	4156	3933	3714	104
13600	4844	4145	3922	3704	4942	4236	4011	3792	105
14000	4927	4224	4000	3781	5027	4317	4091	3870	107
14400	5006	4298	4073	3852	5108	4394	4166	3943	108
14800	5085	4374	4147	3925	5189	4471	4242	4018	110
15200	5166	4450	4222	3998	5289	4549	4319	4094	111
15600	5281	4527	4297	4073	5415	4629	4397	4170	113
16000	5407	4605	4374	4148	5547	4709	4476	4247	114
16400	5530	4678	4446	4218	5677	4784	4549	4320	115
16800	5658	4760	4519	4290	5812	4891	4624	4393	117
17200	5792	4869	4592	4362	5953	5006	4720	4467	118
17600	5934	4983	4697	4436	6103	5126	4833	4553	119
18000	6079	5100	4806	4527	6257	5249	4948	4661	121
18387	6223	5215	4914	4627	6410	5371	5061	4767	122

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE				V _{REF} (KIAS)
	8300 ft				
	WIND				
	-10 kt	0 kt	10 kt	20 kt	
12400	4721	4026	3805	3589	101
12800	4803	4104	3881	3664	102
13200	4887	4183	3959	3740	104
13600	4971	4264	4038	3818	105
14000	5058	4346	4119	3897	107
14400	5139	4423	4194	3971	108
14800	5221	4501	4271	4047	110
15200	5329	4580	4349	4123	111
15600	5457	4660	4427	4200	113
16000	5591	4741	4507	4278	114
16400	5722	4820	4581	4351	115
16800	5860	4931	4657	4425	117
17200	6003	5048	4760	4500	118
17600	6155	5170	4875	4593	119
18000	6312	5296	4991	4702	121
18387	6468	5419	5107	4810	122

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.

FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON (SWPS ICE SPEED) – FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	-1000 ft				0 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4348	3675	3462	3253	4416	3740	3526	3316	108
12800	4424	3748	3533	3323	4494	3815	3599	3388	110
13200	4497	3817	3601	3390	4569	3886	3669	3456	111
13600	4571	3887	3670	3458	4645	3958	3739	3526	113
14000	4646	3958	3740	3526	4722	4031	3811	3596	115
14400	4719	4027	3807	3593	4797	4102	3880	3664	116
14800	4794	4099	3877	3661	4874	4175	3952	3735	118
15200	4869	4170	3947	3730	4951	4248	4024	3805	119
15600	4941	4238	4014	3796	5025	4318	4093	3873	121
16000	5013	4306	4081	3861	5119	4388	4161	3940	122
16400	5123	4377	4151	3929	5234	4461	4233	4010	124
16800	5233	4445	4217	3995	5349	4530	4301	4077	125
17200	5347	4514	4285	4061	5468	4601	4370	4145	127
17600	5466	4586	4353	4128	5592	4695	4440	4214	128
18000	5582	4680	4417	4190	5714	4794	4516	4278	130
18387	5706	4781	4503	4256	5845	4899	4615	4346	131

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	1000 ft				2000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4489	3810	3594	3383	4565	3881	3664	3452	108
12800	4570	3886	3669	3456	4648	3960	3741	3527	110
13200	4647	3959	3740	3527	4726	4035	3814	3599	111
13600	4725	4033	3813	3598	4806	4110	3889	3672	113
14000	4804	4108	3886	3670	4887	4187	3964	3746	115
14400	4880	4180	3958	3740	4966	4262	4037	3818	116
14800	4959	4255	4031	3812	5047	4339	4113	3892	118
15200	5038	4331	4105	3885	5129	4416	4189	3967	119
15600	5120	4403	4176	3954	5233	4490	4261	4038	121
16000	5232	4474	4246	4023	5351	4563	4333	4109	122
16400	5352	4549	4319	4095	5476	4640	4409	4182	124
16800	5472	4620	4389	4164	5602	4715	4481	4253	125
17200	5597	4705	4461	4233	5733	4822	4554	4325	127
17600	5728	4811	4534	4304	5870	4933	4650	4397	128
18000	5856	4915	4631	4370	6006	5042	4752	4476	130
18387	5994	5025	4735	4458	6150	5158	4860	4578	131

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION ON – WINGSTAB ON
 (SWPS ICE SPEED) – FLAP 3**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	3000 ft				4000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4643	3956	3737	3523	4724	4032	3812	3597	108
12800	4728	4036	3816	3600	4811	4115	3893	3676	110
13200	4809	4113	3891	3675	4895	4194	3970	3752	111
13600	4891	4191	3968	3750	4979	4274	4049	3829	113
14000	4974	4270	4045	3826	5064	4355	4129	3907	115
14400	5055	4346	4120	3899	5147	4433	4206	3983	116
14800	5138	4425	4198	3975	5233	4515	4285	4061	118
15200	5232	4505	4276	4052	5352	4596	4365	4140	119
15600	5352	4580	4350	4125	5477	4674	4442	4215	121
16000	5475	4656	4424	4198	5606	4752	4518	4290	122
16400	5607	4735	4502	4273	5744	4843	4597	4368	124
16800	5739	4832	4576	4346	5883	4956	4676	4442	125
17200	5877	4945	4663	4420	6028	5074	4786	4518	127
17600	6021	5061	4773	4497	6180	5196	4901	4619	128
18000	6164	5175	4879	4597	6330	5316	5013	4724	130
18387	6316	5297	4993	4704	6492	5445	5132	4836	131

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	5000 ft				6000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4808	4111	3890	3673	4916	4211	3986	3767	108
12800	4897	4196	3973	3754	5007	4297	4071	3850	110
13200	4983	4277	4052	3833	5095	4381	4153	3931	111
13600	5069	4359	4133	3912	5184	4465	4236	4012	113
14000	5157	4443	4215	3992	5274	4551	4320	4094	115
14400	5242	4523	4294	4070	5361	4633	4401	4174	116
14800	5344	4607	4376	4150	5490	4719	4485	4257	118
15200	5476	4691	4458	4231	5629	4805	4570	4340	119
15600	5608	4771	4537	4308	5767	4888	4651	4419	121
16000	5743	4850	4615	4385	5910	4989	4731	4498	122
16400	5888	4967	4697	4465	6063	5115	4828	4580	124
16800	6034	5086	4799	4542	6217	5240	4945	4664	125
17200	6187	5209	4915	4635	6379	5370	5067	4779	127
17600	6348	5338	5035	4748	6549	5506	5194	4898	128
18000	6507	5465	5153	4858	6719	5640	5319	5014	130
18387	6678	5600	5279	4975	6901	5784	5453	5139	131

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.

FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON (SWPS ICE SPEED) – FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	7000 ft				8000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	5016	4304	4077	3856	5119	4400	4171	3948	108
12800	5110	4393	4165	3942	5215	4492	4261	4036	110
13200	5200	4479	4249	4024	5308	4580	4348	4121	111
13600	5291	4565	4334	4108	5401	4669	4435	4207	113
14000	5383	4653	4420	4192	5496	4759	4524	4294	115
14400	5492	4739	4504	4275	5633	4847	4610	4378	116
14800	5632	4827	4590	4359	5780	4937	4699	4465	118
15200	5777	4915	4677	4445	5933	5028	4788	4553	119
15600	5923	5006	4760	4526	6087	5146	4874	4637	121
16000	6073	5129	4843	4608	6246	5276	4983	4721	122
16400	6235	5261	4967	4692	6416	5415	5113	4825	124
16800	6398	5393	5090	4802	6589	5554	5243	4947	125
17200	6570	5530	5219	4923	6771	5699	5379	5074	127
17600	6750	5674	5353	5048	6963	5851	5520	5206	128
18000	6930	5816	5485	5171	7155	6002	5660	5336	130
18387	7124	5968	5626	5302	7362	6164	5810	5476	131

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP 3

PW535E ENGINES

Weight (lb)	ALTITUDE				V _{REF} (KIAS)
	8300 ft				
	WIND				
	-10 kt	0 kt	10 kt	20 kt	
12400	5150	4430	4200	3976	108
12800	5247	4522	4291	4065	110
13200	5341	4611	4378	4150	111
13600	5435	4701	4466	4237	113
14000	5537	4792	4556	4325	115
14400	5677	4880	4642	4410	116
14800	5826	4971	4732	4498	118
15200	5982	5063	4822	4586	119
15600	6138	5190	4908	4671	121
16000	6299	5321	5026	4755	122
16400	6473	5462	5158	4868	124
16800	6648	5604	5290	4992	125
17200	6834	5751	5428	5121	127
17600	7029	5906	5572	5256	128
18000	7225	6060	5715	5388	130
18387	7436	6224	5867	5530	131

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP FULL**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	-1000 ft				0 ft				
	WIND								
-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt		
12400	3898	3257	3054	2856	3959	3315	3111	2912	100
12800	3959	3315	3111	2912	4022	3375	3169	2969	101
13200	4025	3378	3173	2972	4089	3439	3232	3031	103
13600	4088	3437	3231	3029	4154	3499	3292	3089	104
14000	4151	3497	3289	3087	4218	3560	3352	3148	106
14400	4215	3557	3348	3145	4284	3622	3412	3207	107
14800	4275	3614	3404	3199	4345	3680	3469	3263	109
15200	4340	3676	3464	3258	4412	3743	3531	3324	110
15600	4401	3733	3521	3314	4474	3803	3589	3381	111
16000	4460	3789	3575	3367	4535	3860	3645	3435	113
16400	4523	3848	3634	3425	4599	3921	3705	3494	114
16800	4582	3904	3688	3478	4659	3978	3761	3549	115
17200	4641	3960	3743	3532	4721	4035	3817	3604	117
17600	4717	4021	3803	3590	4817	4097	3878	3663	118
18000	4808	4078	3859	3645	4911	4156	3936	3720	119
18387	4904	4139	3919	3703	5011	4218	3997	3780	121

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP FULL

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	1000 ft				2000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4024	3376	3171	2971	4091	3439	3233	3031	100
12800	4088	3437	3231	3029	4157	3502	3294	3091	101
13200	4158	3503	3295	3092	4228	3569	3360	3156	103
13600	4223	3565	3356	3152	4295	3633	3422	3217	104
14000	4290	3628	3418	3212	4363	3697	3485	3279	106
14400	4357	3691	3480	3273	4432	3762	3549	3341	107
14800	4419	3751	3538	3330	4496	3823	3609	3400	109
15200	4488	3815	3601	3393	4566	3889	3674	3464	110
15600	4552	3876	3661	3451	4632	3952	3735	3523	111
16000	4614	3934	3718	3507	4695	4011	3794	3581	113
16400	4680	3997	3779	3567	4763	4076	3857	3643	114
16800	4742	4055	3837	3623	4835	4135	3915	3700	115
17200	4823	4114	3894	3680	4930	4196	3975	3758	117
17600	4923	4178	3957	3741	5034	4261	4038	3821	118
18000	5022	4239	4016	3799	5137	4323	4100	3881	119
18387	5126	4302	4079	3860	5246	4389	4163	3943	121

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP FULL**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	3000 ft				4000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4160	3505	3297	3094	4232	3572	3363	3159	100
12800	4228	3569	3360	3155	4301	3638	3427	3222	101
13200	4301	3638	3427	3222	4376	3709	3496	3290	103
13600	4370	3703	3491	3284	4446	3775	3562	3354	104
14000	4439	3769	3556	3348	4518	3843	3628	3419	106
14400	4510	3836	3621	3412	4590	3911	3695	3485	107
14800	4576	3898	3682	3472	4658	3975	3758	3546	109
15200	4648	3966	3749	3537	4732	4045	3827	3613	110
15600	4715	4030	3811	3598	4801	4110	3891	3676	111
16000	4780	4091	3872	3657	4867	4173	3952	3736	113
16400	4850	4157	3936	3721	4954	4241	4019	3801	114
16800	4942	4218	3997	3780	5054	4304	4081	3862	115
17200	5041	4280	4057	3839	5158	4368	4143	3923	117
17600	5150	4347	4123	3904	5272	4436	4210	3989	118
18000	5258	4411	4186	3965	5385	4512	4275	4052	119
18387	5372	4496	4251	4029	5505	4610	4342	4118	121

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP FULL

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	5000 ft				6000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4305	3642	3431	3225	4400	3729	3515	3307	100
12800	4376	3709	3497	3290	4473	3798	3583	3374	101
13200	4453	3781	3568	3359	4552	3872	3656	3445	103
13600	4525	3850	3635	3425	4626	3942	3725	3512	104
14000	4599	3919	3703	3492	4701	4013	3794	3581	106
14400	4673	3989	3772	3559	4777	4085	3865	3650	107
14800	4742	4055	3836	3623	4848	4152	3931	3715	109
15200	4818	4127	3906	3691	4925	4226	4003	3785	110
15600	4889	4194	3972	3756	4998	4295	4070	3851	111
16000	4959	4258	4036	3818	5091	4361	4135	3915	113
16400	5067	4328	4104	3885	5204	4432	4205	3984	114
16800	5172	4392	4167	3947	5313	4498	4270	4047	115
17200	5280	4458	4231	4010	5427	4565	4336	4112	117
17600	5399	4530	4300	4077	5552	4659	4407	4181	118
18000	5518	4626	4366	4142	5677	4760	4484	4248	119
18387	5644	4728	4453	4210	5810	4868	4585	4317	121

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP FULL**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	7000 ft				8000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4489	3811	3595	3385	4579	3895	3678	3465	100
12800	4563	3881	3665	3453	4656	3968	3749	3535	101
13200	4644	3958	3739	3526	4738	4046	3825	3610	103
13600	4720	4030	3810	3596	4816	4120	3898	3681	104
14000	4796	4102	3882	3666	4895	4194	3971	3753	106
14400	4874	4176	3954	3737	4975	4270	4045	3826	107
14800	4947	4245	4022	3803	5049	4341	4115	3894	109
15200	5027	4321	4096	3876	5131	4418	4191	3969	110
15600	5110	4391	4165	3943	5239	4490	4262	4038	111
16000	5219	4459	4231	4009	5353	4560	4330	4105	113
16400	5337	4532	4303	4079	5476	4635	4404	4178	114
16800	5452	4600	4370	4145	5597	4707	4472	4245	115
17200	5572	4680	4437	4211	5723	4810	4542	4313	117
17600	5703	4787	4511	4282	5862	4922	4639	4386	118
18000	5835	4894	4611	4350	6000	5033	4743	4467	119
18387	5974	5006	4716	4440	6148	5152	4854	4571	121

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.

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FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
FLAP FULL

PW535E ENGINES

Weight (lb)	ALTITUDE				V _{REF} (KIAS)
	8300 ft				
	WIND				
	-10 kt	0 kt	10 kt	20 kt	
12400	4607	3921	3703	3490	100
12800	4684	3994	3774	3560	101
13200	4767	4073	3852	3636	103
13600	4845	4147	3925	3707	104
14000	4925	4222	3999	3780	106
14400	5005	4299	4073	3853	107
14800	5081	4370	4143	3922	109
15200	5164	4448	4220	3997	110
15600	5279	4521	4291	4067	111
16000	5394	4591	4361	4135	113
16400	5520	4667	4435	4208	114
16800	5642	4745	4504	4276	115
17200	5770	4849	4574	4344	117
17600	5911	4963	4678	4418	118
18000	6051	5077	4784	4506	119
18387	6201	5197	4897	4611	121

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.

FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON (SWPS ICE SPEED) – FLAP FULL

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	-1000 ft				0 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4182	3523	3313	3109	4249	3586	3376	3170	106
12800	4254	3590	3380	3175	4322	3656	3444	3237	108
13200	4322	3655	3443	3237	4392	3722	3509	3301	109
13600	4391	3721	3508	3300	4463	3789	3575	3366	111
14000	4461	3787	3572	3363	4535	3857	3641	3431	112
14400	4528	3851	3635	3425	4604	3922	3705	3494	114
14800	4598	3917	3700	3488	4675	3990	3772	3559	115
15200	4668	3983	3765	3552	4747	4058	3838	3624	117
15600	4734	4045	3826	3612	4815	4122	3901	3686	118
16000	4800	4108	3887	3672	4883	4186	3964	3747	119
16400	4869	4173	3951	3735	4968	4253	4030	3811	121
16800	4962	4235	4012	3794	5068	4316	4092	3873	122
17200	5062	4298	4074	3855	5173	4381	4155	3935	124
17600	5166	4361	4136	3916	5281	4446	4219	3997	125
18000	5278	4429	4202	3981	5399	4524	4287	4064	126
18387	5386	4508	4263	4040	5511	4616	4349	4125	128

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP FULL

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	1000 ft				2000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4321	3654	3442	3235	4394	3723	3510	3302	106
12800	4396	3725	3512	3304	4472	3796	3582	3372	108
13200	4468	3793	3578	3369	4545	3866	3650	3439	109
13600	4540	3862	3646	3435	4619	3937	3719	3507	111
14000	4614	3931	3714	3502	4695	4008	3789	3576	112
14400	4684	3998	3780	3567	4767	4077	3857	3642	114
14800	4758	4068	3848	3633	4843	4148	3926	3710	115
15200	4831	4137	3916	3700	4918	4219	3996	3779	117
15600	4901	4203	3981	3764	4989	4287	4063	3844	118
16000	4971	4269	4045	3827	5081	4354	4129	3908	119
16400	5077	4337	4112	3892	5191	4424	4197	3976	121
16800	5182	4402	4176	3955	5301	4491	4263	4040	122
17200	5291	4468	4241	4019	5415	4559	4330	4106	124
17600	5405	4535	4306	4083	5534	4646	4397	4171	125
18000	5528	4635	4376	4151	5663	4751	4476	4241	126
18387	5646	4731	4455	4214	5787	4851	4570	4306	128

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP FULL**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	3000 ft				4000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4471	3795	3581	3371	4550	3870	3654	3443	106
12800	4550	3870	3654	3443	4631	3947	3729	3516	108
13200	4625	3942	3724	3512	4708	4020	3801	3587	109
13600	4701	4014	3795	3581	4786	4094	3874	3658	111
14000	4779	4087	3867	3652	4866	4169	3947	3730	112
14400	4853	4158	3936	3720	4942	4241	4018	3800	114
14800	4930	4231	4008	3790	5021	4316	4092	3872	115
15200	5008	4304	4079	3860	5100	4391	4165	3944	117
15600	5087	4373	4147	3927	5203	4463	4235	4013	118
16000	5196	4442	4215	3993	5316	4533	4305	4081	119
16400	5312	4514	4286	4063	5437	4607	4377	4152	121
16800	5426	4583	4353	4129	5557	4678	4446	4220	122
17200	5546	4662	4422	4196	5682	4779	4516	4289	124
17600	5670	4763	4490	4263	5813	4885	4605	4358	125
18000	5805	4873	4592	4335	5955	5000	4713	4439	126
18387	5935	4977	4690	4416	6092	5110	4816	4536	128

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.

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FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP FULL

PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	5000 ft				6000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4631	3946	3729	3516	4736	4043	3823	3607	106
12800	4714	4025	3806	3592	4821	4124	3902	3685	108
13200	4793	4100	3880	3664	4902	4201	3977	3759	109
13600	4874	4176	3954	3737	4985	4279	4054	3834	111
14000	4955	4253	4030	3811	5068	4358	4131	3910	112
14400	5033	4328	4103	3883	5148	4434	4206	3983	114
14800	5115	4405	4178	3957	5231	4513	4283	4059	115
15200	5208	4482	4254	4031	5349	4592	4361	4135	117
15600	5324	4555	4326	4101	5471	4667	4435	4208	118
16000	5442	4628	4397	4171	5595	4741	4508	4279	119
16400	5569	4703	4471	4244	5728	4828	4584	4354	121
16800	5695	4795	4543	4314	5861	4936	4657	4426	122
17200	5826	4902	4623	4385	5999	5049	4762	4498	124
17600	5963	5013	4727	4456	6144	5165	4871	4591	125
18000	6112	5134	4840	4560	6301	5292	4990	4702	126
18387	6256	5249	4948	4661	6454	5414	5103	4808	128

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.


FACTORED LANDING DISTANCE (ft)
**ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP FULL**
PW535E ENGINES

Weight (lb)	ALTITUDE								V _{REF} (KIAS)
	7000 ft				8000 ft				
	WIND								
	-10 kt	0 kt	10 kt	20 kt	-10 kt	0 kt	10 kt	20 kt	
12400	4833	4134	3911	3694	4933	4227	4002	3782	106
12800	4921	4217	3992	3773	5023	4312	4086	3864	108
13200	5004	4296	4070	3849	5108	4393	4165	3942	109
13600	5088	4376	4148	3926	5195	4475	4246	4021	111
14000	5174	4457	4228	4004	5282	4558	4327	4101	112
14400	5256	4535	4305	4080	5367	4638	4406	4179	114
14800	5358	4615	4384	4157	5496	4721	4487	4259	115
15200	5487	4697	4463	4235	5630	4805	4569	4339	117
15600	5614	4774	4539	4310	5764	4884	4647	4415	118
16000	5744	4850	4614	4383	5901	4981	4724	4491	119
16400	5884	4961	4693	4460	6048	5101	4815	4570	121
16800	6024	5075	4789	4534	6195	5221	4927	4647	122
17200	6170	5193	4899	4619	6349	5345	5043	4756	124
17600	6322	5316	5014	4727	6510	5474	5164	4869	125
18000	6489	5449	5139	4843	6687	5615	5295	4992	126
18387	6651	5578	5258	4955	6858	5751	5421	5109	128

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



FACTORED LANDING DISTANCE (ft)

ENGINE ICE PROTECTION ON – WINGSTAB ON
(SWPS ICE SPEED) – FLAP FULL

PW535E ENGINES

Weight (lb)	ALTITUDE				V _{REF} (KIAS)
	8300 ft				
	WIND				
	-10 kt	0 kt	10 kt	20 kt	
12400	4964	4256	4030	3810	106
12800	5054	4341	4114	3892	108
13200	5140	4423	4194	3971	109
13600	5227	4506	4275	4050	111
14000	5316	4589	4358	4131	112
14400	5407	4670	4437	4209	114
14800	5538	4754	4519	4290	115
15200	5675	4838	4601	4370	117
15600	5810	4918	4680	4447	118
16000	5949	5023	4758	4524	119
16400	6099	5145	4856	4603	121
16800	6248	5266	4970	4688	122
17200	6405	5392	5088	4798	124
17600	6569	5523	5210	4913	125
18000	6748	5666	5344	5038	126
18387	6923	5804	5472	5157	128

CAUTION: • SHADED AREAS REPRESENT CONDITIONS WHERE THE MAXIMUM LANDING WEIGHT OR CLIMB LIMITED WEIGHT WAS EXCEEDED.

- TO CALCULATE THE UNFACTORED LANDING DISTANCE, USE THE LANDING DISTANCE CORRECTION TABLE IN THIS BLOCK OR THE OPERA SOFTWARE.



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CORRECTED LANDING DISTANCES – ABNORMAL LANDINGS

In order to determine the landing distances on abnormal landings, the Landing Distance Correction table should be used.

TABLE USE

Obtain the reference factored landing distance, considering flap FULL and Ice Protection OFF for the selected configuration (weight, altitude and wind) on the Factored Landing Distance tables.

Enter this reference distance in the Factored Dry Column and go to the right horizontally until intercepting the Minimum Required Runway Length according to the Abnormal Landing Factor for the abnormal condition.

NOTE: Do not interpolate between distances. Use the next highest value of factored landing distance available in Landing Distance Correction table.

EXAMPLE

Given:

Airplane Landing Weight	16800 lb
Airport Pressure Altitude	2000 ft
Engine Ice Protection	ON
Wingstab.....	ON
Flap.....	3
Wind	10 kt

Determine:

Required Landing Distance for Abnormal Landing with Brake Failure (factor 1.85):

- 1) Factored Landing Distance
 Flap FULL (dry) 3915 ft
- 2) Next Highest Value..... 4000 ft
- 3) Minimum Required
 Runway Length (factor 1.85)..... 4934 ft



LANDING DISTANCE CORRECTION

PW535E ENGINES

MINIMUM REQUIRED RUNWAY LENGTH (ft)							
DRY FACTORED	DRY UNFACTORED	ABNORMAL LANDING FACTORS					
		1.00	1.15	1.20	1.25	1.30	1.35
2400	1600	1600	1840	1920	2000	2080	2160
2800	1867	1867	2147	2240	2334	2427	2520
3200	2133	2133	2453	2560	2666	2773	2880
3600	2400	2400	2760	2880	3000	3120	3240
4000	2667	2667	3067	3200	3334	3467	3600
4400	2933	2933	3373	3520	3666	3813	3960
4800	3200	3200	3680	3840	4000	4160	4320
5200	3467	3467	3987	4160	4334	4507	4680
5600	3733	3733	4293	4480	4666	4853	5040
6000	4000	4000	4600	4800	5000	5200	5400
6400	4267	4267	4907	5120	5334	5547	5760
6800	4533	4533	5213	5440	5666	5893	6120
7000	4667	4667	5367	5600	5834	6067	6300

MINIMUM REQUIRED RUNWAY LENGTH (ft)							
DRY FACTORED	DRY UNFACTORED	ABNORMAL LANDING FACTORS					
		1.40	1.45	1.50	1.55	1.60	1.65
2400	1600	2240	2320	2400	2480	2560	2640
2800	1867	2614	2707	2801	2894	2987	3081
3200	2133	2986	3093	3200	3306	3413	3519
3600	2400	3360	3480	3600	3720	3840	3960
4000	2667	3734	3867	4001	4134	4267	4401
4400	2933	4106	4253	4400	4546	4693	4839
4800	3200	4480	4640	4800	4960	5120	5280
5200	3467	4854	5027	5201	5374	5547	5721
5600	3733	5226	5413	5600	5786	5973	6159
6000	4000	5600	5800	6000	6200	6400	6600
6400	4267	5974	6187	6401	6614	6827	7041
6800	4533	6346	6573	6800	7026	7253	7479
7000	4667	6534	6767	7001	7234	7467	7701



LANDING DISTANCE CORRECTION

PW535E ENGINES

MINIMUM REQUIRED RUNWAY LENGTH (ft)							
DRY FACTORED	DRY UNFACTORED	ABNORMAL LANDING FACTORS					
		1.70	1.80	1.85	1.90	2.00	2.15
2400	1600	2720	2880	2960	3040	3200	3440
2800	1867	3174	3361	3454	3547	3734	4014
3200	2133	3626	3839	3946	4053	4266	4586
3600	2400	4080	4320	4440	4560	4800	5160
4000	2667	4534	4801	4934	5067	5334	5734
4400	2933	4986	5279	5426	5573	5866	6306
4800	3200	5440	5760	5920	6080	6400	6880
5200	3467	5894	6241	6414	6587	6934	7454
5600	3733	6346	6719	6906	7093	7466	8026
6000	4000	6800	7200	7400	7600	8000	8600
6400	4267	7254	7681	7894	8107	8534	9174
6800	4533	7706	8159	8386	8613	9066	9746
7000	4667	7934	8401	8634	8867	9334	10034

MINIMUM REQUIRED RUNWAY LENGTH (ft)							
DRY FACTORED	DRY UNFACTORED	ABNORMAL LANDING FACTORS					
		2.20	2.30	2.70	3.10	-	-
2400	1600	3520	3680	4320	4960	-	-
2800	1867	4107	4294	5041	5788	-	-
3200	2133	4693	4906	5759	6612	-	-
3600	2400	5280	5520	6480	7440	-	-
4000	2667	5867	6134	7201	8268	-	-
4400	2933	6453	6746	7919	9092	-	-
4800	3200	7040	7360	8640	9920	-	-
5200	3467	7627	7974	9361	10748	-	-
5600	3733	8213	8586	10079	11572	-	-
6000	4000	8800	9200	10800	12400	-	-
6400	4267	9387	9814	11521	13228	-	-
6800	4533	9973	10426	12239	14052	-	-
7000	4667	10267	10734	12601	14468	-	-

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CORRECTED LANDING DISTANCES – WET RUNWAY AND DRY OVERSPEED RUNWAY

In order to determine the landing distances on wet runway or dry overspeed runways, the Corrected Landing Distance table should be used.

USE

Obtain the factored landing distance for the selected configuration (weight/ice protection setting/altitude and wind) on the Factored Landing Distance tables.

Enter this distance in the Factored Dry Column and go to the right horizontally until intercepting the Minimum Required Landing Distance according to the condition desired.

NOTE: Do not interpolate between distances. Use the next highest value of factored landing distance available in Corrected Landing Distance table.



EXAMPLE

Given:

Airplane Landing Weight	16800 lb
Airport Pressure Altitude	2000 ft
Engine Ice Protection	OFF
Wingstab.....	OFF
Flap.....	FULL
Wind	10 kt

Determine:

- Required Factored Landing Distance (wet):
 - 1) Factored Landing Distance
 Flap FULL (dry) 3915 ft
 - 2) Next Highest Value 4000 ft
 - 3) Minimum Required
 Runway Length (wet) 4602 ft
- Required Dry Overspeed ($V_{REF} + 10$ KIAS) Landing Distance:
 - 1) Factored Landing Distance (dry)..... 3915 ft
 - 2) Next Highest Value 4000 ft
 - 3) Minimum Required
 Runway Length (dry)..... 3124 ft



CORRECTED LANDING DISTANCE WET RUNWAY AND DRY OVERSPEED RUNWAY

PW535E ENGINES

MINIMUM REQUIRED RUNWAY LENGTH (ft)				
FACTORED (V _{REF})		UNFACTORED		
		(V _{REF})		DRY OVERSPEED (V _{REF} + 10 KIAS)
DRY	WET	DRY	WET	DRY
2400	2762	1603	1922	1893
2800	3222	1870	2415	2201
3200	3682	2137	2908	2509
3600	4142	2404	3400	2817
4000	4602	2671	3893	3124
4400	5061	2939	4385	3432
4800	5521	3206	4878	3740
5200	5981	3473	5370	4048
5600	6441	3740	5863	4356
6000	6901	4007	6356	4663
6400	7361	4274	6848	4971
6800	7821	4542	7341	5279
7000	8051	4675	7587	5433

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CORRECTED LANDING DISTANCES – CONTAMINATED RUNWAYS

Corrected landing distances data shown is calculated for runways contaminated with standing water, slush or wet snow at the following depths: 0.125 in, 0.250 in and 0.375 in. They are based on the factored landing distance for a given airplane configuration.

NOTE: For different contaminants and depths refer to OPERA software.

TABLE USE

Obtain the dry factored landing distance for the selected configuration (weight/ice protection setting/altitude and wind) on the Factored Landing Distance tables.

Enter this distance in the Factored Landing Distance Dry Column and go to the right horizontally until intercepting the Corrected Landing Distance according to the runway condition.

NOTE: Do not interpolate between distances. Use the dry factored landing distance next highest value in Corrected Landing Distance tables.

EXAMPLE

Given:

Airplane Landing Weight	16800 lb
Airport Pressure Altitude	2000 ft
Ice Protection.....	OFF
Flap.....	FULL
Wind	10 kt

Determine:

Factored Landing Distance (dry)	3915 ft
Next Highest Value.....	4000 ft
Landing Distance (standing water 0.250 in)	5468 ft
Landing Distance (slush 0.250 in)	5580 ft
Landing Distance (wet snow 0.250 in)	5722 ft

**CORRECTED LANDING DISTANCE (ft)**

**STANDING WATER COVERED RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF/ON
LANDING FLAP 3**

PW535E ENGINES

FACTORED LANDING DISTANCE DRY (ft)	STANDING WATER DEPTH (in)		
	0.125	0.250	0.375
2800	3719	3554	3417
3200	4471	4200	3984
3600	5223	4846	4552
4000	5975	5492	5119
4400	6727	6138	5687
4800	7479	6784	6254
5200	8231	7430	6822
5600	8983	8076	7389
6000	9735	8722	7957

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CORRECTED LANDING DISTANCE (ft)
**SLUSH COVERED RUNWAY
 ICE PROTECTION (WINGSTAB+ENG) OFF/ON
 LANDING FLAP 3**
PW535E ENGINES

FACTORED LANDING DISTANCE DRY (ft)	SLUSH DEPTH (in)		
	0.125	0.250	0.375
2800	3744	3596	3463
3200	4514	4268	4060
3600	5285	4941	4658
4000	6055	5613	5255
4400	6825	6286	5853
4800	7596	6959	6450
5200	8366	7631	7048
5600	9136	8304	7645
6000	9907	8976	8242

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**CORRECTED LANDING DISTANCE (ft)**

**WET SNOW COVERED RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF/ON
LANDING FLAP 3**

PW535E ENGINES

FACTORED LANDING DISTANCE DRY (ft)	WET SNOW DEPTH (in)		
	0.125	0.250	0.375
2800	4330	4109	3999
3200	4947	4663	4526
3600	5564	5218	5053
4000	6182	5772	5579
4400	6799	6327	6106
4800	7416	6881	6633
5200	8033	7436	7159
5600	8650	7990	7686
6000	9267	8545	8213

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CORRECTED LANDING DISTANCE (ft)
**STANDING WATER COVERED RUNWAY
 ICE PROTECTION (WINGSTAB+ENG) OFF/ON
 LANDING FLAP FULL**
PW535E ENGINES

FACTORED LANDING DISTANCE DRY (ft)	STANDING WATER DEPTH (in)		
	0.125	0.250	0.375
2800	3738	3569	3439
3200	4460	4202	4003
3600	5183	4835	4567
4000	5906	5468	5130
4400	6629	6101	5694
4800	7351	6734	6258
5200	8074	7367	6821
5600	8797	8000	7385
6000	9519	8633	7949
6400	10242	9266	8513

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**CORRECTED LANDING DISTANCE (ft)**

**SLUSH COVERED RUNWAY
ICE PROTECTION (WINGSTAB+ENG) OFF/ON
LANDING FLAP FULL**

PW535E ENGINES

FACTORED LANDING DISTANCE DRY (ft)	SLUSH DEPTH (in)		
	0.125	0.250	0.375
2800	3764	3612	3476
3200	4501	4268	4067
3600	5239	4924	4657
4000	5977	5580	5247
4400	6715	6236	5837
4800	7452	6892	6428
5200	8190	7547	7018
5600	8928	8203	7608
6000	9665	8859	8198
6400	10403	9515	8789

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CORRECTED LANDING DISTANCE (ft)
**WET SNOW COVERED RUNWAY
 ICE PROTECTION (WINGSTAB+ENG) OFF/ON
 LANDING FLAP FULL**
PW535E ENGINES

FACTORED LANDING DISTANCE DRY (ft)	WET SNOW DEPTH (in)		
	0.125	0.250	0.375
2800	4215	4018	3921
3200	4843	4586	4461
3600	5472	5154	5002
4000	6100	5722	5542
4400	6728	6290	6083
4800	7356	6857	6623
5200	7985	7425	7164
5600	8613	7993	7704
6000	9241	8561	8245
6400	9869	9129	8786

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SECTION 6

WEIGHT AND BALANCE

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INTRODUCTION

The Weight and Balance Section provides instructions referring to the weighting and loading of the PHENOM 300 airplane model.

The instructions and data herein presented are approved by the airworthiness authority to comply with the applicable regulations.

The basic empty weight value obtained during the airplane weighing procedures should be used as point of departure for each loading operation.

Based on the presented information, the operator can determine airplane weight and CG at any time of flight.



WEIGHT AND CENTER OF GRAVITY LIMITS

To comply with the performance and operating limitations of the regulations, the maximum allowable takeoff and landing operational weights may be equal to, but not greater than design limits.

Refer to Section 2, Limitations, for further information.

WARNING: IT IS RESPONSIBILITY OF THE OPERATOR TO ENSURE THAT THE AIRPLANE IS LOADED SO AS TO REMAIN WITHIN THE CENTER OF GRAVITY LIMITS THROUGHOUT THE FLIGHT.



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BALANCE REFERENCE SYSTEM

BALANCE ARMS

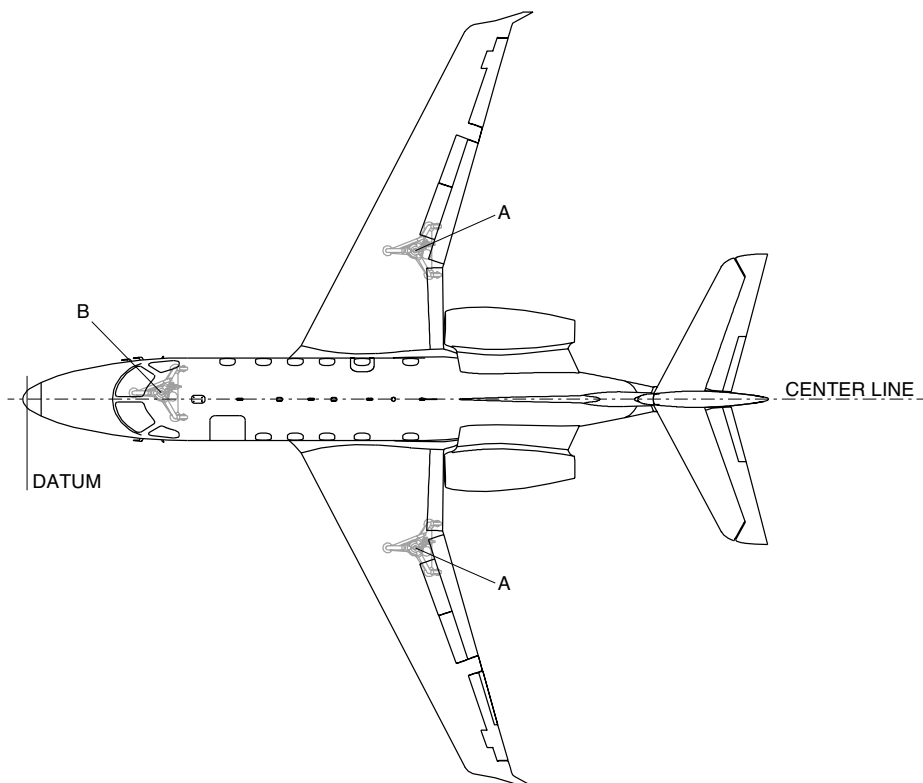
Longitudinal location of the Centers of Gravity (CG) identified throughout this manual regarding airplane and components will be referred to as Balance Arms. Balance Arms are the distance in inches from Airplane Datum.

AIRPLANE DATUM

The Airplane Datum is a plane, perpendicular to the fuselage centerline.

For external reference, the Datum is located at 90.00 inches ahead of the Forward Jack Point.

POINT	BALANCE ARM (in)	CENTERLINE DISTANCE (in)
A	313.26	120.16
B	90.00	6.06



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**WING MEAN AERODYNAMIC CHORD (MAC)**

MAC length = 80.71 in.

LEMAC balance arm = 264.51 in.

Percentage of MAC is obtained using the following formula:

$$\%MAC = \frac{(B.A. - 264.51) \times 100}{80.71}$$

Where B.A. = Balance arm of airplane CG measured in inches.



CONFIGURATION CHECKLIST/EQUIPMENT LIST

The balance arms are shown in the applicable Interior arrangement. Herein, the Standard Configuration is presented as an illustrative example, including the plan view and the Balance Arms.

For other interior configuration options, the respective Balance Arms are supplied together with the "Airplane Weighing Form", inserted in the "FINAL INSPECTION REPORT", by the time of the airplane's delivery.

STANDARD CONFIGURATION – CREW AND PASSENGERS (6 PASSENGER SEATS)

Balance Arm (in)				
Pilot & Copilot	Passengers 1 & 2	Passengers 3 & 4	Passengers 5 & 6	Lavatory
104.33	196.06	239.76	281.89	318.11

NOTE: Balance arms above refer to occupants. Seats balance arms are 5.91 inches towards the back of the seat, from occupant arm.

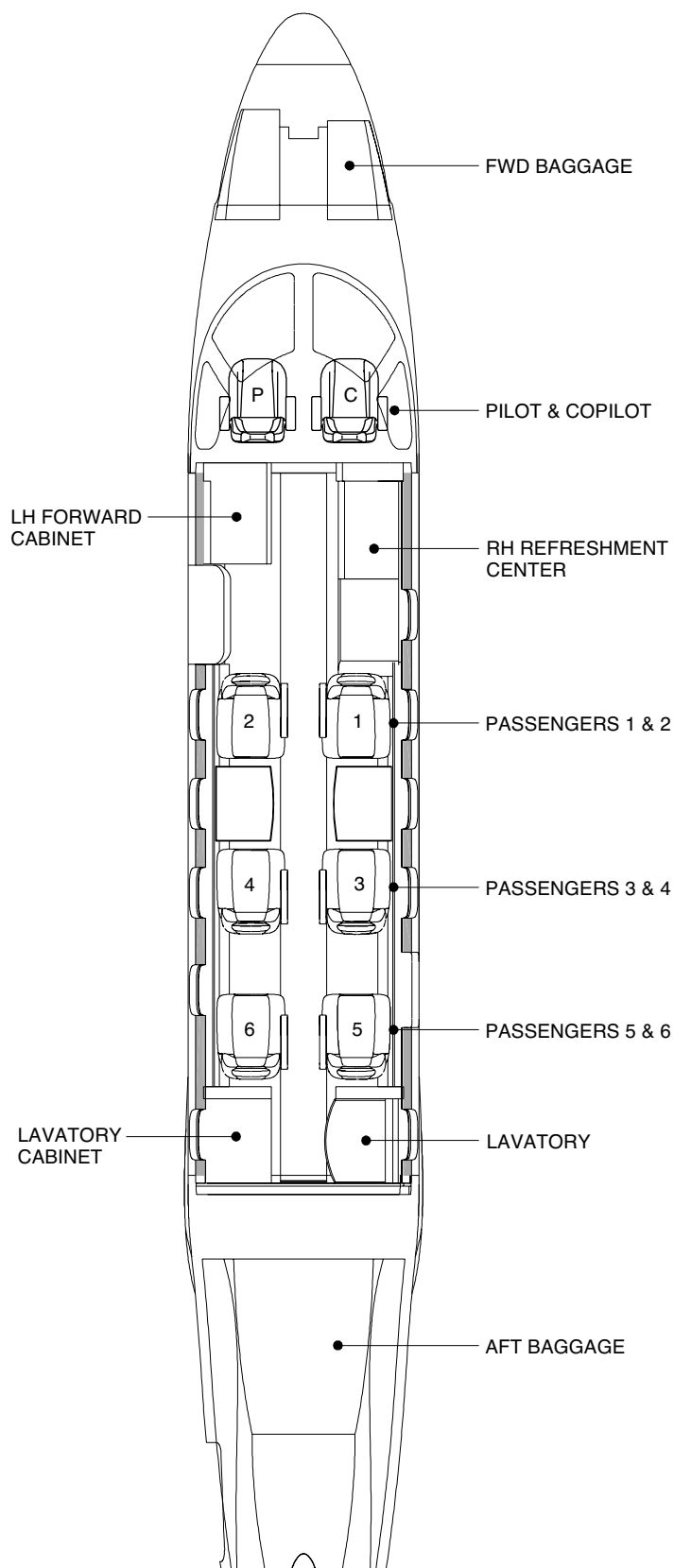
NAVIGATION KIT

Weight (lb)	Balance Arm (in)
22.04	118.11

STANDARD CONFIGURATION – BAGGAGE COMPARTMENTS

Balance Arm (in)				
Forward Baggage Compartment	Aft Baggage Compartment	LH Forward Cabinet	RH Refreshment Center	Lavatory Cabinet
39.37	396.85	135.83	146.06	318.11

Refer to Section 2, Limitations, for further information on baggage compartments load capacity.



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STANDARD CONFIGURATION



FUEL DATA

FUEL QUANTITIES

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

FUEL CATEGORY	VOLUME (US Gal)	WEIGHT (lb)	CG BALANCE ARM (in)
UNUSABLE	7.50	50.26	256.22
USABLE	798.87	5353.20	275.59

NOTE: The values specified above have been determined for an adopted fuel density of 6.701 lb/US Gal.



FUEL DISTRIBUTION TABLE

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

FUEL DISTRIBUTION ON THE LEFT AND RIGHT WING TANKS	
WEIGHT (lb)	CG BALANCE ARM (in)
50	256.38
100	256.02
150	255.64
200	255.32
250	255.07
300	254.84
350	254.66
400	254.51
450	254.38
500	254.27
550	254.22
600	254.20
650	254.19
700	254.25
750	254.30
800	254.38
850	254.50
900	254.64
950	254.80
1000	254.94
1050	255.11
1100	255.29
1150	255.46
1200	255.61
1250	255.83
1300	256.02
1350	256.19
1400	256.38
1450	256.53
1500	256.71
1550	256.90



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FUEL DISTRIBUTION ON THE LEFT AND RIGHT WING TANKS	
WEIGHT (lb)	CG BALANCE ARM (in)
1600	257.06
1650	257.26
1700	257.40
1750	257.61
1800	257.80
1850	257.96
1900	258.20
1950	258.37
2000	258.62
2050	258.83
2100	259.04
2150	259.32
2200	259.55
2250	259.78
2300	260.00
2350	260.24
2400	260.48
2450	260.76
2500	261.01
2550	261.28
2600	261.48
2650	261.77
2700	262.05
2750	262.26
2800	262.54
2850	262.76
2900	263.06
2950	263.28
3000	263.58
3050	263.80
3100	264.05
3150	264.36
3200	264.59

**FUEL DISTRIBUTION ON THE LEFT AND RIGHT WING TANKS**

WEIGHT (lb)	CG BALANCE ARM (in)
3250	264.82
3299	265.05
3350	265.39
3400	265.62
3450	265.86
3500	266.08
3550	266.34
3600	266.57
3650	266.83
3700	267.08
3750	267.32
3800	267.56
3850	267.79
3900	268.05
3950	268.29
4000	268.54
4050	268.79
4100	269.03
4150	269.28
4200	269.52
4250	269.76
4300	269.92
4350	270.16
4400	270.31
4450	270.32
4500	270.88
4550	271.12
4600	271.35
4650	271.60
4700	271.83
4750	272.08
4800	272.34
4850	272.57


FUEL DISTRIBUTION ON THE LEFT AND RIGHT WING TANKS

WEIGHT (lb)	CG BALANCE ARM (in)
4900	272.80
4950	273.12
5000	273.36
5050	273.68
5100	273.99
5150	274.29
5200	274.67
5250	275.03
5300	275.39
5352	275.73

NOTE: The values specified above have been determined for an adopted fuel density of 6.701 lb/US Gal.



MISCELLANEOUS FLUIDS

FLUID	WEIGHT (lb)	BALANCE ARM (in)
WASTE TANK FLUID	7.71	318.03

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WEIGHING AND BALANCE COMPUTATION

The BEW (Basic Empty Weight) is the weight of the empty airplane in its delivered configuration plus the weight of fluids (engine oil and hydraulic fluid serviced full, and the unusable fuel). The BEW and its respective balance arm is obtained from the airplane weighting record.

In order to determine the airplane weight and CG arm, it is necessary to add to the BEW all the weight and moment variations referent to the loaded items. The total moment divided by the total weight gives the final CG arm. The CG arm must be converted into %MAC. The pair Weight/CG must then be checked against the Weight/CG envelope given in Section 2, Limitations.

Example: Computation of Weight and CG arm.

For a BEW equal 11354 lb and BEW moment equal 3432995 lbf.in:

Item	Weight (lb)	Arm (in)	Moment (lbf.in)
BEW	11354	302.36	3432995.4
Pilot and Copilot	388	104.33	40480.0
Pax 1 and 2	388	196.06	76071.3
Pax 3	194	239.76	46513.4
Catering	53	142.13	7532.9
Fuel (328 US Gal)	2200	259.55	571010.0
Aft Baggage	264	396.85	104768.4
Airplane Weight & CG	14841	288.35	4279371.4

$$\%MAC = \frac{(288.35 - 264.51) \times 100}{80.71} = 29.54$$

NOTE: - The fuel weight specified above have been determined for an adopted fuel density of 6.701 lb/US Gal.

- The values above are for exemplification purpose. Use the airplane actual BEW for determining the Weight and CG arm.



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BAGGAGE LOADING

BAGGAGE WEIGHT AND LOCATION

The baggage weight limits, location and the respective balance arm may be obtained from the applicable interior arrangement.

The data shown in this Section are applicable to the airplane's Standard Configuration. For other interior configuration options, the weight limits, location and the respective balance arm are supplied together with the "Airplane Weighing Form", inserted in the "FINAL INSPECTION REPORT".

BAGGAGE COMPARTMENT

The baggage should be evenly distributed in each compartment to avoid load concentration.

Baggage/Cargo must not become a hazard to the airplane structure or systems as a result of shifting under operational loads. Therefore, sharp edge volumes and/or dense cargo (objects significantly more dense than typical passenger baggage) must be arranged with adjacent soft volumes or protections thus preventing airplane damage in case of baggage/cargo shifting due to operational loads.



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SUPPLEMENTS

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| ☐ | SUPPLEMENT 8 | AUTOTHROTTLE |

NOTE: Check (✓) the supplement applicable to the airplane configuration.

- ✓ **NON-EMBRAER SUPPLEMENT (STC FAA.SA01727WI/EASA10048066)** NO SMOKING SIGNS MODIFICATION & Installation of a Coffee machine(DUNCAN AVIATION EDOC-0000015627_A)
- ✓ **NON-EMBRAER SUPPLEMENT (Minor change)** , Installation of Aviator 200 Satellite Data System AV200.(AV200-NY-11276-08).
- ✓ **NON-EMBRAER SUPPLEMENT (Major change)** , F20-001-EO24-0047 - AFMS for EMB-505 (GOGO HDX Antenna System Integration - EASA STC 10086739)
- ✓ **NON-EMBRAER SUPPLEMENT (STC FAA SA01760WI/EASA10054450),** Installation of Aircell Universal Cabin System(UCS5000) with WLAN. (EDOC-0000017304 A)

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SUPPLEMENT 1

RVSM OPERATION

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GENERAL

INTRODUCTION

This Supplement is provided to present the data required for operation in the RVSM (Reduced Vertical Separation Minimum) airspace. The RVSM operation reduces the PHENOM 300 model minimum vertical separation from 2000 ft to 1000 ft between FL 290 and FL 410.

Airworthiness approval alone does not authorize flight into airspace for which an RVSM operational approval is required by an ICAO Regional Navigation Agreement.

The information herein presented must replace or complement the equivalent data in the basic AFM.

LIMITATIONS

KINDS OF OPERATION

- Reduced Vertical Separation Minimum (RVSM)

KINDS OF OPERATION EQUIPMENT LIST (KOEL)

OPERATION: RVSM	
1) INSTALLATIONS	
System	Function/Equipment
Automatic Flight Control System	1 Autopilot with Altitude Hold Mode Operative
Flight Instruments/Navigation	1 Altitude Alerter
Flight Instruments/Navigation	2 RVSM Compliant Air Data System (ADS)
Flight Instruments/Navigation	1 Transponder

NOTE: - The ADS 1 and ADS 2 are compliant with RVSM operation.

- The IESI must not be used for RVSM operation.
- Should any of the required equipment fail prior to the airplane entering RVSM airspace, the pilot should request a new clearance.
- The local authority determines the requirements for an operational transponder in each area where operations are intended.



NORMAL PROCEDURES

The actions listed below must complement the procedures contained in the basic AFM. The remaining Normal Procedures Section remains unchanged.

EXTERNAL SAFETY INSPECTION

NOSE SECTION

Air Data Smart Probes..... CONDITION, NO OBSTRUCTION

Particular attention should be paid to the condition of static sources and to the marked area on the fuselage skin near each Air Data Smart Probe.

BEFORE TAKEOFF

Altimeters..... SET TO THE AIRFIELD QNH

Altitude Indications..... CHECK

NOTE: - For airplanes with G3000 system version 3305.00 and on, an alternative procedure using QFE may also be used.

- The maximum difference between both primary altimeters indications must not exceed 23 m (75 ft).

CRUISE

Be sure that all required equipment is in proper operating condition.

Ensure that the airplane is flown at the cleared flight level and that ATC clearances are fully understood and followed. Do not depart from cleared flight level without a positive clearance from ATC except for a contingency or emergency situation.

While changing flight levels do not overshoot or undershoot the cleared flight level by more than 45 m (150 ft).

The autopilot should be operative and engaged during level cruise, except for circumstances such as the need to re-trim the airplane or when it must be disengaged due to turbulence.



AFTER LANDING

In case of failure or malfunction, the following information should be recorded when appropriate:

- ADS 1, ADS 2 altimeter readings;
- Altitude selector setting;
- Baro Set value and Baro Set unit (InHg/HPa);
- Flight Director used with the Autopilot to control the airplane and any differences when the other Flight Director was coupled;
- Use of air data system reversion for fault diagnosis procedure;
- The transponder selected to provide altitude information to ATC and any difference noted when an alternative transponder was selected, if applicable.

EMERGENCY AND ABNORMAL PROCEDURES

The actions listed below must complement the procedures contained in the basic AFM. The remaining Emergency and Abnormal Procedures Section remains unchanged.

- In case of emergency or abnormal situation or contingencies (equipment failures, weather, etc.) which affect the ability to maintain the cleared flight level, notify ATC and co-ordinate an action plan that is appropriate to the airspace concerned;
- Notify ATC when encountering greater than moderate turbulence;
- If unable to notify ATC and obtain an ATC clearance prior to deviating from the cleared flight level, follow any established contingency procedures and obtain ATC clearance as soon as possible.

PERFORMANCE

Performance Data presented in the basic AFM remains unchanged.



SUPPLEMENT 4

**OPERATION WITH GARMIN G3000
AVIONICS SYSTEM**

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GENERAL

INTRODUCTION

This Supplement presents the required information to operate the airplane with Garmin G3000 avionics system.

The information contained herein supplements or supersedes the basic airplane flight manual only in those areas listed herein. For limitations, procedures, and performance information not contained in this document, consult the basic airplane flight manual.

GARMIN G3000 GLOBAL NAVIGATION SATELLITE SYSTEM (GPS/SBAS)

SYSTEM APPROVALS

Refer to Section 1 – General.



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LIMITATIONS

ADS-B OUT

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

Applicable to airplanes with ADS-B Out installed:

- Airplanes Post-Mod. SB 505-34-0010 or with an equivalent modification factory-incorporated; or
- Post-Mod. SB 505-34-0011 or with an equivalent modification factory-incorporated; or
- Equipped with G3000 system version 3305.00.

SYSTEM APPROVALS

The installed ADS-B Out system has been shown to meet compliance with the following standards:

- AC 20-165A (airplanes with G3000 system version prior to 3305.00);
- AC 20-165B (airplanes with G3000 system version 3305.00 and on);
- CS ACNS.D.ADSB (1090 MHz Extended Squitter ADS-B Out); and
- 14 CFR 91.227.

ELECTRONIC DISPLAY SYSTEM

ELECTRONIC DOCUMENT VIEWER

The document viewer alone will not eliminate the need for other means of accessing the airplane manuals.

ELECTRONIC CHARTS

The own-ship position symbol presented in the electronic charts is intended to improve pilot positional awareness. Do not use it as the basis for maneuvering.

The airplane must be dispatched with adequate paper or electronic backup documentation.

The electronic charts function airworthiness approval alone does not constitute operational approval for paperless cockpit. Make sure that any required operational approval is granted from the Regulatory Authority prior to the intended use.

FLIGHT STREAM 510 (IF APPLICABLE)

The use of FS510 in flight is prohibited.

For single pilot operation the use of FS510 is prohibited during taxi.



FIS-B

Applicable to airplanes with FIS-B installed (optional).

The Garmin FIS-B has been demonstrated compliance to AC 20-149B.

FIS-B information may be used for pilot planning decisions focused on updating the pilot's awareness of the dynamic flight environment, including avoiding areas of inclement weather that are beyond visual range and pilot near-term decisions where poor visibility precludes visual acquisition of inclement weather. FIS-B weather and NAS status information may be used as follows:

- To promote pilot awareness of own ship location with respect to reported weather, including hazardous meteorological conditions, NAS status indicators to enhance pilot planning decisions, and pilot near-term decision making.
- To cue the pilot to communicate with air traffic control, a Flight Service Station specialist, operator dispatch, or an airline operations control center for general and mission critical meteorological information, NAS status conditions, or both. FIS-B information, including weather information, NOTAMs, and TFR areas, are intended for the sole purpose of assisting in long-/near-term planning and decision making. The system lacks sufficient resolution and updating capability necessary for aerial maneuvering associated with immediate decisions. In particular, in extreme scenarios, the oldest weather radar data on the display can be up to 15 to 20 minutes older than the display's age indication for that weather radar data. Therefore, do not attempt to use FIS-B weather information to maneuver the airplane at minimum safe distances from hazardous weather.

FIS-B information must not be used in place of a standard preflight briefing.



GARMIN G3000 AVIONICS SYSTEM

Refer to Garmin – Embraer Prodigy Touch Flight Deck 300 Pilot's Guide for further information on system characteristics. The GARMIN G3000 avionics system has the following limitations:

DEAD RECKONING MODE

Dead Reckoning Mode use is allowed only in en-route (ENR or RNP-2) or oceanic (OCN or RNP-4) phases of flight. The estimated navigation data supplied by the system in DR Mode must not be used as a sole means of navigation.

PERF – PERFORMANCE MANAGEMENT (IF APPLICABLE)

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

- PERF provides only supplemental information and it shall not be the only means to take flight/planning decisions.
- PERF does not provide support to OEI (One Engine Inoperative) operations or ANTI-ICE WING/STABILIZER failure.

QFE

For airplanes with G3000 system version prior to 3305.00, the use of QFE altimeter setting is prohibited.

(Continues on the next page)



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TOLD – TAKEOFF AND LANDING DATA (IF APPLICABLE)

- The FMS TOLD performance information is advisory only and shall be crosschecked against approved performance data prior to use. After this check, the information used by the flight crew during a specific phase of flight must be obtained from only one source of information (approved performance data or TOLD). No intermixing of these data sources is permitted during a specific phase of flight.
- TOLD must not be used for abnormal landing performance calculations.
- Obstacle clearance calculations are not supported.

TRIP PLANNING AND WEIGHT AND FUEL FUNCTIONS

The trip statistics information, fuel statistics information and other kinds of statistics presented in the G3000 Trip Planning and Weight and Fuel screens on the Touchscreen Controller are supplemental information only and must be confirmed by the flight crew prior to use.

VNAV

VNAV path guidance is not available for legs that do not have a defined leg distance on the horizontal plane.



GARMIN G3000 GLOBAL NAVIGATION SATELLITE SYSTEM (GPS/SBAS)

SYSTEM LIMITATIONS

The pilot must confirm at system initialization that the Navigation database is current.

Navigation database is expected to be current for the duration of the flight. If the AIRAC cycle will change during flight, the pilot must ensure the accuracy of navigation data, including suitability of navigation facilities used to define the routes and procedures for flight. If an amended chart affecting navigation data is published for the procedure, the database must not be used to conduct the procedure.

GPS/SBAS based IFR en-route, oceanic, and terminal navigation is prohibited unless the pilot verifies and uses a valid, compatible, and current Navigation database or verifies each waypoint for accuracy by reference to current approved data.

Discrepancies that invalidate a procedure shall be reported to Garmin International. The affected procedure is prohibited from being flown using data from the Navigation database until a new Navigation database is installed in the airplane and verified that the discrepancy has been corrected.

For flight planning purposes, in areas where SBAS coverage is not available, the pilot must check RAIM availability. This requirement is not necessary if SBAS coverage is confirmed to be available along the entire route of flight.

For flight planning purposes, operations within the U.S. National Airspace System on RNP and RNAV procedures when SBAS signals are not available, the availability of GPS integrity RAIM shall be confirmed for the intended route of flight. In the event of a predicted continuous loss of RAIM of more than five minutes for any part of the intended route of flight, the flight should be delayed, canceled, or rerouted on a track where RAIM requirements can be met.

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For flight planning purposes for operations within European B-RNAV and P-RNAV airspace, if more than one satellite is scheduled to be out of service, then the availability of GPS integrity RAIM shall be confirmed for the intended flight (route and time). In the event of a predicted continuous loss of RAIM of more than five minutes for any part of the intended flight, the flight should be delayed, canceled, or rerouted on a track where RAIM requirements can be met.

For flight planning purposes, operations where the route requires Class II navigation, the airplane's operator or pilot-in-command must use the Garmin WFDE Prediction program to demonstrate that there are no outages on the specified route that would prevent the Garmin GNSS navigation system to provide primary means of Class II navigation in oceanic and remote areas of operation that requires RNP-10 or RNP-4 capability. If the Garmin WFDE Prediction program indicates fault exclusion (FDE) unavailability will exceed 34 minutes in accordance with FAA Order 8400.12C for RNP-10 requirements, or 25 minutes in accordance with FAA Order 8400.33 for RNP-4 requirements, then the operation must be rescheduled when FDE is available.

Both Garmin GPS navigation receivers must be operating and providing GPS navigation guidance to their respective PFD for operations requiring RNP-4 performance.

North Atlantic (NAT) Minimum Navigational Performance Specifications (MNPS) Airspace operations per AC 91-49 and AC 120-33 require both GPS/SBAS receivers to be operating and receiving usable signals except for routes requiring only one Long Range Navigation sensor. Each display computes an independent navigation solution based on the on-side GPS sensor. However, either display will automatically revert to the cross-side sensor if the on-side sensor fails or if the cross-side sensor is determined to be more accurate. A "BOTH ON GPS1" or "BOTH ON GPS2" message does not necessarily mean that one GPS has failed. Refer to the GPS STATUS page to determine the state of the unused GPS.

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Whenever possible, RNP and RNAV routes including Standard Instrument Departures (SIDs) and Obstacle Departure Procedures (ODPs), Standard Terminal Arrival (STAR), and en-route RNAV "Q" and RNAV "T" routes should be loaded into the flight plan from the database in their entirety, rather than loading route waypoints from the database into the flight plan individually. Selecting and inserting individual named fixes from the database is permitted, provided all fixes along the published route to be flown are inserted. Manual entry of waypoints using latitude/longitude or place/bearing is prohibited.

"GPS", "or GPS", and "RNAV (GPS)" instrument approaches using the Garmin navigation systems are prohibited unless the pilot verifies and uses the current Navigation database. GPS based instrument approaches must be flown in accordance with an approved instrument approach procedure that is loaded from the Navigation database.

Not all published Instrument Approach Procedures (IAP) are in the Navigation database. Pilots planning on flying an RNAV instrument approach must ensure that the Navigation database contains the planned RNAV Instrument Approach Procedure and that approach procedure must be loaded from the Navigation database into the FMS flight plan by its name.

IFR non-precision approach approval using the GPS/SBAS sensor is limited to published approaches within the U.S. National Airspace System. Approaches to airports in other airspace are not approved unless authorized by the appropriate governing authority.

The GPS/SBAS receivers cannot be used to provide navigation guidance during final approach segments of procedures not approved for GPS navigation. The final course segment must be flown with the NAV receiver (e.g. ILS, LOC, LOC-BC, LDA, SDF, MLS, VOR) tuned to the proper frequency and selecting that NAV receiver on the CDI of the pilot flying.

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For airplanes equipped with G3000 system version 3305.00 and on, the GPS/SBAS receivers can be used to provide navigation guidance to fly a VOR or NDB non-precision approach procedure, even if GPS is not approved as a substitute means of navigation. In this case, the NAV receiver must be tuned and valid, and the pilot must monitor the VOR preview or NDB bearing pointer on the Horizontal Situation Indicator (HSI).

Navigation information is referenced to the WGS-84 reference system, and should only be used where the Aeronautical Information Publication (including electronic data and aeronautical charts) conform to WGS-84 or equivalent.

When using GPS/NAV mode to intercept the final approach course, monitor the flight director transition to LOC/GS. If the transition does not occur automatically, manually select HDG mode and CDI source to LOC in order to intercept the final approach course.

Localizer Performance (LP) approach procedures are not allowed.

OPERATIONAL LIMITATIONS

The Garmin – Embraer Prodigy Touch Flight Deck 300 Cockpit Reference Guide must be available to the flight crew whenever navigation is predicated on the use of the system.

Advisory vertical guidance deviation information is only an aid to help pilots comply with altitude restrictions. When using advisory vertical guidance, the pilot must use the primary barometric altimeter to ensure compliance with all altitude restrictions, particularly during instrument approach operations.

FMS support for Visual approaches (lateral and vertical advisory guidance) is provided only to aid the pilot when performing a visual approach in V_{MC} . The flight crew must still use primary visual cues as primary reference on the approach. The pilot remains responsible for final approach execution and monitoring of airspace, traffic and obstacles.

The Flight Director must be engaged to perform Baro-VNAV operations (Climb, Cruise, Descent and Approach).

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The use of Baro-VNAV approach function is not allowed for airports located in altitudes above 5000 ft.

When using the Baro-VNAV function, the barometric altimeter must be used as the primary altitude reference for all operations; including instrument approach procedure step-down fixes.

For flight planning purposes in European airspace, flight crew should ensure sufficient means are available to navigate and land at the destination or at an alternate aerodrome in the case of loss of RNP APCH airborne capability. In particular, the flight crew should check that:

- a non-RNP APCH procedure is available at the alternate, where a destination alternate is required.
- at least one non-RNP APCH procedure is available at the destination aerodrome, where a destination alternate is not required.

For flight planning purposes in European airspace, flight crew should ensure sufficient means are available to navigate and land at the destination or at an alternate aerodrome in the case of loss of LPV airborne capability. In particular, the flight crew should check that:

- a non-RNAV GNSS based procedure is available at the alternate (where a destination alternate is required) unless the airspace authority does not require such requirement.
- at least one non-RNAV GNSS based procedure is available at the destination aerodrome (where a destination alternate is not required) unless the airspace authority does not require such requirement.

For flight planning purposes, operations within the Brazilian airspace using GNSS requires that the availability of GPS integrity RAIM be confirmed before departure and before entrance to each phase of flight. When the RAIM is not available, other type of navigation must be used or the schedule of the flight must be changed until the RAIM is available again.

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En-route operations within the Brazilian airspace under instrument flight rules require that the airplane has the basic equipment for air navigation appropriate for the route to be flown. Such equipment shall be used compulsorily, when there is an integrity alarm and on the portion of the route where the function RAIM is supposed to be unavailable.

Airplane accomplishing GNSS SIDs or GNSS STARs within the Brazilian airspace must have their basic navigation equipment tuned on the appropriate frequencies, so as to provide fast and safe transition in the case of occurrence of RAIM alarm. In case that there is prevision of unavailability of the RAIM function during the flight period, only the basic equipment of air navigation shall be used.

For airplane accomplishing GNSS approach procedures within the Brazilian airspace the pilot must assure that the air navigation aids needed to the aerodrome operations at the alternative aerodrome are available. Pilot must designate an alternative aerodrome that offers an approach procedure in operation based in conventional air navigation aids.

ICE AND RAIN PROTECTION

OPERATION IN ICING CONDITIONS

Minimum Airspeed:

Wing and Stabilizer Anti-Ice System Armed 165 KIAS

NOTE: - The A-I WINGSTB INHB caution message will be presented if pilots command the WINGSTAB Switch to ON position outside Wing and Stabilizer Anti-Ice System operational envelope below 30000 ft. Follow its associated procedure (Section 4, Emergency and Abnormal Procedures).

- The A-I WINGSTB ARM advisory message will be presented if pilots command the WINGSTAB Switch to ON position above 30000 ft. Follow its associated procedure (Supplement 4, Emergency and Abnormal Procedures).

ROAAS (IF APPLICABLE)

- Runway Overrun Awareness and Alerting System (ROAAS) is intended to be used as an aid to situational awareness only.
- Flight crew landing assessments must not be based solely on ROAAS alerts.
- Compliance with ROAAS alerts is required unless the pilot considers it unsafe to do so.
- ROAAS must be inhibited for landing in contaminated runways.
- ROAAS must be inhibited for QFE operation.

STABILIZED APPROACH (IF APPLICABLE)

- The Stabilized Approach function is an aid to increase situational awareness only. Flight crew should not rely solely on the Stabilized Approach information to determine if a go-around is required due to unstable approach conditions.

SURFACEWATCH™ (IF APPLICABLE)

- Takeoff/Landing Awareness Function of the SurfaceWatch™ is intended to be used as an aid to situational awareness only.
- Flight crew takeoff and landing assessments should not be based on SurfaceWatch™ alerts.
- SurfaceWatch™ must be inhibited for abnormal landings.

SYNTHETIC VISION SYSTEM (SVS) (IF APPLICABLE)

SVS DATABASES

The terrain data has a resolution of 5 arc-seconds, this means that the terrain elevation contours in the database are stored broken down into squares 5 arc-seconds on each side.

The other SVS limitations remain unchanged.



NORMAL PROCEDURES

COCKPIT/CABIN SAFETY INSPECTION

ICE PROTECTION Panel..... CHECK

Check ENG 1 & 2 Switches and WINGSTAB Switch in the OFF position.

BEFORE TAKEOFF

Transponder/TCAS II (if applicable)..... SET

Verify transponder mode as required by local rules.

If airplane is equipped with TCAS II, make sure that it is not in STBY mode.



TAKEOFF

Thrust Levers TO/GA

NOTE: After checking thrust levers in TO/GA, check N1 equal to N1 target and green ATR indication presented on MFD if ATR ON is selected.

Engine Parameters MONITOR

At V_R rotate the airplane according to the table below.

FLAPS POSITION	1	2
PITCH ANGLE	10.5°	8°

If active, the flight director provides the guidance to maintain the pitch angle.

With positive rate of climb:

LDG GEAR Lever UP

Airspeed $V_2 + 10$ KIAS
MINIMUM

After the airplane reaches $V_2 + 10$ KIAS, adjust pitch attitude to the value shown in the table below.

RECOMMENDED PITCH ANGLE	
PHENOM 300	15°

For airplanes with G3000 system version prior to 3305.07:

- If the flight director is active, press CWS Button to synchronize flight director with the recommended pitch angle.

For airplanes with G3000 system version 3305.07 and on:

- If the flight director is active, once the airspeed is greater than the value selected by the speed bug (default: $V_2 + 10$ KIAS), the flight director automatically synchronizes with the recommended pitch angle.



SHUTDOWN

ICE PROTECTION Panel..... CHECK

Check ENG 1 & 2 Switches and WINGSTAB Switch in the OFF position.

OPERATION IN ICING CONDITIONS

The A-I WINGSTB ARM advisory message may be presented during operation in icing conditions. If the message is presented, refer to the associated procedure in Supplement 4 Emergency and Abnormal Procedures.

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EMERGENCY AND ABNORMAL PROCEDURES

NON-ANNUNCIATED PROCEDURES

EMERGENCY PROCEDURES

The non-annunciated emergency procedures of the basic AFM remain the same.

ABNORMAL PROCEDURES

BLANK DISPLAY UNIT WITH AUTOMATIC REVERSION

Cycle the PFD 1 (2) MODE Switch (REV then AUTO).

If the display does not revert to the normal condition or an abnormal behavior occurs, set PFD 1 (2) MODE Switch as required.

NOTE: Only one attempt to return to a normal condition is allowed.

BLANK DISPLAY UNIT WITHOUT AUTOMATIC REVERSION

PFD 1 (2) MODE Switch.....AS REQUIRED

GTC 1 (2) FAILURE

In case one GTC fails, the remaining GTC will provide interface to the system.

In case both GTCs fail, the FMS function becomes unavailable and communication is switched to emergency frequency.

LAND AS SOON AS PRACTICAL.



ONE ENGINE INOPERATIVE APPROACH AND LANDING

During descent:

ROAAS..... INHIBIT

ANNUNCIATED PROCEDURES

ELECTRICAL

ELECTRICAL EMERGENCY

Communication System 2 (VHF and Audio) is inoperative.

GTC 2 is inoperative.

DC BUS 2 OFF

Communication System 2 (VHF and Audio) is inoperative.

GTC 2 is inoperative.

EMERGENCY BUS OFF

Audio 1 is inoperative.

GTC 1 is inoperative.

FLIGHT CONTROLS

FLAP FAILURE

For landing:

- If applicable, press the Stabilized APPR Flap Override.



FLIGHT INSTRUMENTS, COMMUNICATION AND NAVIGATION

FMS LANDING DISAGREE

Check FMS/TOLD landing input parameters against the current airplane configuration.

FMS TAKEOFF DISAGREE

Check FMS/TOLD takeoff input parameters against the current airplane configuration.

GTC 1 (2) OVERHEAT

If associated with smoke from the affected GTC:

- Crew Oxygen Masks DON,
EMERGENCY
THEN 100%
- EMERGENCY may be selected during 2 minutes
maximum then set to 100%.
- Crew Oxygen Masks Auto
- Dilution Valve CLOSED
- Smoke Goggles (if available) DON
- Communication ESTABLISH
- GTC 1 (B8) or GTC 2 (B26)
- Circuit Breaker PULL
- SMOKE EVACUATION Procedure AS REQUIRED

MFD CONFIGURATION

On ground, report to the maintenance personnel.

In-flight:

- PFD 1 (2) MODE Switch REV OR SPLIT



MFD FAULT

On ground, report to the maintenance personnel.

In-flight:

PFD 1 (2) MODE Switch REV OR SPLIT

PFD 1 (2) CONFIGURATION

On ground, report to the maintenance personnel.

In-flight and associated with flying pilot:

PFD 1 (2) MODE Switch REV OR SPLIT

PFD 1 (2) FAULT

On ground, report to the maintenance personnel.

In-flight and associated with flying pilot:

PFD 1 (2) MODE Switch REV OR SPLIT

PFD 1 (2) OVERHEAT

If associated with smoke from the affected PFD:

Crew Oxygen Masks DON,
EMERGENCY
THEN 100%

EMERGENCY may be selected during 2 minutes
maximum then set to 100%.

Crew Oxygen Masks Auto

Dilution Valve CLOSED

Smoke Goggles (if available) DON

Crew Communication ESTABLISH

PFD 1 (A7) or PFD 2 (B24)

Circuit Breaker PULL

PFD 1 (2) MODE Switch AS REQUIRED

SMOKE EVACUATION Procedure AS REQUIRED

If required during approach:

Associated Circuit Breaker PUSH



WARNING

SURFACEWATCH NO DATA

If the use of SurfaceWatch is considered:

If Before Takeoff:

Takeoff Data..... SET/CHK

If Before Landing:

Landing Data..... SET/CHK

If the use of SurfaceWatch is not considered:

SurfaceWatch Alerts..... INHIBIT

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PERFORMANCE

The airplane performance data presented in the basic AFM remains unchanged.

WEIGHT AND BALANCE

The airplane weight and balance data presented in the basic AFM remains unchanged.

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SUPPLEMENT 5

DATA LINK COMMUNICATION SYSTEM

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GENERAL

This Supplement presents the required information to operate the airplane data link communication system.

The information herein presented must replace or complement the equivalent data or procedures in the basic AFM.

These data link types are optionally installed in the airplane. The data link type FANS 1/A – CPDLC is applicable only to airplanes equipped with G3000 avionics system.

This design approval does not constitute operational authorization.

DATA LINK COMMUNICATION SYSTEM

The airplane data link communication system has been approved for the following data link capabilities:

DATA LINK TYPE	AIRPLANE-ALLOCATED PERFORMANCE	SUB-NETWORKS
ATN B1 (1) (3)	CPDLC at Continental Performance	VDL M2
FANS 1/A – CPDLC (2) (3) (with automation)	CPDLC at RCP 130	VDL MA/2

NOTE: 1) The airplane data link communication system has been approved for the criteria contained in AC 20-140A for ATN B1 over VDL M2 and the performance requirements for continental applications.

The airplane data link communication system has been demonstrated to comply with the applicable safety, performance and interoperability requirements of RTCA DO-290/EUROCAE ED-120 (including change 1 and 2), RTCA DO-280B/EUROCAE ED-110B, and the Commission Regulation (EC) No 29/2009 for continental ATN-CPDLC over VDL M2.

The airplane data link communication system supports multi-frequency operation as defined in ARINC Spec. 631-5.

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NOTE: 2) The airplane data link communication system has been approved for the criteria contained in AC 20-140C for FANS 1/A+ over VDL MA/2 for AFN and CPDLC applications, excluding the ADS-C application. This partial build of FANS 1/A+ without the ADS-C application is referred as FANS 1/A – CPDLC. The system is not eligible for operations in oceanic or remote airspaces where ADS-C is required.

The airplane data link communication system has been demonstrated to comply with the applicable safety, performance and interoperability requirements of RTCA DO-306/EUROCAE ED-122 (including change 1), RTCA DO-258A/EUROCAE ED-100A.

The airplane data link communication system supports multi-frequency operation as defined in ARINC Spec. 631-6.

NOTE: 3) The airplane data link communication system has been demonstrated to comply with the interoperability requirements of RTCA DO-305 A/EUROCAE ED-154A for a dual stack system (i.e., integrating both FANS 1/A – CPDLC and ATN B1 CPDLC applications).

The airplane data link communication system has been demonstrated to comply with the CS-ACNS Book 1 and Book 2, Subpart B, Section 2 for ATN B1 over VDL M2, and for a dual stack system.



LIMITATIONS

The ATN-CPDLC system shall be used for routine data link communication with the ATC during en route operations in upper airspace only.

The use of ATN-CPDLC is restricted to supplementary means of communication for non-time-critical messages. Voice shall remain the primary means of communication.

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NORMAL PROCEDURES

AFTER TAKEOFF/CLIMB

LDG GEAR Lever	CHECK UP
Flaps	ZERO
Retract flaps according to the Maximum Flap Extended Speed (V _{FE}).	
Thrust Levers	CON/CLB
Altimeters (pilots and IESI)	SET & X-CHECK
Yaw Damper	ON
Icing Conditions	VERIFY
Flight Director	AS REQUIRED
-----ABOVE 10000 FT-----	
Fasten Seat Belts Signs	AS REQUIRED
External Lights	AS REQUIRED
Weather Radar (if installed)	AS REQUIRED
Airspeed	AS REQUIRED
CPDLC	AS REQUIRED

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DESCENT

Windshield Heating ON
Pressurization CHECK LFE
Landing Speeds SET

Set V_{REF} , V_{AC} and V_{FS} .

Icing Conditions VERIFY
CPDLC LOGOFF

-----BELOW 10000 FT-----

Fasten Seat Belts Signs..... ON

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EMERGENCY AND ABNORMAL PROCEDURES

The Emergency and Abnormal Procedures presented in the basic AFM remain unchanged.

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PERFORMANCE

The Performance data presented in the basic AFM remain unchanged.

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WEIGHT AND BALANCE

The Weight and Balance data presented in the basic AFM remain unchanged.

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SUPPLEMENT 6

STEEP APPROACH LANDING

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GENERAL

INTRODUCTION

This Supplement presents the required information to Phenom 300 airplanes equipped for Steep Approach Operation. It is applicable to airplanes Post-Mod. SB 505-27-0018 or with an equivalent modification factory incorporated.

The information presented herein must replace or complement the equivalent data in the basic AFM.

This AFM Supplement does not constitute approval to conduct Steep Approach Operations. Approval must be obtained from the appropriate regulatory authority prior to conducting these operations.

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LIMITATIONS

OPERATIONAL LIMITATIONS

The Steep Approach Landing must not be initiated with one engine inoperative or in emergency/abnormal situations.

The Steep Approach Landing must not be performed with any failure leading to the loss of longitudinal or lateral primary flight controls.

The Steep Approach procedure may be performed with an approach glide path angle up to 5.5°.

The Steep Approach operation in glide path angles from 4.5° to 5.4° is prohibited in contaminated runways.

The Steep Approach Landing must not be performed with Current Speed Control (CSC) engaged.

The maximum airport altitude for Steep Approach Landing is 8000 ft.

AIRPLANE CONFIGURATION

The steep approach landing must be performed with Flaps surfaces at FLAP FULL position and Speed Brake Switch set to the OPEN position.

AIRSPEEDS

The landing procedure must be performed at the Landing Reference Speed specific to Steep Approach ($V_{REF \text{ STEEP}}$). Refer to Factored Landing Distance Tables in Performance Section (block S6-15) or OPERA software.

WIND COMPONENTS

Maximum Tailwind Component 5 kt



OPERA SOFTWARE AND TOLD

OPERA Software and TOLD (if applicable) only provides performance information for Steep Approach Landing with Glide Path Angles of 5.5°.

For different Steep Approach Glide Path Angles, refer to "Glide Path Angles from 4.5° to 5.4°" in Performance Section (block S6-15).



NORMAL PROCEDURES

APPROACH

External Lights AS REQUIRED

Passengers ADVISE

XFEED Knob OFF

Altimeters (pilots and IESI) SET & X-CHECK

Icing Conditions VERIFY

Icing conditions may exist whenever the Total Air Temperature (TAT) in flight is 10°C or below and visible moisture in any form is present (such as clouds, fog with visibility of one mile or less, rain, snow, sleet or ice crystals).

ICING PROTECTION Panel..... AS REQUIRED

BEFORE LANDING

Yaw Damper OFF

LDG GEAR Lever DN AND
CHECK

Flap..... FULL

NOTE: Wait until Flaps surfaces reach the FLAP FULL position before setting the SPEED BRAKE Switch to the OPEN position.

Speed Brake OPEN

Airspeed..... V_{REF} STEEP

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GO-AROUND

TO/GA Button.....PRESS

Thrust LeversTO/GA

Rotate the airplane following flight director guidance or 6° nose up.

Flap Lever2

With positive rate of climb:

LDG GEAR LeverUP

Airspeed.....V_{AC} MINIMUM

At the acceleration altitude:

Speed BrakeCLOSE

Proceed as in a normal takeoff.

DEMONSTRATED CROSSWIND

The maximum demonstrated crosswind component for steep approach landing is 28 kt.

This value is not considered to be limiting.

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EMERGENCY AND ABNORMAL PROCEDURES

NON-ANNUNCIATED PROCEDURES

EMERGENCY PROCEDURES

ENGINE FAILURE DURING STEEP APPROACH

Above DA (DH) or MDA (MDH), whichever is applicable:

GO-AROUND Procedure ACCOMPLISH

Perform a missed approach procedure with OEI and divert to a suitable airport.

NOTE: The demonstrated height loss during go-around after engine failure is 63 ft.

At or below DA (DH) or MDA (MDH), whichever is applicable, the approach may be continued provided the existing configuration is maintained.

ABNORMAL PROCEDURES

The AFM abnormal procedures remain the same.

ANNUNCIATED PROCEDURES

FLIGHT CONTROLS

STEEP FAILURE

GO-AROUND Procedure ACCOMPLISH

Perform a missed approach procedure and divert to an airport that does not require steep approach.

NOTE: The demonstrated height loss during go-around is 37 ft.

STEEP NOT AVAILABLE

Divert to an airport that does not require steep approach.



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PERFORMANCE

APPLICABLE TO AIRPLANES EQUIPPED WITH PW535E ENGINES

STALL SPEED

This table presents the stall speed for steep approach configuration (FLAP FULL and speed brake set to the ON position) for Maximum Landing Weight:

STALL SPEEDS(KIAS) ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF					
MLW (lb)	BANK ANGLE				
	0°	15°	30°	45°	60°
16865	94	95	101	111	132
17042	94	96	101	112	133

STALL SPEEDS(KIAS) ENGINE ICE PROTECTION ON – WINGSTAB ON					
MLW (lb)	BANK ANGLE				
	0°	15°	30°	45°	60°
16865	99	101	107	118	141
17042	100	102	107	119	141



LANDING DATA

MAXIMUM LANDING WEIGHT – CLIMB LIMITED

The Maximum Landing Weight – Climb Limited tables present the maximum landing climb weight in compliance with the airworthiness climb requirements according to the temperature, altitude and ice protection system configuration.

The landing climb weights shown may be above the Maximum Landing Weight and are codified as follows:

- (A) Approach Climb Limited;
- (L) Landing Climb Limited;
- (E) Enroute Climb Limited;
- (O) Overweight Landing Limited.

NOTE: - The overweight landing notation (O) is used to indicate that the airplane is not climb limited. Always check the MLW structural limitation in Section 2.

- For more detailed information regarding Maximum Landing Weight – Climb Limited, refer to the OPERA software.



MAXIMUM LANDING WEIGHT – CLIMB LIMITED
STEEP APPROACH MODE
(APPROACH FLAP 2 – LANDING FLAP FULL)
ENGINE ICE PROTECTION OFF – WINGSTAB OFF
PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)			
	ALTITUDE (ft)			
	0	1000	2000	3000
-40	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-35	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-30	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-25	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-20	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-15	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-10	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-5	18387 (O)	18387 (O)	18387 (O)	18387 (O)
0	18387 (O)	18387 (O)	18387 (O)	18387 (O)
5	18387 (O)	18387 (O)	18387 (O)	18387 (O)
10	18387 (O)	18387 (O)	18387 (O)	18387 (O)
15	18387 (O)	18387 (O)	18387 (O)	18387 (O)
20	18387 (O)	18387 (O)	18387 (O)	18387 (O)
25	18387 (O)	18387 (O)	18387 (O)	18387 (O)
30	18387 (O)	18387 (O)	18387 (O)	18387 (O)
35	18387 (O)	18387 (O)	18387 (O)	17698 (A)
40	18387 (O)	18269 (A)	17339 (A)	16415 (A)
45	17899 (A)	16975 (A)	16065 (A)	-
50	16600 (A)	-	-	-

NOTE: The values above Maximum Landing Weight must not be used for dispatch.

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**MAXIMUM LANDING WEIGHT – CLIMB LIMITED
STEEP APPROACH MODE
(APPROACH FLAP 2 – LANDING FLAP FULL)
ENGINE ICE PROTECTION OFF – WINGSTAB OFF
PW535E ENGINES**

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)			
	ALTITUDE (ft)			
	4000	5000	6000	7000
-40	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-35	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-30	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-25	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-20	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-15	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-10	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-5	18387 (O)	18387 (O)	18387 (O)	18387 (O)
0	18387 (O)	18387 (O)	18387 (O)	18387 (O)
5	18387 (O)	18387 (O)	18387 (O)	18387 (O)
10	18387 (O)	18387 (O)	18387 (O)	18387 (O)
15	18387 (O)	18387 (O)	18387 (O)	18362 (A)
20	18387 (O)	18387 (O)	18387 (O)	17465 (A)
25	18387 (O)	18387 (O)	17390 (A)	16312 (A)
30	18110 (A)	17185 (A)	16157 (A)	15053 (A)
35	16801 (A)	15921 (A)	14951 (A)	13935 (A)
40	15555 (A)	14733 (A)	-	-

NOTE: The values above Maximum Landing Weight must not be used for dispatch.



MAXIMUM LANDING WEIGHT – CLIMB LIMITED
STEEP APPROACH MODE
(APPROACH FLAP 2 – LANDING FLAP FULL)
ENGINE ICE PROTECTION ON – WINGSTAB ON
PW535E ENGINES

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)			
	ALTITUDE (ft)			
	0	1000	2000	3000
-40	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-35	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-30	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-25	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-20	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-15	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-10	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-5	18387 (O)	18387 (O)	18387 (O)	18387 (O)
0	18387 (O)	18387 (O)	18387 (O)	18387 (O)
5	18387 (O)	18387 (O)	18387 (O)	18387 (O)
8	18387 (O)	18387 (O)	18387 (O)	18387 (O)

TEMP (°C)	MAXIMUM LANDING WEIGHT – CLIMB LIMITED (lb)			
	ALTITUDE (ft)			
	4000	5000	6000	7000
-40	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-35	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-30	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-25	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-20	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-15	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-10	18387 (O)	18387 (O)	18387 (O)	18387 (O)
-5	18387 (O)	18387 (O)	18387 (O)	18387 (O)
0	18387 (O)	18387 (O)	18387 (O)	18011 (E)
5	18387 (O)	18387 (O)	17654 (E)	16755 (E)
8	18387 (O)	17824 (E)	16865 (E)	15954 (E)

NOTE: The shaded area and values above Maximum Landing Weight must not be used for dispatch.

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FACTORED LANDING DISTANCES

Factored landing distance is the actual distance to land the airplane from a point 40 ft above runway threshold to complete stop, factored according to operational rules.

Factored Landing Distance tables are presented for the conditions below:

- Approach type: Steep;
- Glide path (GP) angle of 5.5 degrees;
- Dry and wet runways;
- Zero runway slope;
- Ice Protection System:
For Wingstab OFF, Engine Ice Protection OFF or ON,
For Wingstab ON, Engine Ice Protection ON;
- Temperature: ISA (1);
- No SAT effect;
- No V_{REF} Overspeed;
- No Drag Index.

NOTE: (1) Wing and Stabilizer Ice Protection system is autoinhibited outside its operational envelope. Refer to its operational envelope in Section 2 for further information.

NOTE: (2) Refer to the OPERA software configuration to check if any special setting is required in addition to the choices above.



FACTORED LANDING DISTANCE (ft) – ISA
STEEP APPROACH MODE – 5.5° GP
ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
ZERO RUNWAY SLOPE

PW535E ENGINES

ALT (ft)	WEIGHT (lb)	SPEEDS			5 kt TAILWIND		NO WIND	
		V _{REF} STEEP (KIAS)	V _{AC} STEEP (KIAS)	V _{FS} (KIAS)	DRY (ft)	WET (ft)	DRY (ft)	WET (ft)
Sea Level	12400	112	112	113	3681	4233	3379	3886
	12800	113	113	115	3730	4289	3427	3941
	13200	114	114	117	3775	4341	3470	3991
	13600	116	116	119	3824	4397	3518	4046
	14000	117	117	121	3869	4450	3562	4096
	14400	117	117	122	3915	4502	3606	4147
	14800	118	118	124	3967	4562	3650	4198
	15200	119	119	126	4020	4623	3691	4245
	15600	120	120	127	4074	4685	3735	4295
	16000	121	121	129	4130	4749	3787	4355
	16400	122	122	130	4186	4814	3839	4415
	16800	123	123	132	4242	4879	3891	4475
	17200	124	124	133	4296	4940	3941	4532
1000	12400	112	112	113	3761	4325	3456	3975
	12800	113	113	115	3811	4382	3505	4031
	13200	114	114	117	3857	4435	3550	4082
	13600	116	116	119	3907	4494	3599	4138
	14000	117	117	121	3954	4547	3644	4190
	14400	117	117	122	4003	4603	3689	4242
	14800	118	118	124	4061	4670	3734	4295
	15200	119	119	126	4116	4734	3777	4343
	15600	120	120	127	4173	4798	3828	4403
	16000	121	121	129	4230	4864	3882	4464
	16400	122	122	130	4288	4932	3936	4527
	16800	123	123	132	4347	4999	3990	4589
	17200	124	124	133	4402	5063	4042	4648
2000	12400	112	112	114	3843	4419	3536	4066
	12800	113	113	115	3894	4479	3586	4123
	13200	114	114	117	3942	4533	3631	4176
	13600	116	116	119	3994	4593	3682	4234
	14000	117	117	121	4041	4648	3728	4287
	14400	117	117	122	4099	4714	3775	4341
	14800	118	118	124	4159	4783	3821	4394
	15200	119	119	126	4216	4849	3871	4452
	15600	120	120	127	4275	4916	3925	4514
	16000	121	121	129	4334	4984	3981	4578
	16400	122	122	130	4395	5054	4037	4643
	16800	123	123	132	4455	5124	4093	4707
	17200	124	124	133	4513	5190	4147	4769

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**FACTORED LANDING DISTANCE (ft) – ISA
STEEP APPROACH MODE – 5.5° GP
ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
ZERO RUNWAY SLOPE
PW535E ENGINES**

ALT (ft)	WEIGHT (lb)	SPEEDS			5 kt TAILWIND		NO WIND	
		V _{REF} STEEP (KIAS)	V _{AC} STEEP (KIAS)	V _{FS} (KIAS)	DRY (ft)	WET (ft)	DRY (ft)	WET (ft)
3000	12400	112	112	114	3928	4517	3618	4160
	12800	113	113	115	3981	4578	3669	4219
	13200	114	114	117	4030	4634	3716	4274
	13600	116	116	119	4083	4695	3768	4333
	14000	117	117	121	4137	4758	3815	4388
	14400	117	117	122	4199	4829	3864	4443
	14800	118	118	124	4261	4901	3915	4502
	15200	119	119	126	4321	4969	3970	4565
	15600	120	120	127	4381	5038	4026	4630
	16000	121	121	129	4443	5109	4084	4696
	16400	122	122	130	4506	5181	4142	4763
	16800	123	123	132	4568	5254	4200	4830
4000	17200	124	124	134	4628	5323	4256	4894
	12400	112	112	114	4016	4619	3703	4258
	12800	113	113	115	4070	4681	3755	4319
	13200	114	114	117	4120	4738	3804	4374
	13600	116	116	119	4176	4802	3857	4436
	14000	117	117	121	4239	4875	3906	4492
	14400	117	117	122	4303	4948	3955	4548
	14800	118	118	124	4367	5023	4015	4618
	15200	119	119	126	4429	5093	4073	4684
	15600	120	120	127	4492	5165	4131	4751
	16000	121	121	129	4556	5239	4191	4819
	16400	122	122	130	4621	5314	4251	4889
5000	16800	123	123	132	4686	5389	4312	4958
	17200	124	124	134	4748	5460	4369	5025
	12400	112	112	114	4107	4723	3790	4359
	12800	113	113	115	4163	4787	3845	4421
	13200	114	114	117	4214	4846	3894	4478
	13600	116	116	119	4279	4921	3949	4541
	14000	117	117	121	4344	4996	3999	4599
	14400	117	117	122	4410	5072	4058	4666
	14800	118	118	124	4477	5149	4120	4738
	15200	119	119	126	4541	5222	4179	4806
	15600	120	120	127	4606	5297	4240	4876
	16000	121	121	129	4673	5373	4302	4947
	16400	122	122	130	4740	5451	4365	5019
	16800	123	123	132	4808	5529	4427	5091
	17200	124	124	134	4873	5604	4487	5160

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FACTORED LANDING DISTANCE (ft) – ISA
STEEP APPROACH MODE – 5.5° GP
ENGINE ICE PROTECTION OFF/ON – WINGSTAB OFF
ZERO RUNWAY SLOPE

PW535E ENGINES

ALT (ft)	WEIGHT (lb)	SPEEDS			5 kt TAILWIND		NO WIND	
		V _{REF} STEEP (KIAS)	V _{AC} STEEP (KIAS)	V _{FS} (KIAS)	DRY (ft)	WET (ft)	DRY (ft)	WET (ft)
6000	12400	112	112	114	4223	4856	3901	4486
	12800	113	113	115	4280	4922	3957	4550
	13200	114	114	117	4336	4986	4007	4608
	13600	116	116	119	4407	5068	4063	4673
	14000	117	117	121	4474	5145	4118	4736
	14400	117	117	122	4543	5224	4182	4810
	14800	118	118	124	4612	5304	4247	4884
	15200	119	119	126	4678	5380	4309	4955
	15600	120	120	127	4746	5458	4371	5027
	16000	121	121	129	4815	5537	4435	5101
	16400	122	122	131	4885	5618	4501	5176
	16800	123	123	132	4956	5699	4566	5251
	17200	124	124	134	5023	5776	4628	5322
76000	12400	112	112	114	4331	4981	4005	4606
	12800	113	113	115	4390	5048	4062	4671
	13200	114	114	117	4455	5123	4114	4731
	13600	116	116	119	4529	5208	4171	4797
	14000	117	117	121	4599	5288	4236	4871
	14400	117	117	122	4670	5370	4303	4948
	14800	118	118	124	4742	5453	4370	5025
	15200	119	119	126	4810	5532	4433	5098
	15600	120	120	127	4880	5612	4498	5173
	16000	121	121	129	4952	5695	4565	5250
	16400	122	122	131	5025	5779	4633	5328
	16800	123	123	132	5098	5863	4701	5406
	17200	124	124	134	5168	5944	4765	5480

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**FACTORED LANDING DISTANCE (ft) – ISA
STEEP APPROACH MODE – 5.5° GP
ENGINE ICE PROTECTION ON – WINGSTAB ON
ZERO RUNWAY SLOPE
PW535E ENGINES**

ALT (ft)	WEIGHT (lb)	SPEEDS			5 kt TAILWIND		NO WIND	
		V _{REF} STEEP (KIAS)	V _{AC} STEEP (KIAS)	V _{FS} (KIAS)	DRY (ft)	WET (ft)	DRY (ft)	WET (ft)
Sea Level	12400	115	115	126	3801	4371	3494	4018
	12800	116	116	128	3855	4433	3547	4079
	13200	117	117	130	3905	4491	3596	4135
	13600	118	118	132	3956	4549	3645	4191
	14000	119	119	134	4012	4614	3700	4254
	14400	120	120	136	4065	4675	3745	4306
	14800	121	121	138	4129	4749	3795	4365
	15200	122	122	140	4201	4831	3856	4434
	15600	124	124	142	4296	4940	3945	4537
	16000	126	126	143	4390	5049	4034	4639
	16400	127	127	145	4486	5159	4123	4742
	16800	128	128	147	4582	5269	4213	4844
1000	17200	130	130	149	4679	5381	4304	4949
	12400	115	115	126	3883	4466	3574	4110
	12800	116	116	128	3939	4530	3628	4172
	13200	117	117	130	3990	4589	3678	4230
	13600	118	118	132	4042	4649	3728	4288
	14000	119	119	134	4103	4718	3785	4352
	14400	120	120	136	4162	4786	3831	4405
	14800	121	121	138	4229	4863	3883	4466
	15200	122	122	140	4302	4948	3953	4545
	15600	124	124	142	4400	5060	4044	4651
	16000	126	126	143	4498	5173	4136	4756
	16400	127	127	145	4597	5287	4229	4863
2000	16800	128	128	147	4696	5400	4321	4969
	17200	130	130	149	4797	5517	4416	5078
	12400	115	115	126	3968	4564	3656	4205
	12800	116	116	129	4026	4630	3712	4269
	13200	117	117	130	4079	4690	3763	4328
	13600	118	118	132	4132	4752	3815	4387
	14000	119	119	134	4201	4831	3873	4453
	14400	120	120	136	4263	4902	3920	4508
	14800	121	121	138	4332	4981	3981	4578
	15200	122	122	140	4408	5069	4053	4661
	15600	124	124	142	4509	5186	4148	4770
	16000	126	126	144	4611	5302	4243	4879
	16400	127	127	145	4713	5420	4339	4989
	16800	128	128	147	4815	5538	4434	5099
	17200	130	130	149	4920	5658	4532	5212

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FACTORED LANDING DISTANCE (ft) – ISA
STEEP APPROACH MODE – 5.5° GP
ENGINE ICE PROTECTION ON – WINGSTAB ON
ZERO RUNWAY SLOPE

PW535E ENGINES

ALT (ft)	WEIGHT (lb)	SPEEDS			5 kt TAILWIND		NO WIND	
		V _{REF} STEEP (KIAS)	V _{AC} STEEP (KIAS)	V _{FS} (KIAS)	DRY (ft)	WET (ft)	DRY (ft)	WET (ft)
3000	12400	115	115	127	4057	4665	3741	4303
	12800	116	116	129	4116	4733	3799	4369
	13200	117	117	131	4170	4795	3851	4429
	13600	118	118	132	4229	4863	3905	4490
	14000	119	119	134	4304	4949	3964	4558
	14400	120	120	136	4367	5023	4016	4619
	14800	121	121	138	4439	5105	4083	4696
	15200	122	122	140	4518	5196	4157	4781
	15600	124	124	142	4623	5316	4256	4894
	16000	126	126	144	4728	5437	4354	5007
	16400	127	127	145	4834	5559	4453	5121
	16800	128	128	147	4940	5681	4552	5235
4000	17200	130	130	149	5049	5806	4653	5351
	12400	115	115	127	4148	4770	3830	4404
	12800	116	116	129	4209	4840	3888	4472
	13200	117	117	131	4264	4904	3942	4534
	13600	118	118	132	4333	4983	3997	4597
	14000	119	119	134	4410	5072	4058	4667
	14400	120	120	136	4476	5148	4120	4738
	14800	121	121	138	4550	5233	4189	4817
	15200	122	122	140	4632	5327	4266	4905
	15600	124	124	142	4741	5452	4367	5022
	16000	126	126	144	4849	5576	4469	5139
	16400	127	127	145	4959	5703	4572	5258
5000	16800	128	128	147	5069	5830	4675	5376
	17200	130	130	149	5182	5959	4780	5497
	12400	115	115	127	4242	4878	3920	4508
	12800	116	116	129	4304	4950	3981	4578
	13200	117	117	131	4368	5024	4036	4642
	13600	118	118	133	4440	5106	4093	4706
	14000	119	119	134	4520	5198	4163	4787
	14400	120	120	136	4589	5277	4227	4861
	14800	121	121	138	4666	5366	4299	4944
	15200	122	122	140	4751	5463	4378	5035
	15600	124	124	142	4863	5592	4484	5156
	16000	126	126	144	4976	5722	4589	5277
5000	16400	127	127	145	5090	5853	4696	5400
	16800	128	128	147	5204	5984	4802	5522
5000	17200	130	130	149	5321	6119	4911	5648

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**FACTORED LANDING DISTANCE (ft) – ISA
STEEP APPROACH MODE – 5.5° GP
ENGINE ICE PROTECTION ON – WINGSTAB ON
ZERO RUNWAY SLOPE
PW535E ENGINES**

ALT (ft)	WEIGHT (lb)	SPEEDS			5 kt TAILWIND		NO WIND	
		V _{REF} STEEP (KIAS)	V _{AC} STEEP (KIAS)	V _{FS} (KIAS)	DRY (ft)	WET (ft)	DRY (ft)	WET (ft)
6000	12400	115	115	127	4362	5016	4035	4641
	12800	116	116	129	4427	5091	4097	4712
	13200	117	117	131	4500	5175	4154	4777
	13600	118	118	133	4574	5261	4214	4846
	14000	119	119	134	4657	5356	4292	4936
	14400	120	120	136	4728	5437	4358	5012
	14800	121	121	138	4808	5529	4433	5097
	15200	122	122	140	4896	5630	4515	5192
	15600	124	124	142	5012	5764	4624	5318
	16000	126	126	144	5129	5898	4733	5443
	16400	127	127	145	5248	6035	4844	5571
	16800	128	128	147	5366	6171	4955	5698
7000	17200	130	130	149	5488	6311	5068	5828
	12400	115	115	127	4474	5145	4143	4765
	12800	116	116	129	4550	5232	4207	4838
	13200	117	117	131	4625	5319	4265	4904
	13600	118	118	133	4702	5407	4335	4985
	14000	119	119	134	4788	5506	4416	5078
	14400	120	120	136	4861	5590	4484	5157
	14800	121	121	138	4944	5685	4561	5245
	15200	122	122	140	5035	5790	4647	5344
	15600	124	124	142	5156	5929	4760	5474
	16000	126	126	144	5277	6069	4873	5604
	16400	127	127	145	5401	6211	4989	5737
	16800	128	128	147	5524	6352	5103	5869
	17200	130	130	149	5650	6498	5221	6004

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GLIDE PATH ANGLES FROM 4.5° TO 5.4°

To perform a steep approach landing with glide path angles from 4.5° to 5.4°, the landing distance provided by OPERA software, TOLD function or Factored Landing Distance Tables must be corrected with the landing distance increment/penalty according to the calculation type as follows:

- In case of direct calculation of the required landing distance for a given weight, an increment must be **added** to the landing distance output obtained from OPERA software, TOLD function or Factored Landing Distance Tables.
- If the performance is obtained from OPERA software in "Airport Analysis" calculation type, a length penalty must be **subtracted** from the Landing Distance Available.
- Calculate the steep approach landing performance for glide path angle of 5.5°.

The landing distance increment to be added or the length penalty to be subtracted is the same for each landing condition, and it is presented in the table below:

LND DISTANCE INCREMENT/LENGTH PENALTY (FEET)
 ICE PROTECTION ON/OFF

	V _{REF}	V _{REF} OVERSPEED			
		5 KIAS	10 KIAS	20 KIAS	30 KIAS
DRY	1339	1359	1385	1444	1565
WET	1539	1562	1592	1661	1802

NOTE: The performance information obtained from this table is not valid for contaminated runways.

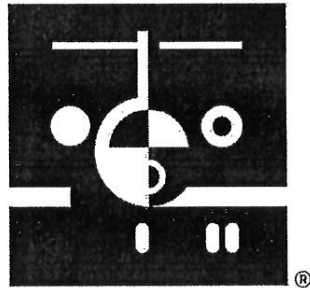


WEIGHT AND BALANCE

The Weight and Balance information presented in the basic AFM remains unchanged.

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**DUNCAN
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FAA APPROVED

AIRPLANE FLIGHT MANUAL SUPPLEMENT

Document No: EDOC-0000015627

Revision A

NO SMOKING SIGNS MODIFICATION

In

EMBRAER EMB-505

This supplement must be attached to the approved Airplane Flight Manual when the No Smoking signs are modified in accordance with Duncan Aviation STC SA01727WI.

The information contained herein supplements or supersedes the basic manual only in those areas listed herein. For Limitations, Procedures, and Performance information not contained in this supplement, consult the basic Airplane Flight Manual.

FAA APPROVED

P. Mike Chick, Jr.
ODA Administrator
ODA-501013-CE
Duncan Aviation, Inc.
Lincoln, NE 68524

STC No. SA01727WI

Interior Alteration

for use on

Embraer EMB-505 s/n 50500189

FAA APPROVED DATE: AUG 28 2013

Page 1 of 4



No Smoking Signs Modification in Embraer EMB-505 Airplanes

LOG OF REVISIONS

REV	PAGES	DESCRIPTION	APPROVED BY	DATE
A	All	Initial Release	N. Grzebielski Prepared by  P. Mike Chick, Jr. ODA STC Administrator	6/27/2013 AUG 28 2013

Note: When this document is revised, it will be revised in its entirety. The latest revision letter or number will be shown in the upper right-hand corner of each page.



No Smoking Signs Modification in Embraer EMB-505 Airplanes

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No Smoking Signs Modification in Embraer EMB-505 Airplanes

1.0 GENERAL

The existing No Smoking signs have been modified to turn off when the SIGNS/OUTLET switch is set to the BELTS/ON or the OFF/ON positions. Otherwise the No Smoking signs will be on when the SIGNS/OUTLET switch is set to the PED-BELTS/OFF position. This will allow the crew to allow smoking during flight.

2.0 LIMITATIONS

No limitations are added due to this modification.

3.0 NORMAL PROCEDURES

Below 10,000 ft MSL, the crew is to set the SIGNS/OUTLET switch (Figure 3-1) located on the pilot's side instrument panel to the PED-BELTS/OFF position. This will turn on the No Smoking ordinance signs. Above 10,000 ft MSL, the crew is allowed to set the SIGNS/OUTLET switch to the BELTS/ON or OFF/ON positions. This will turn off the No Smoking ordinance signs.

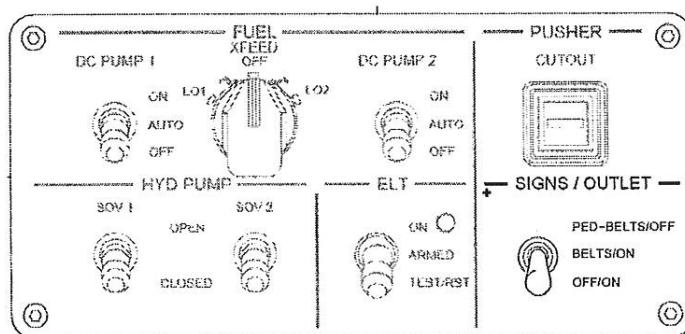


Figure 3-1

4.0 EMERGENCY AND ABNORMAL PROCEDURES

4.1 Circuit Breaker

The following circuit breaker provides 28VDC to the No Smoking signs:

Label	Value	Location
PAX SIGNS	5A	Circuit breaker panel behind pilot seat

Table 4-1

In an emergency or abnormal situation the crew is to set the SIGNS/OUTLET switch (Figure 3-1) to the PED-BELTS/OFF position.

5.0 PERFORMANCE

No change to Basic Airplane Flight Manual.

6.0 WEIGHT AND BALANCE

No change to Basic Airplane Flight Manual.

Installation of AVIATOR 200 Satellite Data System in EMB-505

APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT (AFMS)

Doc.-No.: AV200-NY-11276-08

Rev.00, dated 03.02.2016

This document must mandatory be attached to the airplane flight manual if equipment is installed in accordance with minor design change NAYAK project number AV200-NY-11276-08. This supplement must mandatory be carried on the airplane specified below at all times.

AIRCRAFT MANUFACTURER: EMBRAER

AIRCRAFT MODEL/TYPE: EMB-505

AIRCRAFT SERIAL NO.: 50500203

The information herein supplements or supersedes the information made available to the operator by the aircraft manufacturer in the form of clearly stated placards or markings, or in the form of an approved airplane flight manual, only in those areas listed herein. For limitations, procedures and performance information not contained in this document, consult the basic placards or markings, or the basic approved airplane flight manual.

Revision No. 00 to AFM supplement ref. AV200-NY-11276-08 is approved under the authority of DOA ref. EASA.21J.077.

AFM SUPPLEMENT LOG OF REVISIONS

Rev.	Revision Date	Description	Approved under authority of
00	03.02.201	Initial Release	DOA ref. EASA.21J.077

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GENERAL

This Supplement is provided to present the data required for operation of the AVIATOR 200 INMARSAT satellite datalink system. It is a compact, light-weight aeronautical SATCOM system that uses Inmarsat's Swift Broadband services to provide internet connectivity to passengers.

The system consists of the following units:

- TT-5040A SBU (Swift Broadband unit including configuration module)
- TT-5016A HLD (High Power Amplifier, Low Noise Amplifier and Diplexer in one unit).
- TT-3002A Low Gain Antenna (LGA).
- TT-5019A Iridium Band Reject Filter (to minimize interference with adjacent Iridium frequency bands).
- Inmarsat Reject Filter (installed on the existing GARMIN Iridium system to minimize interference with adjacent Inmarsat frequency bands)

The information herein must replace or complement the equivalent data in the basic AFM.

LIMITATIONS

No change to basic AFM.

EMERGENCY PROCEDURES

Emergency Procedures

No change to airplane flight manual.

Abnormal Procedures

The AVIATOR 200 system is considered non-essential. If it becomes necessary to reduce electrical load (for example, during loss of generators or alternators), power to the system may be removed manually by pulling the following circuit breaker.

“SBU” circuit breaker.....PULL

NOTE: This guidance is supplementary to any guidance provided in the POH or AFM for load shedding.

NORMAL PROCEDURES

To supply electrical power the Aviator 200 System:

“IFE/ Refreshment Center” Switch.....ON

To remove electrical power from the Aviator 200 System::

“IFE/ Refreshment Center” Switch.....OFF

NOTE: This guidance is supplementary to any guidance provided in the POH or AFM for equipment related to the IFE switch. Switching also affects all other equipment connected to the IFE switch.

PERFORMANCE

The Performance Data presented in the basic AFM remains unchanged.

WEIGHT AND BALANCE

The Weight and Balance presented in the basic AFM remains unchanged.

AIRCRAFT FLIGHT MANUAL SUPPLEMENT

FOR

EMB-505

Aircraft Manufacturer: Embraer

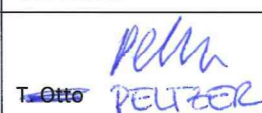
Aircraft Model(s): EMB-505

Aircraft Serial Number: 50500203

Aircraft Registration Number: CS-PHA

Type Certificate Number: EASA.IM.A 158 Issue 9, 14th May 2020

Iss.	Reason for Issue	Affected pp.	Date of issue
01	Initial Issue	all	10.02.2025

Issued (Design Engineer)	 M. Brunner	Checked (Compliance Verification Engineer)	 A. Berthold
Date	10.02.2025	Date	11.02.2025
Released (Head of Office of Airworthiness)	 T. Otto	Approved (Head of Office of Airworthiness)	 T. Otto
Date	13.12.2025 13. FEB. 2025	Date	13.12.2025 20.03.25

The technical content of this document is approved under the authority of the DOA ref. EASA. 21J.405

For Continued Airworthiness Operation please look at www.ftigroup.net

This document must be carried in the aircraft at all times. It describes the operating procedures for GoGo System Implementation when it has been installed in accordance with Installation Manual Supplement, Doc. N° F23-001-EO24-0047. The information contained herein supplements or supersedes the basic Airplane Flight Manual only in those areas listed herein. For limitations, procedures, and performance information not contained in this document, consult the basic Airplane Flight Manual.

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Abbreviations

Abbrev.	Description
A/C	Aircraft
AUX	Auxiliary
CB	Circuit Breaker
FWD	Forward
GOGO	Gogo Galileo broadband internet system for business jets based on Low Earth Orbit satellites
HDX	Gogo antenna assembly
IAW	In Accordance With
IFE	In-flight Entertainment
LGA	Low Gain Antenna
LHS	Left Hand Side
RHS	Right Hand Side
SCS	Smart Cabin System
STC	Supplemental Type Certificate

Section 01 – General

The EMB-505 business jet project is equipped with a GOGO HDX antenna system. An LGA antenna and cables (if installed) was removed, and the Iridium Satcom antenna (if installed) was moved.

The HDX antenna is mounted to the upper pressurized section of the central fuselage.

Section 02 – Limitations

Before using De-Ice the IFE system must be turned off by the pilot through the IFE Switch on the LHS Lateral Console in the cockpit.

Section 03 – Emergency Procedures

No Change

Section 04 – Abnormal Procedures

In case of any malfunction or abnormal behavior of any IFE equipment turn off the IFE system through the IFE Switch on the LHS Lateral Console in the cockpit.

Section 05 – Normal Procedures

No Change

Section 06 – Performance

There are no changes in flight performance and flight characteristics. Nevertheless, it is recommended to include a reserve of approx. 2% when calculating the range.

Section 07 – Weight and Balance

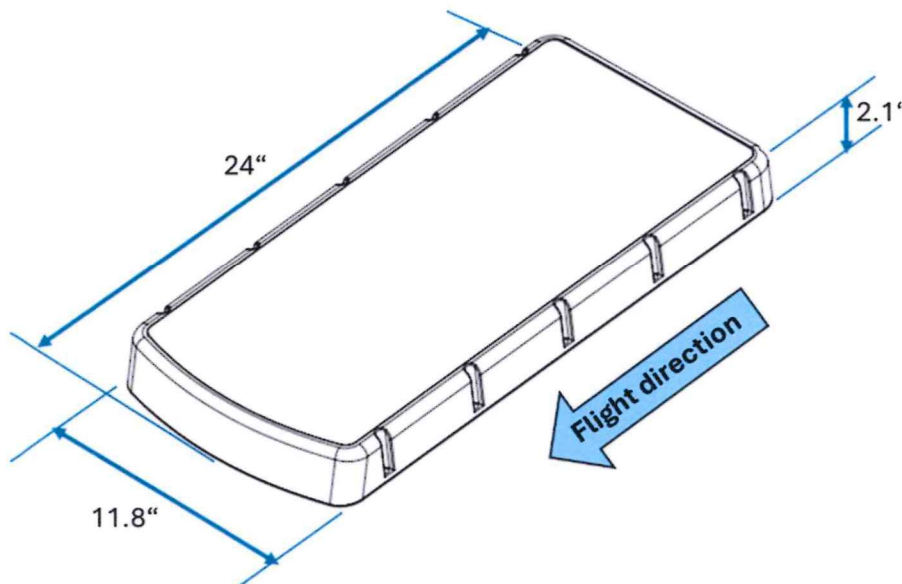
No Change

For current A/C weight and balance data see appropriate section in basic AFM.

Section 08 – Airplane & Systems Descriptions

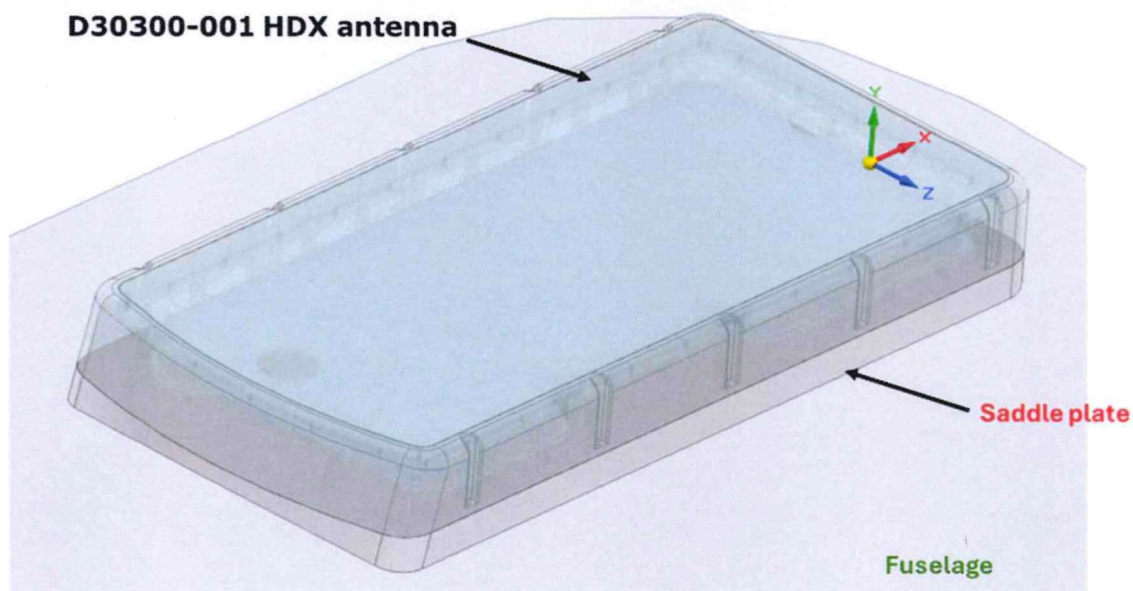
HDX Antenna

The HDX antenna is a rectangular item of length 24 inch, width 11.8 inch and height 2.1 inch, with a curved leading edge, and mass 9.8kg (see following figure).



D30300-001 HDX antenna

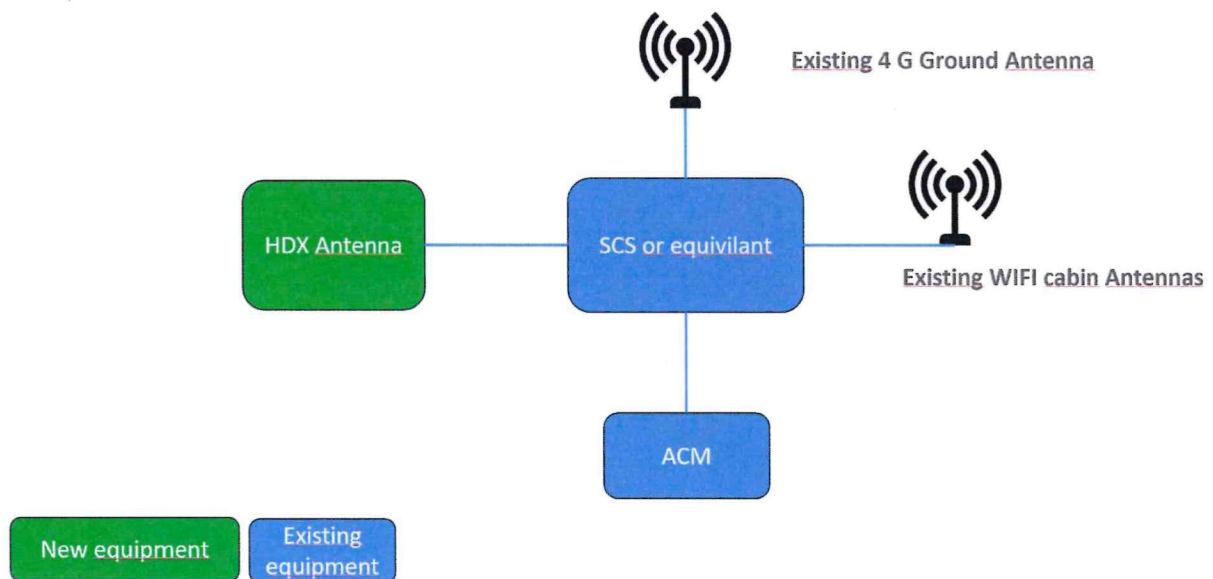
The antenna is installed on a saddle plate that provides the attachment to the fuselage (see following figure).



Installation of HDX showing saddle plate

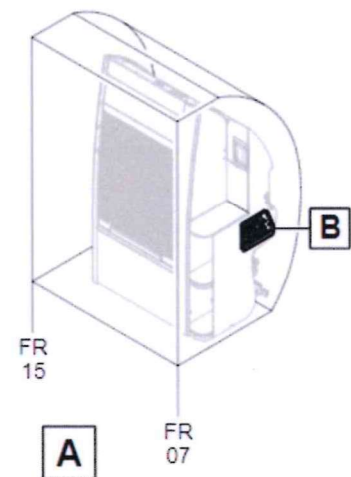
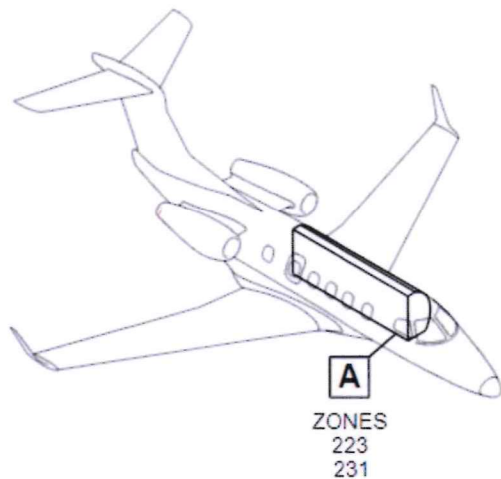
System overview

The following figure is showing the general system configuration. Pre-mod existing equipment is shown in blue and the to be installed hardware in green.



Power supply

The HDX antenna is supplied via the LHS AUX CB panel through the dedicated CB "HDX".



Appendix

N/A

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FAA APPROVED

Airplane Flight Manual Supplement

Document No: EDOC-0000017304
Revision A

Universal Cabin System (UCS) 5000 with Wireless Local Area Network (WLAN)
STC No. SA01760WI
Aircell UCS 5000 with WLAN
for use on
Embraer S.A. EMB-505 s/n 50500203

This supplement must be attached to the approved Airplane Flight Manual when an Aircell High Speed Internet with Wireless Local Area Network System is installed in accordance with Duncan Aviation STC ST01760WI

The information contained herein supplements or supersedes the basic manual only in those areas listed herein. For Limitations, Procedures, and Performance information not contained in this supplement, consult the basic Airplane Flight Manual.

FAA APPROVED

P. Mike. Chick, Jr.
ODA STC administrator
Duncan Aviation
3701 Aviation Road
Lincoln, Nebraska 68524

FAA APPROVED DATE: AUG 1 2014



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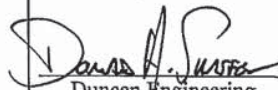
Airplane Flight Manual Supplement

Document No.: EDOC-0000017304

Revision: A

Universal Cabin System 5000 with Wireless Local Area Network

LOG OF REVISIONS

REV	PAGES	DESCRIPTION	APPROVED BY	DATE
A	All	Initial Release	J.Hall Prepared by  Duncan Engineering	July 17, 2014

Note: When this document is revised, it will be revised in its entirety. The latest revision letter is shown in the upper right-hand corner of each page.

FAA APPROVED DATE: AUG 1 2014

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DUNCAN
AVIATION

Airplane Flight Manual Supplement
Document No.: EDOC-0000017304
Revision: A

Universal Cabin System 5000 with Wireless Local Area Network

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Universal Cabin System 5000 with Wireless Local Area Network

1.0 GENERAL

The Universal Cabin System (UCS) 5000 with Wireless Local Area Network (WLAN) is for use with Transmitting Portable Electronic Devices (TPEDs). The UCS provides bi-directional wireless communication, characterized by IEEE 802.11b/g/n (commonly referred to as Wi-Fi). The UCS can be used with an existing Aircell Gogo Biz, which provides ground based broadband internet access, but the UCS does not require a Gogo Biz installation for it to function.

The UCS can provide:

- Voice over Wi-Fi enabled Aircell devices via the Gogo Biz Voice networks.
- Audio and video streaming to TPEDs from the internal solid state hard drive (SDD), if installed, via Wi-Fi.

2.0 LIMITATIONS

No change to basic Airplane Flight Manual.

3.0 NORMAL PROCEDURES

- The WLAN is intended to provide wireless bi-directional communication with TPED.
- Wi-Fi connectivity remains active at all altitudes including ground operation.
- Operational approval for the use of TPED in the passenger cabin and on the flight deck is the responsibility of the airplane operator in accordance with 14 CFR §91.21 and §135.144. Refer to FAA InFO 13010 "Expanding Use of Passenger Portable Electronic Devices (PED)" dated Oct-31-2013 for general guidance and for additional guidance for the operation of cellular telephones and cabin briefing.

4.0 EMERGENCY AND ABNORMAL PROCEDURES

In the case that interference to essential or critical airplane systems is detected or suspected, the transmitting capability of the WLAN must be terminated using the existing IFE ON/OFF switch located on the pilot's instrument panel.

The UCS-5000 with Wireless Local Area Network (WLAN) System is protected by the following circuit breakers:

UNIT	LABEL	RATING	LOCATION
Aircell UCS w/WLAN	UCS	7.5A	Behind pilot's seat

5.0 PERFORMANCE

No change to basic Airplane Flight Manual.

6.0 WEIGHT AND BALANCE

No change to basic Airplane Flight Manual.



APPENDICES

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APPENDIX 1

CONFIGURATION DEVIATION LIST

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ATA 23 Communications	AP1-23	01
ATA 33 Lights	AP1-33	01
ATA 53 Fuselage	AP1-53	01
ATA 57 Wings	AP1-57	01

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INTRODUCTION

This Configuration Deviation List contains additional certificate limitations for operation of the airplanes without certain secondary airframe and engine parts as listed herein. When the airplane is operated using the CDL, the limitations specified in the AFM must still be complied with, as amended in this Appendix. All the items that are related to the airworthiness of the airplane and not included on the list are automatically required to be installed.

The associated limitations must be listed on a placard affixed in the cockpit in clear view of the pilots and other appropriate crewmember(s). The pilot in command should be notified of each operation with a missing part(s) by listing the missing part(s) in the flight or dispatch release. The operator should list in the airplane logbook an appropriate notation covering the missing part(s) on each flight.

If an additional part is lost, the airplane may not depart the airport at which it landed following this event, until it complies with the limitation of the CDL. This, of course, does not preclude the issuance of a ferry permit to allow the airplane to be flown to a point where the necessary repairs or replacement can be made.

No more than one part for any one system may be missing, unless specific combinations of parts are included in the CDL. Unless otherwise specified, parts from different systems may be missing. The performance penalties are cumulative, unless specifically designated penalties are indicated for the combination of missing parts.

Where performance penalties are listed as negligible, no more than three negligible items may be missing without taking further penalty. For each missing item more than three, reduce the takeoff, landing and enroute climb limits by 41 kg (90 lb). Where performance penalties are listed as no penalty, any accumulative number of items listed as no penalty may be missing without further penalty.



Takeoff performance penalties should be applied to the takeoff weights that are limited by performance considerations (i.e., takeoff field length, first, second, or final segment climb, or takeoff flight path).

If the performance-limited takeoff weight is greater than the maximum certified takeoff weight, the takeoff performance penalties should be applied to the maximum certified takeoff weight to ensure compliance with the noise requirements.

Landing performance penalties should be applied to the landing weights that are limited by performance considerations (i.e., landing field length, landing climb, or approach climb). If the performance-limited landing weight is greater than the maximum certified landing weight, the landing performance penalties should be applied to the maximum certified landing weight to ensure compliance with the noise requirements.

En route performance penalties apply only to operations that are limited by the one-engine inoperative en route climb performance.

To correctly apply the speed tape, perform Aircraft Maintenance Manual TASK 51-26-00-910-801-A. The use of speed tape is time limited to 10 days.

If the CDL item refers to a MMEL item, the MMEL repair interval must be applied.

COMPONENT LOCATION

The numbering and designation of each system in this Appendix is based on ATA Spec. 2200. The parts within each system are identified by its functional description and, when necessary, by door or panel identification. See Aircraft Maintenance Manual, Chapter 06, for panel identification.



CONFIGURATION DEVIATION LIST					
System & Sequence Number	ITEM	1.	2. Number installed	3. Number required for dispatch	4. Remarks and/or exceptions
23 COMMUNICATIONS					
61-00	Static Dischargers (For airplanes with 2 static dischargers on each aileron)	14	8		The airplane is allowed to be dispatched with 1 (one) static discharger per control surface and winglet, provided the total amount of installed static dischargers is equal to or greater than 8 (eight).
	(For airplanes with 3 static dischargers on each aileron)	16	8		The airplane is allowed to be dispatched with 1 (one) static discharger per control surface and winglet, provided the total amount of installed static dischargers is equal to or greater than 8 (eight).

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CONFIGURATION DEVIATION LIST				
System & Sequence Number		1.	2. Number installed	
ITEM			3. Number required for dispatch	
			4. Remarks and/or exceptions	
33 LIGHTS				
44-01	Wing Inspection Light Lens	1	0	May be missing with no penalty provided: a) Cavity is covered with speed tape, and b) Affected light is deactivated and considered inoperative. Refer to MMEL 33-44-01.
45-01	Red Beacon Light Assembly	1	0	May be missing with no penalty provided: a) Cavity is covered with speed tape, and b) Affected light is deactivated and considered inoperative. Refer to MMEL 33-45-01.
48-02	Aft Navigation Light Assembly	2	0	One or both may be missing with negligible penalty provided: a) Cavity is covered with speed tape, and b) Affected light is deactivated and considered inoperative. Refer to MMEL 33-48-00.
49-01	Upper Anti-Collision (Strobe) Light Assembly	1	0	May be missing with no penalty provided: a) Cavity is covered with speed tape, and b) Affected light is deactivated and considered inoperative. Refer to MMEL 33-49-00.

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CONFIGURATION DEVIATION LIST				
System & Sequence Number		1.	2. Number installed	
ITEM			3. Number required for dispatch	
			4. Remarks and/or exceptions	
53 FUSELAGE				
03-00	Forward Fuselage Jack Point Cover (122 AR)	1	0	May be missing with no penalty provided cavity is covered with speed tape.

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CONFIGURATION DEVIATION LIST			
System & Sequence Number	ITEM	1. 2. Number installed	3. Number required for dispatch 4. Remarks and/or exceptions
57 WINGS			
03-00	Wing Jack Point Covers (541 VB/641 VB)	2	0 One or both may be missing with no penalty provided cavity is covered with speed tape.

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